Lab Introduction to AWS Identity Access Management

Task 1: Sign in to the AWS Management Console

1. Click on the Open Console button, and you will get redirected to AWS Console in a new browser tab.
2. On the AWS sign-in page,
   - Leave the Account ID as default. Never edit/remove the 12-digit Account ID present in the AWS Console. Otherwise, you cannot proceed with the lab.
   - Now copy your Username and Password in the Lab Console to the IAM Username and Password in the AWS Console and click on the Sign-in button.
3. Once Signed In to the AWS Management Console, make the default AWS Region as US East (N. Virginia) us-east-1.

Task 2: Create IAM Users

In this task, we are going to create new IAM users by providing the name, password access, permissions, and tags. These users will be added to their respective groups in the next task.

1. Click on Services and select IAM under the Security, Identity, & Compliance section.
2. In the IAM dashboard, select the Users option in the left panel and click on the Create User button to create a new IAM user.
3. In the Add User page, fill in the User Details section as follows:
   - User name: Enter John (or the desired name for the user)
   - Check the Provide user access to the AWS Management Console - optional checkbox
   - Select Custom password under Console Password and Enter whizlabs@123 (or the desired password for the user)
   - Uncheck the Users must create a new password at the next sign-in (recommended) checkbox.

- Click on the Next button.

- In the Set permissions section, keep things as default. Click on the Next button.
- Scroll down and Under Tags, Click on the Add new tag button:
  - Key: Enter *Dev-Team*

- Value: Enter *Developers*

- Click on the Create User button

- Note: Ignore the above error if it appears while creating Users and click on Close.
- Click on the Return to users list button and then on the Continue button.
- Repeat the same steps and tags for the IAM user by the name Sarah.
- Repeat the steps to create IAM users by the name *Ted and **Rita*** with the following details,
  - Custom password: *whizlabs@123*
  - Key: *HR-Team*
  - Value: *HR*

- We have created 4 IAM users.

Task 3: Create IAM Groups and add IAM Users

In this task, we are going to create new IAM groups and will add the users to their respective groups. Moreover, we will be adding permissions to the group so that users within the group have access to the services allocated to them using the permission policies.

1. Select the User groups in the left panel and click on the Create group
2. Set Group Name:
    - User group name: Enter *Dev-Team*
    - Scroll down and select *John* and *Sarah* under Add Users to the group.

    - Scroll down to the Attach permissions Policies section and search for AmazonEC2ReadOnlyAccess and AmazonS3ReadOnlyAccess policies. These policies provide read access for EC2 and S3 to the added users in the group.
    - Note: Do not add other policies than the ones mentioned above. You will get an error while creating a group
    - Review all details and click on the Create group button.
    - Repeat the same steps to create an HR-Team group.
        - Click on the Create group
        - User group name: *HR-Team*
        - Scroll down and select *Ted* and *Rita* under Add Users to the group.
        - Under Attach permissions Policies, select the Billing policy.
        - Note: Do not add other policies than the ones mentioned above. You will get an error while creating a group
        - Review all details and click on the Create group button.

Lab **Build Amazon VPC with Public and Private Subnets**

Task 1: Sign in to AWS Management Console

1. Click on the Open Console button, and you will get redirected to AWS Console in a new browser tab.
2. On the AWS sign-in page,
   - Leave the Account ID as default. Never edit/remove the 12 digit Account ID present in the AWS Console. otherwise, you cannot proceed with the lab.
   - Now copy your User Name and Password in the Lab Console to the IAM Username and Password in AWS Console and click on the Sign in button.
3. Once Signed In to the AWS Management Console, Make the default AWS Region as US East (N. Virginia) us-east-1.

Task 2 : Creating New VPC

1. Make sure you are in the US East (N. Virginia) us-east-1 Region.
2. Navigate to VPC by clicking on Services on the top of AWS Console.
3. Click on VPC (under Networking & Content Delivery section) or you can also search for VPC.
4. Click on Your VPCs from the left menu.
5. Here you can see the list of all VPC, No need to do anything with the existing and default VPCs, we will create a new VPC for this lab.
6. Click on Create VPC button.
   - Select VPC Only
   - Name tag: Enter a VPC name for identification to your VPC. Ex: *MyVPC*
   - IPv4 CIDR block: Enter *10.0.0.0/16*
   - IPv6 CIDR block: No need to change this, make sure No IPv6 CIDR Block is checked.
   - Tenancy: No need to change this, make sure Default is selected.
   - Now click on Create VPC button.

7. Once VPC is created, it will appear with details as shown below:

Task 3 : Creating Subnets

In this lab, we will create one public subnet and a private subnet in us-east-1a and us-east-1b Availability Zones respectively as follows:

1. For the Public Subnet, click on Subnets from the left menu and click on Create subnet button.
   - VPC ID: Select MyVPC from the list you created earlier.
   - Subnet Name: Enter Name *MyPublicSubnet*
   - Availability Zone: Select us-east-1a
   - IPv4 CIDR block: Enter the range *10.0.1.0/24*
   - Click on Create subnet button.

2. For the Private Subnet, click on Create subnet again.
   - VPC ID : Select MyVPC from the list you created earlier.

- Subnet Name : Enter Name *MyPrivateSubnet*
- Availability Zone : Select us-east-1b
- IPv4 CIDR block : Enter the range *10.0.2.0/24*
- Click on Create subnet button.

Task 4: Create and configure Internet Gateway

In this task, we are going to create an internet gateway and configure it with the VPC.

1. Click on Internet Gateways from the left menu and click on Create internet gateway button.
    - Name Tag: Enter *MyInternetGateway*
    - Click on Create internet gateway button.

2. Select the Internet gateway you created from the list
    - Click on Actions.
    - Click on Attach to VPC.

    - Select MyVPC which you created from the list and click on Attach internet gateway button.

Task 5: Create Route Tables

In this task, we are going to create two route tables and associate them with their respective subnets.

1. Go to Route Tables from the left menu and click on Create route table button.
    - Name: Enter *PublicRouteTable*
    - VPC: Select MyVPC from the list.
    - Click on Create route table button.

2. Repeat the same steps to create a route table for the Private subnet.
    - Name: Enter *PrivateRouteTable*
    - VPC: Select MyVPC from the list.
    - Click on Create route table button.

3. Now we will associate the subnets to the route tables.
4. Select the PublicRouteTable and go to the Subnet Associations tab.
    - Click on Edit subnet associations.
      Select MyPublicSubnet from the list.
    - Click on Save associations button.

5. Select the PrivateRouteTable and go to the Subnet Associations tab.

- Click on Edit subnet associations.

- Select MyPrivateSubnet from the list.
- Click on Save associations button.

6. Make sure not to associate any subnets with the Main Route Table.
7. PublicRouteTable: Add a route to allow Internet traffic to the VPC.
    - Select PublicRouteTable.
    - Go to Routes tab, click on Edit routes.

    - Click on Add route button, Specify the following values:
        - Destination: Enter *0.0.0.0/0*
        - Target: Select Internet Gateway from the dropdown menu to select MyInternetGateway.
        - Click on Save changes button.

Lab Steps

Introduction to Amazon Elastic Compute Cloud (EC2)

Task 1: Sign in to AWS Management Console

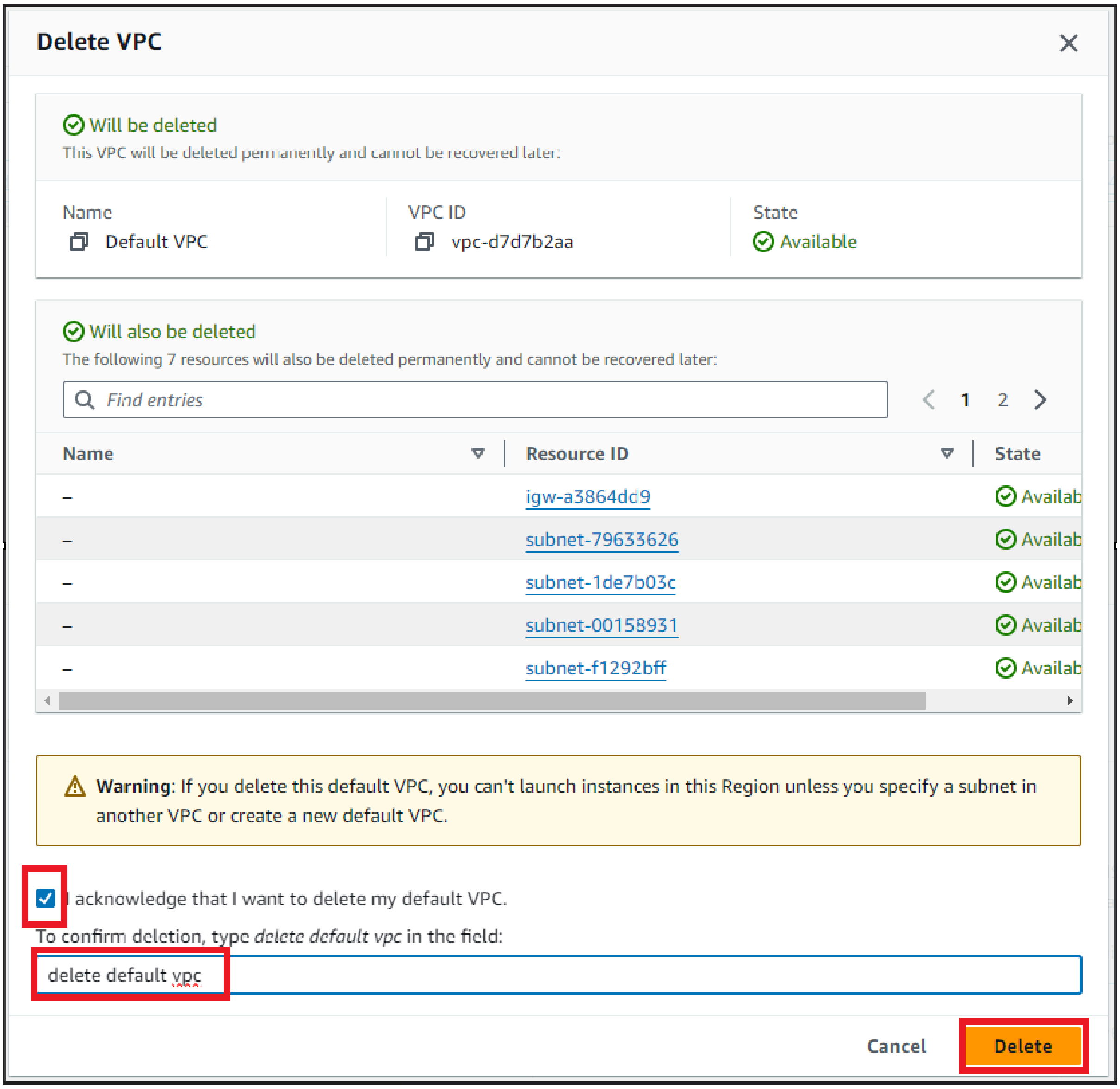
1. Click on the Open Console button, and you will get redirected to AWS Console in a new browser tab.
2. On the AWS sign-in page,
    - Leave the Account ID as default. Never edit/remove the 12 digit Account ID present in the AWS Console. otherwise, you cannot proceed with the lab.
    - Now copy your User Name and Password in the Lab Console to the IAM Username and Password in AWS Console and click on the Sign in button.
3. Once Signed In to the AWS Management Console, Make the default AWS Region as US East (N. Virginia) us-east-1.

Task 2: Provision Default VPC

1. Navigate to VPC by clicking on the Services menu in the top, then click on VPC or Open the Amazon VPC console via https://console.aws.amazon.com/vpc/.
2. Delete the default VPC by following the below steps:
    - In the navigation pane, choose Your VPCs.
    - Select the VPC with value as yes in default VPC column.

- Go to actions button and click on delete VPC button.

- Check I acknowledge that I want to delete my default VPC option.
- Type default delete VPC and click on Delete button.



3. Now to provision Default VPC again, Refresh your console go to actions and click Create default VPC

4. Click Create default VPC button and your default VPC will get created

Task 3 : Launch an EC2 Instance with desired specifications

1. Ensure you are in the US East (N. Virginia) us-east-1 Region to begin launching an EC2 instance in the Amazon cloud.

2. Navigate to EC2 by clicking on the Services menu in the top, then click on EC2 in the Compute section.
3. Click on the Instances option on the left panel, and then click on the Launch Instances button.
4. Name : Enter MyEC2Server

5. Select Amazon Linux 2023 AMI from the dropdown.

   Note: if there are two AMI's present for **Amazon Linux 2o23 kernel-6.1 AMI**.
6. An instance type in AWS refers to a virtual server configuration that determines the computing resources, such as CPU, memory, and storage, available to an instance. It is the basic building block for creating an EC2 instance in the AWS cloud.
   - For Instance Type: Select t2.micro

- t2.micro is an instance type in AWS that comes with 1 vCPU, and 1GB memory and is suitable for low-traffic web servers, small development environments, and other lightweight applications.

7. AWS key pair is a secure pair of keys used for login and access to EC2 instances. It includes a public key placed on the instance and a private key kept on the user's local computer, used for authentication to prevent unauthorized access.

- For Key pair(login): Select Create a new key pair Button
  - Key pair name: WhizKey
  - Key pair type: RSA
  - Private key file format: .pem

8. In Network Settings, Click on Edit Button:

- Auto-assign public IP: Enable
- Select Create security group
- Security group name: Enter MyEC2Server_SG
- Description: Enter Security Group to allow traffic to EC2.

9. We will now add the security group rules. SSH will already be present there.

- For HTTP, Select Add security group rule Button
  - Choose Type: Select HTTP
  - Source: Select Anywhere

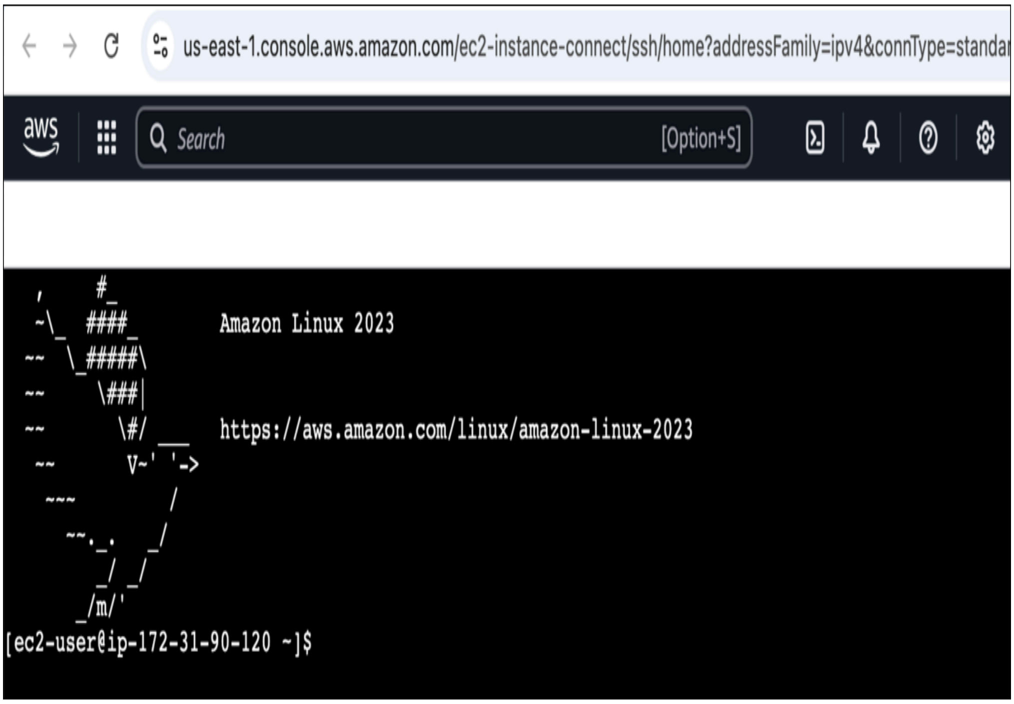
10. A security group is a virtual firewall that controls the inbound and outbound traffic for instances in a particular network in a cloud computing environment. Here we have selected SSH and HTTP rules that will allow incoming SSH and HTTP traffic to instances that are associated with the security group.
11. Proceed with launching the instance while leaving all other settings as default. Simply click on the Launch Instance without modifying any other configuration.
12. To view the instance that you have created, choose the View all Instances option.
13. **Launch Status:** Once you have initiated the instance launch process, Go to the Instances page from the left menu and wait for your EC2 instance to become "Running" while ensuring the health check status is 2/2 checks passed for optimal performance.

14. Select the instance that you have created and copy the public IPv4 address within the details section and paste it into the editor for later use. An example of this process is depicted in the screenshot provided.

Task 4 : SSH into EC2 Instance using the key pair

1. Select your EC2 instance (MyEC2Server) and click on the Connect button.

2. Select EC2 Instance Connect option and click on Connect button. Keep everything else as default.
3. A new tab will open in the browser where you can execute the Linux Commands.
4. Please follow the steps in SSH into EC2 Instance for more options to SSH.



Task 5: Install an Apache Server on the instance

In this task, our goal is to configure an Amazon EC2 instance to run an Apache Web Server and verify its functionality by accessing the web server via a web browser using the instance's public IPv4 address.

1. Switch to root user:

```
sudo su
```

2. Now run the updates using the following command:

```
dnf update -y
```
3. Once completed, lets install and run an apache server

4. Install the Apache web server:

```
dnf install httpd -y
```
5. Start the web server:

```
systemctl start httpd
```
6. Now Enable httpd:

```
systemctl enable httpd
```
7. Check the webserver status

```
systemctl status httpd
```
8. You can see Active status is running.

9. You can test that your web server is properly installed and started by entering the public IPv4 address of your EC2 instance in the address bar of a web browser. If your web server is running, then you see the Apache test page. If you don't see the Apache test page, then verify whether you followed the above steps properly and check your inbound rules for the security group that you created.

Task 6 : Create a web page and publish it on the instance

In this task, you will add content to the index.html file using the "echo" command and restart the webserver. Then, you can view the content by entering the public IPv4 address followed by "/index.html" in a web browser, ensuring that the URL protocol is HTTP.

1. To add the contents into index.html file using echo, copy and paste the below command to shell.

```
echo "<html>Hi Whizlabs, I am a public page</html>" >
/var/www/html/index.html
```

2. Restart the webserver by using the following command:

```
systemctl restart httpd
```
3. Now enter the file name, */index.html* after the public IPv4 Address which you got when you created the ec2 instance in the browser, and you can see your HTML content.

- Make sure URL Protocol is http not https.
- Syntax: http://<Your_Public_IPv4_Address>/index.html
- Sample URL: http://52.87.50.168/*index.html*
- Note: If the index.html page is not loading, try removing s from the link, it should be HTTP.

4. If you can see the above text in the browser, then you have successfully completed the lab.

Lab Creating NAT Gateways in AWS

Task 1: Sign in to AWS Management Console

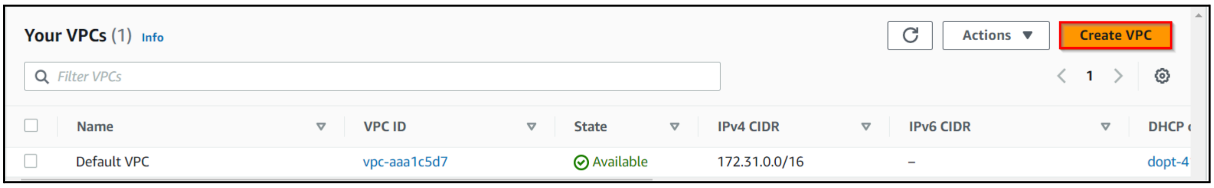
1. Click on the Open Console button, and you will get redirected to AWS Console in a new browser tab.
2. On the AWS sign-in page,
   - Leave the Account ID as default. Never edit/remove the 12 digit Account ID present in the AWS Console. otherwise, you cannot proceed with the lab.
   - Now copy your User Name and Password in the Lab Console to the IAM Username and Password in AWS Console and click on the Sign in button.
3. Once Signed In to the AWS Management Console, Make the default AWS Region as US East (N. Virginia) us-east-1.

Note : If you face any issues, please go through FAQs and Troubleshooting for Labs.

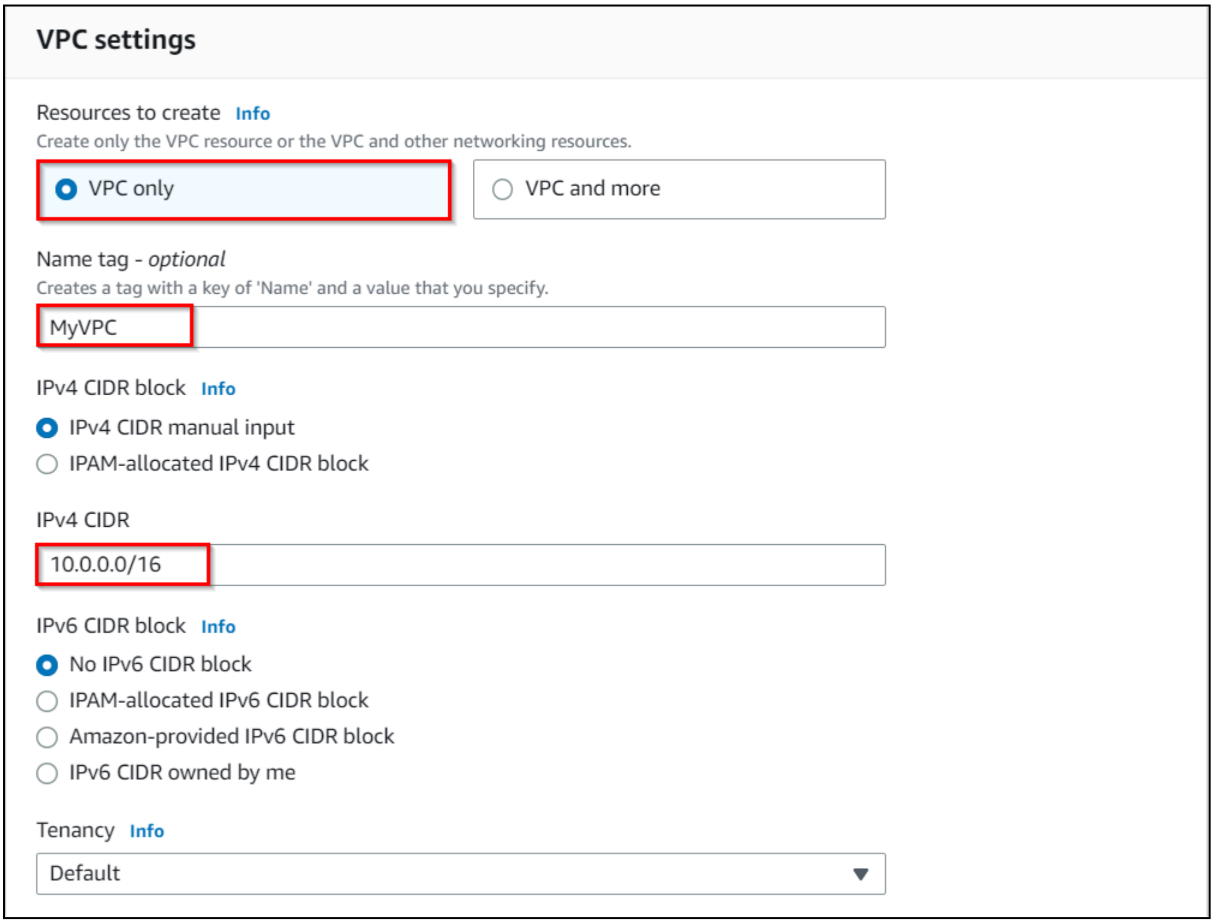
Task 2 : Create a VPC

In this task, we are going to create a Virtual Private Cloud (VPC), which provides a logically isolated section of the AWS cloud where users can launch resources such as instances, subnets, and gateways.

1. Make sure you are in the US East (N. Virginia) us-east-1 region.
2. Navigate to VPC under the services menu. Click on Your VPCs.
3. Click on Create VPC button.



- Resources to create : Select VPC Only
- Name Tag : Enter MyVPC
- IPv4 CIDR block : Enter 10.0.0.0/16 (You can also put any other CIDR range)
- IPv6 CIDR block : Select No IPv6 CIDR Block
- Tenancy : Default



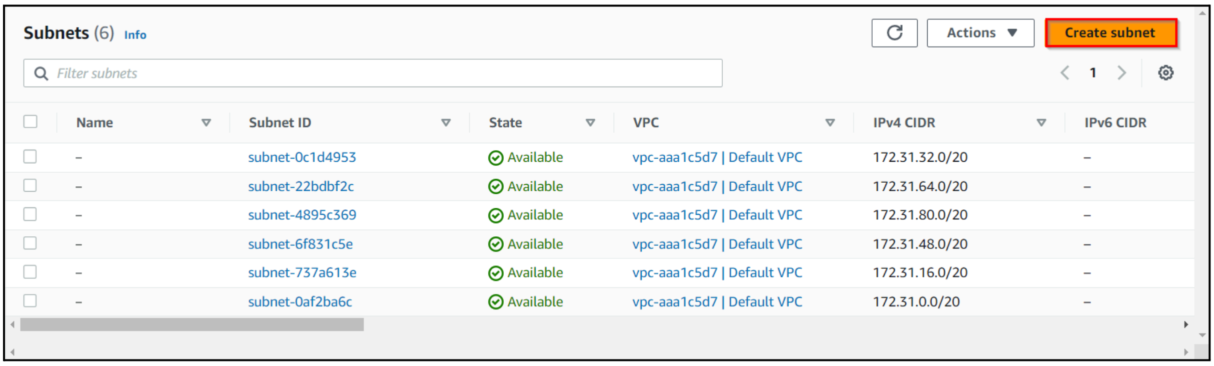
- Click on Create VPC.

4. The VPC is now created.

Task 3 : Create Public and Private Subnets

In this task, we are going to create both public and private subnets within the VPC. Public subnets have internet connectivity, while private subnets do not have direct internet access.

1. Navigate to Subnets in the left panel of the VPC page.
2. Let's create a Public subnet. Click on Create Subnet button.



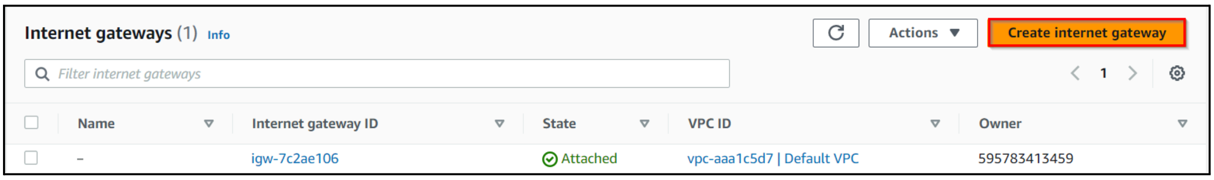
- VPC ID : select MyVPC
- Subnet Name : Enter MyPublicSubnet
- Availability Zone : No Preference
- IPv4 CIDR block : Enter 10.0.0.0/24
- Click on Create subnet button.

3. Let's enable Auto Assign public IP to Instances created within this subnet,
    - Select MyPublicSubnet , Click on Actions.
    - Click on Edit subnet settings.
    - Enable auto-assign public IPv4 address : Check
    - Click on Save.
4. Now, the Instances launched inside the MyPublicSubnet will have Public IPs assigned to them by default.
5. Let's create a private subnet. Click on Create subnet.
    - VPC ID : select MyVPC
    - Subnet Name : Enter MyPrivateSubnet
    - Availability Zone : No Preference
    - IPv4 CIDR block : Enter 10.0.1.0/24
    - Click on Create subnet button.
6. Now, two subnets are created.

Task 4 : Create Internet Gateway

In this task, we are going to create an Internet Gateway, which acts as a bridge between the VPC and the internet, allowing instances in the VPC to communicate with the internet.

1. Navigate Internet Gateways in the left panel of the VPC page.
2. Click on Create Internet gateway button.



Name	Internet gateway ID	State	VPC ID	Owner
–	igw-7c2ae106	Attached	vpc-aaa1c5d7 | Default VPC	595783413459

- Name tag : Enter MyIGW
- Click on Create internet gateway button.

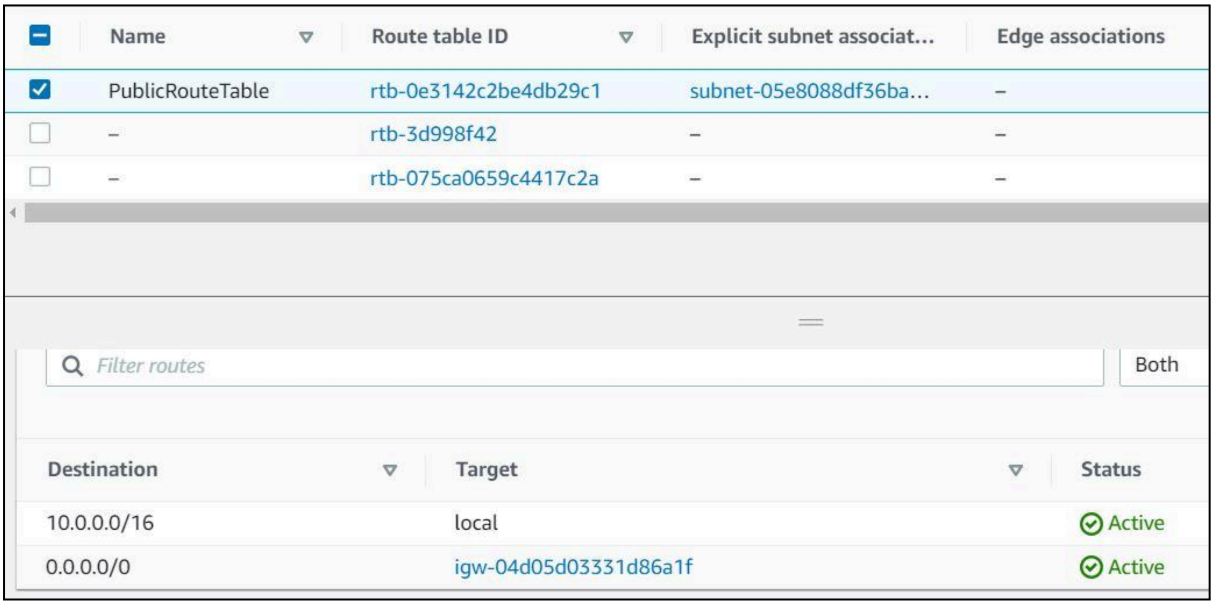
3. An Internet Gateway is now created.
4. To attach an Internet Gateway to a VPC,
    - Click on Actions, Select Attach to VPC.
    - VPC : Select MyVPC
    - Click on Attach internet gateway.
5. Now MyIGW is attached to MyVPC.

Name	Internet gateway ID	State	VPC ID
MyIGW	igw-04d05d03331d86a1f	Attached	vpc-070606d8d0c3b038a | MyVPC

Task 5 : Create Public Route Table and Configure

In this task, we are going to create a public route table and configure it to associate with the public subnet. The public route table defines how traffic is routed between the VPC and the internet.

1. Navigate to Route Table in the left panel of the VPC page.
2. Click on Create route table button
    - Name tag : Enter PublicRouteTable
    - VPC : Select MyVPC
    - Click on Create route table button.
3. A route table by name PublicRouteTable will be created.
4. To attach an Internet Gateway, select PublicRouteTable.
5. In the Routes tab below:
    - Click on Edit routes.
    - On the next page, click on Add route
    - Destination : Enter 0.0.0.0/0
    - Target : Select Internet Gateway, and once the internet gateways have been created, select MyIGW
    - Click on Save changes.
6. To associate the Public Subnet to the route table, Select PublicRouteTable.
    - Click on the Subnet Associations tab.
    - Click on Edit subnet associations.
    - On the next page, select MyPublicSubnet from the list displayed.
    - Click on Save associations.
7. Once all the configurations are completed, it should look like below:

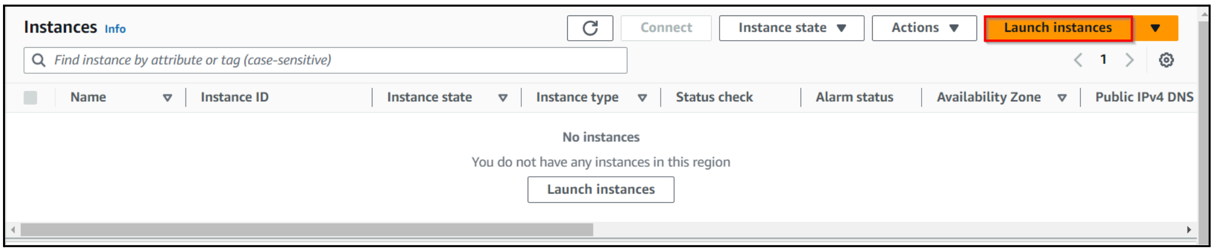


8. Now the Instances launched within MyPublicSubnet will have access to the Internet.
9. As you can see, there is another existing route table already available for MyVPC. It is a main route table created at the time the VPC was created. We will use it while creating the NAT Gateway.

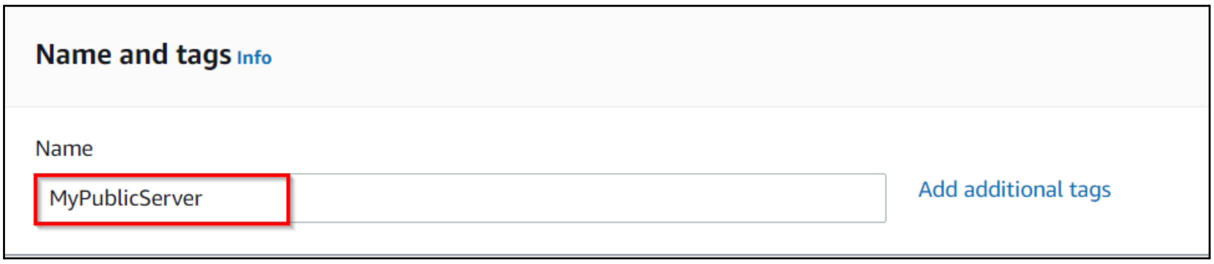
Task 6 : Launch an EC2 Instance in Public Subnet

In this task, we are going to launch an EC2 instance in the public subnet. This allows users to have a publicly accessible instance that can directly communicate with the internet.

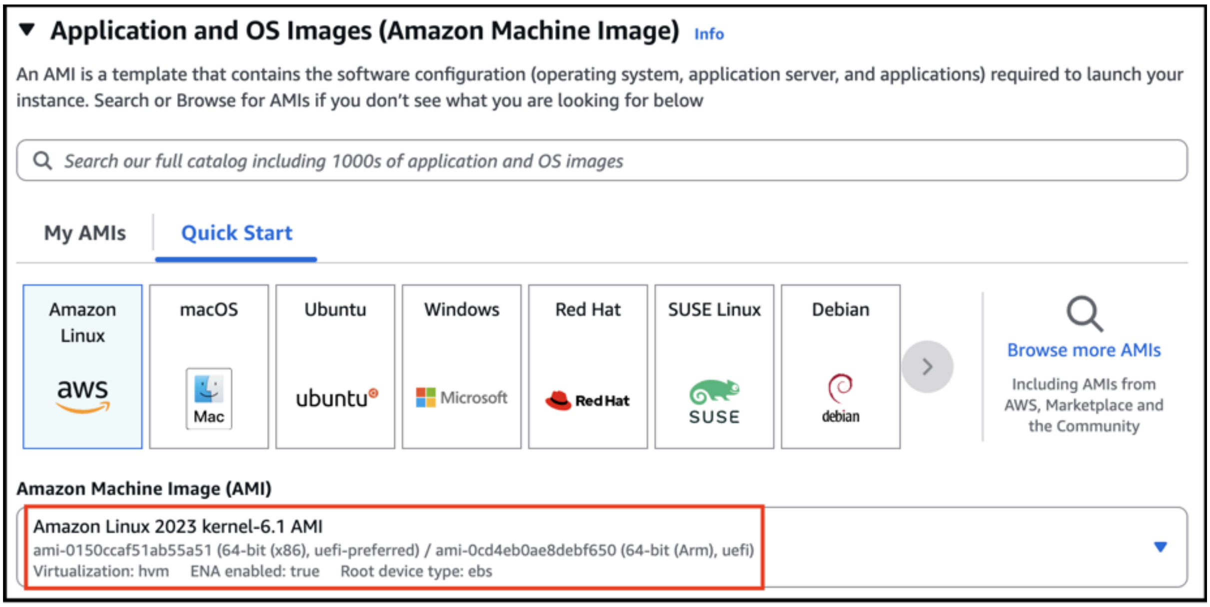
1. Make sure you are in the N.Virginia region.
2. Navigate to the Services menu in the top, click on EC2 in the Compute section.
3. Navigate to Instances on the left panel and click on Launch instances button



4. Name : Enter MyPublicServer



5. For Amazon Machine Image (AMI): Search for Amazon Linux 2023 AMI in the search box and click on the select button.



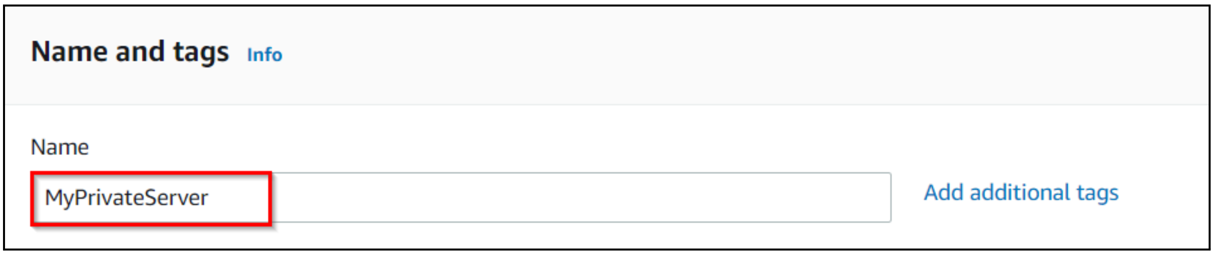
6. For Instance Type: select t2.micro
7. For Key pair: Select Create a new key pair Button
    - Key pair name: MyKey
    - Key pair type: RSA
    - Private key file format: .pem
8. Select Create key pair Button.
9. In Network Settings Click on Edit:
10. VPC : Select MyVPC
11. Subnet : Select MyPublicSubnet

12. Auto-assign public IP: Enable
13. Select Create new Security group
14. Security group name : Enter MyEC2Server_SG
15. Description : Enter Security Group to allow traffic to EC2
    - To add SSH
        - Choose Type:
        - Select SSH
        - Source: Select Anywhere
16. Keep Rest thing Default and Click on Launch Instance Button.
17. Select View all Instances to View Instance you Created
18. Launch Status: Your instance is now launching, Select the instance and wait for it to change status to Running.

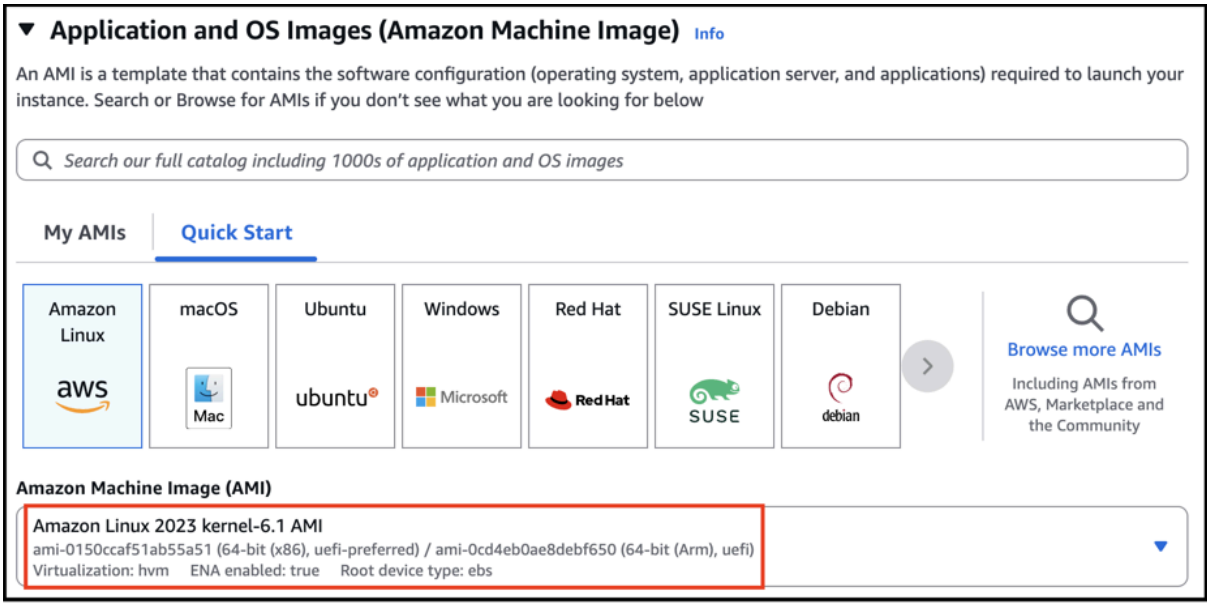
Task 7 : Launch an EC2 Instance in Private Subnet

In this task, we are going to launch an EC2 instance in the private subnet. This demonstrates the concept of a private subnet, which does not have direct internet access.

1. Click on Launch instances.
2. Name : Enter MyPrivateServer



3. For Amazon Machine Image (AMI): Search for Amazon Linux 2023 AMI in the search box and click on the select button.



4. For Instance Type: select t2.micro
5. For Key pair: Select the key pair you created before
6. In Network Settings Click on Edit:

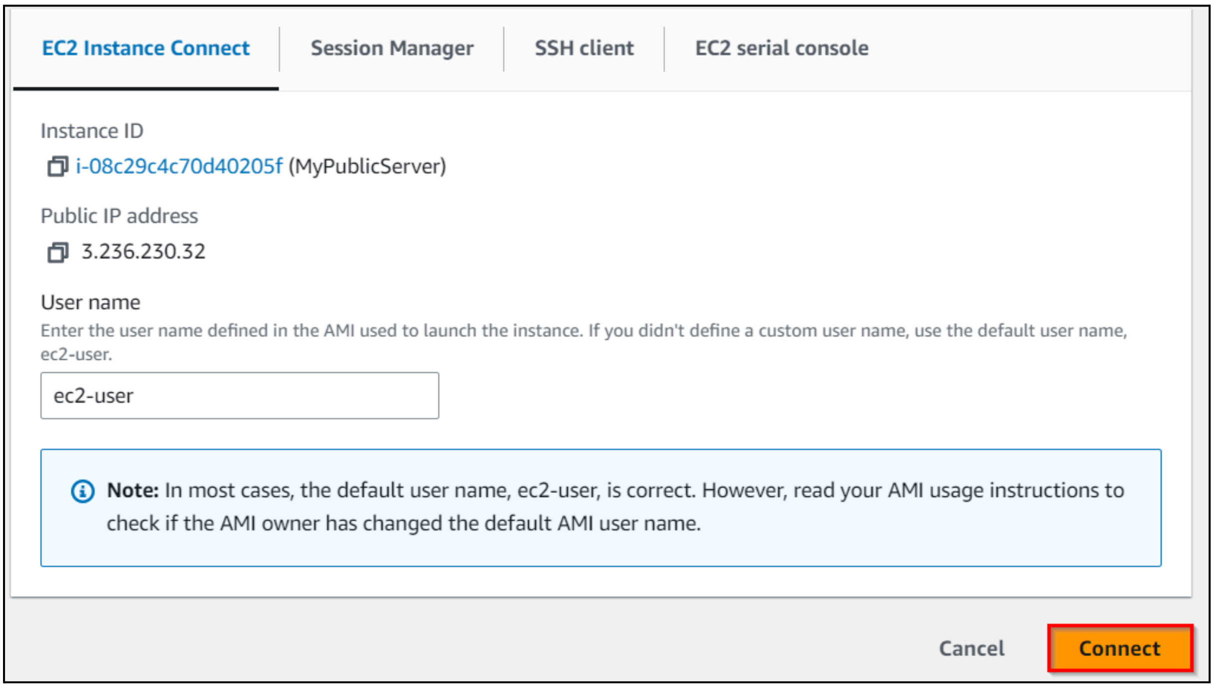
- VPC : Select MyVPC
- Subnet : Select MyPrivateSubnet
- Auto-assign public IP: Disable
- Select Select existing security group
- Select MyEC2Server_SG

7. Keep Rest thing Default and Click on Launch Instance Button.
8. Select View all Instances to View Instance you Created
9. Launch Status: Your instance is now launching, Select the instance and wait for it to change status to Running.
10. Note the Private IP Address of MyPrivateServer : Example 10.0.1.45

Task 8 : SSH into Public and Private EC2 Instance and Test Internet Connectivity

In this task, we are going to establish SSH connections to both the public and private EC2 instances. By testing internet connectivity from both instances, users can verify if the public instance has internet access and the private instance does not.

1. SSH into MyPublicServer Instance. Follow the below steps
2. Once the instance is created. Select the instance MyPublicServer.Click on connect button
3. Now select the EC2 instance connect option. You can see there are four options for connecting to EC2. You can use any of the given options to get in the console but for our lab, we are using EC2 instance connect.



4. A new tab would be opened in your browser where we can see the console.

```
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-0-0-59 ~]$
```

5. Switch to root user:

```
sudo su
```

7. Run the updates using the following command:

```
yum -y update
```

7. Since the Internet Gateway MyIGW is connected to MyPublicSubnet, updates will be completed successfully.

```
Updated:
  cloud-utils-growpart.noarch 0:0.31-1.amzn2          dracut.x86_64 0:033-535.amzn2.1.3
  dracut-config-generic.x86_64 0:033-535.amzn2.1.3    glib2.x86_64 0:2.56.1-4.amzn2
  kernel-tools.x86_64 0:4.14.143-118.123.amzn2        mariadb-libs.x86_64 1:5.5.64-1.amzn2
  python.x86_64 0:2.7.16-3.amzn2.0.1                  python-devel.x86_64 0:2.7.16-3.amzn2.0.1
  python-libs.x86_64 0:2.7.16-3.amzn2.0.1

Complete!
[root@ip-10-0-1-45 ec2-user]#
```

8. Let's SSH into MyPrivateEC2Server from MyPublicEC2Server.
9. In order to SSH into MyPrivateEC2Server, first, we need to create the PEM file in the public EC2 ie, MyPublicEC2Server, and copy the data from our MyKey.pem in the local machine.
10. We need MyKey.pem in order to SSH. We shall copy key details from the MyKey.pem from your local machine (which was downloaded earlier while launching EC2 instances).
11. To copy the contents of the MyKey.pem, open the file in a text editor and copy the whole content.
12. To create the MyKey.pem in MyPublicEC2Server, run

```
vi MyKey.pem
```

13. Now press the following for inserting the data(You can see insert popped at the last then you can paste the key).
    - Click i

```
~
~
~
~
~
-- INSERT --
```

Note: In the editor, copy and paste the key that looks similar to the example below:

```
-----BEGIN RSA PRIVATE KEY-----
MIIEpAIBAAKCAQEAmsHG3lriaWEHo93izlRP4Ll0WE8SdGlxabrnIe3BUyRy5A05
ciELGww06TJihcogH30lMZvMXKxA+9fEXYMnwVGMYh0zELisZqmh5A7nD12H+0sc
wcMmahaj4tQsL3fK0aZp0jKMWsLm3XTDkmMnR/IrSUGGS5b8uY0d/91Ew550+e5p
dgWShnYxvG+lPHdIRrJqymbmGqwzQoZlsqTv+rMqMVLZB8J3JZoOgRib0cqH13iX
n9BWoYANw4xDivBL0RjDTC6freArAMZ2hvYLYrnSY0qFM+8Kic/AgZ0GNDLt2B0m
6R34oQmmR2GsGRawabLGwyIFIdILAxaC40aLDwIDAQABAoIBAQCKxw6hQ3sZ6MtD
Ja1IPc9LxKD/DBm0jygHGnY+tMjS2DK3Sqe4Mu2UmaGdeXNkareXvMr8y6vdxxKX
RQxwlzbFY22yZmDus8zWW3Qz4TFPVB6nONpW37vJEvhwZrgg8RnOG591waKfmXwS
R61Bzuzg1cf/b2UQ0Btd+Sts0W3ixNys8tLPbWgZTYXSoq6MstCd0utQ+gqSduGR
I7wyLIQAiwAI5d7TRKr3pGV6mdTxQdRfiXnLh0xNzH6qDYt8TqFMipAGTMnCK+ET
zY2EgXY3TX/9CE1o85wiOho2BtL7eDp4ZcksEmws1YzIAaXL01HZZEn7N+J850YS
Xlz9GlHhAoGBAPd6+11ay4keXx0w9hS9WGnVkaIBS8bLtf0J0Ig2RA5YFL30FxVW
MIrHslY5QC1SHofST7RGeSjOrc5cWpw4/byByro/jvEn1S2y8sBFz7SPa+jRK2dE
CKn9IGkFDXnl5fQ9/u2gCVfl0dbh5sFD6mwYolJeX5oGcRHo8ScbcnypAoGBAKAV
ohIrNBRxB74LHHI5rjoHu7KLYn4MZ689I5y7AWFmVi57KmBqj7EowacNhaaLkOYd
F5Aem9Itj60c/OMwai9cN2zpDuFqHTNfIoXit4dtCUhshz+qCxEqy/MmlkB52KmN
2b2+A4rtu0V95db0++k0R4r8WzWe+93uzVy01KT3AoGBANHdoLwR+vhUoSQIGSaN
SySOz8MqhjAXIEVsxUI9gokHeM9crYaSiKenoTwtYJcORIW5e43NFYHdgta0ysIg
MkxhZKF8tKMcvFL8pY8XpIPh/Ah7GOuBbjd5SLrGZh8xLkhpKn4ocMPxSHmJKD5M
cccpPL3M+hWL0Ob300mGpRfBAoGATcx1qXLJqq2L37YyquiRXa5ob8sBoNSShz2l
UwQnFZjQhWJiI91l2+k8w9z0eQ4W6150FxxpCeoOi0ktNZRP5phgb1OMYcxaSJ8R
fhPCsdQx0Lq9uBkBq+Qqng7AsrEEtNI1Yds4iYoic5WN37vLuXgo1Mjz4slTCnaF
42oWdd0CgYB0y+PpIIZNJjW6nIYvqDQoKnlAjICzbXv4YlHWYuVrH92TVUurpISe
7XN9nUJdztBemkzEL5SZtSeXuISmdqPjb+/2vg07xjIhYGEmwMDMuKLrFAlgA5/I
ENX7o9RuYJ+EySq9W23XveQuFMohJnA1RGeEbAt76BTidYZQCg0sfQ==
-----END RSA PRIVATE KEY-----
-- INSERT --
```

14. Save the File
    - click esc

```
:wq
```

```
fhPCsdQx0Lq9uBkBq+Qqng7AsrEEtNI1Yds4iYoic5WN37vLuXgo1Mjz4slTCnaF
42oWdd0CgYB0y+PpIIZNJjW6nIYvqDQoKnlAjICzbXv4YlHWYuVrH92TVUurpISe
7XN9nUJdztBemkzEL5SZtSeXuISmdqPjb+/2vg07xjIhYGEmwMDMuKLrFAlgA5/I
ENX7o9RuYJ+EySq9W23XveQuFMohJnA1RGeEbAt76BTidYZQCg0sfQ==
-----END RSA PRIVATE KEY-----
:wq
```

15. Check that the file was created correctly.

```
ls
```

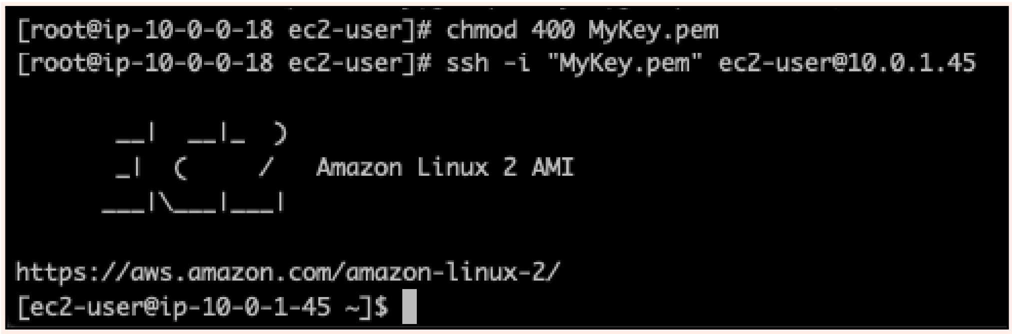
16. Update Permissions for the MyKey.pem

```
chmod 400 MyKey.pem
```

17. Use the Private IP address of MyPrivateEC2Server to SSH.

```
ssh  ec2-user@<Private IP of MyPrivateEC2Server> -i
MyKey.pem
```

Note: Incase if this message shows Are you sure you want to continue connecting (yes/no)? : Enter yes



18. Switch to root user

```
sudo su
```

19. Run the updates using the following command:

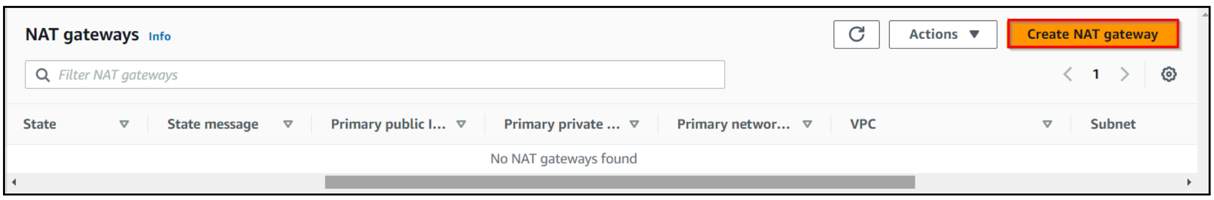
```
yum -y update
```

20. Since no internet access is provided for EC2 instances in a private subnet, you will not be able to get updates.

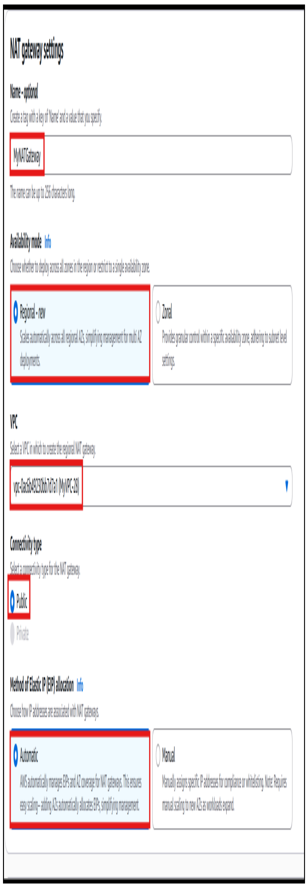
Task 9 : Create a NAT Gateway

In this task, we are going to create a NAT Gateway, which provides internet access to instances in the private subnet. The NAT Gateway acts as a middleman to forward traffic between the private subnet and the internet.

1. Navigate to the VPC Page.
2. Make sure you are still in the N.Virginia Region.
3. In the Left Panel, click on NAT Gateways.
4. Click on Create NAT gateway button.



- Name : Enter MyNATGateway
- VPC: Choose MyVPC
- Connectivity type: **Public** and Method of Elastic IP: **Automatic**.
- Once the new Elastic IP is allocated, click on Create NAT gateway.



5. Note that NAT Gateway is always created in a public subnet.
6. NAT Gateway will be created in a few minutes. Once created, the status will change to available.

	Name	NAT gateway ID	State	State message
◉	MyNATGateway	nat-0dd9d8d0fd9481414	Available	–

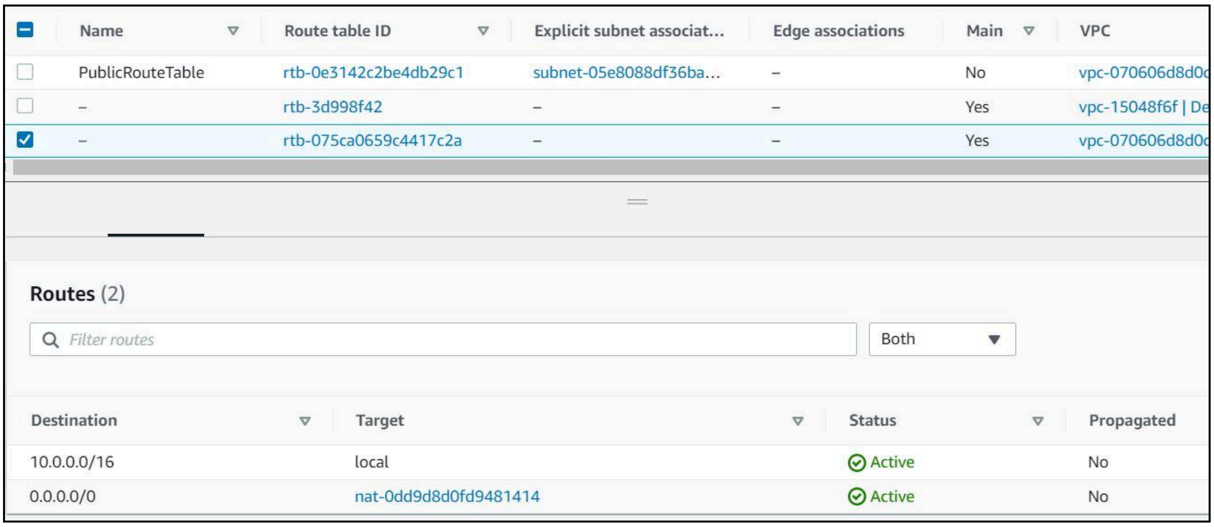
Task 10 : Update Route table and configure NAT Gateway

In this task, we are going to update the route table associated with the private subnet to include the NAT Gateway as the target for internet-bound traffic. This ensures that traffic from the private subnet is directed through the NAT Gateway for internet access.

1. Navigate to Route Tables in the left panel.
2. You can see two Route Tables available for MyVPC

	Name	Route table ID	Explicit subnet associat...	Edge associations	Main	VPC
☐	PublicRouteTable	rtb-0e3142c2be4db29c1	subnet-05e8088df36ba...	–	No	vpc-070606d8d0c3b038a | MyVPC
☐	–	rtb-3d998f42	–	–	Yes	vpc-15048f6f | Default VPC
☐	–	rtb-075ca0659c4417c2a	–	–	Yes	vpc-070606d8d0c3b038a | MyVPC

3. To attach Nat Gateway, select the Main Route Table (which is different from the one created by you).
4. In the Routes tab below,
    - Click on Edit routes.
    - In the next page, Click on Add route
    - Destination: Enter 0.0.0.0/0
    - Target: Select NAT Gateway, and once the internet gateways have loaded, select the NAT Gateway you created.
    - Click on Save changes.
5. Once all the configurations are completed, it should look like below.



6. Now the Instances launched within MyPrivateSubnet can access the Internet through the NAT Gateway.

Task 11 : Test Internet connection from Instance inside Private Subnet

In this task, we are going to validate that the instance in the private subnet can successfully establish an internet connection by accessing the internet through the NAT Gateway.

1. SSH back into MyPublicEC2Server .
2. Switch to root user

```
sudo su
```

3. SSH into MyPrivateEC2Server

```
ssh  ec2-user@<Private IP of MyPrivateEC2Server> -i MyKey.pem
```

4. Switch to root user

```
sudo su
```

5. Run the updates using the following command:

```
yum -y update
```

6. You can see that the updates have been completed successfully in the terminal.

```
Updated:
  cloud-utils-growpart.noarch 0:0.31-1.amzn2          dracut.x86_64 0:033-535.amzn2.1.3
  dracut-config-generic.x86_64 0:033-535.amzn2.1.3    glib2.x86_64 0:2.56.1-4.amzn2
  kernel-tools.x86_64 0:4.14.143-118.123.amzn2        mariadb-libs.x86_64 1:5.5.64-1.amzn2
  python.x86_64 0:2.7.16-3.amzn2.0.1                  python-devel.x86_64 0:2.7.16-3.amzn2.0.1
  python-libs.x86_64 0:2.7.16-3.amzn2.0.1

Complete!
[root@ip-10-0-0-18 ec2-user]#
```

7. This shows that MyPrivateEC2Server has internet access.
8. Use exit command to close the private server connection.

Lab **Understanding and Configuring Layered Security in an AWS**

Task 1: Sign in to AWS Management Console

1. Click on the Open Console button, and you will get redirected to AWS Console in a new browser tab.
2. On the AWS sign-in page,
    - Leave the Account ID as default. Never edit/remove the 12 digit Account ID present in the AWS Console. otherwise, you cannot proceed with the lab.
    - Now copy your User Name and Password in the Lab Console to the IAM Username and Password in AWS Console and click on the Sign in button.
3. Once Signed In to the AWS Management Console, Make the default AWS Region as US East (N. Virginia) us-east-1.

Note: There is no Validation function for this lab.

Task 2: Creating a new VPC

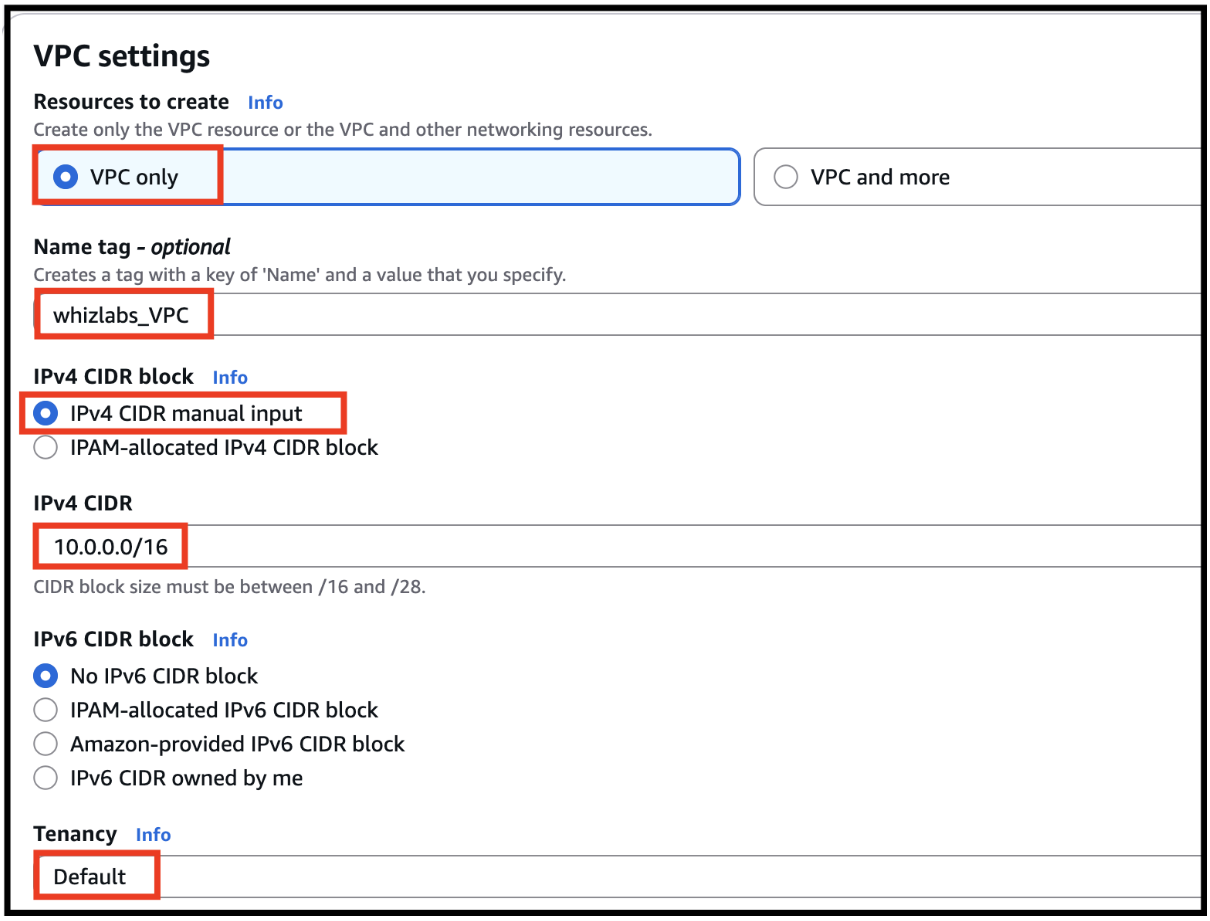
In this task, we are going to create a new VPC with the required configurations such as name, IPv4 CIDR block.

1. Make sure to choose the N.Virginia region in the AWS Management Control dashboard, which is present in the top right corner.
2. Navigate and click on VPC which will be available under the Networking & Content Delivery section of Services
3. Click on Your VPCs from the left menu and Click on Create VPC button.

4. In Create VPC page fill the following details,

- Select VPC Only
- Name tag: Enter *whizlabs_VPC*
- IPV4 CIDR Block: Enter *10.0.0.0/16*
- IPV6 CIDR block: Select No IPV6 CIDR block

- Tenancy: Default



- Click on Create VPC button.

Concept: Amazon Virtual Private Cloud (VPC)

Amazon VPC allows you to create a **logically isolated network** within AWS where you can launch AWS resources such as EC2 instances, databases, and load balancers.

A VPC gives you control over:

- IP address ranges using CIDR blocks
- Subnets within the network
- Route tables to control traffic
- Security layers such as Security Groups and Network ACLs

Creating a custom VPC allows organizations to design **secure and scalable network architectures** for their cloud infrastructure.

Task 3: Creating and attaching an Internet gateway

In this task, we are going to create an internet gateway and will attach it to the created VPC.

1. Click on Internet Gateways from the left menu and click on Create internet gateway button and enter the following details:
  - Name tag: Enter *whizlabs_IGW*
  - Click on Create internet gateway button.

2. Select the Internet gateway you created from the list.
  - Click on Actions button.
  - Click on Attach to VPC button.

  - Select MyVPC from the drop-down and click on Attach internet gateway button.

Pro Tip: Internet Gateway and Public Access

An Internet Gateway (IGW) enables communication between resources in your VPC and the public internet.

However, attaching an Internet Gateway alone does **not make a subnet public.**
A subnet becomes public **only when its route table contains a route (0.0.0.0/0) pointing to the Internet Gateway.**

This routing configuration allows instances in that subnet to send and receive internet traffic.

Task 4: Creating two Subnets

1. You will create 2 Subnets, one for public and another for private resources. First, we will create a public subnet.
2. For the Public Subnet, click on Subnets from the left menu and click on Create subnet button.
  - VPC ID: Select whizlabs_VPC (Select the VPC which you created from the dropdown)
  - Name tag: Enter *public_subnet*
  - Availability zone: Select No Preference
  - IPv4 CIDR block: Enter *10.0.1.0/24*

  - Click on Create subnet button
3. For Private Subnet, click on Create Subnet again.
  - VPC ID: Select whizlabs_VPC (Select the VPC which you created from the dropdown)
  - Name tag: Enter *private_subnet*
  - Availability zone: Select No Preference
  - IPv4 CIDR block: Enter *10.0.2.0/24*
  - Click on Create subnet button.

Concept: Public and Private Subnets

Subnets divide a VPC into smaller network segments.

- **Public Subnet:** Contains resources that must be accessible from the internet, such as web servers. These subnets typically have routes pointing to an Internet Gateway.
- **Private Subnet:** Contains internal resources such as application servers or databases that should not be directly accessible from the internet.

This separation helps implement **multi-tier architectures** and improves security.

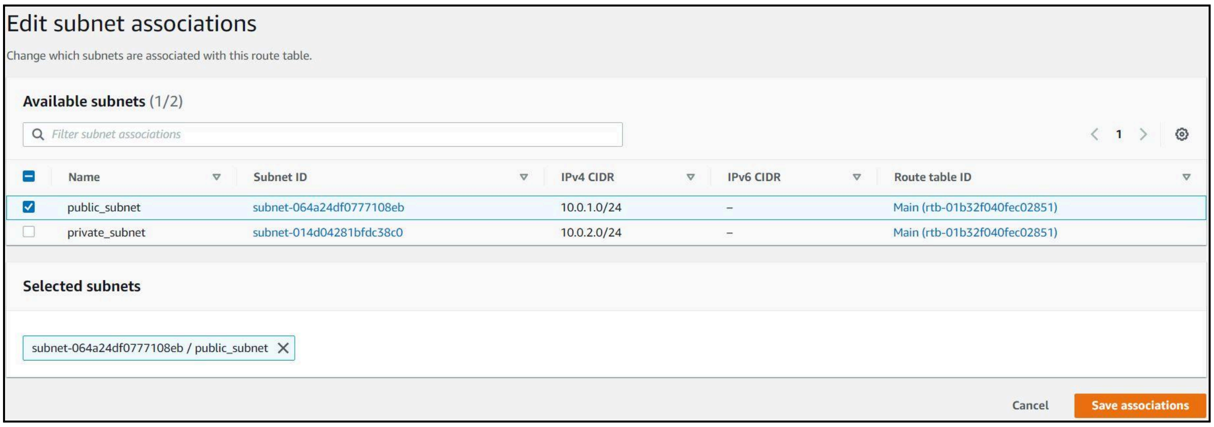
Task 5: Creating Route tables, configuring routes and associating them with Subnets

1. You will create 2 route tables, one for public routes and another for private routes.
2. Go to Route Tables from the left menu and click on Create route table button.
   - Name tag: Enter *public_route*
   - VPC: Select whizlabs_VPC (Select the VPC you created from the dropdown)
   - Click on Create route table button.

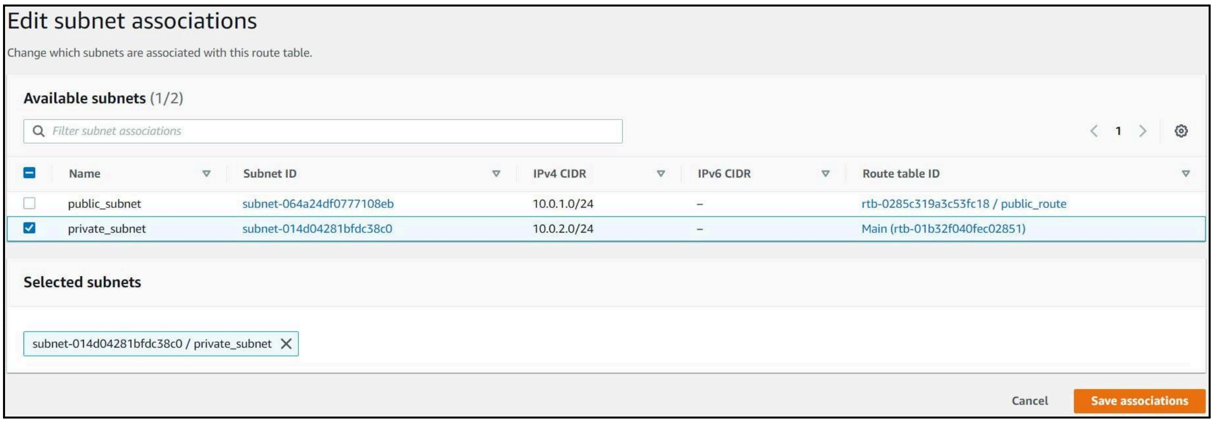
3. Similarly, go to Route Tables from the left menu and click on Create route table.
   - Name tag: Enter *private_route*
   - VPC: Select whizlabs_VPC (Select the VPC you created from the dropdown)
   - Click on Create route table button.

4. Now, you need to add routes to the Route Tables.
   - Select public_route.
   - Go to the Routes tab. Click on Edit routes. On the next page, click on Add route.
   - Specify the following values:
     - Destination: Enter *0.0.0.0/0*
     - Target: Select Internet Gateway from the dropdown menu to select whizlabs_IGW.
     - Click on Save changes button.

5. Next, you need to associate the public_subnet with this public_route. Select the public_route and go to the Actions and in that go to Edit Subnet Associations tab.
   - Click on Edit Subnet Associations.
   - Select public_subnet from the list.
   - Click on Save Associations button.



6. Similarly, you need to associate the private_subnet with this private_route. Select the private_route and go to the Actions and in that go to Edit Subnet Associations tab.
    - Click on Edit Subnet Associations.
    - Select private_subnet from the list.
    - Click on Save Associations button.



Deep Dive: Route Tables in VPC Networking

A Route Table controls how network traffic flows within a VPC.

Each route contains:

- **Destination** – The IP address range for the traffic
- **Target** – Where the traffic should be directed (Internet Gateway, NAT Gateway, etc.)

For example:

- **local route** ? enables communication within the VPC
- **0.0.0.0/0** ? directs internet-bound traffic to the Internet Gateway

Associating route tables with subnets determines whether a subnet behaves as public or private.

Task 6: Creating Security Group

In this task, we are going to create a security group for the EC2 Instance.

1. Go to Security Group from the left menu and click on Create security group button and then provide the following details:
- Security group name: Enter *whizlabs_securitygroup*
- Description: Enter *Security group for multilayered VPC*
- VPC: Select whizlabs_VPC (select from the dropdown)

- Under Inbound Rules, click on Add Rule button.
- To add SSH,
    - Choose Type: Select SSH
    - Source: Anywhere-IPv4
- To add All ICMP - IPv4,
    - Click on Add Rule
    - Choose Type: Select All ICMP - IPv4
    - Source: Anywhere IPv4

- Click on Create security group button.

Concept: Security Groups

Security Groups act as **virtual firewalls** that control inbound and outbound traffic for EC2 instances.

Key characteristics:

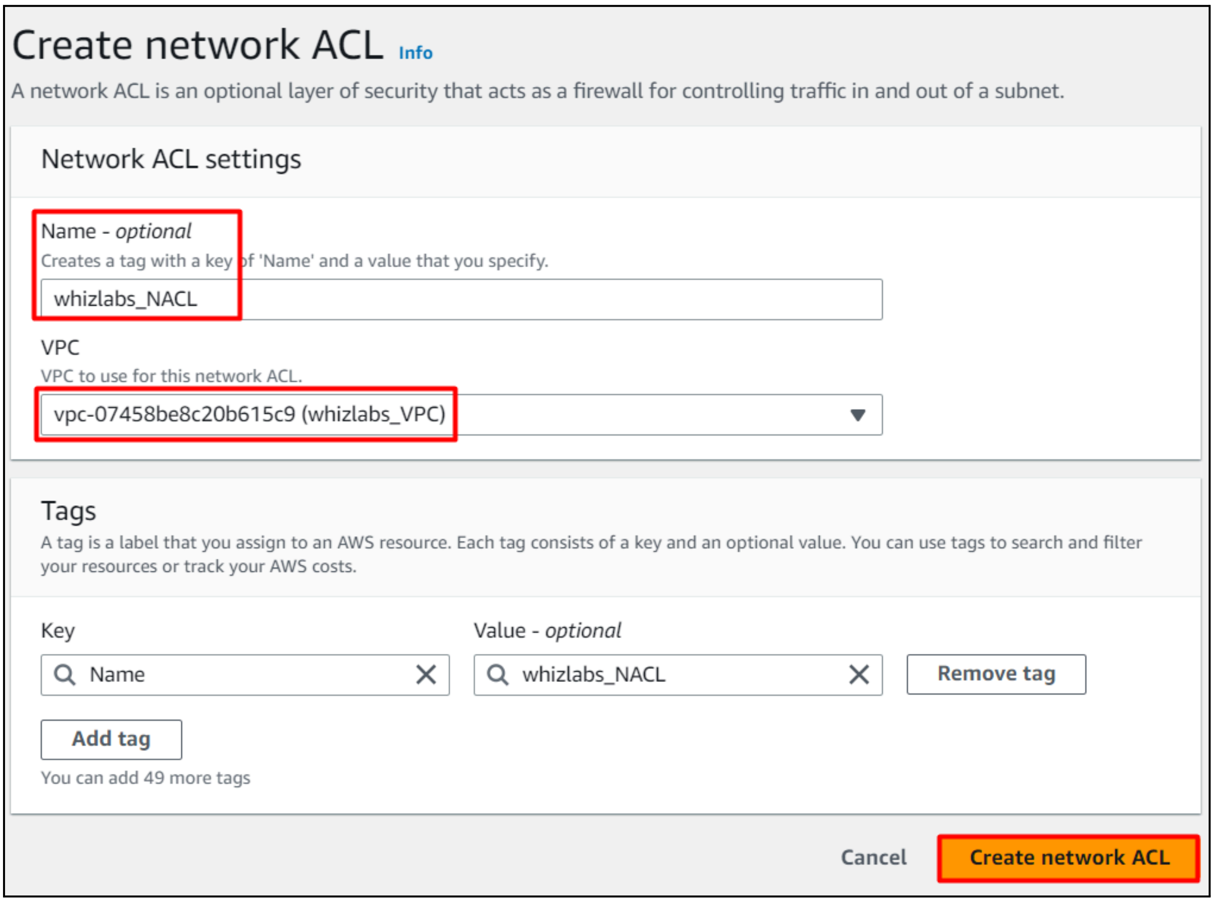
- **Stateful** – If inbound traffic is allowed, the response is automatically allowed.
- **Instance-level security** – Applied directly to EC2 instances.
- **Rule-based filtering** – Controls traffic based on ports, protocols, and IP ranges.

Security groups are commonly used to allow only specific traffic such as **SSH (port 22)** or **HTTP (port 80)**.

Task 7: Creating and configuring Network ACL

Network Access Control Lists (ACLs) in AWS are used to control inbound and outbound traffic at the subnet level. They act as virtual firewalls that provide an additional layer of security for your Amazon Virtual Private Cloud (VPC). In this task, we are going to create and configure Network ACL by adding required inbound and outbound rules.

1. Go to Network ACLs from the left menu and click on Create network ACL. Provide the following details:
- Name tag: Enter *whizlabs_NACL*
- VPC: Select whizlabs_VPC (Select the VPC which you created from the dropdown)
- Click on Create network ACL button.



2. Select whizlabs_NACL and go to Inbound rules tab. Click on Edit inbound rules and then click on the Add new rule.
3. Add the following rules:
- For SSH, click on Add new rule,
    - Rule number : Enter *100*
    - Type: Choose *SSH (22)*
    - Source: Enter *0.0.0.0/0*
    - Allow / Deny: Select Allow
- For ALL ICMP- IPv4, click on Add new rule,
    - Rule number : Enter *200*
    - Type: Choose *ALL ICMP - IPv4*
    - Source: Enter *0.0.0.0/0*
    - Allow / Deny: Select Allow
- Click on Save changes button.

4. NACLs are stateless. You need to add the rules in Outbound rules too.
5. Select whizlabs_NACL and go to Outbound rules tab. Click on Edit Outbound rules and then click on the Add Rule button.
- For ALL ICMP- IPv4, click on Add new rule,
    - Rule# : Enter *100*

    - Type: Choose *ALL ICMP - IPv4*
    - Destination: Enter *0.0.0.0/0*
    - Allow / Deny: Select Allow
- For Custom TCP Rule, click on Add new rule,
    - Rule# : Enter *200*
    - Type: Choose *Custom TCP Rule*
    - Port Range: Enter *1024 - 65535*
    - Destination: Enter *0.0.0.0/0*
    - Allow / Deny: Select Allow
- Click on Save changes button.

6. You need to associate both public and private subnets with the NACL.
7. Select whizlabs_NACL and go to the Subnet associations tab. Click on Edit subnet associations.
8. Choose all the available subnets and then click on Save changes button.

Concept: Network Access Control Lists (NACLs)

Network ACLs provide an **additional layer of security at the subnet level.**

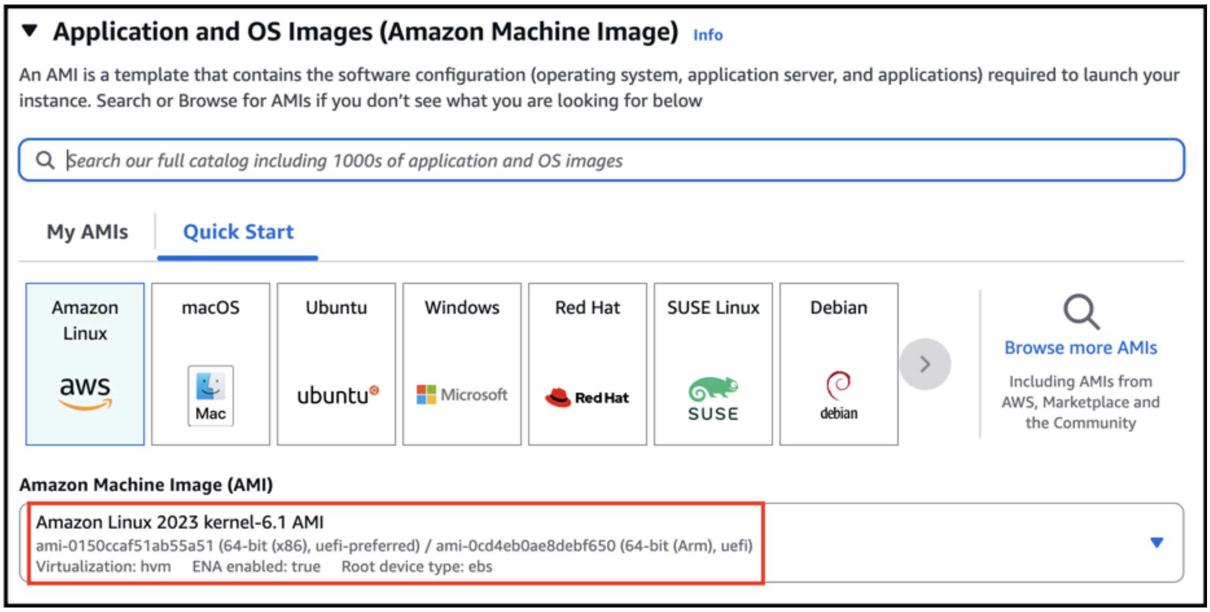
Key characteristics:

- **Stateless filtering** – Return traffic must be explicitly allowed.
- **Subnet-level control** – Applied to entire subnets instead of individual instances.
- **Rule numbering system** – Lower rule numbers are evaluated first.

In most AWS architectures, **Security Groups and NACLs work together** to provide layered security.

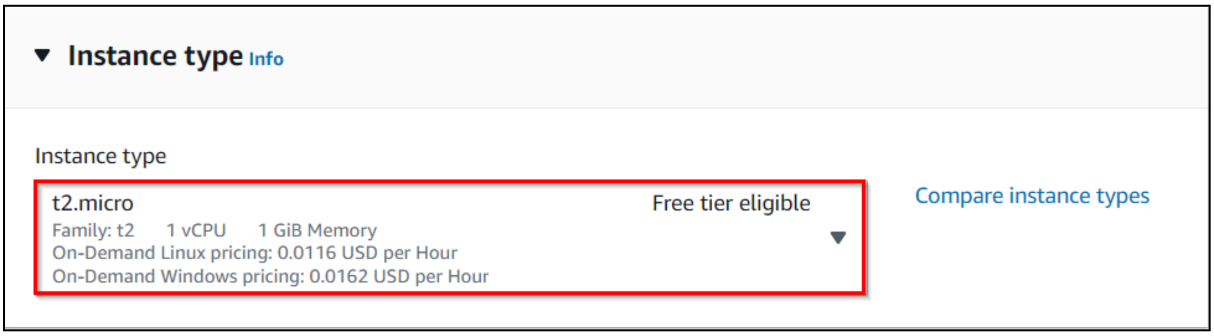
Task 8: Launching 2 EC2 Instances

1. Navigate to Services and choose EC2 under Compute
2. Navigate to Instances on the left panel and click on Launch Instances button.
3. Name : Enter *public_instance*
4. For Amazon Machine Image (AMI): Search for Amazon Linux 2023 AMI in the search box and click on the select button.



Note: if there are two AMI's present for Amazon Linux 2 AMI, choose any of them.

5. For Instance Type: Select *t2.micro*



6. For Key pair: Select Create a new key pair Button
    - Key pair name: WhizKey
    - Key pair type: RSA
    - Private key file format: .pem
7. Select Create key pair Button.
8. In Network Settings Click on Edit button:
    - VPC: Select whizlabs_VPC
    - Subnet: Select public_subnet
    - Auto-assign public IP: Enable
    - Choose Select an existing security group and remove default one then select whizlabs_securitygroup

9. Keep Rest thing Default and Click on Launch Instance Button.
10. Select View all Instances to View Instance you Created
11. Similar to the above, launch another EC2 instance:

- Name the instance as *private_instance*
  Instance type: **t2.micro**
- Key pair : Select the existing one (**WhizKey**)

- VPC: Select whizlabs_VPC
- Subnet: Select private_subnet
- Auto-assign public IP: Disable
- Choose Select an existing security group and remove default one then select whizlabs_securitygroup

12. Keep Rest thing Default and Click on Launch Instance Button.
13. Select View all Instances to View Instance you Created.

Pro Tip: Multi-Tier Architecture

Deploying instances across public and private subnets is a common **three-tier architecture pattern:**

- **Public Layer** – Web servers accessible from the internet
- **Application Layer** – Backend services in private subnets
- **Database Layer** – Databases isolated from direct internet access

This architecture improves both security and scalability of applications hosted in AWS.

Task 9: Testing the EC2 instances

1. Now you'll be able to see both the Instances.

2. Copy the Private IPv4 Address of the private_instance.
   SSH into public_instance EC2 Instance.
- Please follow the steps in SSH into EC2 Instance.
4. Ping the private IP of your private_instance by using the below command:
    - ping <your Private EC2 IPv4 address>
5. Once you execute this command, you will receive a response from the IP.

Lab Resizing Amazon EBS Volume

Task 1: Sign in to AWS Management Console

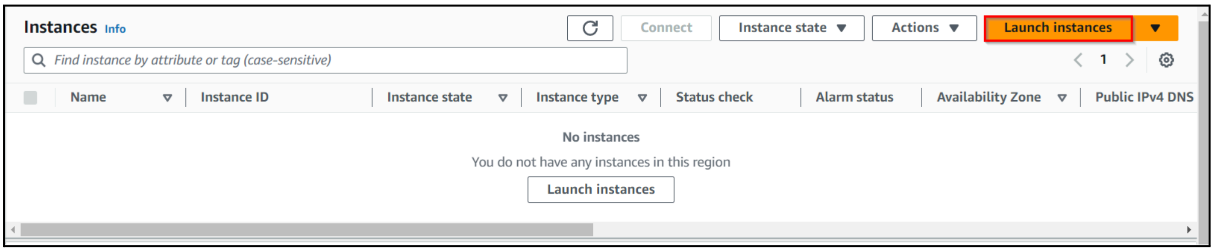
1. Click on the Open Console button, and you will get redirected to AWS Console in a new browser tab.
2. On the AWS sign-in page,
    - Leave the Account ID as default. Never edit/remove the 12-digit Account ID present in the AWS Console. Otherwise, you cannot proceed with the lab.
    - Now copy your Username and Password in the Lab Console to the IAM Username and Password in AWS Console and click on the Sign-in button.
3. Once Signed In to the AWS Management Console, make the default AWS Region as US East (N. Virginia) us-east-1.

Note : If you face any issues, please go through FAQs and Troubleshooting for Labs.

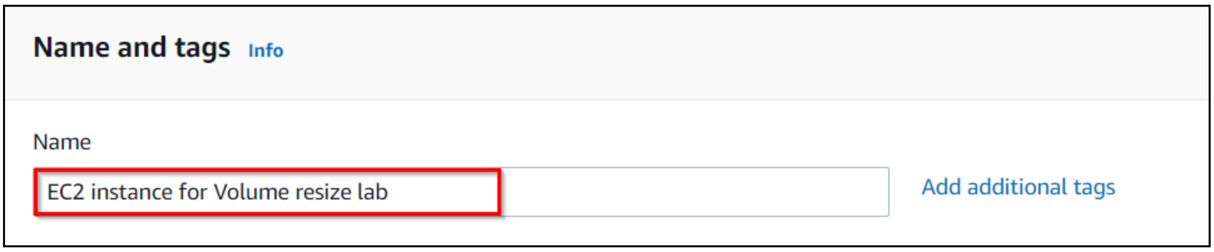
Task 2 : Launching Amazon EC2 instance

In this task, we are going to launch an EC2 Instance.

1. Make sure to choose the US East (N. Virginia) us-east-1 region.
2. Navigate to Services menu in the top, then click on EC2 in the Compute section.
3. Click on Instances from the left sidebar and then click on Launch instances button.



4. Name : Enter EC2 instance for Volume resize lab



5. For Amazon Machine Image (AMI): Search for Amazon Linux 2023 kernel-6.1 AMI in the Quick Start menu.
6. For Instance Type: select **t2.micro**
7. For Key pair: Select Create a new key pair Button
    - Key pair name: WhizKey
    - Key pair type: RSA
    - Private key file format: .pem
8. Select Create key pair Button.
9. In Network Settings, click on Edit:
- Auto-assign public IP: Enable
- Select Create new Security group
- Security group name : Enter MyEC2Server_SG
10. Check Allow SSH from and Select Anywhere from dropdown
- To add SSH,
    - Choose Type: SSH
    - Source: Select Anywhere
11. Keep rest thing Default and Click on Launch Instance Button.
12. Select View all Instances to View the Instance you Created
13. Launch Status: Your instances are now launching, Navigate to Instances page from left menu and wait the status of the EC2 Instance changes to running

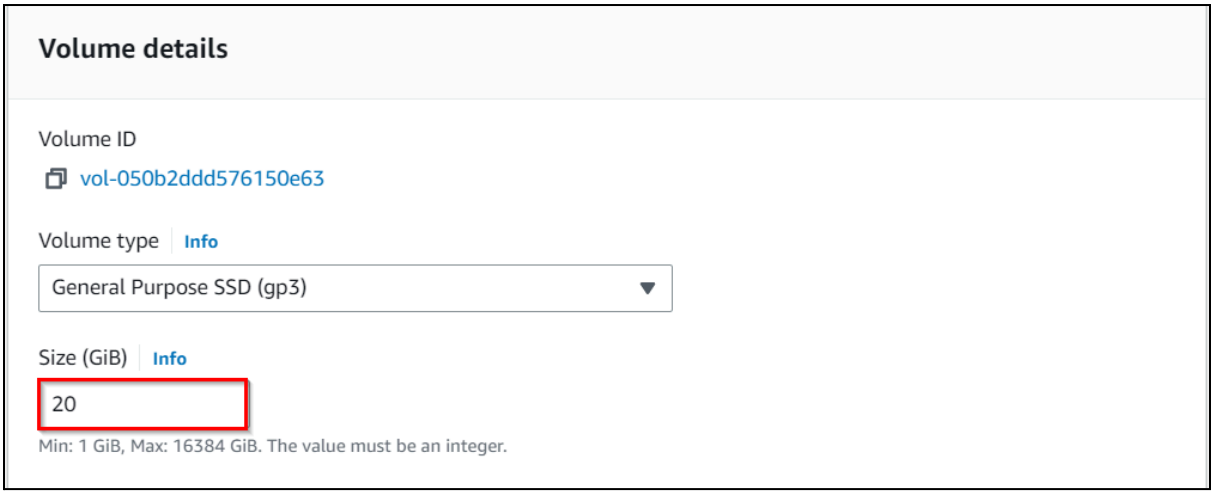
Task 3 : Resizing the EBS Volumes

In this task, we are going to demonstrate how to resize an Amazon EBS volume associated with an EC2 instance.

1. In the EC2 dashboard, select EC2 instance for Volume resize lab and navigate to the Storage tab and then click on Volume ID present.

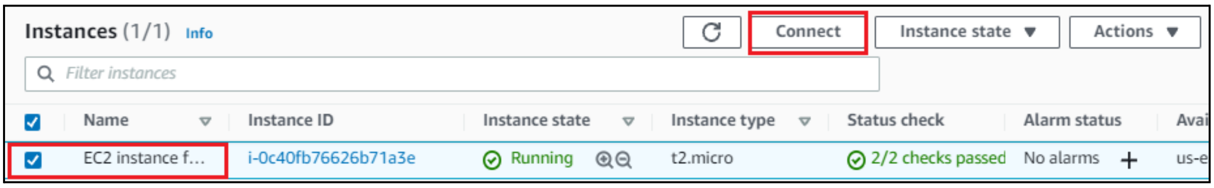
2. Select the Volume and then go to the **Actions** menu dropdown. Click on Modify Volume.

3. Provide the new Volume Size you need to increase to, and also note that we are not able to decrease the volume size. In our case, I'm modifying from 8GiB to 20GiB and then clicking on Modify.

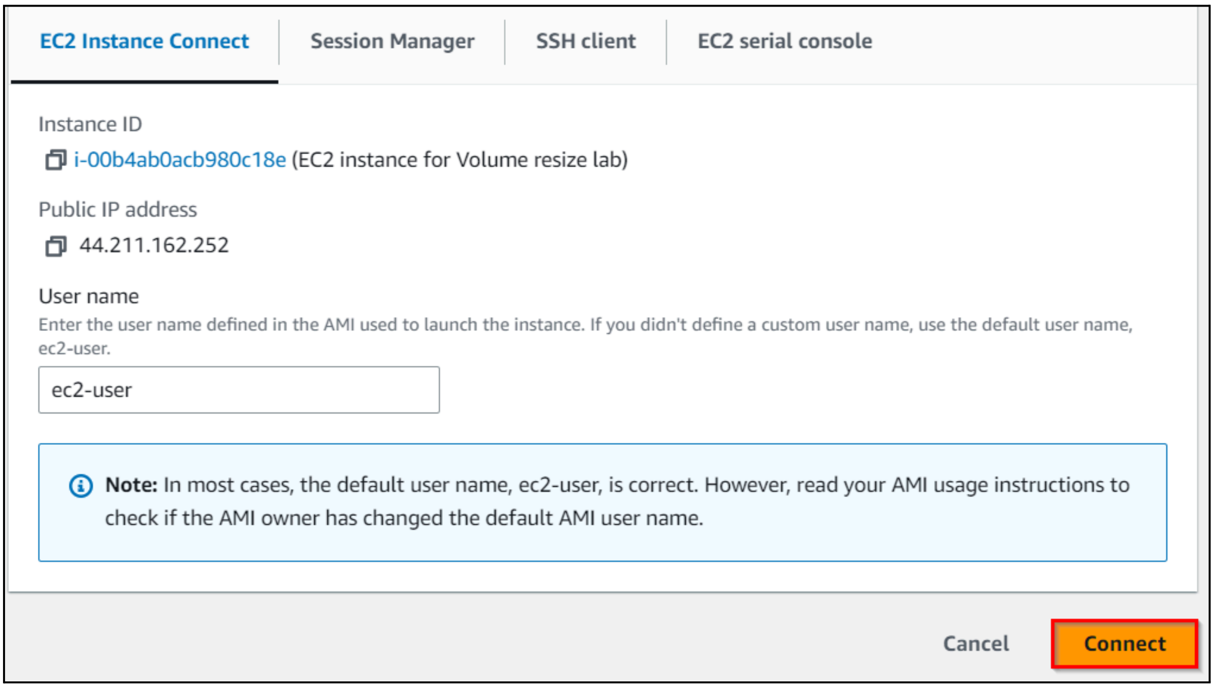


4. Now, after modifying the volume, we need to increase the partition and file system. To do so, we need to SSH into the server with the key we downloaded while launching the server.
5. To SSH into EC2 Instance. Follow the below steps.

- Once the instance is created. Select the instance EC2 instance for Volume resize lab. Click on connect button.



- Now select the EC2 instance connect option. We can see there are four options for connecting to EC2. You can use any of the given options to get in the console but for our lab, we are using EC2 instance connect.



- A new tab would be opened in your browser where we can see the console.

```
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-172-31-87-199 ~]$
```

6. Enter the command

```
sudo su
```

8. Run the following commands.

```
df -h
```

```
[root@ip-172-31-87-199 ec2-user]# df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M     0  4.0M   0% /dev
tmpfs           475M     0  475M   0% /dev/shm
tmpfs           190M  2.8M  188M   2% /run
/dev/xvda1      8.0G  1.5G  6.5G  19% /
tmpfs           475M     0  475M   0% /tmp
tmpfs            95M     0   95M   0% /run/user/1000
[root@ip-172-31-87-199 ec2-user]#
```

The above command will list the amount of free disk space available in the server. You can also see that the drive volume which we have increased will not be added to it.

8. Now run the following command to see the list of all the block devices that are associated with the instance (which will show the increased size of the partition volume):

```
lsblk
```

```
[root@ip-172-31-87-199 ec2-user]# lsblk
NAME        MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
xvda        202:0    0  20G  0 disk
├─xvda1     202:1    0   8G  0 part /
├─xvda127   259:0    0   1M  0 part
└─xvda128   259:1    0  10M  0 part
[root@ip-172-31-87-199 ec2-user]#
```

The root volume has a partition /dev/xvda1 whereas the root volume (/dev/xvda) reflects the new size. This tells us the partition size needs to be extended.

9. Extend the size by using the following command:

```
growpart /dev/xvda 1
```

10. Find the file system of the block devices associated with the EC2 instance by entering the following command:

```
lsblk -f
```

```
[root@ip-172-31-87-199 ec2-user]# growpart /dev/xvda 1
CHANGED: partition=1 start=24576 old: size=16752607 end=16777183 new: size=41918431 end=41943007
[root@ip-172-31-87-199 ec2-user]# lsblk -f
NAME       FSTYPE FSVER LABEL UUID                                 FSAVAIL FSUSE% MOUNTPOINTS
xvda
├─xvda1    xfs          /     55a1aebd-f196-4f84-8afe-075f5d1dda63    6.4G    19% /
├─xvda127
└─xvda128 vfat   FAT16        0383-1543
[root@ip-172-31-87-199 ec2-user]#
```

11. The above command will provide us the file system of the block devices which will be one among the following types: ext2,ext3,ext4 or XFS

Our file system type is ext4, so for this file system, use the below command (also for ext2,ext4):

```
resize2fs /dev/xvda1
```

Note: If the file system is XFS then run the below command

```
xfs_growfs /dev/xvda1
```

```
[root@ip-172-31-87-199 ec2-user]# xfs_growfs /dev/xvda1
meta-data=/dev/xvda1             isize=512    agcount=2, agsize=1047040 blks
         =                       sectsz=4096  attr=2, projid32bit=1
         =                       crc=1        finobt=1, sparse=1, rmapbt=0
         =                       reflink=1    bigtime=1 inobtcount=1
data     =                       bsize=4096   blocks=2094075, imaxpct=25
         =                       sunit=128    swidth=128 blks
naming   =version 2              bsize=16384  ascii-ci=0, ftype=1
log      =internal log           bsize=4096   blocks=16384, version=2
         =                       sectsz=4096  sunit=4 blks, lazy-count=1
realtime =none                   extsz=4096   blocks=0, rtextents=0
data blocks changed from 2094075 to 5239803
[root@ip-172-31-87-199 ec2-user]#
```

12. Now find the effective file system extended volume size,

```
df -h
```

```
[root@ip-172-31-87-199 ec2-user]# df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M     0  4.0M   0% /dev
tmpfs           475M     0  475M   0% /dev/shm
tmpfs           190M  2.8M  188M   2% /run
/dev/xvda1       20G  1.6G   19G   8% /
tmpfs           475M     0  475M   0% /tmp
tmpfs            95M     0   95M   0% /run/user/1000
[root@ip-172-31-87-199 ec2-user]#
```

13. You have successfully launched an EC2 instance and re-sized the root EBS volume of the instance with zero downtime.

Before

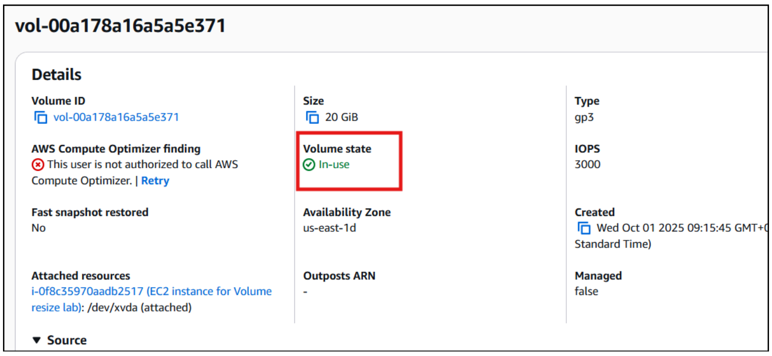
```
[root@ip-172-31-91-202 ec2-user]# df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        482M     0  482M   0% /dev
tmpfs           492M     0  492M   0% /dev/shm
tmpfs           492M  400K  492M   1% /run
tmpfs           492M     0  492M   0% /sys/fs/cgroup
/dev/xvda1      8.0G  1.5G  6.6G  19% /
tmpfs            99M     0   99M   0% /run/user/1000
```

After

```
[root@ip-172-31-91-202 ec2-user]# df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        482M     0  482M   0% /dev
tmpfs           492M     0  492M   0% /dev/shm
tmpfs           492M  400K  492M   1% /run
tmpfs           492M     0  492M   0% /sys/fs/cgroup
/dev/xvda1       20G  1.5G   19G   8% /
tmpfs            99M     0   99M   0% /run/user/1000
```

14. Similarly, we can resize the secondary volume associated with the instance in the same way.

Note: Make sure that the Volume state is <u>In-Use</u> before validating the lab.


vol-00a178a16a5a5e371
Details
Volume ID
vol-00a178a16a5a5e371
Size
20 GiB
Type
gp3
AWS Compute Optimizer finding
This user is not authorized to call AWS Compute Optimizer. | Retry
Volume state
In-use
IOPS
3000
Fast snapshot restored
No
Availability Zone
us-east-1d
Created
Wed Oct 01 2025 09:15:45 GMT+0
Standard Time)
Attached resources
i-0f8c35970aadb2517 (EC2 instance for Volume resize lab): /dev/xvda (attached)
Outposts ARN
-
Managed
false
Source

Lab Introduction to Amazon Simple Storage Service (S3)

Task 1: Sign in to AWS Management Console

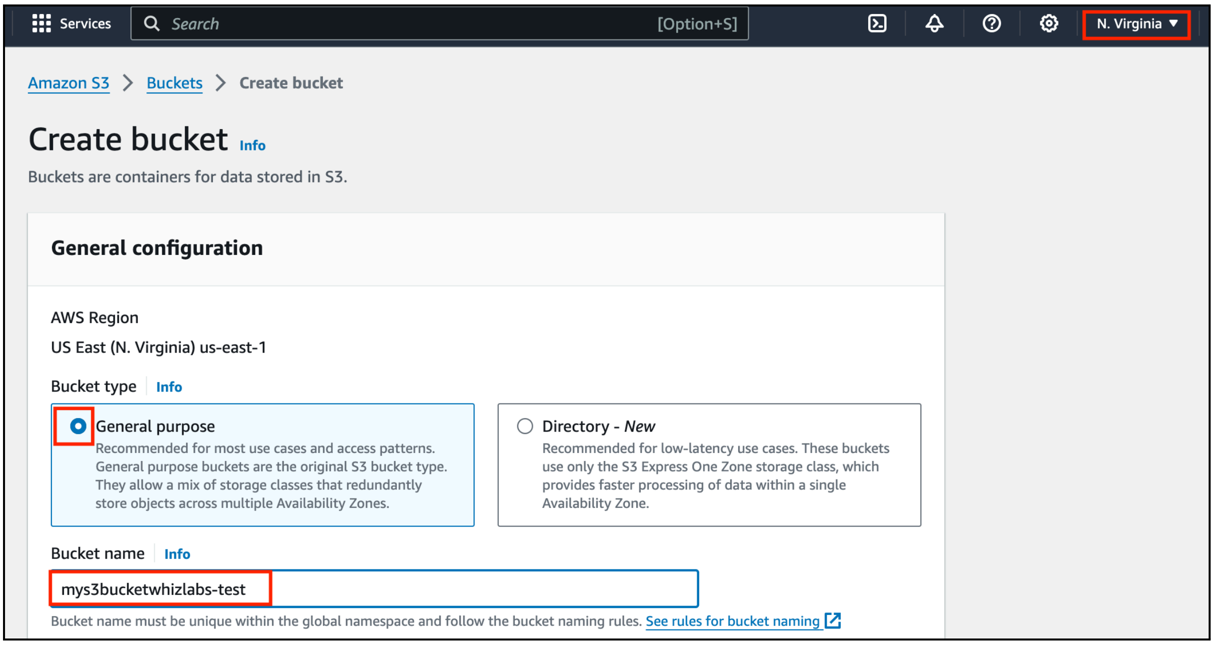
1. Click on the Open Console button, and you will get redirected to AWS Console in a new browser tab.
2. On the AWS sign-in page,
   - Leave the Account ID as default. Never edit/remove the 12 digit Account ID present in the AWS Console. otherwise, you cannot proceed with the lab.
   - Now copy your User Name and Password in the Lab Console to the IAM Username and Password in AWS Console and click on the Sign in button.

3. Once Signed In to the AWS Management Console, Make the default AWS Region as US East (N. Virginia) us-east-1.

Task 2: Creating an S3 Bucket

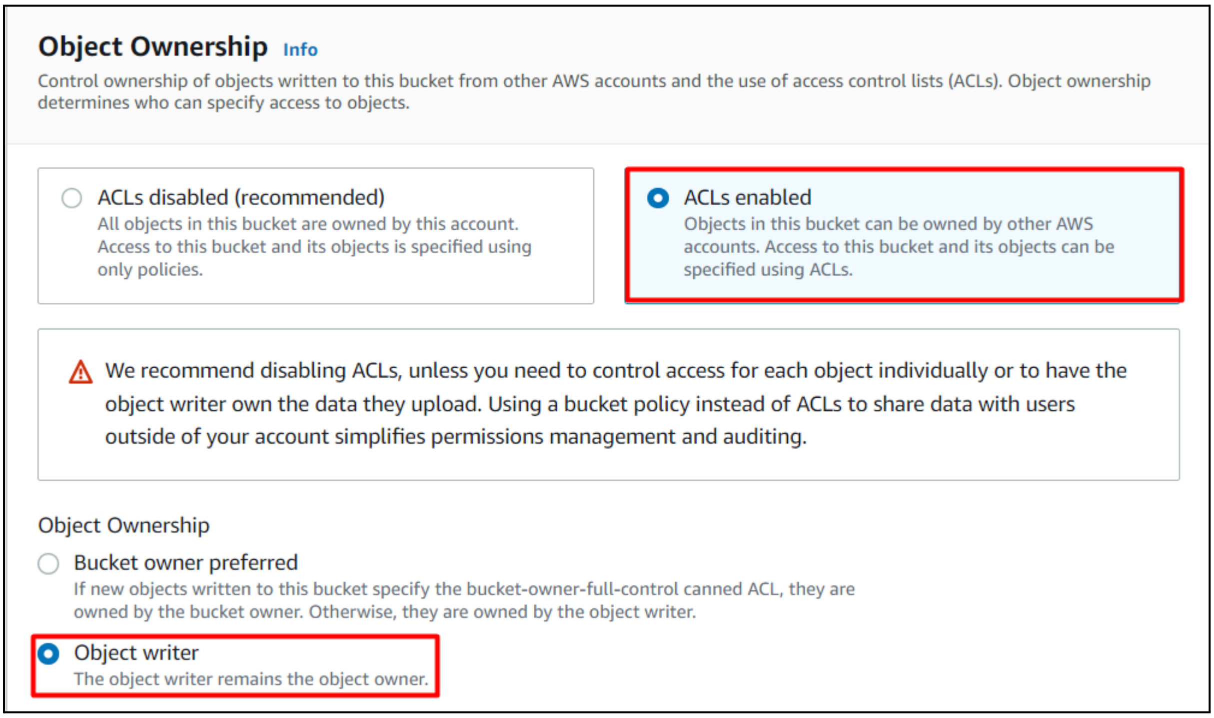
In this task, we are going to create a S3 bucket by providing the required configurations such as name, region and ACLs.

1. Ensure you are in the US East (N. Virginia) us-east-1 Region to begin creating an S3 bucket in the Amazon cloud.
2. Navigate to the Services menu at the top. Click on S3 in the Storage section.



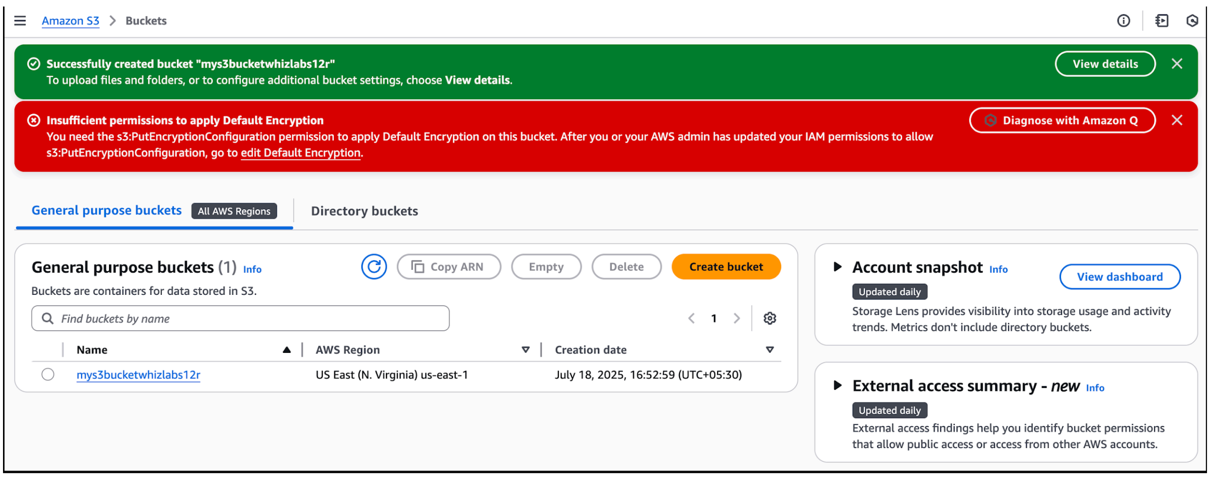
3. On the S3 Page, click on Create bucket and fill in the bucket details.
   - Select Bucket Type : General purpose
   - Bucket name: Enter ***Unique name of your choice***
     - Note: Ensure you provide a unique bucket name of your choice, as **S3 bucket names must be globally unique.**
   - For Object ownership: Select ACLs enabled

- Object ownership option: Choose Object writer



- Note: Please select Object owner, or else it won't allow editing the bucket ACL.
- For Block Public Access settings for this bucket for this bucket section,
    - Uncheck the option, Block all public access,
        - Check the I acknowledge that the current settings might result in this bucket and the objects within becoming public checkbox.

- Leave other settings as default.
- Click on Create bucket button.
- Kindly ignore the permission error if it appears during the process.



4. The S3 bucket will be created successfully.

Name	Region	Access
mys31bucketwhizlabs	US East (N. Virginia) us-east-1	Objects can be public

Task 3: Upload an object to S3 bucket

1. Click on your bucket name.
2. In the Overview, You can see the following message:
   - You don't have any objects in this bucket.

3. You can upload any image from your local machine or download the image from this Link.
4. To upload a file to our S3 bucket,
   - Click on the Upload button.
   - Click on Add files button.
   - Browse for your local image or the image we provided and select it.
   - Click on the Upload button.
   - You can watch the progress of the upload from within the transfer panel at the top of the screen.
   - Once your file has been uploaded, it will be displayed in the bucket.

5. Now click on the Close button on the top right corner of the screen.

Task 4: Change Bucket Permissions

In this task, we are going to change the permissions of the bucket to make the image publicly available.

1. Under Objects, Click on smiley.jpg, You will see image details like owner, size, link, etc.
2. A URL will be listed under the Object URL
3. Copy the Object URL and paste it in the browser in a new tab.
   - You will see an AccessDenied message, which means the object is not publicly accessible.

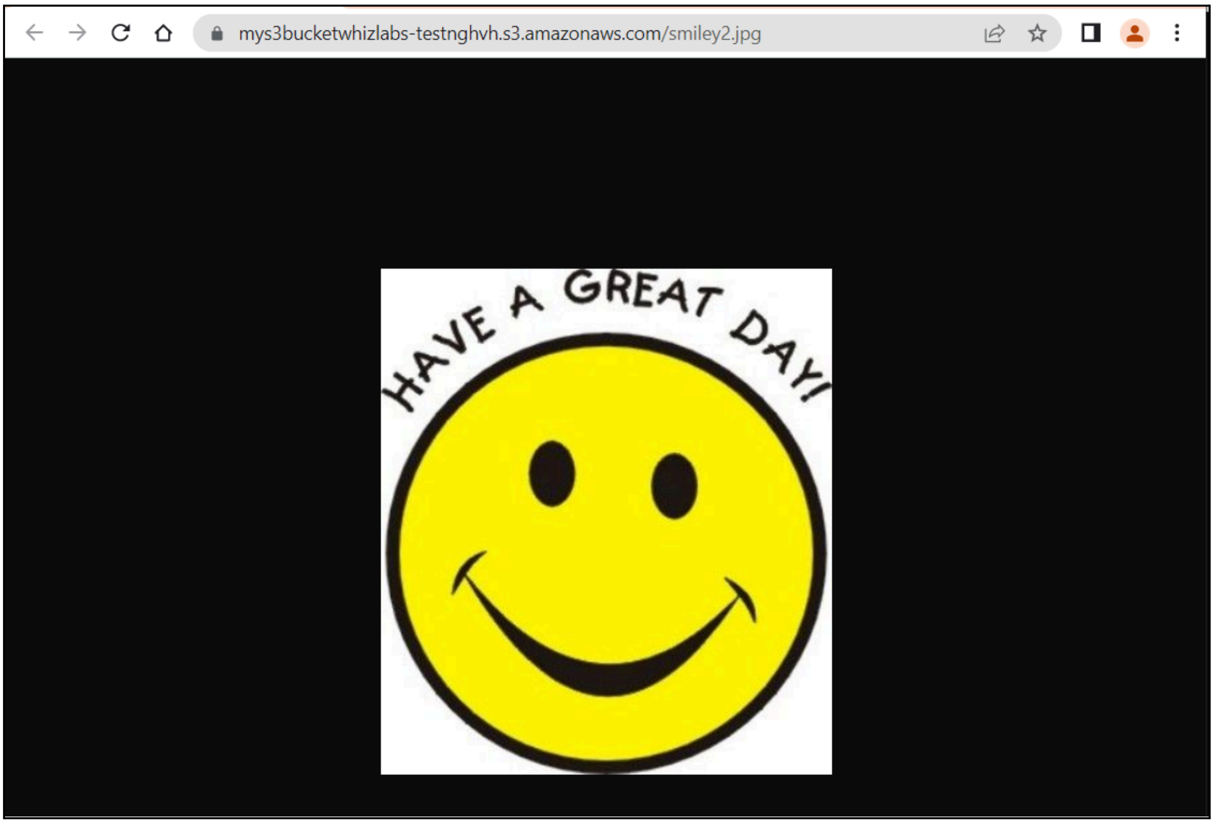
This XML file does not appear to have any style information associated with it. The document tree is shown b

```
▼<Error>
   <Code>AccessDenied</Code>
   <Message>Access Denied</Message>
   <RequestId>N1XBBWP7ZATY7ETY</RequestId>
   <HostId>wsbFnpNWePMkRsfH3GazXXv9F7hx2xJCGYz7cruEG7WLA5OHvdZhAM+vXuC1lniCiEFL+M9emTQ=</HostId>
 </Error>
```

4. Go to your smiley.jpg object and navigate to the Permissions tab.

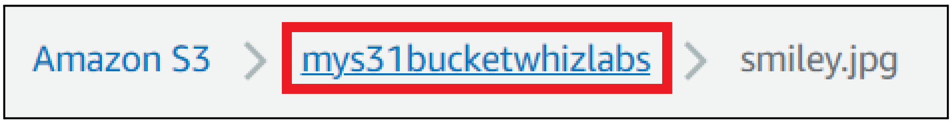
5. Now click on the Edit button on the right side.
6. Note: If the Edit button is disabled, please change the bucket ownership to ACLs enabled and choose the Object writer as the owner.
7. Everyone (public access) : Check the Read checkbox under Objects column.
8. Now scroll a little bit below and check I understand the effects of these changes on this object checkbox.

9. Now, Scroll to the bottom and click on Save changes button.
10. Now again paste the Object URL in the browser and you can see the image you have uploaded.



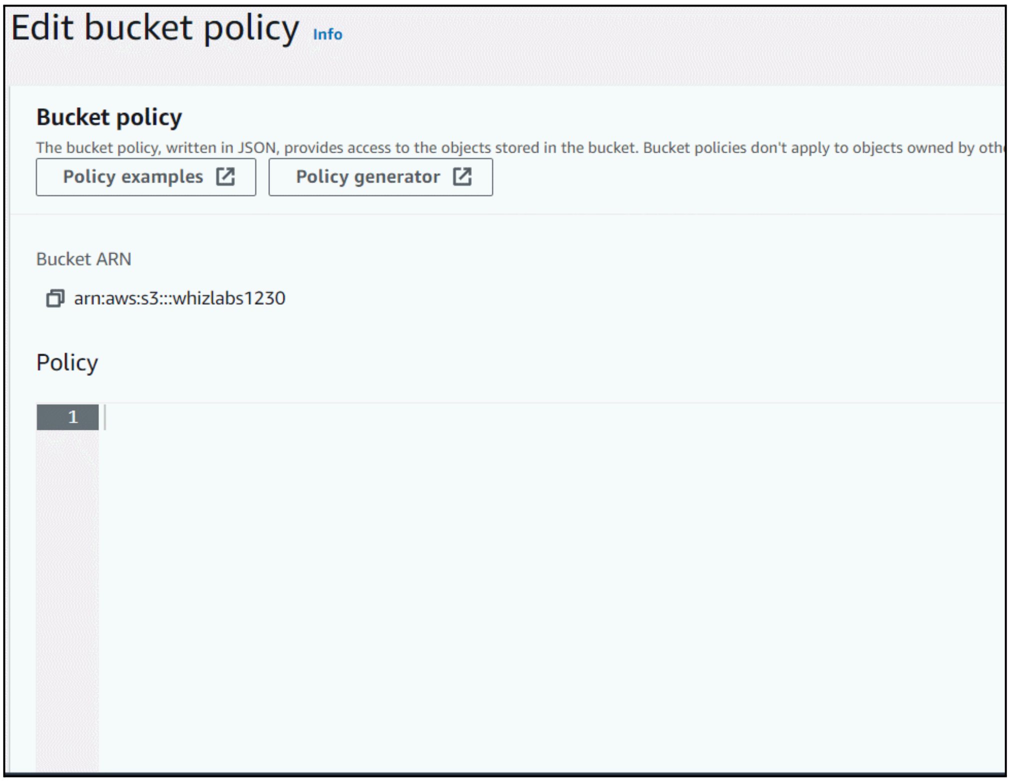
Task 5: Create a Bucket Policy

1. In the previous step, you granted read access only to a specific object. If you wish to make all objects inside a bucket available publicly, you can achieve this by creating a Bucket policy.
2. Go to the bucket by clicking on your bucket name - mys3bucketwhizlabs on the top.



3. Click the Permissions tab, then configure the following
    - Scroll down to Bucket policy, click on Edit button on the Right side.
    - A blank Bucket policy editor is displayed.
    - Copy the ARN of your bucket to the clipboard.
        - arn:aws:s3:::mys3bucketwhizlabs-test
4. Copy the entire policy, paste it into the bucket policy, and replace your bucket ARN with the ARN listed in the JSON below.

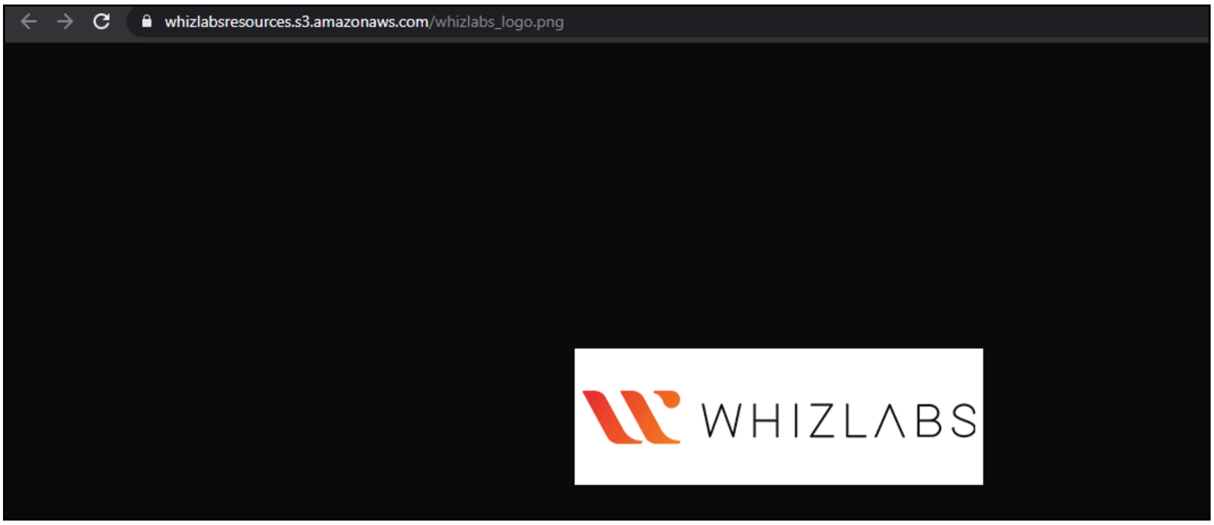
```json
{
  "Id": "Policy1",
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "Stmt1",
      "Action": [
        "s3:GetObject"
      ],
      "Effect": "Allow",
      "Resource": "replace-this-string-with-your-bucket-arn/*",
      "Principal": "*"
    }
  ]
}
```



- Click on Save changes button.

Task 6: Test Public Access

1. Go to the bucket and upload another image; download the Whizlabs logo image from Download Me
    - Click on the Upload button.
    - Click on Add files button.
    - Browse for the downloaded image we provided and select it.
    - Click on the Upload button.
2. Once the image is uploaded successfully, click on the object name (Image name) , copy the Object URL, and open it in a browser.
    - https://whizlabsresources.s3.amazonaws.com/whizlabs-logo.png



3. You can see your image loaded successfully and is publicly accessible.

Lab Introduction to Amazon Lambda

Task 1: Sign in to AWS Management Console

1. Click on the Open Console button, and you will get redirected to AWS Console in a new browser tab.
2. On the AWS sign-in page,
   - Leave the Account ID as default. Never edit/remove the 12 digit Account ID present in the AWS Console. otherwise, you cannot proceed with the lab.
   - Now copy your User Name and Password in the Lab Console to the IAM Username and Password in AWS Console and click on the Sign in button.
3. Once Signed In to the AWS Management Console, Make the default AWS Region as US East (N. Virginia) us-east-1.

Task 2: Create Two Amazon S3 Buckets

In this task, we will create two AWS S3 buckets i.e the source bucket and the destination bucket by providing the required configurations like name, region etc.

1. Navigate to the Services menu in the top, then click on S3 in the Storage section.
2. Click on Create Bucket button.

3. Bucket Name: Enter *mysourcebucket1234567*

Note: Every S3 bucket name is unique globally, so you can enter any unique name of your choice.

4. Leave other settings as default and click on the Create bucket button.

5. Copy the name of the bucket and save it in a text file for later use.

6. Create Destination Bucket

   - Click on Create bucket button.
   - Bucket Name: Enter *mydestinationbucket1234567*
     Note: Every S3 bucket name is unique globally, so you can enter any unique name of your choice.

7. Leave other settings as default and click on the Create bucket button.

8. Copy the name of the bucket and save it in a text file for later use.

9. Now we have two S3 buckets (Source and Destination). We will make use of our AWS Lambda function to copy the content from source bucket to destination bucket.

Task 3: Create a Lambda Function

1. Make sure you are in the US East (N. Virginia) region.
2. Go to the Services menu and click on Lambda under Compute section.
3. Click on the Create a function button.
    - Choose Author from scratch
    - Function name : Enter *mylambdafunction*
    - Runtime : Select **Python 3.13**

- Role: In the permissions section, click on Change default execution role and then select Use an existing role.
- Existing role: Select **Lambda_role** from the list.

4. Click on the Create function button.
5. If you scroll down a little bit, you can see the Code source section. Here we need to write a Python function that copies the object from the source bucket and paste it into the destination bucket.
6. Remove the existing code in AWS lambda lambda_function.py. Copy the below code and paste it into your **lambda_function.py** file.

```
import boto3

def lambda_handler(event, context):
    s3 = boto3.client('s3')
    source_bucket = "Source_Bucket_name"
    destination_bucket = "Destination_Bucket_name"

    # Extract the object key from the event
    object_key =
event['Records'][0]['s3']['object']['key']

    # URL encode the source object key
    copy_source = {'Bucket': source_bucket, 'Key':
object_key}

    # Set up the parameters for the copy operation
    copy_params = {
        'Bucket': destination_bucket,
        'CopySource': copy_source,
        'Key': object_key
    }

    try:
        # Perform the copy operation
        result = s3.copy_object(**copy_params)
        print("S3 object copy successful. Response:",
result)
```

```
        except Exception as e:
            print("Error copying object:", str(e))
```

7. You need to change the source and destination bucket name in the index.js file line number 5 & 6 which we copied earlier.
8. Save the function by clicking on Deploy button.

Task 4: Adding Triggers to Lambda Function

1. Scroll up to go to Function overview and click on + Add trigger button.

2. Scroll down the list and select S3 from the trigger list. Once you select S3, a form will appear. Enter these details:

- Bucket: Select your source bucket - mysourcebucket1234567
- Event type: Select All object create events
- Leave other fields as default.
- And check the option of Recursive invocation to avoid failures in case you upload multiple files at once by Checking the acknowledge option.

- Click on Add button.

Task 5: Test Lambda function

1. If you have a test image on your local machine, you can use that image.
    - Click on Download Me to open the image in new tab, right click the image and save it to your local.

      Note: Please name the image file as "image.jpeg" without using variations like "image (2).jpeg".
2. Go to S3 Bucket list and click on source bucket - mysourcebucket1234567.
3. Upload image to source S3 bucket. To do that:
    - Click on the Upload button.
    - Click on Add files button to add the files.
    - Select the image and click on the Upload to upload the image.

4. Now go back to the S3 list and open your destination bucket - mydestinationbucket1234567.
5. You can see a copy of your uploaded source bucket image in the destination bucket.

Optional: Debugging Lambda Function Using CloudWatch

If you encounter errors such as the source file not being copied to the destination S3 bucket, please debug your Lambda function by following the steps mentioned below:

1. From the AWS Management Console, navigate to the **CloudWatch** service.
2. In the left sidebar, under the **"Logs"** menu, select **"Log Management.**

3. Select your Lambda function's **log group,** which should be in the format **/aws/lambda/function-name.**

4. Choose a **log stream** to view its logs.

5. In the list of **Log events,** select the right arrow next to the Timestamp to expand the details for that event.

Lab Steps How to enable versioning Amazon S3

Task 1: Sign in to AWS Management Console

1. Click on the Open Console button, and you will get redirected to AWS Console in a new browser tab.
2. On the AWS sign-in page,
   - Leave the Account ID as default. Never edit/remove the 12 digit Account ID present in the AWS Console. otherwise, you cannot proceed with the lab.
   - Now copy your User Name and Password in the Lab Console to the IAM Username and Password in AWS Console and click on the Sign in button.

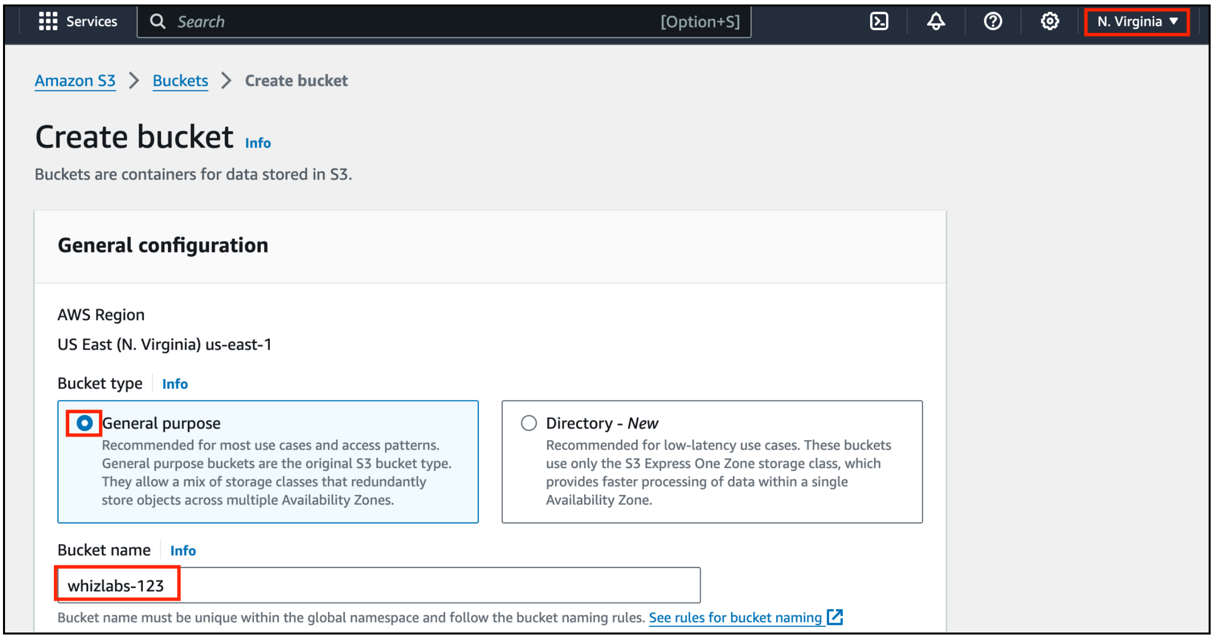
3. Once Signed In to the AWS Management Console, Make the default AWS Region as US East (N. Virginia) us-east-1.

Task 2: Create an S3 Bucket

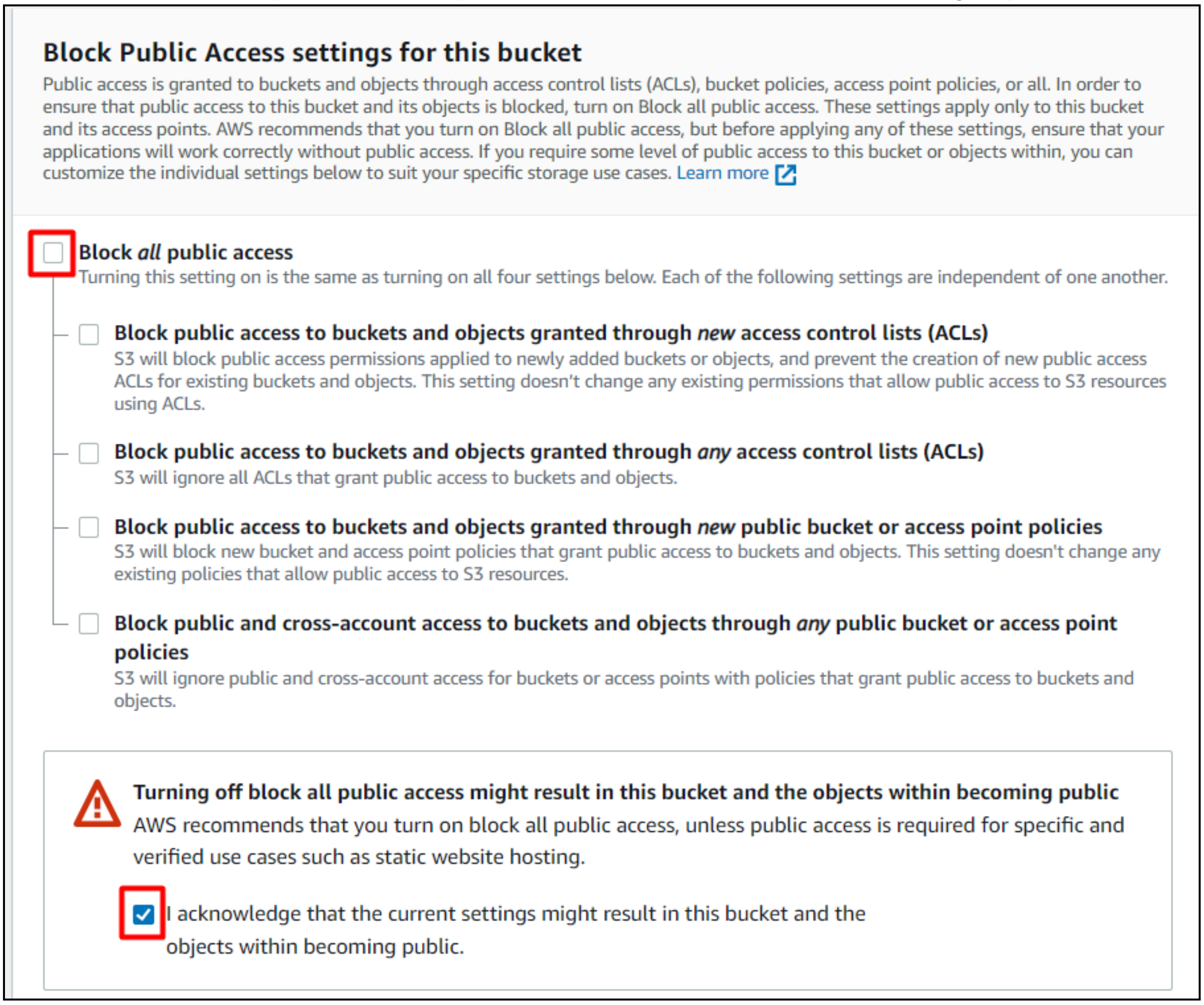
In this task , we are going to create an S3 bucket with the required configurations like name, access and region.

1. Navigate to the Services menu at the top and click on S3 in the Storage section.
2. On the S3 dashboard, click on Create bucket button and fill in the bucket details.
3. Select Region as N.Verginia to create an S3 bucket.
   - Bucket type Select : General purpose
   - Bucket name: Enter *whizlabs-<RANDOM_NUMBER>*

- Note: S3 bucket names are globally unique, choose an available name.

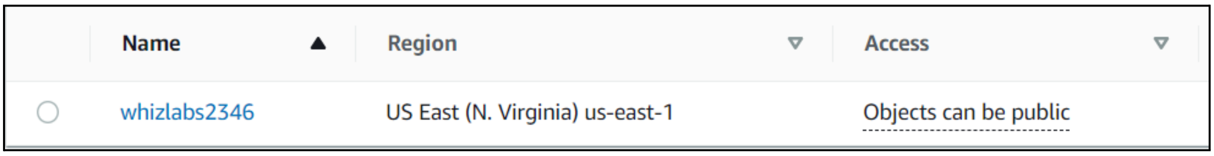


- Object ownership: Select ACLs disabled (recommended) option
- Uncheck the option, Block all public access, and check the acknowledge option.



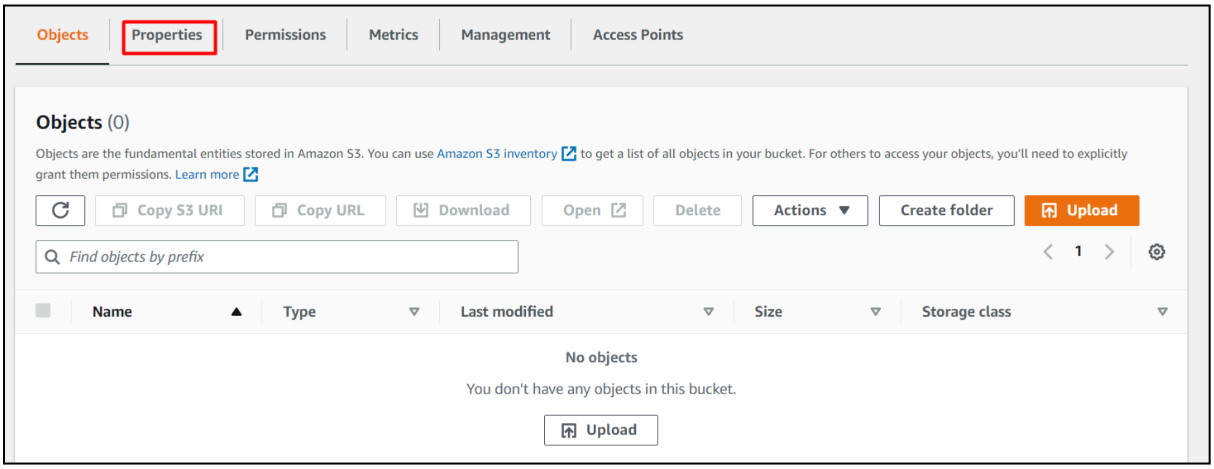
- Leave other settings as default.
- Click on Create bucket button.

4. Your S3 Bucket is now created.



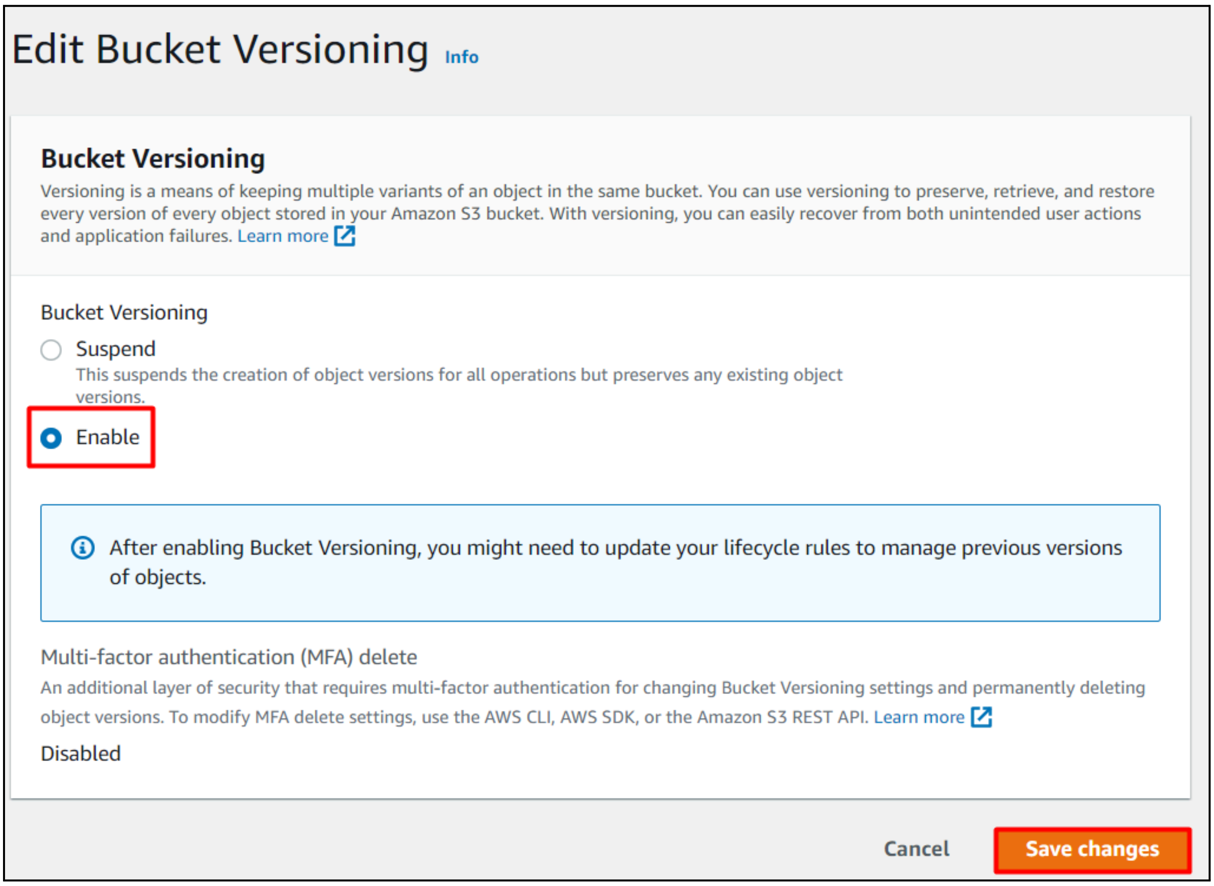
Task 3: Enable Versioning on the S3 bucket

1. Click on your bucket name whizlabs2346.
2. Click on Properties tab.



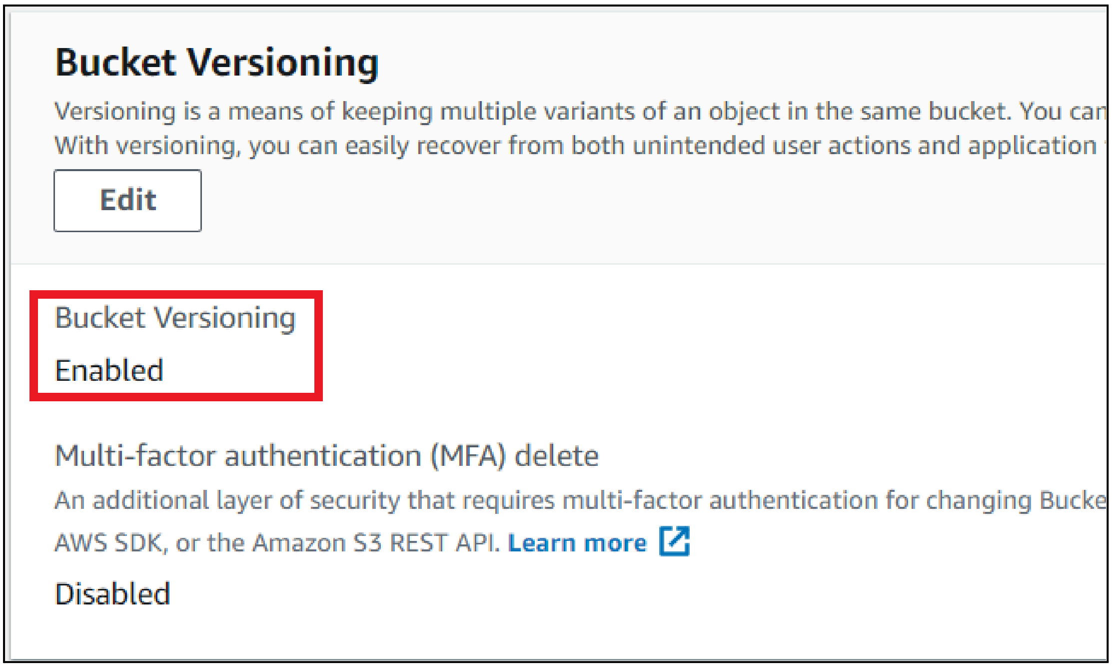
3. Under Bucket Versioning, click on Edit button.

4. Bucket Versioning : Select Enable radio button.



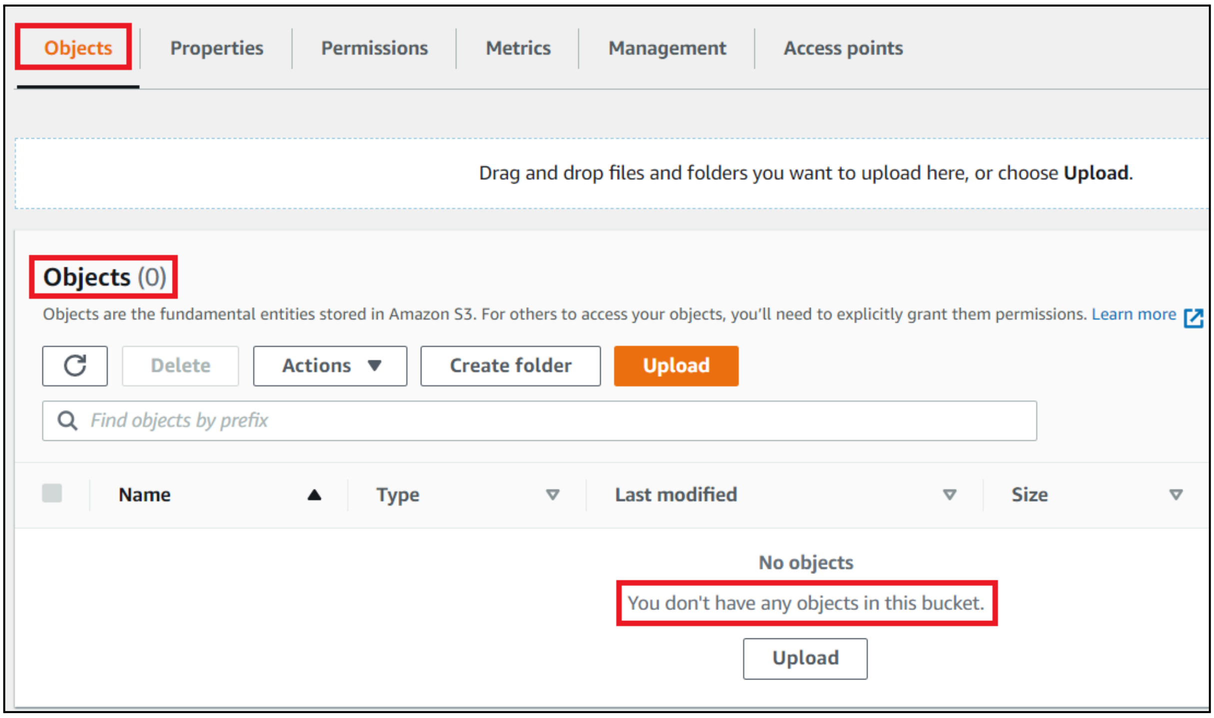
5. Now Click on the Save changes button.

6. Now Bucket versioning is enabled.



Task 4: Upload an object and make the bucket public

1. Click on the Objects tab.
2. In the Overview, you can see the following message:
   - This bucket is empty. Upload new objects to get started.



3. Upload sample.txt file present in zip file download the zip file from our Download Me link.
4. To upload a file to our S3 bucket,

- Click on the Upload button.
- Click on the Add files button.
- Browse for the file you want to upload.
- Click on the Upload button.
- You can watch the progress of the upload from within the transfer panel at the bottom of the screen. If it's a small file, you might not see the transfer. Once your file has been uploaded, it will be displayed in the bucket.

	Name ▲	Type ▽	Last modified
☐	sample.txt	txt	December 15, 2020, 20:07 (UTC+05:30)

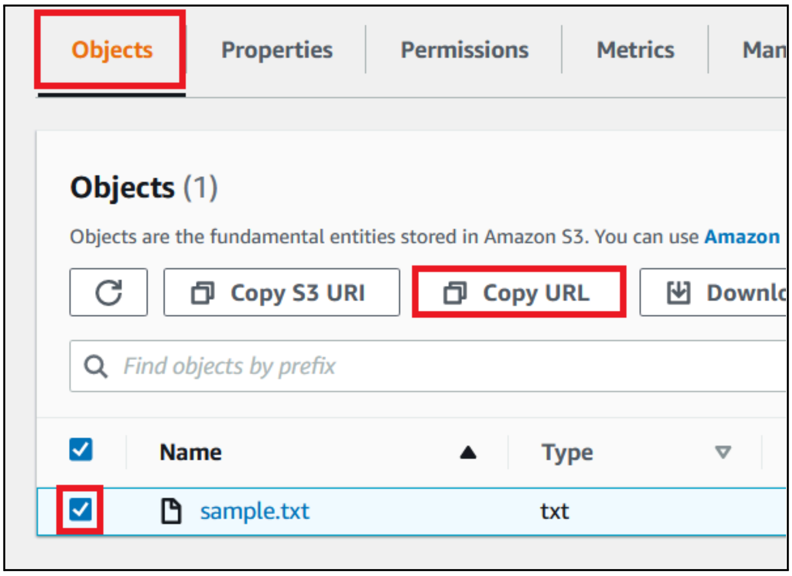
5. Now click on the Close button on the top right corner of the screen.
6. To copy the ARN of your S3 bucket, click on Properties tab and copy the ARN.
7. Make the bucket public with a Bucket Policy:
    - Click the Permissions tab to configure your bucket:
    - Scroll down to Bucket policy , click on Edit button on the Right side.
    - The policy editor will open.
- In the policy below, update your bucket ARN in the Resource key-value and copy the policy code.
    - Paste the bucket policy into the Bucket policy editor.

```
{
"Id": "Policy1",
"Version": "2012-10-17",
"Statement": [
{
"Sid": "Stmt1",
"Action": [
"s3:GetObject"
],
"Effect": "Allow",
"Resource": "replace-this-string-from-your-bucket-arn/*",
```

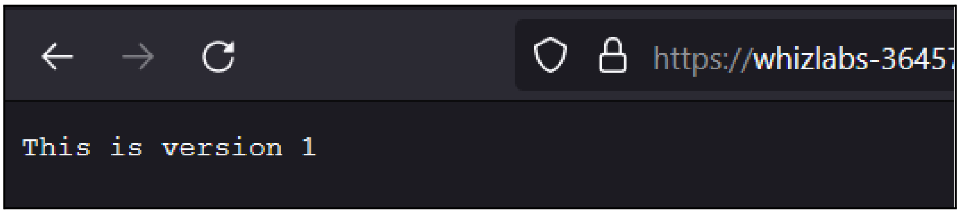
```
"Principal": "*"

}

]

}
```

- Click on Save Changes button

8. Now again open the Objects tab, select the object name and click on Copy URL.

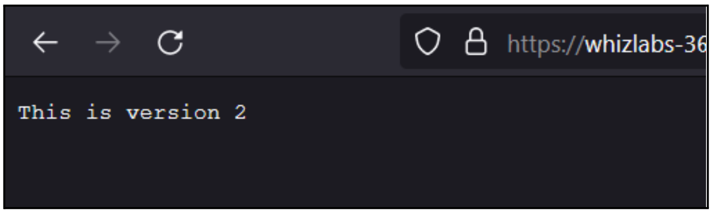


9. Open a new tab, paste the URL and you will get the below output.



Task 5: Upload different versions of the file

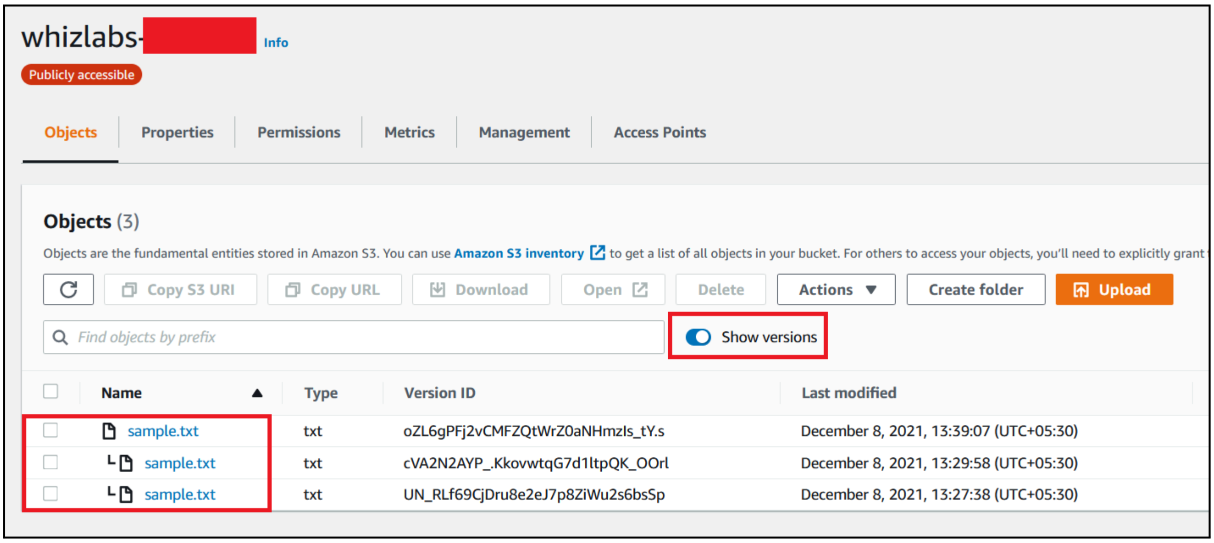
1. Now open the sample.txt file in your local text editor and update the file content as “*This is version 2*”.
2. Click on the Upload button.
3. Click on the Add files button.
4. Browse for the file you want to upload.
5. Click on the Upload button.
6. Now click on the Close button on the top right corner of the screen.
7. Once the file has been uploaded successfully, copy the URL and paste it into your browser.
8. You should see the latest version of the file you uploaded.



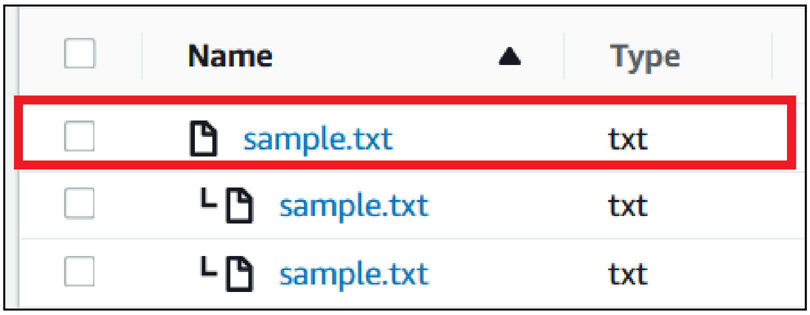
9. Similarly, again edit the sample.txt and update the content as "*This is version 3*".
10. Then upload the file into the bucket and check the output in the browser.

Task 6: Test the Versioning of the object

1. Open your S3 bucket, click on Show versions. You will get the list of versions available for your sample.txt.



2. Now delete the first object and try to see the output in the browser.



3. You will get the output as "This is version 2".
4. Hence, when you delete the object from the bucket, the bucket pointer points to the previously available version.

Lab Mount Elastic File System (EFS) on EC2

Task 1: Sign in to AWS Management Console

1. Click on the Open Console button, and you will get redirected to AWS Console in a new browser tab.
2. On the AWS sign-in page,
    - Leave the Account ID as default. Never edit/remove the 12 digit Account ID present in the AWS Console. otherwise, you cannot proceed with the lab.
    - Now copy your User Name and Password in the Lab Console to the IAM Username and Password in AWS Console and click on the Sign in button.
3. Once Signed In to the AWS Management Console, Make the default AWS Region as US East (N. Virginia) us-east-1.

Note : If you face any issues, please go through FAQs and Troubleshooting for Labs.

Task 2: Launching two EC2 Instances

1. Make sure you are in the N.Virginia Region.
2. Navigate to the Services menu in the top, then click on EC2 in the Compute section.
3. Click on Instances from the left side bar and then click on Launch instances.
4. Number of Instances : Enter *2* on the right side under summary
5. Name : Enter *MyEC2*
6. For Amazon Machine Image (AMI): Search for Amazon Linux 2023 kernel-6.1 AMI in the search box and click on the Select button.

7. **For Instance Type:** select *t2.micro*

8. For Key pair: Select Create a new key pair Button
    - Key pair name: MyEC2Key
    - Key pair type: RSA
    - Private key file format: .pem
9. Select Create key pair Button.
10. In Network Settings Click on Edit:

- Auto-assign public IP: Enable
- Select Create new Security group
- Security Group Name: Enter *EFS Security Group*
    - To add SSH:
        - Choose Type: SSH
        - Source: Anywhere

- For NFS:
    - Click on Add security group rule
    - Choose Type: NFS
    - Source: Anywhere

11. Keep Rest thing Default and Click on Launch Instance Button.
12. Select View all Instances to View Instance you Created
13. Launch Status: Your instance is now launching. Click on the instance ID and wait until the initialization status changes to Running.
14. Click on each instance and enter a names as *MyEC2-1* and *MyEC2-2.*

15. Take note of the IPv4 Public IP Addresses of the EC2 instances and save them for later.

Task 3: Creating an Elastic File System

1. Navigate to EFS by clicking on the Services menu at the top. Click on EFS in the Storage section.
2. Click on Create file system
3. Click on Customize button.
4. Enter the details below, Type the Name as *EFS-Demo* and make sure default VPC and default Regional options are selected.

5. Uncheck the option of Enable automated backups
6. Leave everything by default and click on the Next button present below.
7. Network Access:

- VPC
    - An Amazon EFS file system is accessed by EC2 instances running inside one of your VPCs.
    - Choose the same VPC you selected while launching the EC2 instance (leave as default).
- Mount Targets
    - Instances connect to a file system by using a network interface called a mount target. Each mount target has an IP address, which we assign automatically or you can specify.
    - We will select all the Availability Zones (AZ's) so that the EC2 instances across your VPC can access the file system.
    - Select **all the Availability Zones**, and in the Security Groups, select EFS Security Group instead of the default value.
    - Scroll down to '**Mount Targets**', then click on **'Add Mount Target'**. Select the availability zone and change the security group to **EFS Security Group.**
    - Make sure you remove the default security group and select the EFS Security Group, otherwise you will get an error in further steps.
    - Click on Next button

8. File system policy - optional let it be optional only. Click on Next button
9. Review and Create: Review the configuration below before proceeding to create your file system. Click on Create button.
10. Congratulations on creating the EFS File system, It's time to mount your EC2 Instance with the EFS File system.

Task 4: Mount the File System to MyEC2-1 Instance

1. Select the MyEC2-1 Instance and copy the IPv4 Public IP.
2. SSH into the EC2 Instance
   - Please follow the steps in SSH into EC2 Instance.
3. Switch to root user

```
sudo -s
```

5. Run the updates using the following command:

```
yum -y update
```

7. Install the NFS client as amazon-efs-utils.

```
yum install -y amazon-efs-utils
```

9. Create a directory by the name efs

```
mkdir efs
```

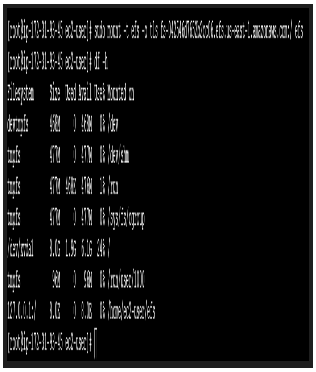
11. We have to mount the file system in this directory.
12. To do so, navigate to the AWS console and click on the created file system. On the top-right corner, click on Attach
    - Copy the command of Using the EFS mount helper.

```
sudo mount -t efs -o tls <DNS Name>:/ efs
```

    - Note: Replace <dns name of EFS> with the DNS of the Elastic File System you've created.

9. To display information for all currently-mounted file systems, we'll use the command bellow:

```
df -h
```



10. Create a directory in our current location:

```
mkdir aws
```

Task 5: Mount the File System to MyEC2-2 Instance

1. Select the MyEC2-2 Instance and copy the IPv4 Public IP.
2. SSH into the EC2 Instance
   - Please follow the steps in SSH into EC2 Instance.
3. Switch to root user

```
sudo -s
```

5. Run the updates using the following command:

```
yum -y update
```

7. Install the NFS client as amazon-efs-utils.

```
yum -y install amazon-efs-utils
```

9. Create a directory with the name efs

```
mkdir efs
```

11. We have to mount the file system in this directory.
12. To do so, navigate to the AWS console and click on the created file system. On the top-right corner, click on Attach
    - Copy the command of Using the EFS mount helper.

    ```
    sudo mount -t efs -o tls <DNS Name>:/ efs
    ```

    - Note: Replace <dns name of EFS> with the DNS of the Elastic File System you've created.

- To display information for all currently mounted file systems, we'll use the command

  ```
  df -h
  ```

```
[root@ip-172-31-89-166 ec2-user]# mkdir efs
[root@ip-172-31-89-166 ec2-user]# sudo mount -t efs -o tls fs-042546d7652b2cc06.efs.us-east-1.amazonaws.com:/ efs
[root@ip-172-31-89-166 ec2-user]# df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        468M     0  468M   0% /dev
tmpfs           477M     0  477M   0% /dev/shm
tmpfs           477M  468K  476M   1% /run
tmpfs           477M     0  477M   0% /sys/fs/cgroup
/dev/xvda1      8.0G  1.9G  6.1G  24% /
tmpfs            96M     0   96M   0% /run/user/1000
127.0.0.1:/     8.0E     0  8.0E   0% /home/ec2-user/efs
[root@ip-172-31-89-166 ec2-user]#
```

Task 6: Testing the File System

1. SSH into both instances in a side-by-side view on your machine, if possible.
2. Switch to root user

```
sudo -s
```

4. Navigate to the efs directory in both the servers using the command

```
cd efs
```

6. Create a file in any one server.

```
touch hello.txt
```

8. Check the file using the command

```
ls -ltr
```

10. Now go to the other server and give the command

```
ls -ltr
```

12. You can see the file created on this server as well. This proves that our EFS is working.
13. You can try creating files (touch command) or directories (mkdir command) on other servers to continue to grow the EFS implementation.

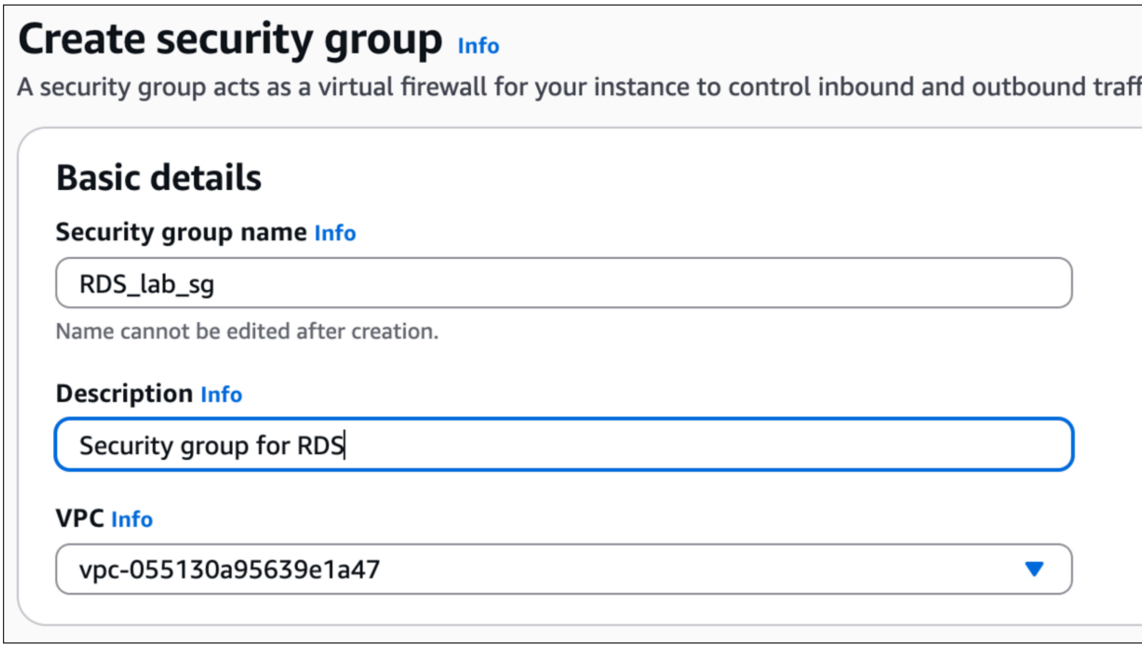
Lab Introduction to AWS Relational Database Service

Task 1: Sign in to AWS Management Console

1. Click on the Open Console button, and you will get redirected to AWS Console in a new browser tab.
2. On the AWS sign-in page,
   - Leave the Account ID as default. Never edit/remove the 12 digit Account ID present in the AWS Console. otherwise, you cannot proceed with the lab.
   - Now copy your User Name and Password in the Lab Console to the IAM Username and Password in AWS Console and click on the Sign in button.
3. Once Signed In to the AWS Management Console, Make the default AWS Region as US East (N. Virginia) us-east-1.

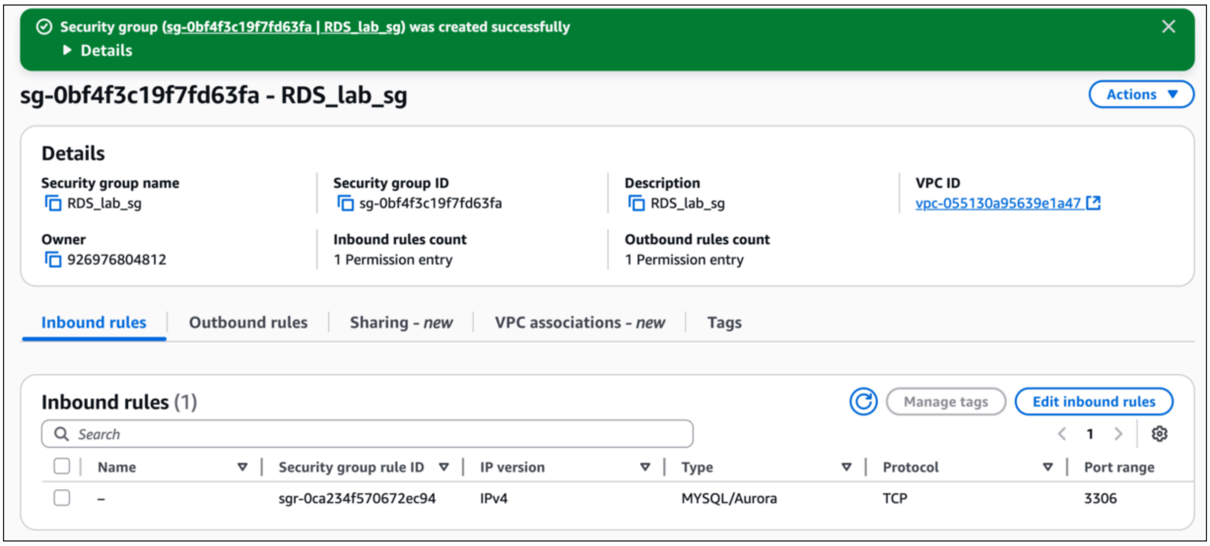
Task 2 : Create a Security Group for RDS instance

1. Make sure you are in the N.Virginia Region.
2. Navigate to EC2 by clicking on the Services menu available under the Compute section.
3. On the left panel menu, select the security group under the Network & Security section.
4. Click on the Create Security Group.
5. We are going to create a Security group for RDS with a 3306 port number enabled.
   - Security group name : Enter *RDS_lab_sg*
   - Description: Enter *Security group for RDS*
   - VPC: Select Default VPC



- Click on the Add Rule button under Inbound rules.
    - Type: Select MYSQL/Aurora
    - Source: Select Anywhere Ipv4

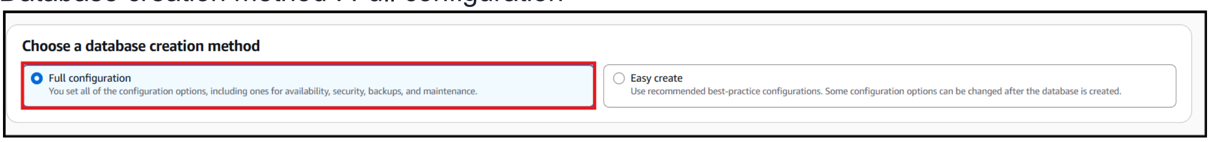
6. Leave everything as default and click on the Create Security Group button.



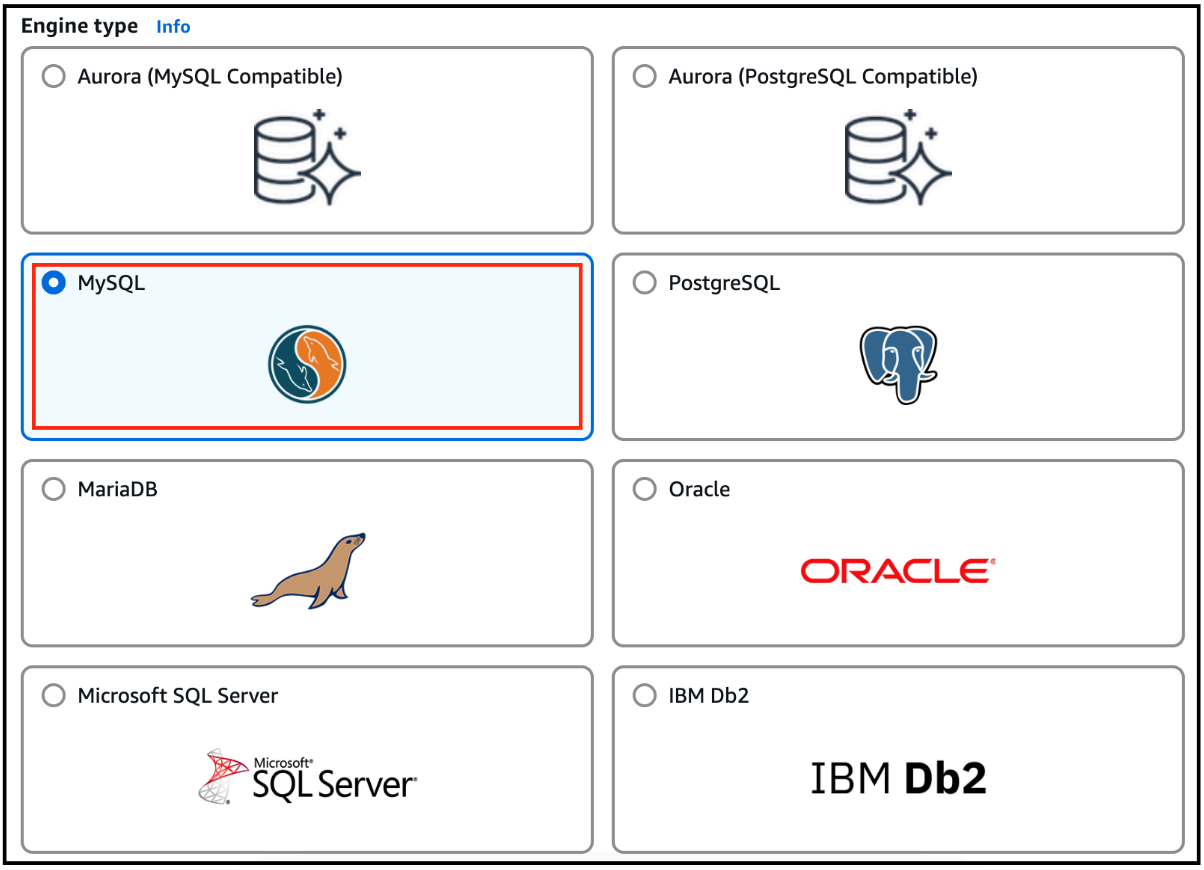
Task 3 : Create RDS Database Instance

1. Make sure you are in the US East (N. Virginia) us-east-1 Region.
2. Navigate to Aurora and RDS by clicking on the Services menu available under the Database section.
3. Click on Create Database.

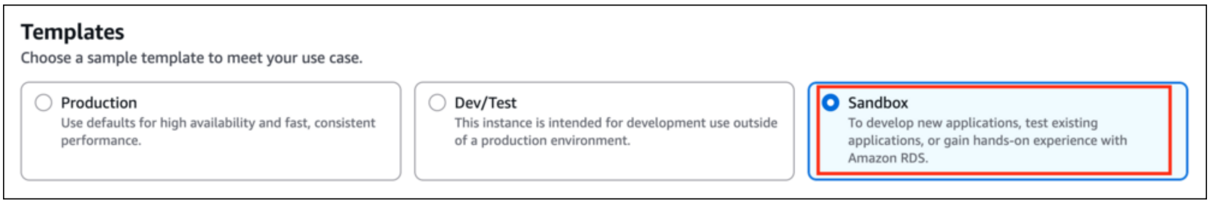
- Database creation method : Full configuration



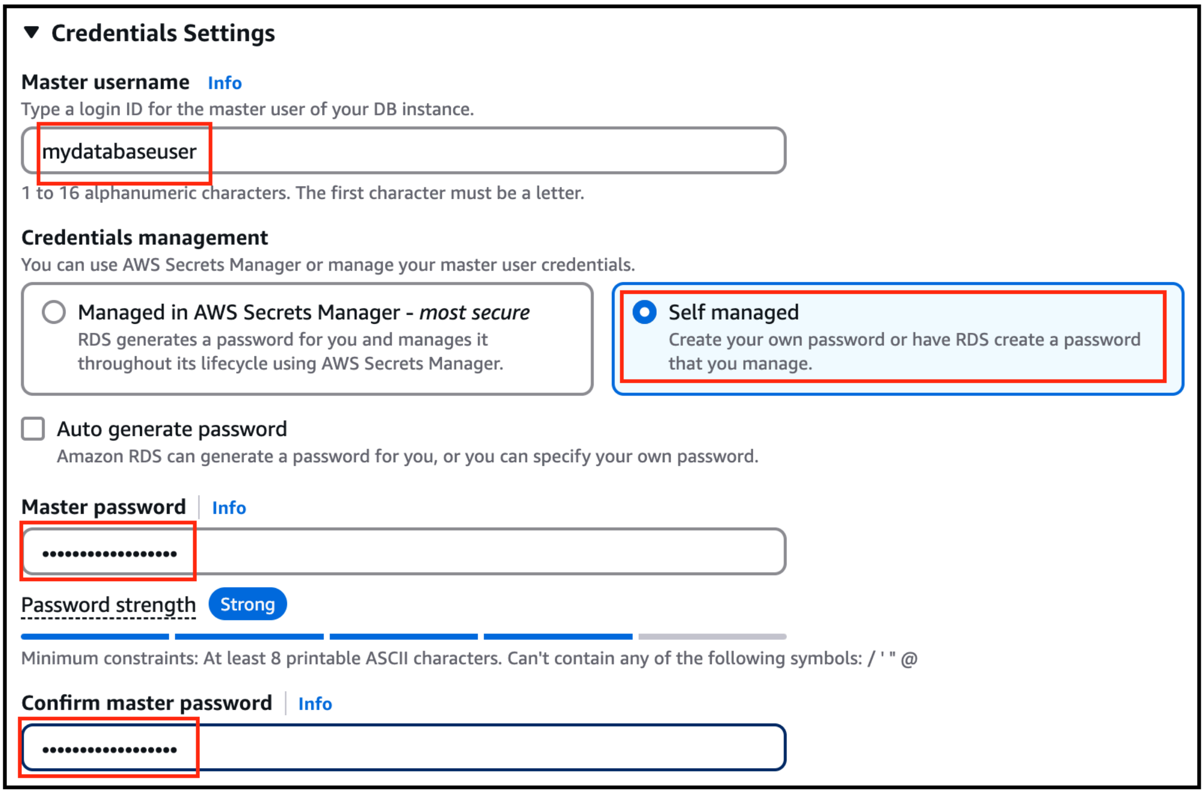
- Engine options : Select MySQL
- Version : Default



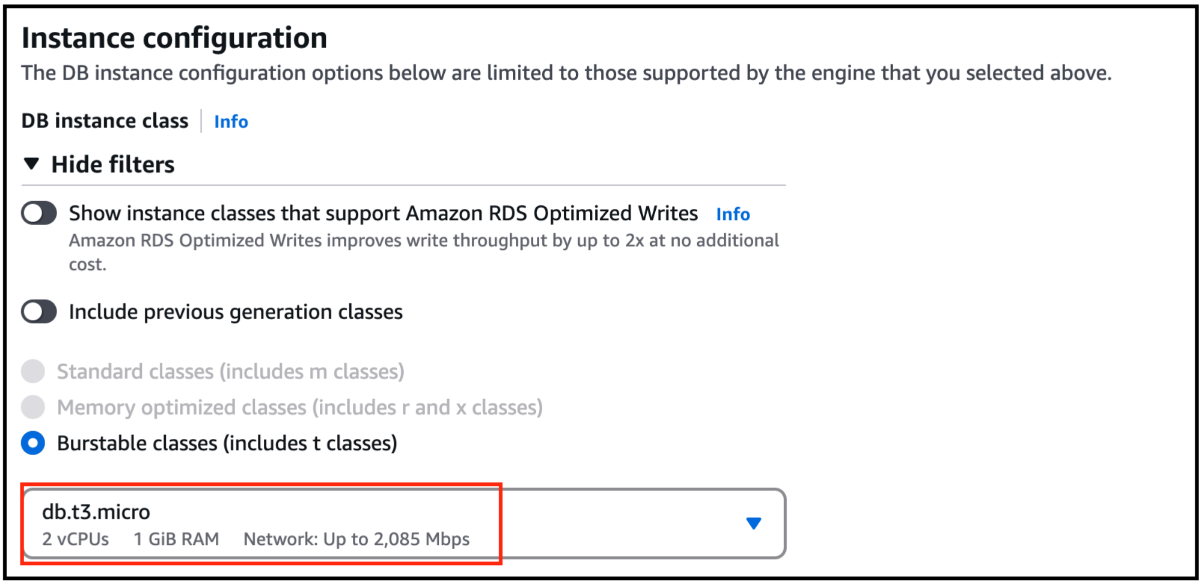
- Templates : Select Sandbox
  Note: Selecting the Sandbox as a template is compulsory, else database won't be created.



- DB instance identifier : *mydatabaseinstance*
- Credentials management: Select Self managed
- Master username : *mydatabaseuser*
- Master password and Confirm password: *mydatabasepassword*
  Note: This is the username/password combo used to log onto your database. Please make note of them somewhere safe.



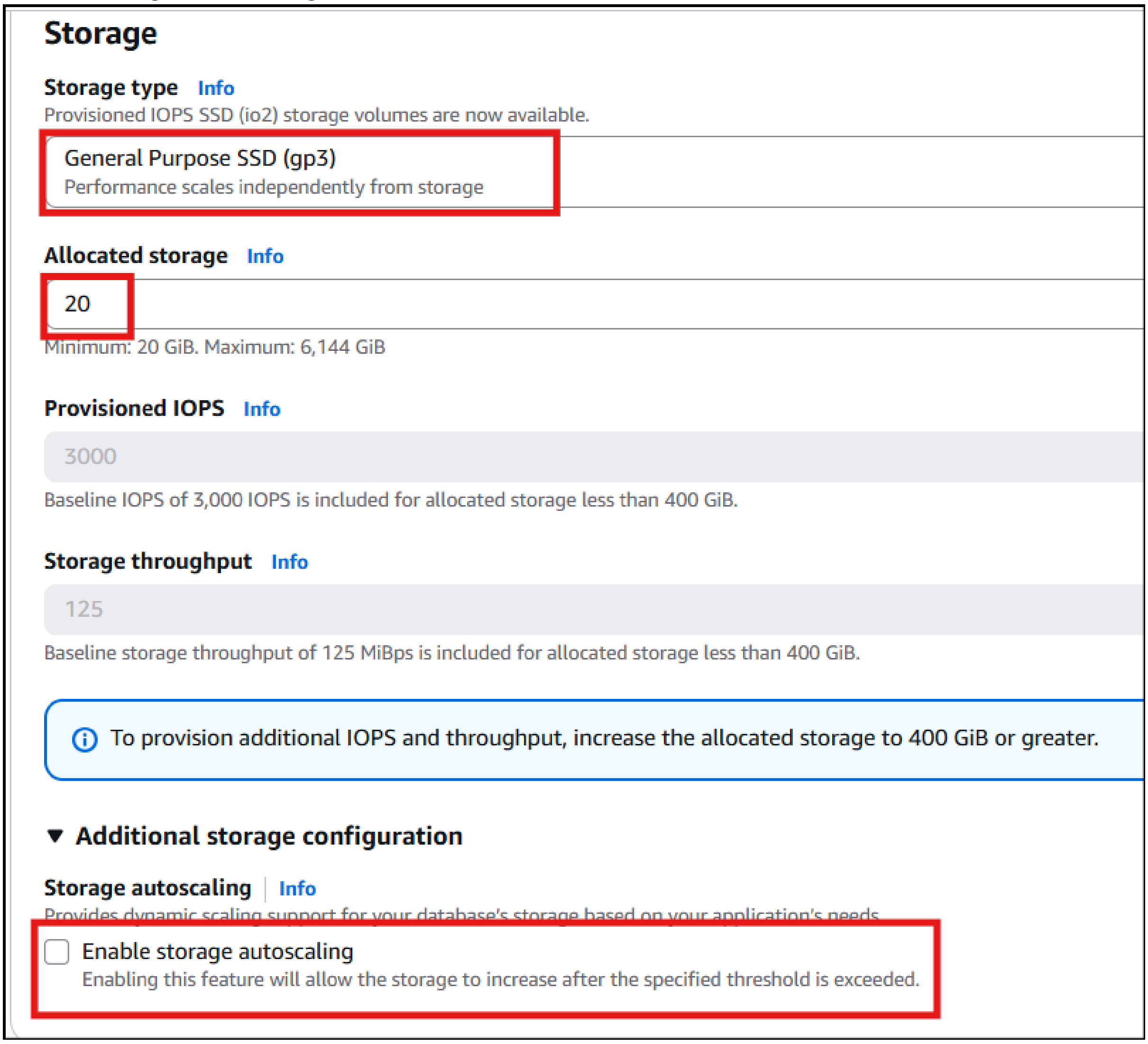
- DB instance class : db.t3.micro — 2 vCPU, 1 GiB RAM



- Storage type : General Purpose SSD (gp3).
  Note: By default, it will be gp2, kindly change to gp3.

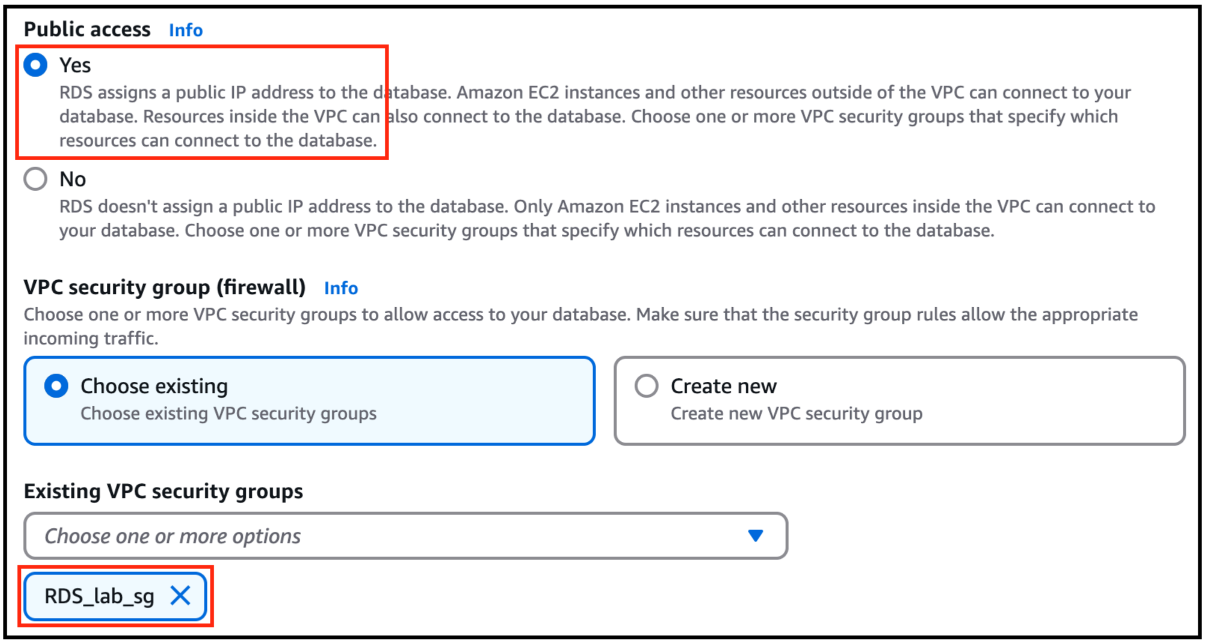
- Allocated storage : 20 (default)
- Click on Additional storage configuration to expand.

- Enable storage autoscaling : Uncheck

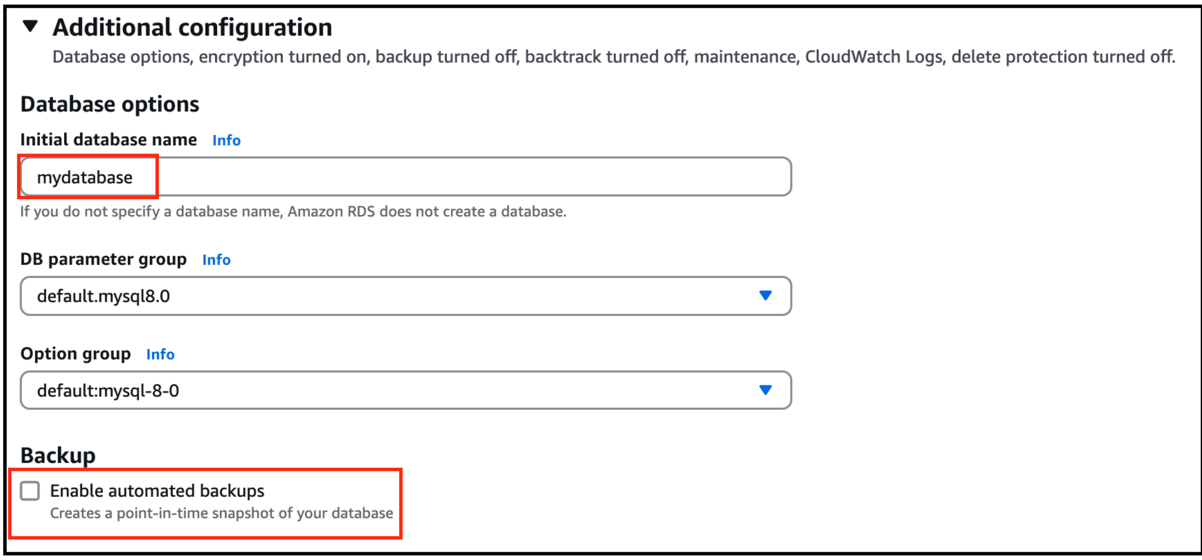


- Public access : Select Yes

- VPC security groups : Select Choose existing
  Note: Remove the default one and select RDS_lab_sg instead



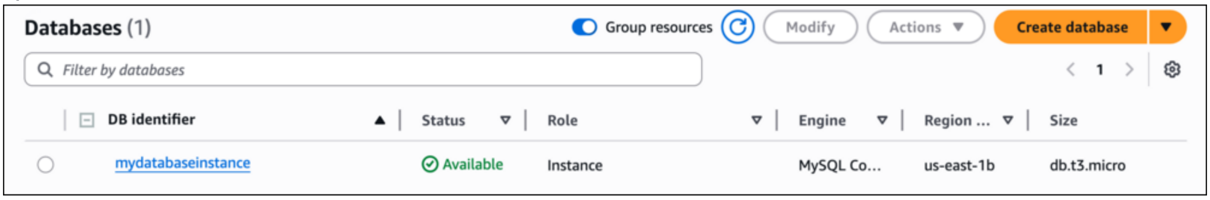
- Click on Additional Configuration in the bottom of the page to expand it:
  - Initial database name: *mydatabase*
  - DB parameter group: default
  - Option group: default
  - Enable automated backups: uncheck



  - Note: Leave all the other setting as default
- Click on Create database

5. Navigate to Databases.
6. On the RDS console, the details for the new DB instance appear. The DB instance has a status of creating until the DB instance is ready to use. When the state changes to Available, you can connect to the DB instance. It can take

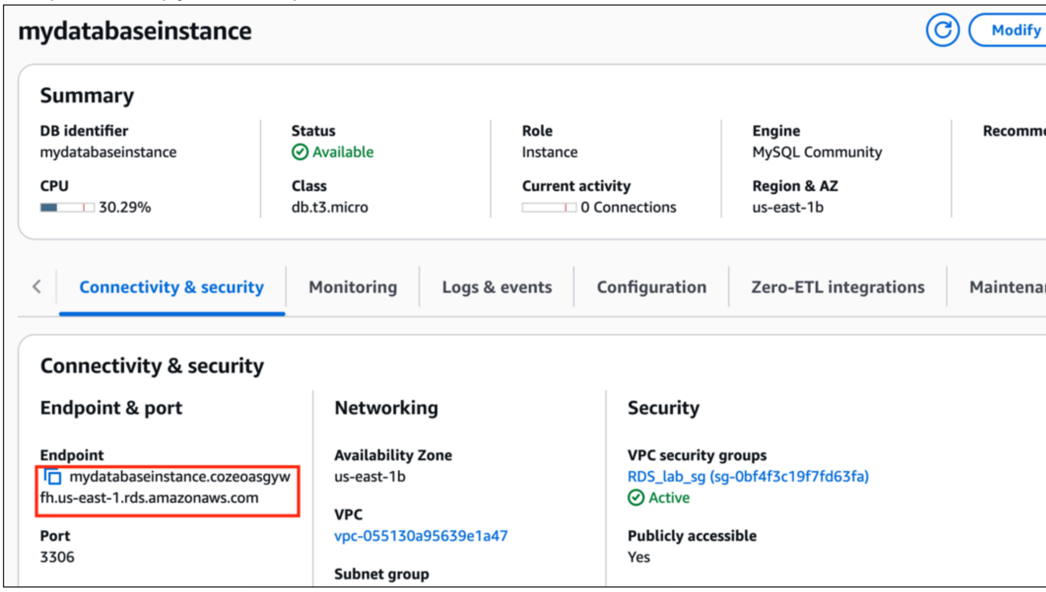
up to 20 minutes before the new instance status becomes Available.



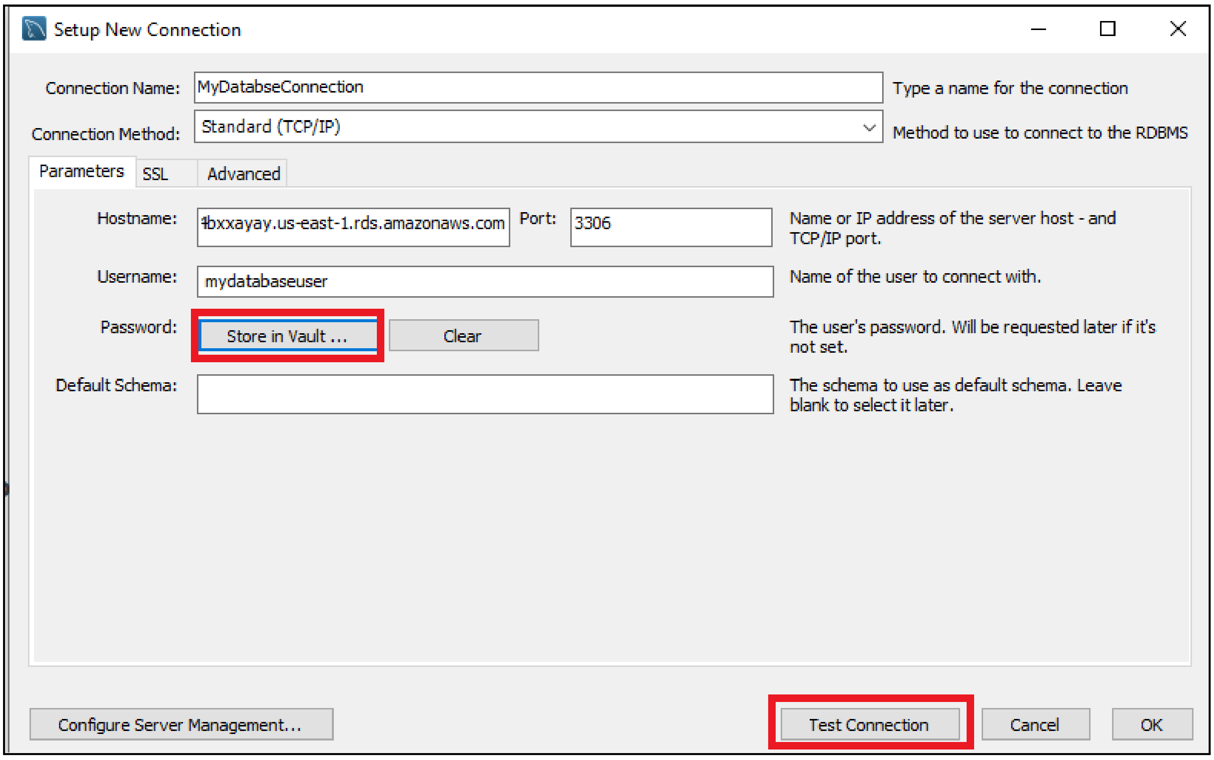
Task 4 : Connecting to RDS Database on a DB Instance using the MySQL Workbench

In this task, we will connect to a database on a MySQL DB instance using MySQL monitor commands. One GUI-based application you can use to connect is MySQL workbench, which you have already downloaded and installed based on instructions in the prerequisite section.

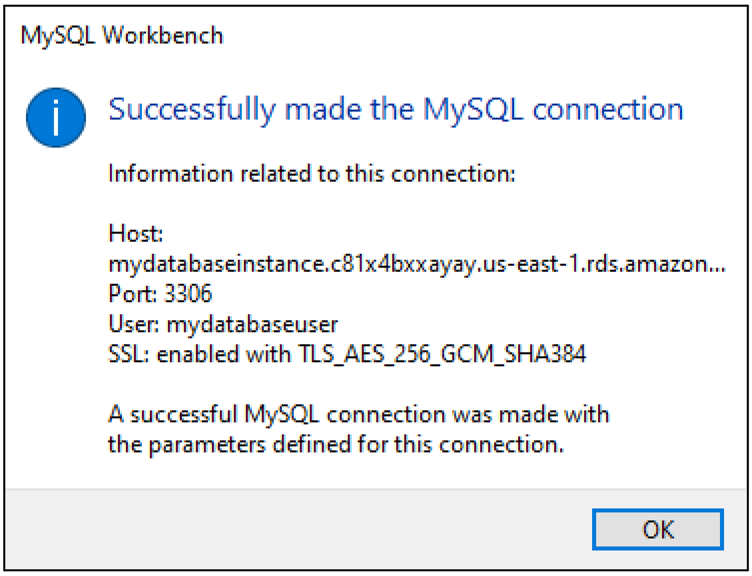
1. To connect to a database on a DB instance using MySQL monitor, find the endpoint (DNS name) and port number for your DB Instance.
    - Navigate to Databases and click on *mydatabaseinstance*.
    - Under Connectivity & security section, copy and note the endpoint and port.
        - Endpoint: Copy the endpoint



        - Port: 3306
        - You need both the endpoint and the port number to connect to the DB instance.
2. Open MySQL Workbench. Click on the plus icon aside on MySQL Connections
    - Connection Name: Enter a sample name *MyDatabseConnection*
    - Host Name: Past the Endpoint that you copied
    - Port : 3306
    - Username : *mydatabaseuser*
    - Password : Click on Store in keychain and enter password *mydatabasepassword* Click on ok.



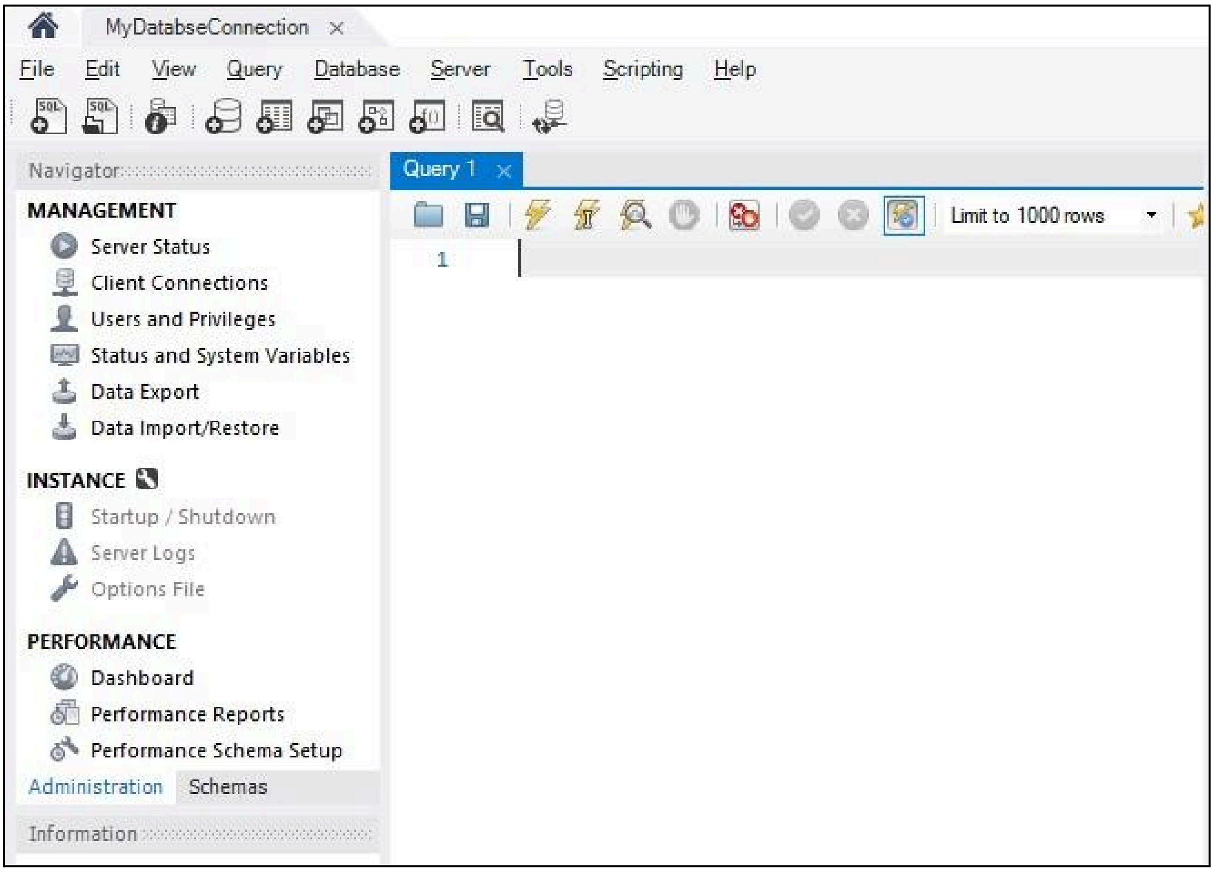
- Click on Test Connection to make sure that you are able to connect to the database properly.



- Click on ok and ok again to save the connection.

3. A database connection will be created in MySQL Workbench

4. Click on it to open the database. Enter the database password if prompted.
5. After successfully connecting and opening the database, you can create tables and perform various queries over the connected database.



6. Navigate to the Schemas tab to see databases available to start doing database operations. More details on database operations are available here.

Lab Steps
Connecting RDS from EC2 Instance

Task 1: Sign in to AWS Management Console

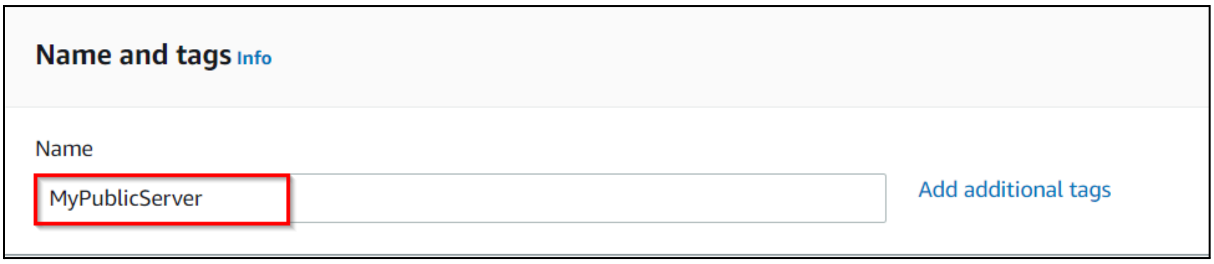
1. Click on the Open Console button, and you will get redirected to AWS Console in a new browser tab.
2. On the AWS sign-in page,
    - Leave the Account ID as default. Never edit/remove the 12-digit Account ID present in the AWS Console. Otherwise, you cannot proceed with the lab.
    - Now copy your Username and Password in the Lab Console to the IAM Username and Password in AWS Console and click on the Sign-in button.
3. Once Signed In to the AWS Management Console, make the default AWS Region as US East (N. Virginia) us-east-1.

Note : If you face any issues, please go through FAQs and Troubleshooting for Labs.

Task 2: Launch EC2 Instance

1. Make sure you are in US East (N.Virginia) us-east-1 Region.
2. Navigate to EC2 by clicking on the Services menu in the top, then click on the EC2 in the Compute section.
3. Navigate to Instances on the left panel and click on Launch instances.

4. Name : Enter MyPublicServer



5. For Amazon Machine Image (AMI): Search for Amazon Linux **2023 kernel-6.1** AMI in Quick Start menu.
6. For Instance Type: select t2.micro

7. For Key pair: Select Create a new key pair Button
    - Key pair name: Enter WhizKey
    - Key pair type: Select RSA
    - Private key file format: Select .pem
8. Select Create key pair Button.
9. In Network Settings, Click on Edit:

- Auto-assign public IP: Enable
- Select Create new Security group
- Security group name : Enter MyEC2Server_SG
- Description : Enter Security Group to allow traffic to EC2

10. To add SSH:
    - Choose Type: SSH
    - Source: Anywhere
11. For RDS:
    - Click on Add security group role
    - Choose Type: MySQL/Aurora
    - Source: Anywhere

12. Click on Advanced Details. Under the User data section, enter the following script, (which installs MySQL):

```
#!/bin/bash
# Download the MySQL repository package
wget https://dev.mysql.com/get/mysql80-community-release-el9-5.noarch.rpm
# Install the MySQL repository package
sudo dnf install mysql80-community-release-el9-5.noarch.rpm -y
# Enable the MySQL community repository
sudo dnf repolist enabled | grep "mysql.*-community.*"
# Install MySQL
sudo dnf install mysql -y
```

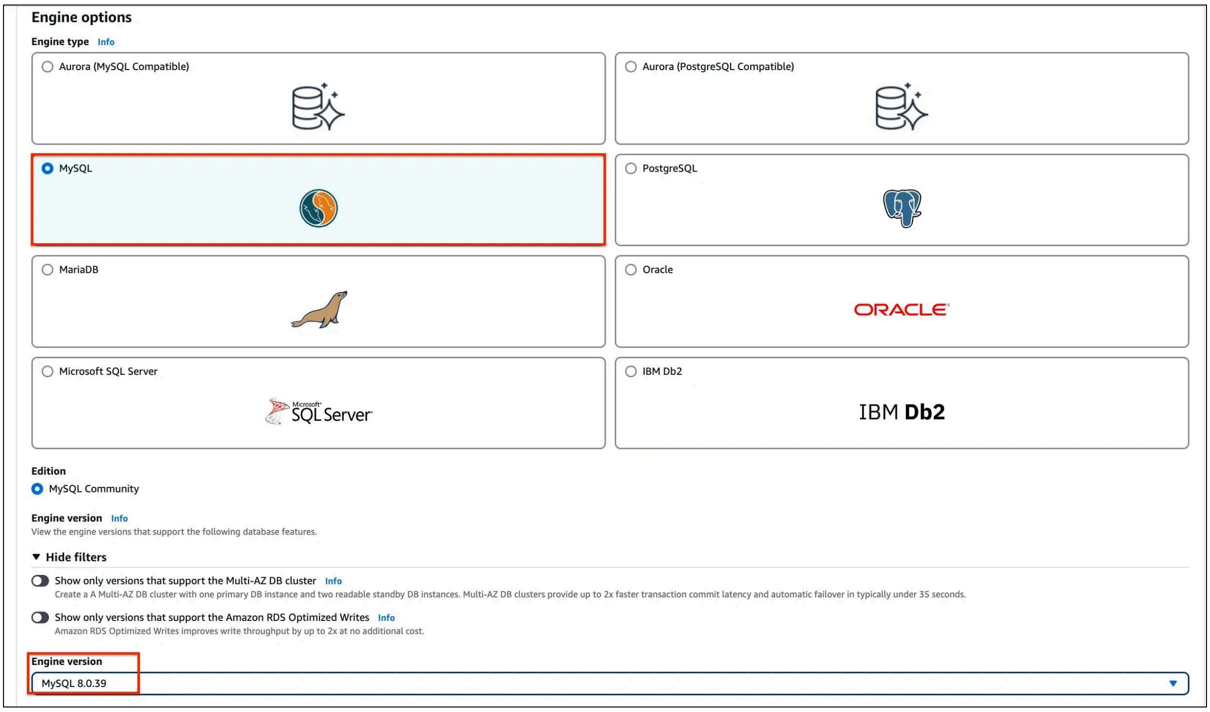
22. Keep rest thing Default and Click on Launch Instance Button.
23. Select View all Instances to View the Instance you Created
24. Launch Status: Your instance is now launching. Click on the instance ID and wait for complete initialization of the instance (until the status changes to running).

Task 3: Create an Amazon RDS Database

1. Navigate to **Aurora and RDS** under Database.
2. Click on Create Database button
3. Specify DB Details:
    - Database creation method : Select Standard create

- Engine options : Select MySQL

- Version : Default



- Templates : Select Sandbox

- DB instance identifier : Enter mydbinstance
- Master username : Enter rdsuser
- Master password and Confirm password: Enter whizlabs123
- Note: This is the username/password combo used to log onto your database. Please make note of them somewhere safe.
- DB instance class : db.t3.micro — 1 vCPU, 1 GiB RAM

- Storage type : General Purpose SSD (gp2)
- Allocated storage : 20
- Enable storage autoscaling : Uncheck

- Public access : Select No
- VPC Security groups : Select Choose existing
- Existing VPC security groups : Remove the default one and Select MyEC2Server_SG
- Go to Additional Configuration options
  - Initial database name: Enter mydbinstance
  - DB parameter group: default
  - Option group: default
  - Enable automated backups: uncheck
  - Log Exports: Not needed for the purpose of this lab.
  - Note: Leave all the other settings as default
- Click on Create database

4. Navigate to databases.
5. On the RDS console, the details for the new DB instance appear. The DB instance has a status of creating until the DB instance is ready to use. When the state changes to Available, you can connect to the DB instance. It can take up to 20 minutes before the new instance status becomes Available.
6. Once the database becomes Available, click on the database name and copy the Endpoint under Connectivity & security tab.

Task 4: Create a connection to the Amazon RDS database from the EC2 instance

In this task, we are going to establish a connection between the EC2 instance and the RDS database.

1. Now navigate to EC2 by clicking on the Services menu in the top, then click on the EC2 in the Compute section.
2. Navigate to Instances on the left panel and select your EC2 instance(MyPublicServer) and click on the Connect button.

3. Select EC2 Instance Connect option and click on Connect button.(Keep everything else as default)

4. A new tab will open in the browser where you can execute the CLI Commands.

```
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-172-31-87-199 ~]$
```

5. Once connected to the server:
   - Switch to the root user using the command:

```
sudo su
```

6. Connect to the MySQL RDS Instance with the following command:
   - Syntax: mysql -h <mysql-instance-dns> -u <username> -p
   - In our case: mysql -h mydbinstance.c1sh5a4j91dp.us-east-1.rds.amazonaws.com -u rdsuser -p
   - Password : Enter whizlabs123
7. You will enter the MYSQL command line.

```
[root@ip-172-31-85-112 ec2-user]# mysql -h mydbinstance.ch51fkqdms8y.us-east-1.rds.amazonaws.com -u rdsuser -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 17
Server version: 8.0.32 Source distribution

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```

Task 5: Create a Database, Table and insert data for testing

Lets create a simple database and table to see if it's working.

- Create a database:

```
CREATE DATABASE SchoolDB;
```

- You can see the created database with following command:

```
show databases;
```

```
mysql> CREATE DATABASE SchoolDB;
Query OK, 1 row affected (0.01 sec)

mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| SchoolDB           |
| information_schema |
| mydbinstance       |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
6 rows in set (0.00 sec)

mysql>
```

- Switch to the database named SchoolDB.

```
use SchoolDB;
```

- Create a sample table consisting of Subjects.

```
CREATE TABLE IF NOT EXISTS subjects (subject_id INT
AUTO_INCREMENT,subject_name VARCHAR(255) NOT NULL,teacher
VARCHAR(255),start_date DATE,lesson TEXT,PRIMARY KEY
(subject_id)) ENGINE=INNODB;
```

- Insert some details into the table:

```
INSERT INTO subjects(subject_name, teacher) VALUES ('English',
'John Taylor');
INSERT INTO subjects(subject_name, teacher) VALUES ('Science',
'Mary Smith');
INSERT INTO subjects(subject_name, teacher) VALUES ('Maths', 'Ted
Miller');
INSERT INTO subjects(subject_name, teacher) VALUES ('Arts',
'Suzan Carpenter');
```

- Let's check the items we added into the table:

```
select * from subjects;
```

```
MySQL [SchoolDB]> select * from subjects;
+------------+--------------+-----------------+------------+--------+
| subject_id | subject_name | teacher         | start_date | lesson |
+------------+--------------+-----------------+------------+--------+
|          1 | English      | John Taylor     | NULL       | NULL   |
|          2 | Science      | Mary Smith      | NULL       | NULL   |
|          3 | Maths        | Ted Miller      | NULL       | NULL   |
|          4 | Arts         | Suzan Carpenter | NULL       | NULL   |
+------------+--------------+-----------------+------------+--------+
4 rows in set (0.01 sec)
```

- Try out some more SQL commands and play around with the table to strengthen your understanding.
- Run below command to exit the mysql command

```
exit;
```

Lab Steps

Deploying Amazon RDS Multi-AZ and Read Replica, Simulate Failover

Task 1: Sign in to AWS Management Console

1. Click on the Open Console button, and you will get redirected to AWS Console in a new browser tab.
2. On the AWS sign-in page,
   - Leave the Account ID as default. Never edit/remove the 12-digit Account ID present in the AWS Console. Otherwise, you cannot proceed with the lab.
   - Now copy your Username and Password in the Lab Console to the IAM Username and Password in AWS Console and click on the Sign in button.
3. Once Signed In to the AWS Management Console, make the default AWS Region as US East (N. Virginia) us-east-1.

Note : If you face any issues, please go through FAQs and Troubleshooting for Labs.

Task 2: Launch an EC2 instance

In this task, we are going to launch an EC2 instance which will be used for connecting to the Amazon Aurora RDS DB instance later in the lab.

1. Navigate to EC2 by clicking on the Services menu in the top, then click on EC2 in the Compute section.
2. Click on Instances from the left sidebar and then click on Launch instances

3. Name : Enter MyRdsEc2server.
4. For Amazon Machine Image (AMI): In Quick Start menu, select Amazon Linux 2023 kernel-6.1 AMI.

5. For Instance Type: select t2.micro

6. For Key pair: Select Create a new key pair
   - Key pair name: MySSHKey
   - Key pair type: RSA
   - Private key file format: .pem
     Note: If you are using putty, select the Private key file format as .ppk
7. Select Create key pair Button.
8. In Network Settings, Click on Edit:

- Auto-assign public IP: Enable
- Select Create new Security group
- Security group name : Enter *MyEC2Server_SG*
- Description : Enter *Security for ec2 server to connect with RDS*
  - To add SSH:

- Choose Type: SSH
- Source: select Anywhere

9. Click on Advanced Details . Under the User data section, enter the following script, (which installs MySQL):

```
#!/bin/bash

# Download the MySQL repository package

wget
https://dev.mysql.com/get/mysql80-community-release-el9-5.noarch.r
pm

# Install the MySQL repository package

sudo dnf install mysql80-community-release-el9-5.noarch.rpm -y

# Enable the MySQL community repository

sudo dnf repolist enabled | grep "mysql.*-community.*"

# Install MySQL

sudo dnf install mysql -y
```

Note: After pasting the user data, make sure to remove extra spacing.

10. Keep rest thing Default and Click on Launch Instance Button.
11. Select View all Instances to View the Instance you Created
12. After 1-2 minutes, the Instance State will become running as shown below:

13. Select the EC2 Instance and copy the Private IPv4 address. Note down IPv4 Private IP address in your text editor, navigate to the EC2 Dashboard and look in the instance details.

Task 3: Create a Security Group for RDS instance

In this task, we will create a security group specifically for the Amazon Aurora RDS instance which ensures that the RDS instance has appropriate network security settings by allowing access only through the MySQL/Aurora port (3306) from any IP address (0.0.0.0/0).

1. Make sure you are in the N.Virginia Region.
2. Navigate to EC2 by clicking on the Services menu available under the Compute section.

3. On the left panel menu, select the security group under the Network & Security section.
4. Click on the Create security group button.

5. We are going to create a Security group for RDS with 3306 port number enabled.
    - Security group name : Enter rds-maz-SG
    - Description : Enter Security group for RDS Aurora
    - VPC : Select Default VPC

- Click on the Add rule button under Inbound rules.
    - Type : Select MYSQL/Aurora
    - Source : Select Anywhere IPv4

6. Leave everything as default and click on the Create security group button.

Task 4: Create an Amazon Aurora database with Multi-AZ enabled

In this task, we will create an aurora database using AWS RDS service. The database is configured with Multi-AZ deployment, which replicates data to a standby instance in a different availability zone, providing high availability and durability.

1. Make sure you are in US East (N.Virginia) us-east-1 Region.
2. Navigate to RDS under the database section of the Services menu.
3. Click on Databases in the left Panel.
4. Click on Created database button.

5. Next, we'll configure the database on the Create Database Page
    - In Engine option:
        - Select **Aurora (MySQL Compatible)**

- Choose a Database Creation Method:
- Select **Full Configuration**

- Choose Template: **Dev/Test**
- Cluster scalability type: Choose **Provisioned**
- **Choosing DB instance size**
- DB instance class: Choose Burstable classes (includes t classes)
  Select **db.t3.medium** instance.

- Fill in the required details for the database (Aurora Cluster Settings)
    - DB cluster identifier: Specify cluster name MyAuroraCluster
    - Give the following details in the credential settings
        - Master Username: Enter WhizlabsAdmin
        - Credentials management: Select **Self managed**
        - Master password: Enter Whizlabs123
        - Confirm password: Enter Whizlabs123
        - Note: This is the username and password used to log into your database. Please make note of them.

        - Availability and Durability: Choose Multi-AZ deployment: Create Aurora Replica or Reader node in a different AZ as shown below:

- Connectivity
    - Choose the Default VPC
- Additional connectivity configuration
    - Subnet group: Default
    - Publicly accessible: Yes {IMPORTANT}

    - Existing VPC security groups :
        - Remove the Default security group, which is selected by default :
        - Select rds-maz-SG for the dropdown.(This is the security group which you have created in the beginning)

- Under Monitoring, **uncheck** monitoring option.

- Select the Additional configuration present at the bottom of the page and enter the following details**:**
    - Database options
        - Initial database name: Enter whizlabsrds
    - Enable Encryption : Uncheck

    - Leave other settings as default

6. Once the details above have been filled in, click on Create database button.
7. It will take around 10-15 minutes for the database to be created. Please wait until database status changes from creating to Available.
8. Once the database has been created, you should see the following page:

- Similar to the screenshot, you should be able to see that our database launched in multiple AZs, namely us-east-1c and us-east-1a

Task 5: Connecting to the Aurora (MySQL) database on RDS

Now we have successfully launched Aurora RDS with Multi-AZ enabled. To connect to the new Aurora database, we need the endpoint.

1. Click on the RDS cluster name and then navigate to Connectivity & security to find the endpoint of your Master (Writer) and Reader instances, with which you can connect to your DB instance.

- The endpoints you see to be similar to these examples:

  Master (Writer):
  myauroracluster.cluster-c2tbqtanuaea.us-east-1.rds.amazonaws.com

  Reader: myauroracluster.cluster-ro-c2tbqtanuaea.us-east-1.rds.amazonaws.com

Note: Please carefully look at the role of the DB instance (reader vs Master (Writer)) and their respective availability zones. (here us-east-1c and us-east-1a)

Task 6: Connecting the EC2 Server to RDS:

In this task, we are going to establish connectivity between the EC2 instance created earlier and the Amazon Aurora RDS database. By modifying the security group of the RDS instance, they allow access from the EC2 instance, enabling communication between the two resources.

1. Now we need to connect the RDS with EC2 instance in order to eventually connect with the Aurora database.
2. Navigate to RDS available under the Database section of the Services menu.
3. Click on Master (writer) database and click on the security group name in this example it is rds-maz-SG under VPC security groups as shown below:

4. It will open the Security Group page. Click on Inbound rules.

- Click on Edit inbound rules. The MySQL rule will already exist.
- Click on Add rule button.
    - Type: Select MYSQL/Aurora
    - Source: Paste the Private IP address of your MyRdsEc2server EC2 instance.
    - Delete the old rule and then Click on Save rules button.

Task 7: Execute Database Operations via SSH

In this task, we are going to execute database operations on the Amazon Aurora RDS database. By SSHing into the EC2 instance and accessing the database using MySQL, they can create tables, insert data, and perform queries.

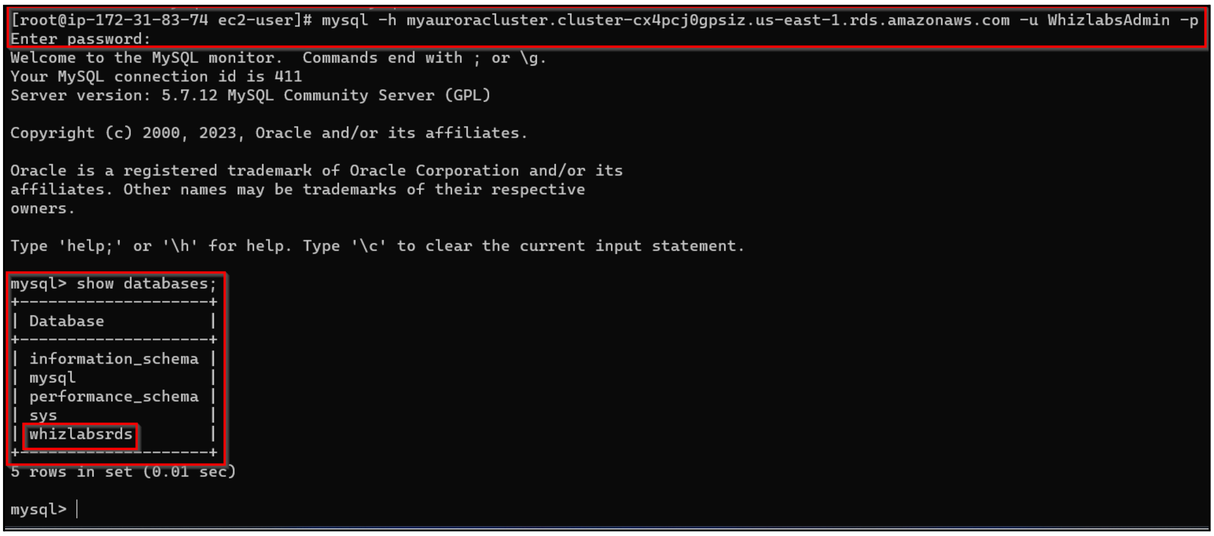
1. Navigate to the EC2 Dashboard and select the instance created.
2. Once the instance is launched, Select **EC2 Instance Connect** option and click on **Connect** button.

3. Please follow the steps in SSH into EC2 Instance for more option to SSH.
4. Switch to the root user using the command :
5.
```
sudo su
```
6. Log into the RDS instance using the below command:

- **mysql -h myauroracluster.cluster-cjp5t9qppjnx.us-east-1.rds.amazonaws.com -u WhizlabsAdmin -p**
- Syntax: mysql -h <Hostname> -u <username> -p
    - Note: Make sure to change the above Master (Writer)Cluster endpoint and Username with yours.
- Host name : myauroracluster.cluster-cpoz6c7903cx.us-east-1.rds.amazonaws.com (Master(Writer)cluster endpoint)
- Username : Enter WhizlabsAdmin
- Password : Enter Whizlabs123 (Use yours incase you changed the password while creating RDS)
- You should now be able to log into the database,

6. List all Databases using the below command. You will see the database whizlabsrds created while launching the RDS cluster.

```
show databases;
```



7. Now create the database in the Master(Writer) RDS as given in the screenshot. We'll create a demo database named auroro_db.

```
Create database auroro_db;
```

8. Select the newly-created database:

```
use auroro_db;
```

9. List all Databases:

```
Show databases;
```

```
mysql> use auroro_db;
Database changed
mysql> Show databases;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| auroro_db          |
| mysql              |
| performance_schema |
| sys                |
| whizlabsrds        |
+--------------------+
6 rows in set (0.01 sec)

mysql>
```

10. Next we'll create a table named students and insert few rows of data using list of commands:

```
CREATE TABLE students ( subject_id INT AUTO_INCREMENT, subject_name
VARCHAR(255) NOT NULL,  teacher VARCHAR(255),start_date DATE, lesson
TEXT,PRIMARY KEY (subject_id));
```

```
Select * from students;
```

```
Show
tables;
```

```
mysql> CREATE TABLE students ( subject_id INT AUTO_INCREMENT, subject_name VARCHAR(255) NOT NULL,  teacher VARCHAR(255),start_date DATE, lesson TEXT,PRIMARY
 KEY (subject_id));
Query OK, 0 rows affected (0.06 sec)

mysql> Select * from students;
Empty set (0.00 sec)

mysql> Show tables;
+--------------------+
| Tables_in_auroro_db |
+--------------------+
| students            |
+--------------------+
1 row in set (0.00 sec)

mysql> |
```

11. Insert data into the table:

```
INSERT INTO students(subject_name, teacher) VALUES ('English',
'John Taylor');
INSERT INTO students(subject_name, teacher) VALUES ('Science',
'Mary Smith');
INSERT INTO students(subject_name, teacher) VALUES ('Maths',
'Ted Miller');
INSERT INTO students(subject_name, teacher) VALUES ('Arts',
'Suzan Carpenter');
```

12. Now you can view the contents of the table student using the below command:

```
select * from students;
```

```
mysql> INSERT INTO students(subject_name, teacher) VALUES ('English', 'John Taylor');
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO students(subject_name, teacher) VALUES ('Science', 'Mary Smith');
Query OK, 1 row affected (0.05 sec)

mysql> INSERT INTO students(subject_name, teacher) VALUES ('Maths', 'Ted Miller');
Query OK, 1 row affected (0.02 sec)

mysql> INSERT INTO students(subject_name, teacher) VALUES ('Arts', 'Suzan Carpenter');
Query OK, 1 row affected (0.02 sec)

mysql> select * from students;
+------------+--------------+-----------------+------------+--------+
| subject_id | subject_name | teacher         | start_date | lesson |
+------------+--------------+-----------------+------------+--------+
|          1 | English      | John Taylor     | NULL       | NULL   |
|          2 | Science      | Mary Smith      | NULL       | NULL   |
|          3 | Maths        | Ted Miller      | NULL       | NULL   |
|          4 | Arts         | Suzan Carpenter | NULL       | NULL   |
+------------+--------------+-----------------+------------+--------+
4 rows in set (0.00 sec)

mysql> |
```

13. Exit from mysql console use the below command:

```
exit;
```

Task 8: Forcing a Failover to Test Multi-AZ

In this task, we are going to simulate a failover scenario in which the master instance fails, and the read replica takes over as the new master.

1. Navigate to RDS Database and in Database section Select Master(Writer) Instance and in Actions dropdown, click on failover

2. To test if Multi-AZ is working, we will create a situation where master fails and the read replica has to become the new Master(Writer).

2. On the next screen, confirm the Failover.

3. Wait for a few minutes for the RDS instances to failover.
   - (i.e Master (Writer) becomes Reader and Reader becomes Master (Writer) as shown below)

Task 9: Testing the Failover Condition

In this task, we are going to the new master instance and verify the presence of the database, table, and data created earlier. This step ensures that the failover was successful, and the new master instance is functioning as expected.

1. Now connect to RDS with new Master endpoint
   - Copy the endpoint of the new Master(Writer)cluster and replace it with your endpoint link.

   Mysql -h <endpoint> -u <username> -p and press [Enter]

   - **mysql -h myauroracluster.cluster-ro-cjp5t9qppjnx.us-east-1.rds.amazonaws.com -u WhizlabsAdmin -p**
   - Password: Enter Whizlabs123
2. You will be able to Log into MySQL and check for the database and table created in the master DB instance before the failover.

- You can notice the resources created on the original master db are present, implying that the Failover worked successfully.

```
Show databases;
```

```
mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| auroro_db          |
| mysql              |
| performance_schema |
| sys                |
| whizlabsrds        |
+--------------------+
6 rows in set (0.03 sec)

mysql>
```

3. Now check the existence of table named students and data (that we created earlier in the lab):

```
use auroro_db;
select * from students;
```

```
mysql> use auroro_db;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> select * from students;
+------------+--------------+-----------------+------------+--------+
| subject_id | subject_name | teacher         | start_date | lesson |
+------------+--------------+-----------------+------------+--------+
|          1 | English      | John Taylor     | NULL       | NULL   |
|          2 | Science      | Mary Smith      | NULL       | NULL   |
|          3 | Maths        | Ted Miller      | NULL       | NULL   |
|          4 | Arts         | Suzan Carpenter | NULL       | NULL   |
+------------+--------------+-----------------+------------+--------+
4 rows in set (0.02 sec)

mysql>
```

Lab Introduction to AWS Elastic Load Balancing

Task 1: Sign in to AWS Management Console

1. Click on the Open Console button, and you will get redirected to AWS Console in a new browser tab.
2. On the AWS sign-in page,
   - Leave the Account ID as default. Never edit/remove the 12 digit Account ID present in the AWS Console. otherwise, you cannot proceed with the lab.
   - Now copy your User Name and Password in the Lab Console to the IAM Username and Password in AWS Console and click on the Sign in button.
3. Once Signed In to the AWS Management Console, Make the default AWS Region as US East (N. Virginia) us-east-1.

Task 2: Provision Default VPC

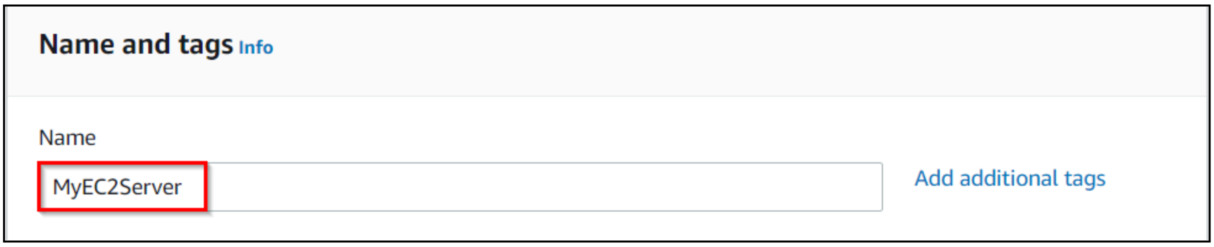
1. Navigate to VPC by clicking on the Services menu in the top, then click on VPC or Open the Amazon VPC console via https://console.aws.amazon.com/vpc/.
2. Delete the **default VPC** by following the **below steps:**
   - In the navigation pane, choose **Your VPCs.**
   - Select the **VPC** with value as **yes** in **default VPC column.**
   - Go to **Actions button** and click on **delete VPC button**
   - Check **I acknowledge that I want to delete my default VPC** option.
   - Type **delete default VPC** and click on **Delete button**

3. Now to provision **Default VPC** again
   - Refresh your console go to **Actions** and click **Create default VPC**

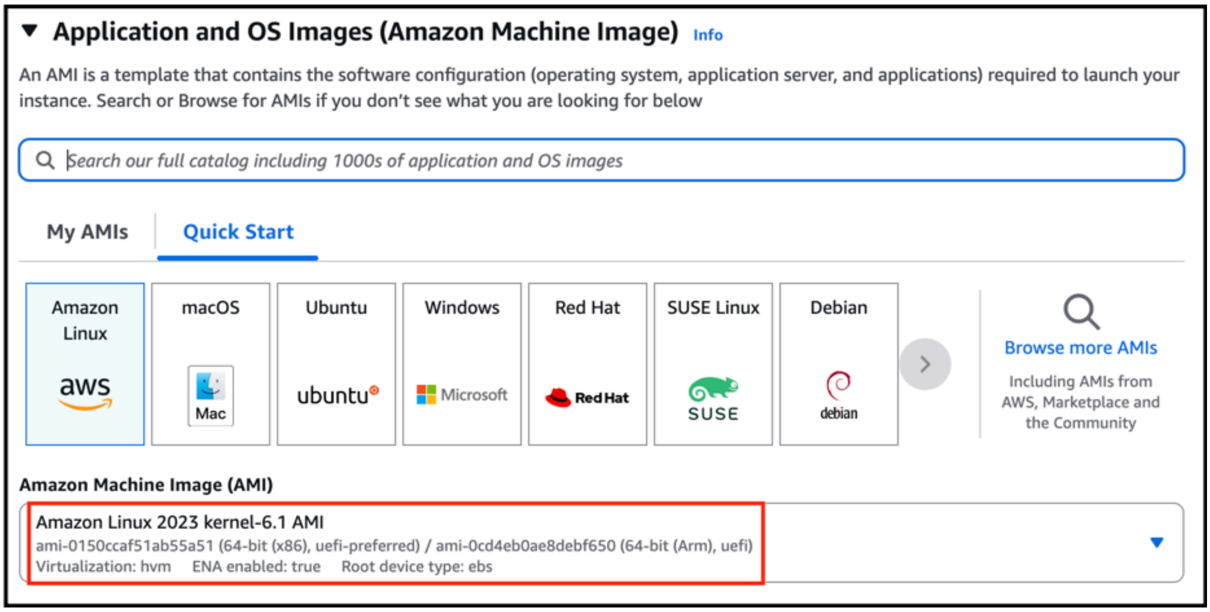
   - Click **Create default VPC button** and your default VPC will get created

Task 3: Launching First EC2 Instance

1. Make sure you are in the US East (N. Virginia) us-east-1 Region.
2. Navigate to EC2 by clicking on the Services menu in the top, then click on EC2 in the Compute section.
3. Navigate to Instances on the left panel and click on Launch Instances button.
4. Name: Enter *MyEC2Server*

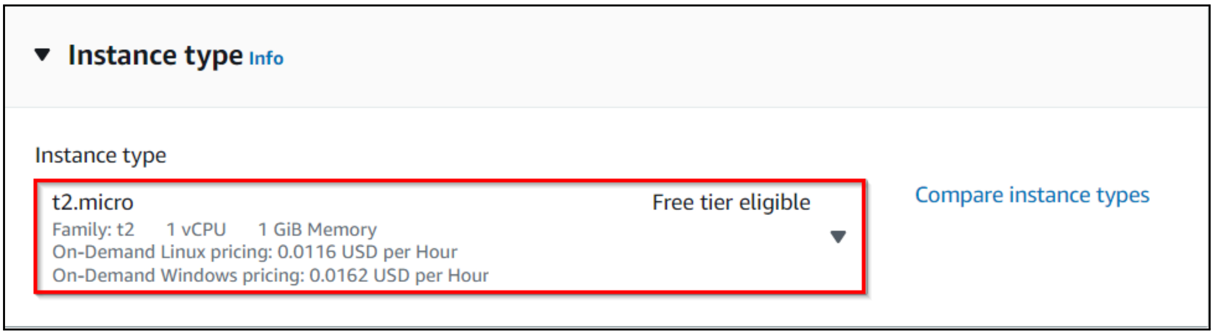


5. For Amazon Machine Image (AMI): Search for Amazon Linux 2023 AMI in the search box and click on the Select button.



6. **Note: if there are two AMI's present for Amazon Linux 2023 AMI, choose kernel-6.1.**

7. For Instance Type: Select t2.micro

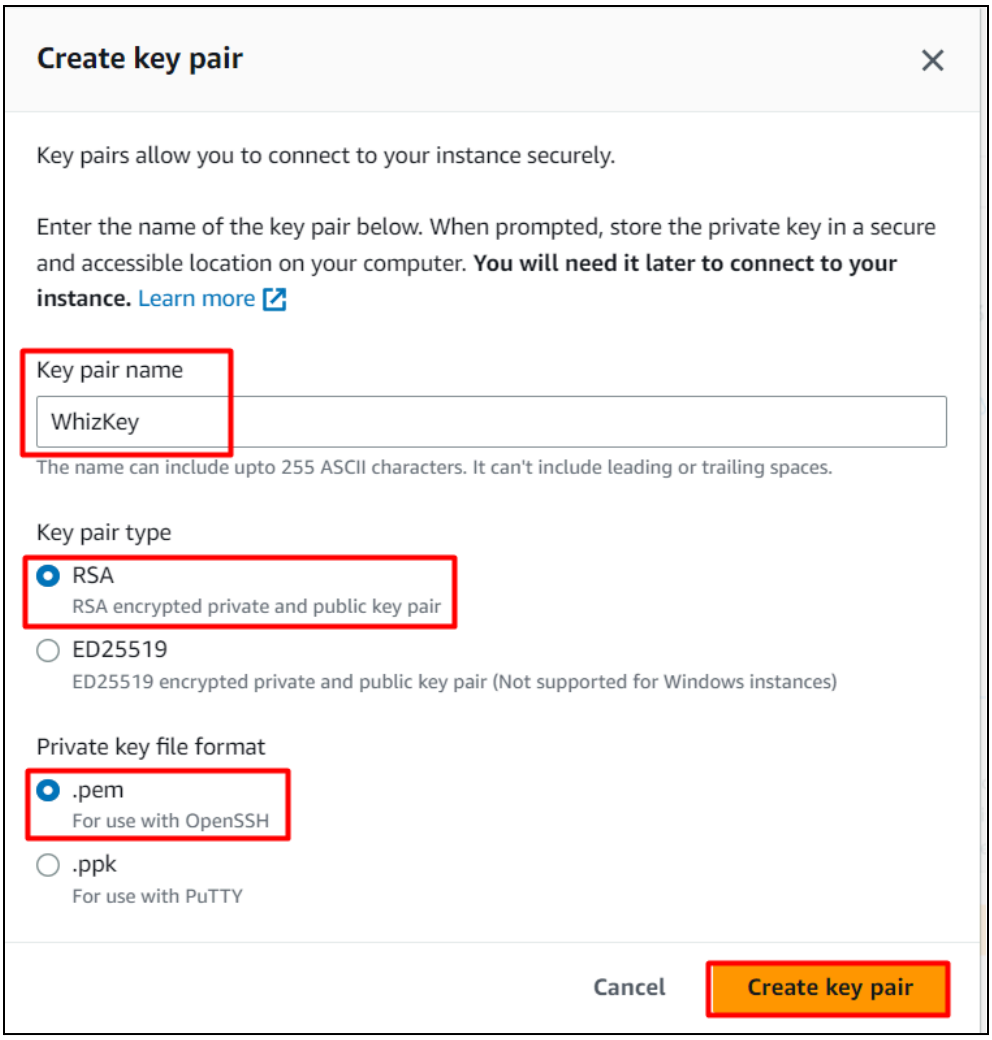


8. For Key pair: Select Create a new key pair Button

- Key pair name: WhizKey
- Key pair type: RSA

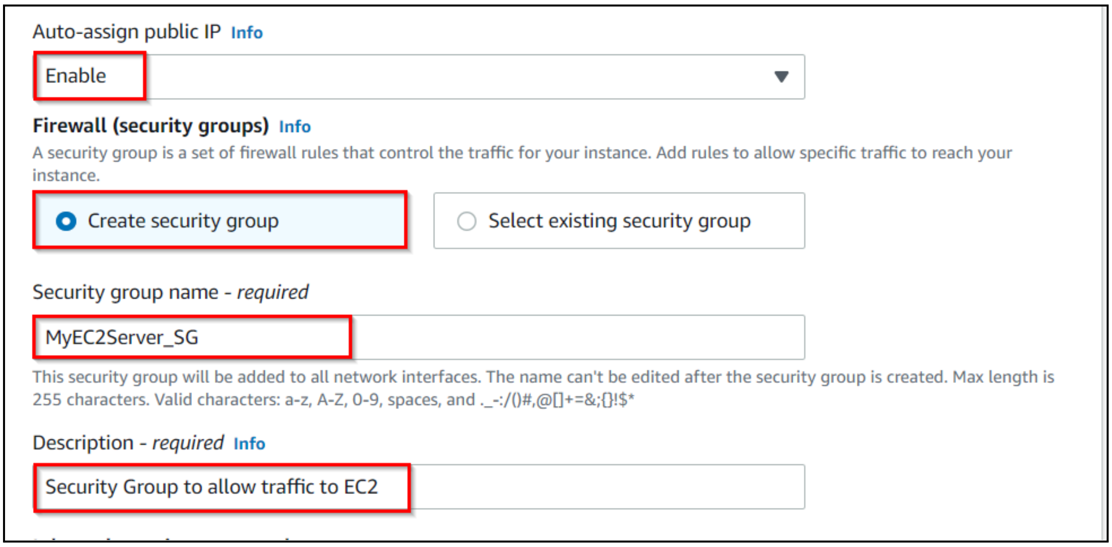
- Private key file format: .pem

9. Select Create key pair Button.



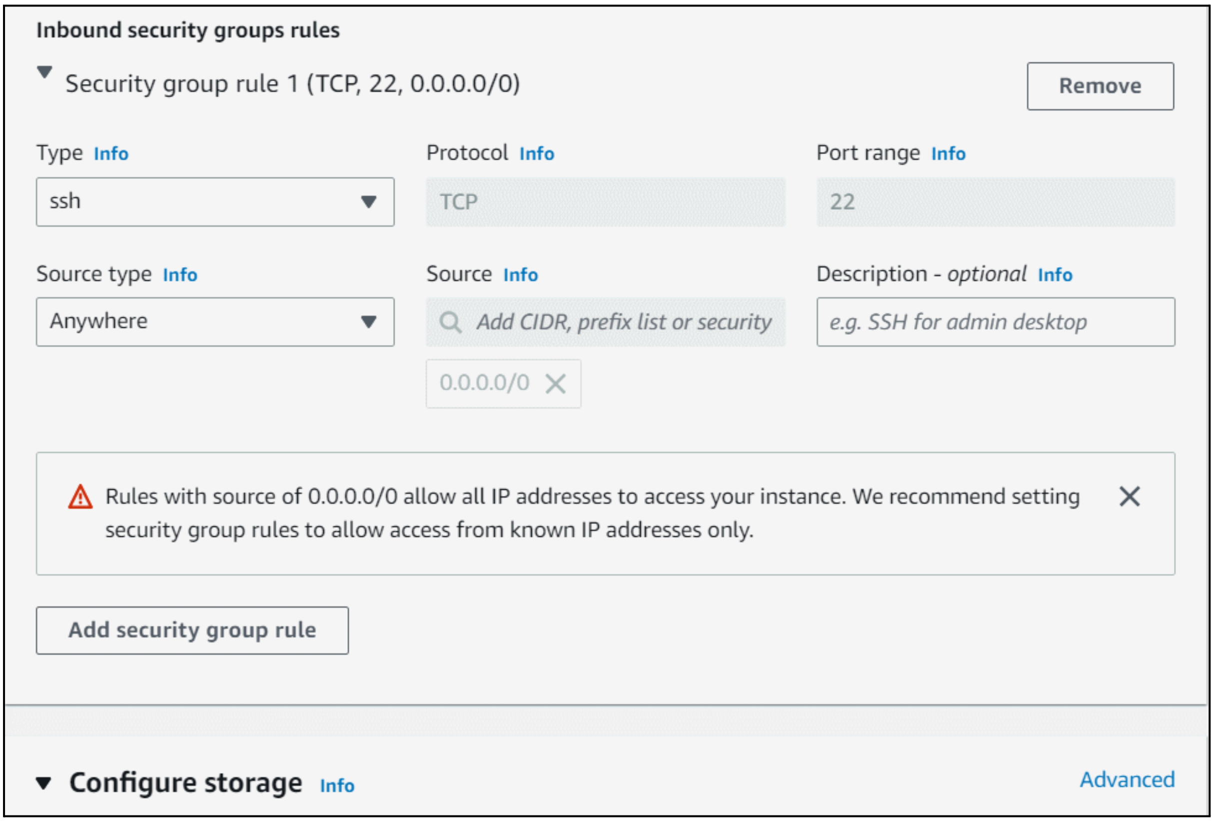
10. In Network Settings Click on Edit button:

- Auto-assign public IP: Enable
- Select Create new Security group
- Security group name : Enter *MyEC2Server_SG*
- Description : Enter *Security Group to allow traffic to EC2*



11. Check Allow SSH from and Select Anywhere from dropdown

- To add SSH,
  - Choose Type: SSH
  - Source: Select Anywhere
- For HTTP, Click on Add security group rule
  - Choose Type: HTTP
  - Source: Select Anywhere
- For HTTPS, Click on Add security group rule
  - Choose Type: HTTPs
  - Source: Select Anywhere



12. Click on Advanced details and under the User data: section, Enter the following script to create an HTML page served by an Apache httpd web server.

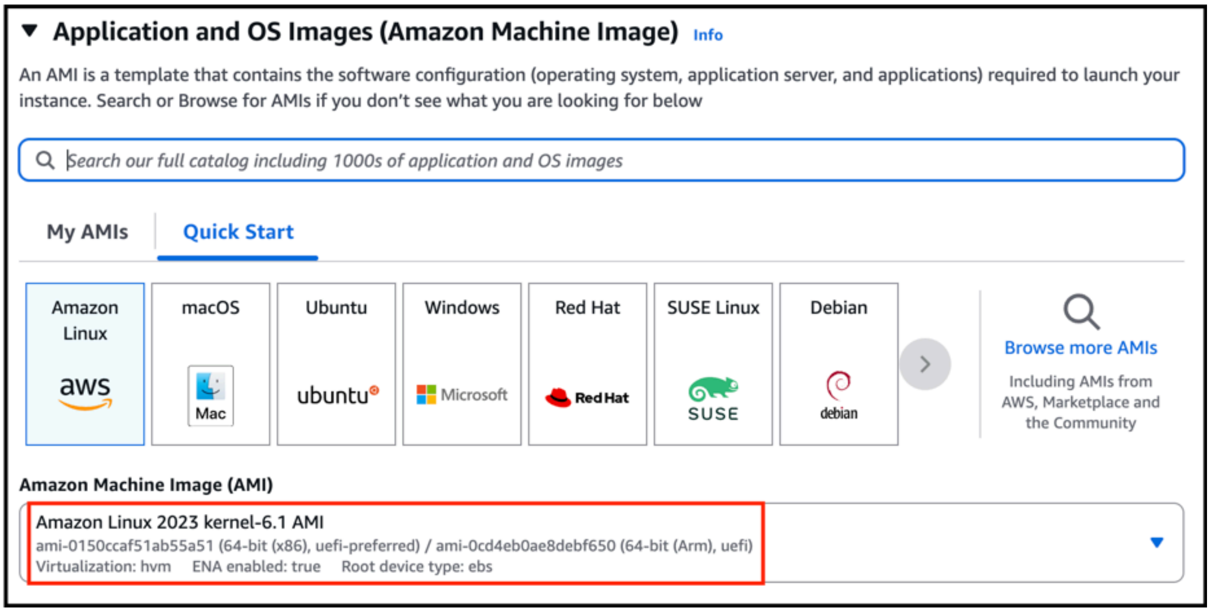
```
#!/bin/bash
sudo su
dnf update -y
dnf install httpd -y
echo "<html><h1> Welcome to Whizlabs Server 1 </h1></html>" >
/var/www/html/index.html
systemctl start httpd
systemctl enable httpd
```

13. Keep Rest thing Default and Click on Launch Instance Button.
14. Select View all Instances to View Instance you Created.
15. Launch Status: Your instance is now launching, Click on the instance ID and wait for complete initialization of the instance till status changes to Running.

Task 4 : Launching Second EC2 Instance

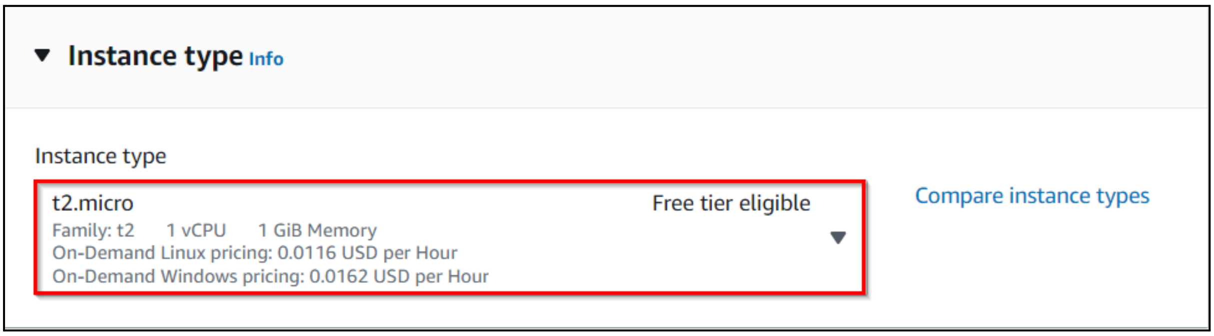
1. Make sure you are in the US East (N. Virginia) us-east-1 Region.
2. Navigate to EC2 by clicking on the Services menu in the top, then click on EC2 in the Compute section.
3. Navigate to Instances on the left panel and click on Launch Instances button.
4. Name : Enter ***MyEC2Server2***

5. For Amazon Machine Image (AMI): Search for **Amazon Linux 2023 AMI** in the search box and click on the Select button.

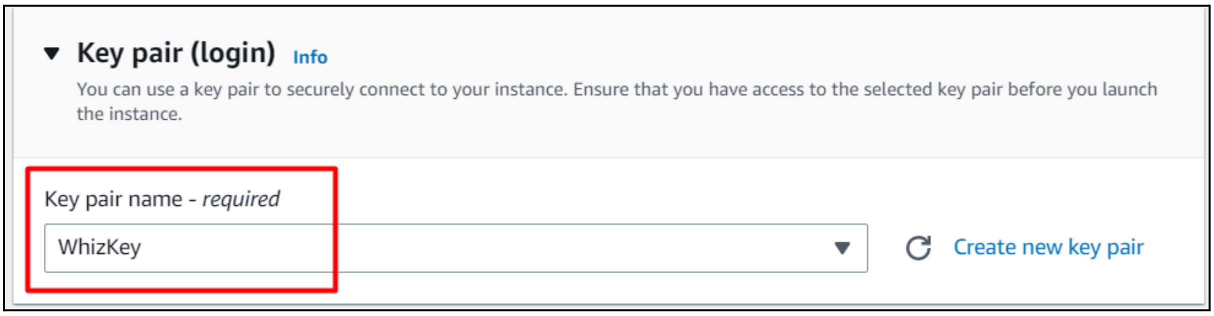


6. Note: if there are two AMI's present for Amazon Linux 2023 AMI, choose **kernel-6.1 AMI.**

7. For Instance Type: select **t2.micro**



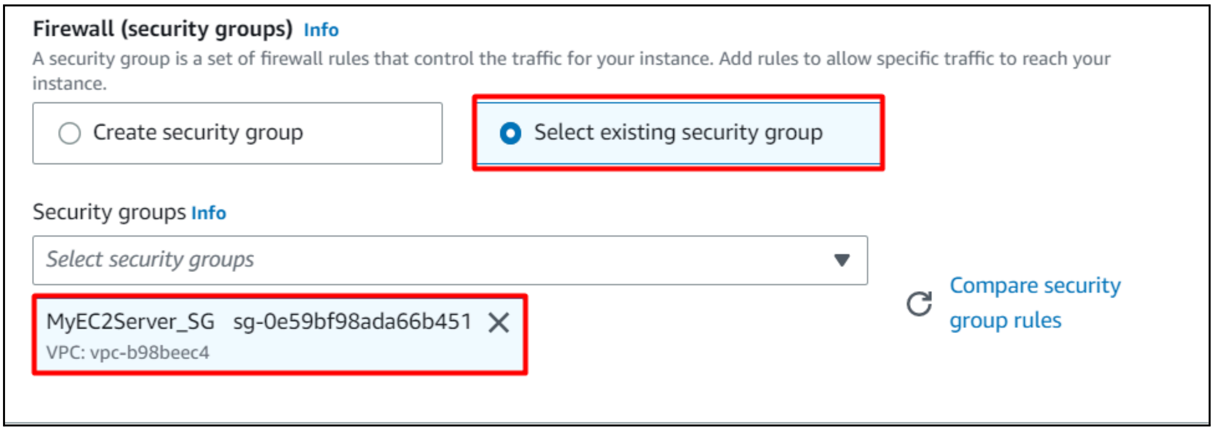
8. For Key pair: Select the key you created previously.



9. In Network Settings Click on Edit:

- Auto-assign public IP: **Enable**

- Select Select existing security group and Select the group created earlier.



10. Click on Advanced details and under the User data: section, Enter the following script to create an HTML page served by Apache httpd web server:

```
11. #!/bin/bash
12. sudo su
13. dnf update -y
14. dnf install httpd -y
15. echo "<html><h1> Welcome to Whizlabs Server 2 </h1><html>"
    > /var/www/html/index.html
16. systemctl start httpd
17. systemctl enable httpd
```

11. Keep Rest thing Default and Click on Launch Instance Button.

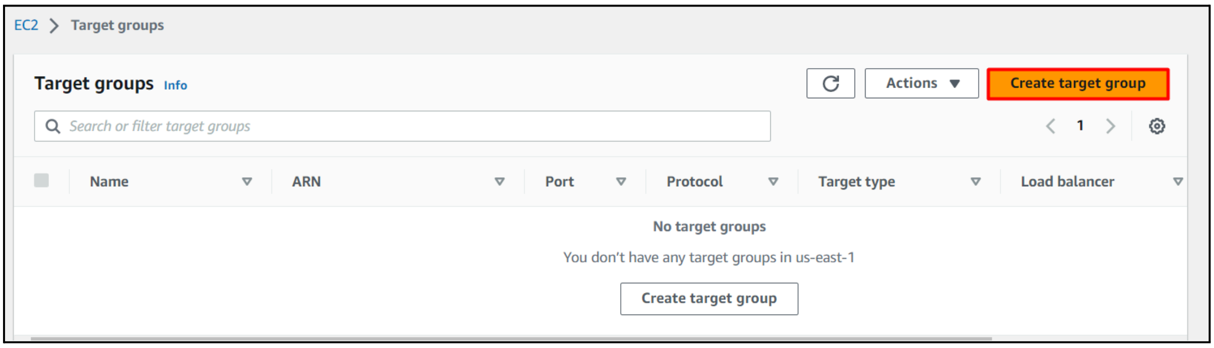
12. Select View all Instances to View Instance you Created

13. Launch Status: Your instance is now launching. Click on View Instances. In the dashboard find your instance and wait for complete initialization of the instance until the instance state changes to running.

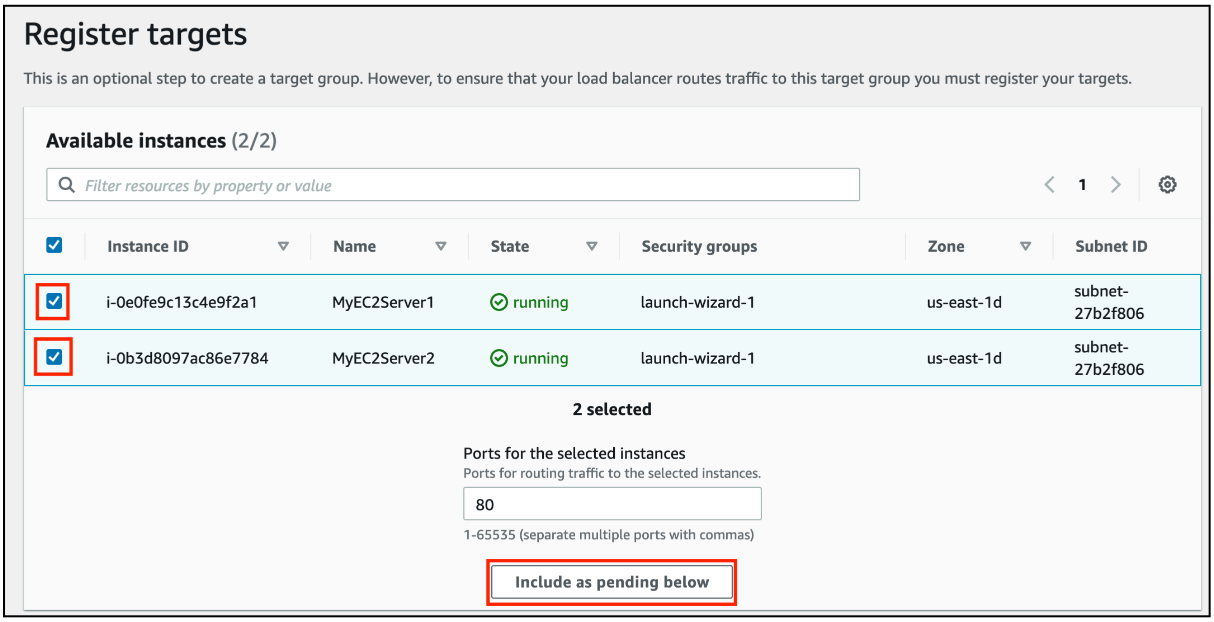
Task 5: Creating the Target Group and Load Balancer

1. In the EC2 Console, Navigate to Target Groups, present in the left panel under Load Balancing.

2. Click on the Create target group button.



3. For Step 1, Specify group details
    - Under Basic configurations,
        - Choose a target group: Choose Instances
        - Target group name: Enter ***MyTargetGroup***
    - Keep all the settings as default.
    - Scroll to the end of the page and click on the Next button.
4. For Step 2, Register targets
    - Select both instances and click on the Include as pending below button.



- Instances will be present in the Review targets part, having health status as Pending.
- Click on the **Create target group** button.

5. The Target group is now created.

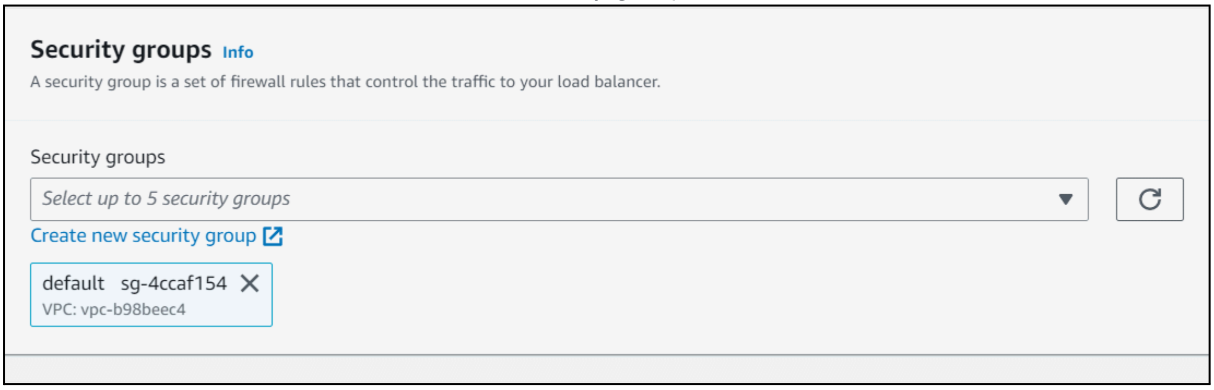
6. In the EC2 console, navigate to Load balancers in the left-side panel.

7. Click on Create Load Balancer button at the top-left to create a new load balancer for our web servers.

8. Select Load Balancer Type: Under the Application load balancer, click on the **Create** button

9. To create an Application load balancer, configuring the load balancer as below

- For the Basic configuration section,
    - Name: Enter ***MyLoadBalancer***
    - Scheme: Select Internet-facing
    - IP address type: Choose IPv4
- For the Network mapping section:
    - VPC: Select Default
    - Mappings: Select all the Availability zone present
- For the Security groups section,
    - Select the ***MyEC2Server_SG*** Security group from the dropdown and remove the default security group.



- For the Listeners and routing section,
    - The listener is already present with Protocol HTTP and Port 80.
        - Select the target group **MyTargetGroup** for the Default action forwards to option.

10. Keep the tags as default and click on the **Create load balancer** button.

11. You have successfully created the Application Load Balancer. Click on the View load balancers button.

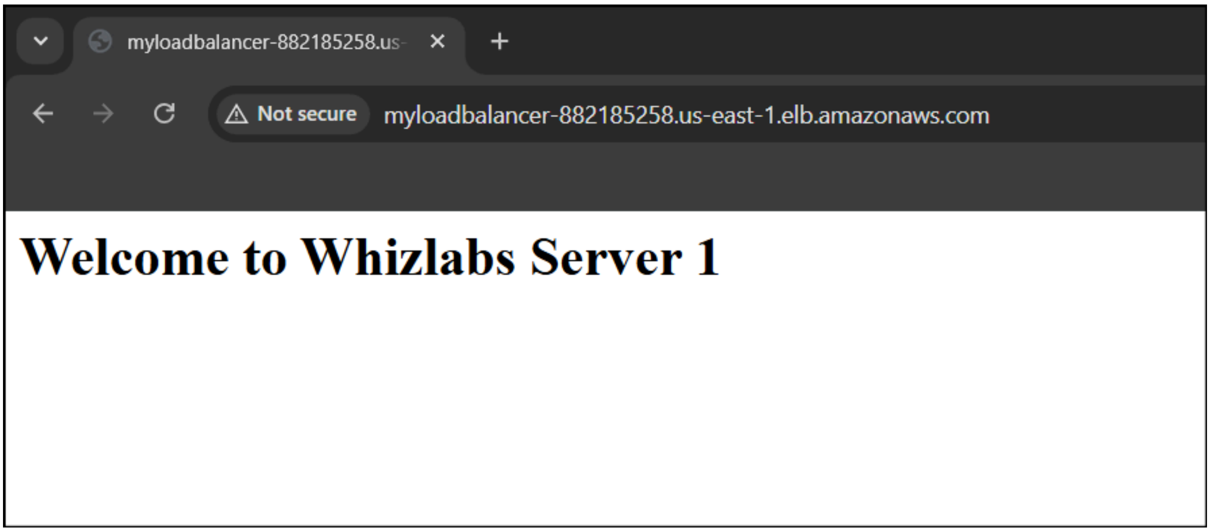
12. Wait for 2 to 3 minutes for the load balancer to become Active.

Task 6: Testing the Elastic Load Balancer

1. Click on Target groups from the left menu section.
2. Select MyTargetGroup and navigate to the Targets tab below.
3. Wait until the status column of the instances changes to healthy (this means both web servers have passed ELB health check)

4. Next, navigate to Load Balancers and notice the state of ELB is active.
5. Copy the DNS name of the ELB and enter the address in the browser.
   - DNS Example: MyLoadBalancer-882185258.us-east-1.elb.amazonaws.com

6. You should see the index.html page content of Web Server 1 or Web Server 2

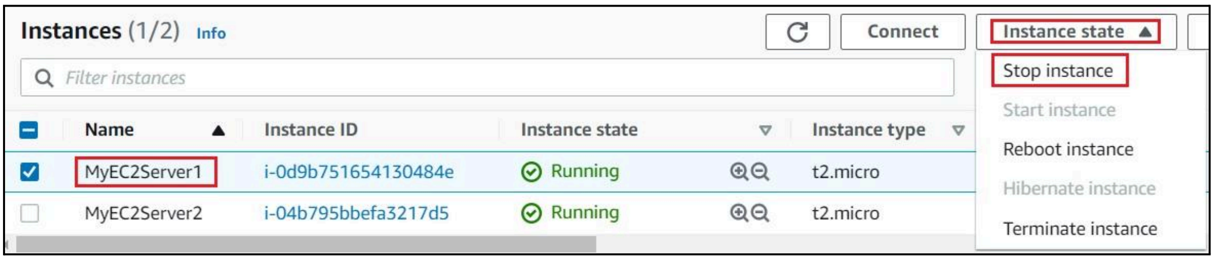


7. Now Refresh the page a few times. You will observe that the index pages change each time you refresh.

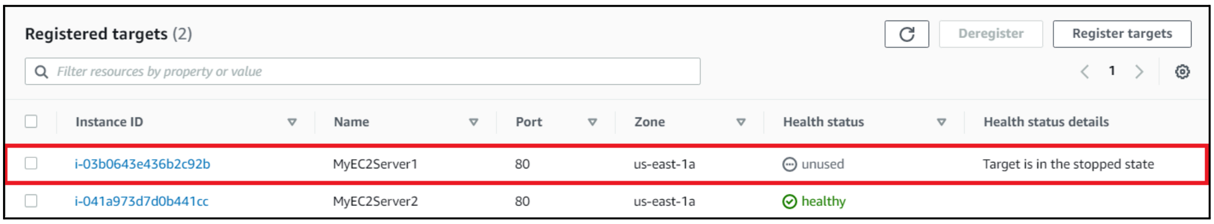
- Note: The ELB is equally dividing the incoming traffic to both servers in a Round Robin manner.

8. For testing, if ELB is working properly,

- In the left side menu, scroll up and navigate back to the Instances page.
- Select MyEC2Server1, click on Instance State and click on Stop instance to stop the EC2 instance.



- Once MyEC2Server1 is stopped, navigate to Target Groups. Select the MyTargetGroup, Click on the Targets.
- It will say that the stopped instance MyEC2Server1 is unused.



- Refresh the ELB domain name URL in Browser, and notice the HTML webpage remains visible. The ELB is only rendering the HTML page from the MyEC2Server2 instance.

Lab Introduction to Amazon Auto Scaling

Task 1: Sign in to AWS Management Console

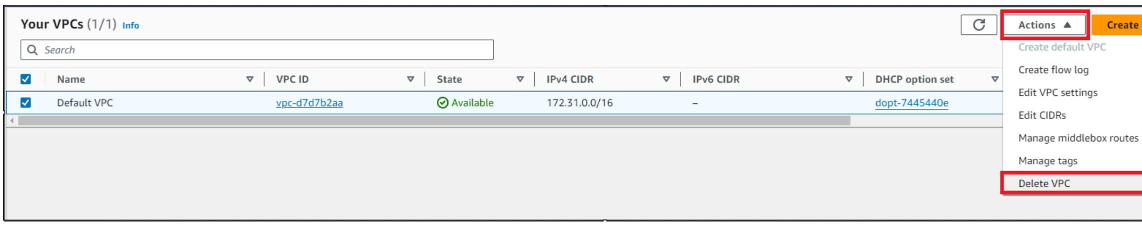
1. Click on the Open Console button, and you will get redirected to AWS Console in a new browser tab.
2. On the AWS sign-in page,
   - Leave the Account ID as default. Never edit/remove the 12-digit Account ID present in the AWS Console. Otherwise, you cannot proceed with the lab.
   - Now copy your Username and Password in the Lab Console to the IAM Username and Password in AWS Console and click on the Sign-in button.
3. Once Signed In to the AWS Management Console, make the default AWS Region as US East (N. Virginia) us-east-1.

Note : If you face any issues, please go through FAQs and Troubleshooting for Labs.

Task 2: Provision Default VPC

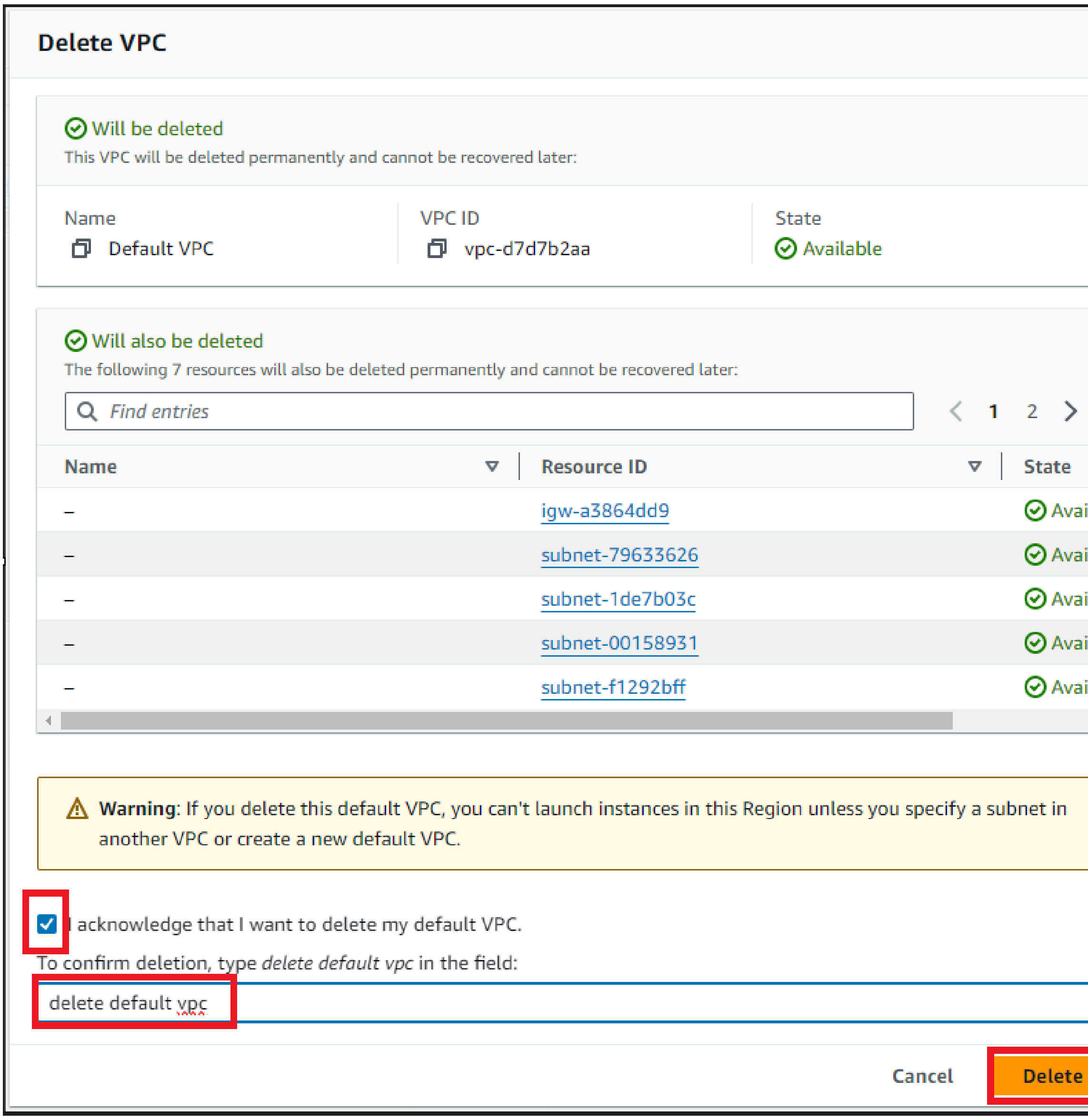
1. Navigate to VPC by clicking on the Services menu in the top, then click on VPC or Open the Amazon VPC console via https://console.aws.amazon.com/vpc/.
2. Delete the default VPC by following the below steps:
   - In the navigation pane, choose Your VPCs.
   - Select the VPC with value as yes in default VPC column.

   - Go to Actions button and click on delete VPC button

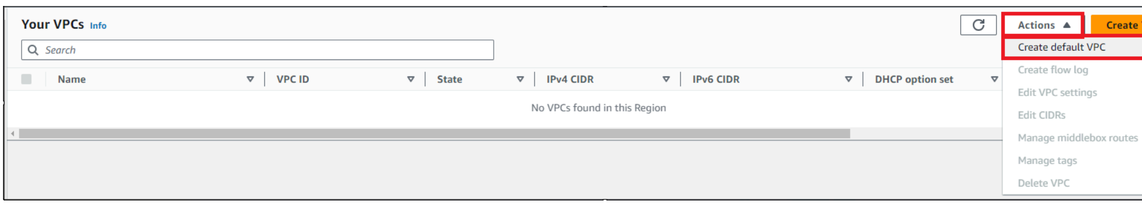


   - Check I acknowledge that I want to delete my default VPC option.

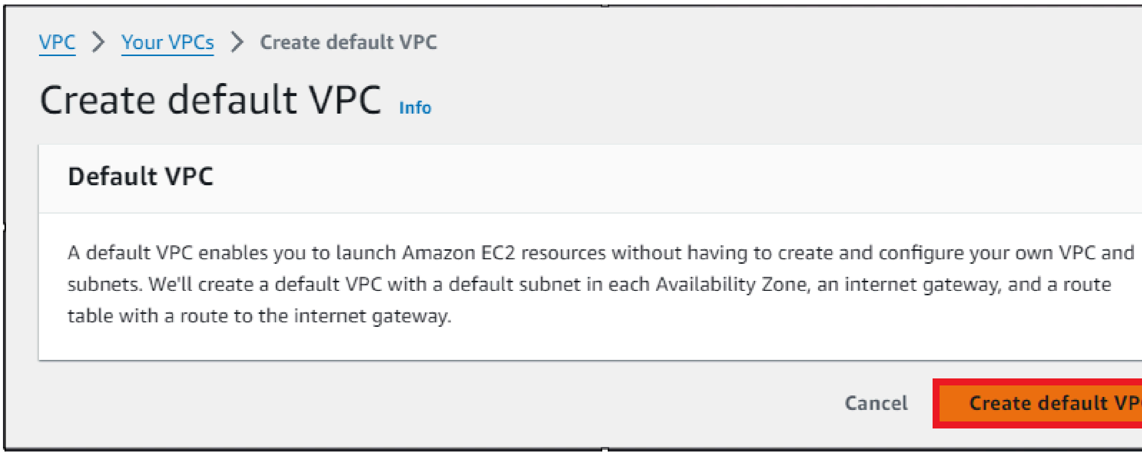
- Type delete default VPC and click on Delete button



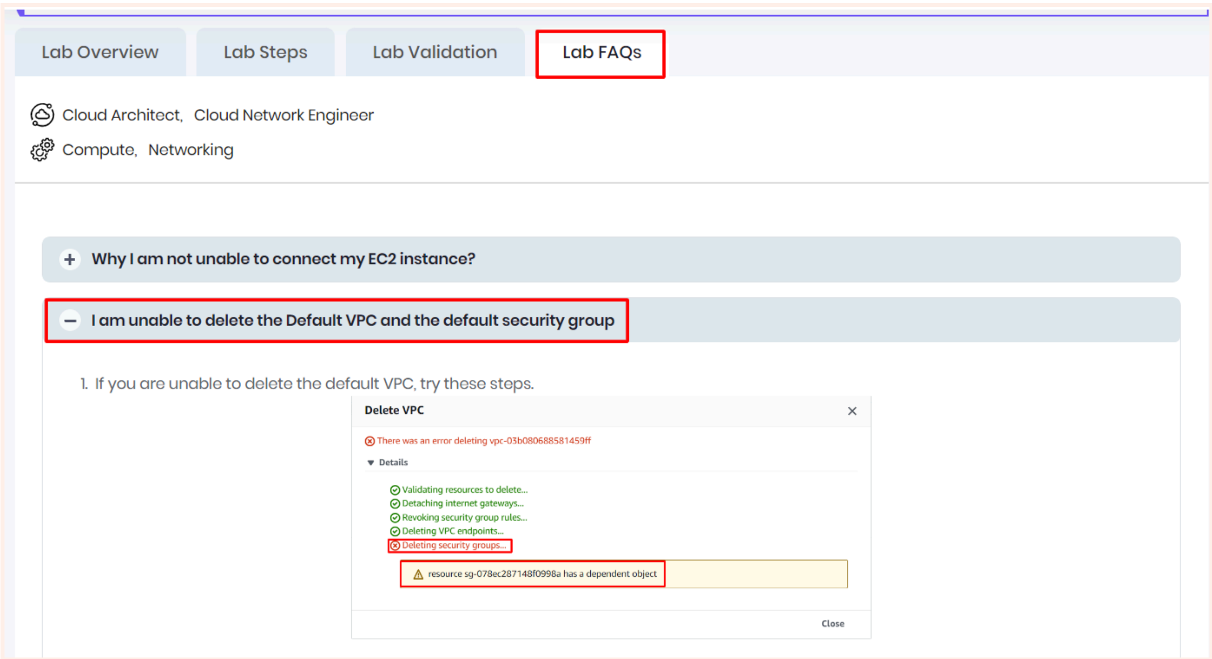
3. Now to provision Default VPC again
   - Refresh your console go to Actions and click Create default VPC



   - Click Create default VPC button and your default VPC will be created



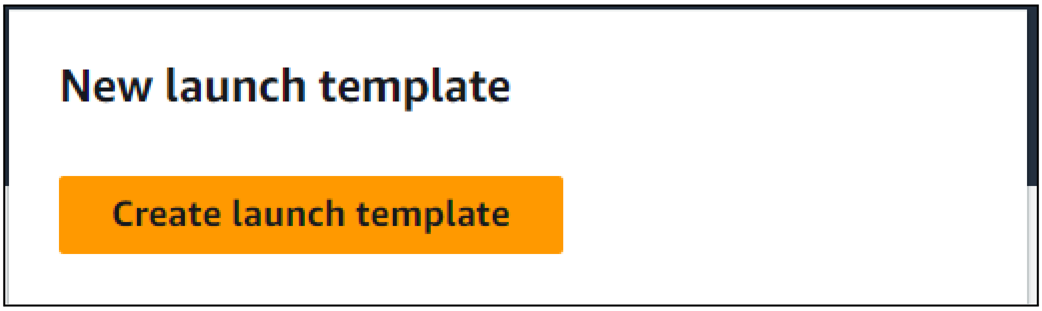
Note: If you are unable to delete the default VPC, please visit our Lab FAQ page for step-by-step instructions.



Task 3: Creating Launch Template

In this task, we are going to create Launch Templates, which are a recommended approach for creating instances in Auto Scaling groups. By creating a Launch Template, we define the configuration settings for their EC2 instances, including the Amazon Machine Image (AMI), instance type, network settings, and security groups.

1. Make sure you are in the US East (N. Virginia) us-east-1 Region before proceeding with the Lab.
2. Navigate to EC2 by clicking on the Services menu in the top, then click on EC2 in the Compute section.
3. In the left navigation menu, scroll down to Launch Templates and click on Create launch template button.

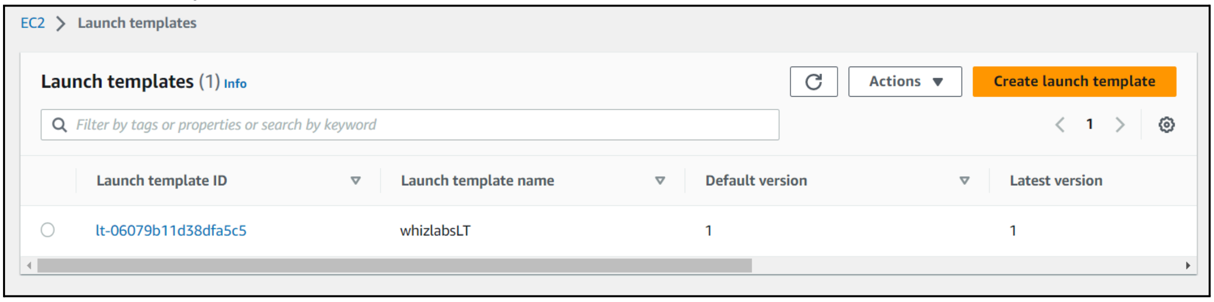


4. Launch template name: Enter whizlabsLT
5. Template version description: Enter Launch template version 1
6. Launch template contents:
    - Amazon machine image (AMI): Select Amazon Linux 2023 kernel-6.1 AMI

    - Instance type: Select t2.micro

- Key pair (login):
    - Key pair name: Don't include in the launch template

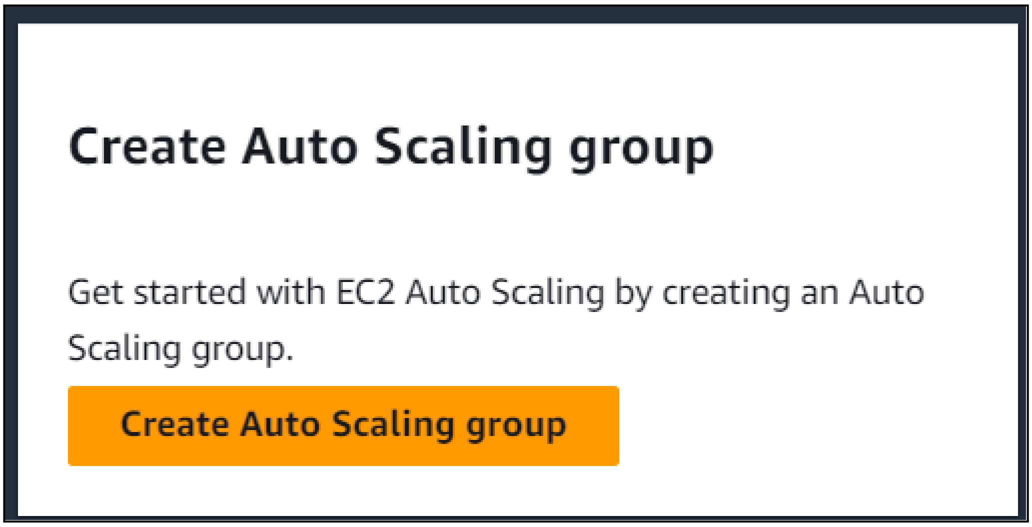
7. Network settings:
    - Security groups: Select the Default security group of Default VPC
8. Keep all the settings as default.
9. Now, click on the Create launch template button
10. The launch template is now created.
11. Click on the View Launch template button.
12. The launch template is now listed.



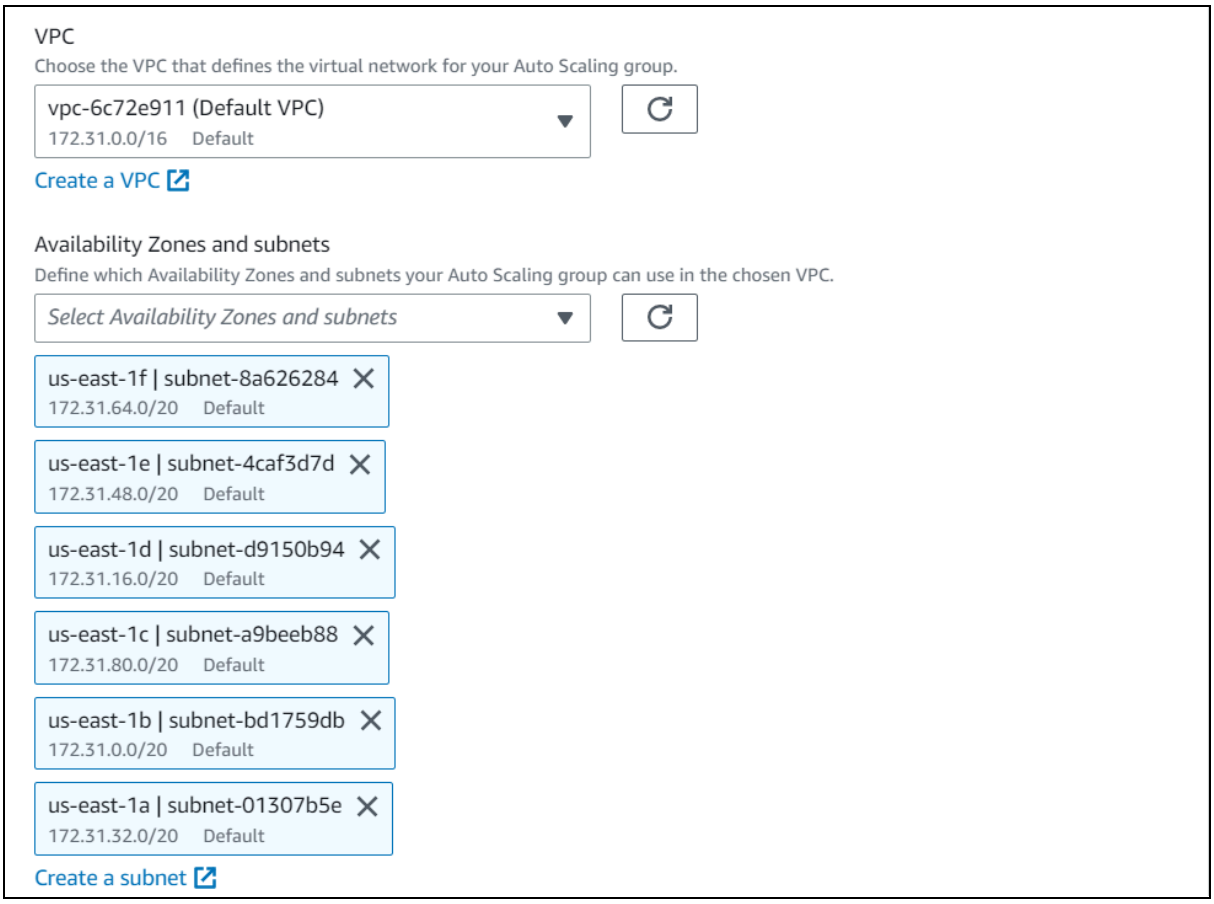
Task 4: Create an Auto Scaling Group

In this task, we are going to create an Auto Scaling group, which is a collection of EC2 instances that can automatically scale based on predefined policies and conditions.

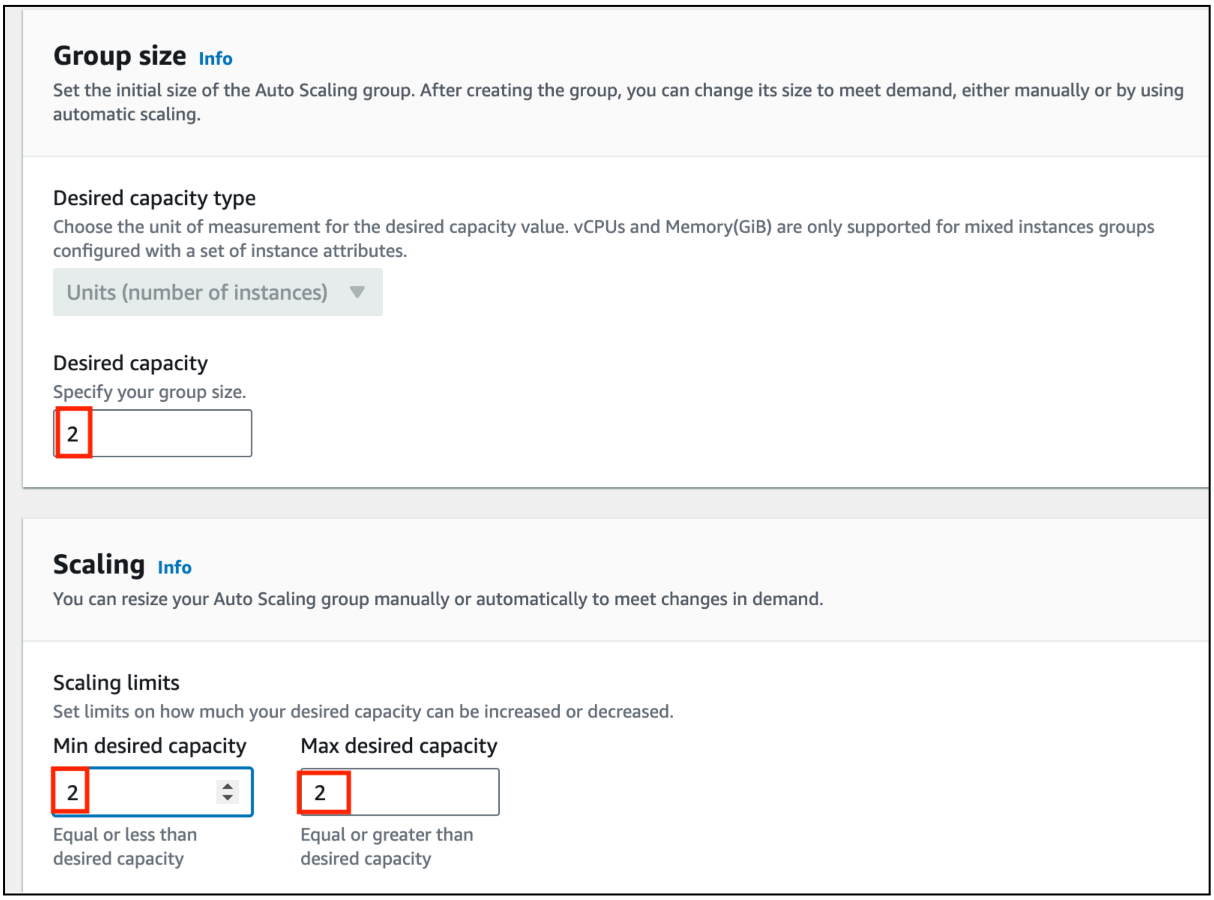
1. An Auto Scaling group is a scalable collection of EC2 instances. When you create an Auto Scaling group, you include information such as the subnets for the instances and the number of instances the group must maintain at all times.
2. Go to the left menu under EC2 and choose Auto Scaling Group under Auto Scaling
3. Click on the Create Auto Scaling group button.



4. Step 1: Choose launch template or configuration
    - Auto Scaling group name: Enter whiz-ASG
    - Launch template: Select whizlabsLT
    - Click on the Next button.
5. Step 2: Choose instance launch options
    - VPC: Select the Default VPC from the list.
    - Subnet: Select all the subnets.
    - Click on the Next button.



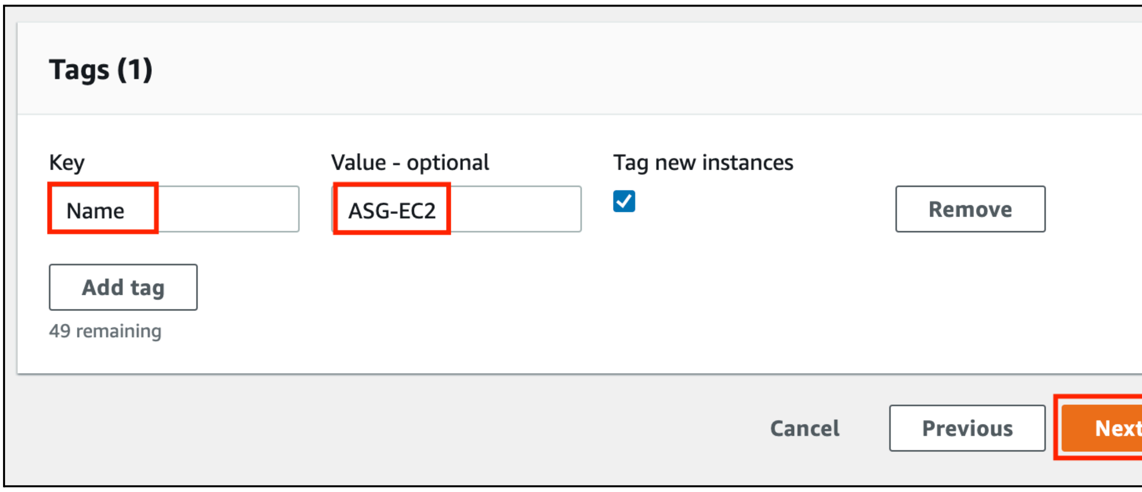
6. Step 3: Integrate with other services
    - No changes are needed on this page, click on the Next button.
7. Step 4: Configure group size and scaling
    - Under Group size – optional
        - Desired capacity: Enter 2
        - Minimum capacity: Enter 2
        - Maximum capacity: Enter 2



- For Instance maintenance policy – is optional, leave it as it is.
- No changes are needed, click on the Next button

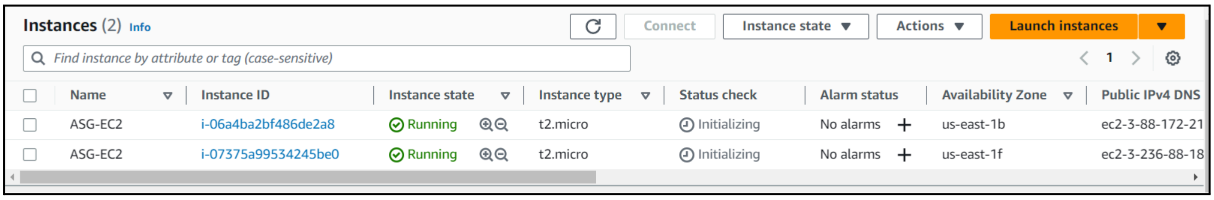
7. Step 5: Add notifications
    - No changes are needed on this page, click on the Next button.
8. Step 6: Add tags
    - Enter tags in key-value pairs to identify your auto-scaling group.
    - Click on the Add tag button.
    - Key: Name

- Value: ASG-EC2



- Click on the Next button

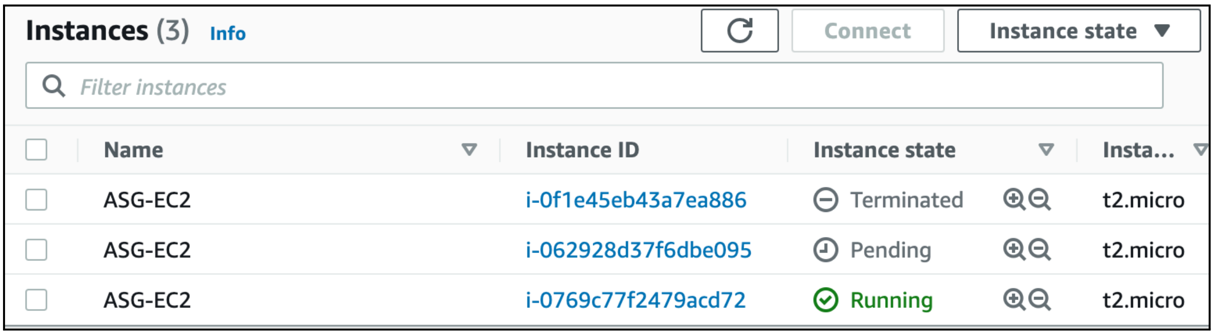
9. Now scroll down and click on the Create Auto Scaling group button.
10. Whiz-ASG Auto scaling is created successfully.
11. You will be redirected to the autoscaling group page, you will be able to see that two instances are launched by the autoscaling group.
12. Now go to the EC2 instances list. You will see that there are two new running instances (which were created by your autoscaling group) You can confirm this from their tag name, which you gave at the time of creating the autoscaling group.
13. You have successfully created an autoscaling group with a policy of a minimum of 2 and a maximum of 2 instances.



Task 5: Test Auto Scaling Group

In this task, we are going to validate the functionality of the Auto Scaling group

1. For testing the auto-scaling policy, go to the EC2 instance list and select one of your instances.
2. Next, select an instance and click on Instance state and then Stop instance.
3. Click the Stop button on the pop-up window to stop your instance.
4. Once your instance is stopped (after 1-2 minutes) you can see that your stopped instance will be terminating automatically, and a new instance will be launched to fulfill the policy condition. A sample screenshot is provided below:



- Note: Launching a new instance may take a few minutes, you can refresh the page to view the new instance.

Lab Access S3 from Private EC2 instance using VPC Endpoint

Task 1: Sign in to AWS Management Console

1. Click on the Open Console button, and you will get redirected to AWS Console in a new browser tab.
2. On the AWS sign-in page,
   - Leave the Account ID as default. Never edit/remove the 12 digit Account ID present in the AWS Console. otherwise, you cannot proceed with the lab.
   - Now copy your User Name and Password in the Lab Console to the IAM Username and Password in AWS Console and click on the Sign in button
3. Once Signed In to the AWS Management Console, Make the default AWS Region as US East (N. Virginia) us-east-1.

Task 2: Create a VPC

1. Make sure you are in the N.Virginia Region.
2. Navigate to VPC by clicking on the Services menu at the top, then click on VPC in the Network and Content Delivery section.
3. To create a VPC click on Your VPCs the present in the VIRTUAL PRIVATE CLOUD section on the left sidebar.
4. Create a new VPC by clicking on the Create VPC button.

- Select VPC only.
- Name tag - optional:Enter *MyVPC*
- IPv4 CIDR block: Enter *192.168.0.0/26*
- IPv6 CIDR block: No IPv6 CIDR block
- Tenancy: Default
- Click on the Create VPC button to create the MyVPC.
- VPC is now created.

Task 3: Create and attach an Internet Gateway with custom VPC

1. By default, instances that are launched in a VPC cannot communicate with the Internet. To enable Internet access, an Internet gateway needed to be attached to the VPC.
2. Click on Internet Gateways from the left menu and click on Create internet gateway.

   - Name Tag : Enter *MyInternetGateway*
   - Click on Create internet gateway.

3. Select the Internet gateway you created from the list.

- Click on Actions.
- Select the Attach to VPC.

4. Available VPCs: Select the MyVPC

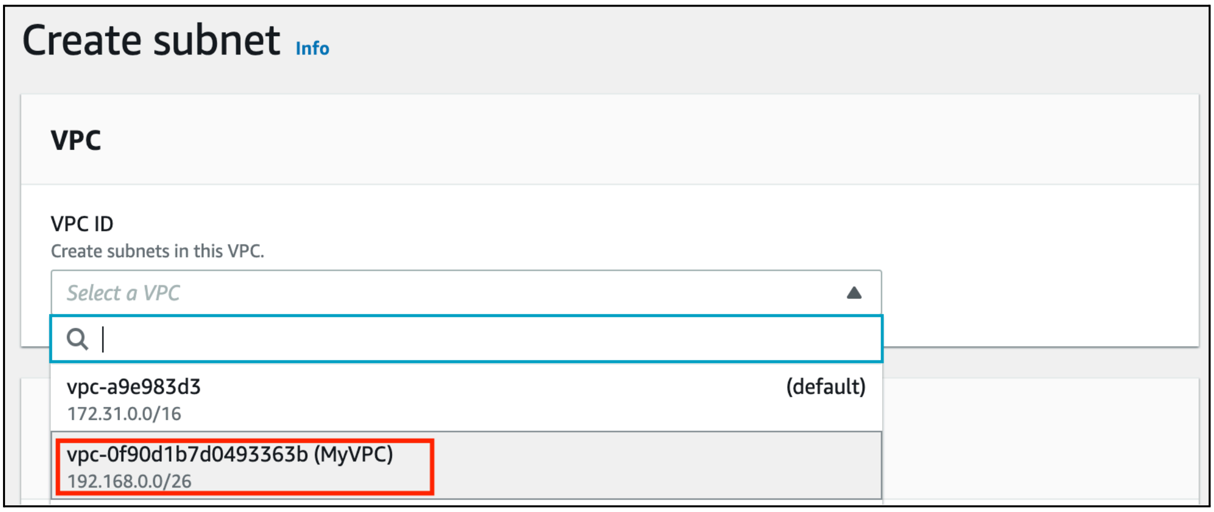
5. And click on the Attach internet gateway button.

6. The Internet gateway is now attached with MyVPC.

Task 4: Create a Public and Private Subnet

1. To create a subnet click on Subnets the present in the VIRTUAL PRIVATE CLOUD section on the left sidebar.
2. Click on the Create Subnet button.

3. In the VPC ID, select MyVPC.



4. Create the first subnet, you will use this subnet to launch public instances, this subnet will be associated with the main route table of the VPC:

- Select the MyVPC from the drop-down.
- Subnet name: Enter *Public subnet*
- Availability Zone: Select US East (N. Virginia) / us-east-1a
- IPV4 CIDR block: Enter *192.168.0.1/27*

- Click on the Create subnet button to create the subnet.

5. Create another subnet, click on the Create subnet button.

6. The second will be called Private subnet, you will use this subnet to launch private instances, this subnet will be associated with a custom route table of the same VPC:

- Select the MyVPC from the drop-down.
- Subnet name: Enter *Private subnet*
- Availability Zone: Select US East (N. Virginia) / us-east-1b
- IPV4 CIDR block: Enter *192.168.0.32/27*

7. Finally, click on the Create Subnet button.

8. Both the subnets are now created.

Task 5: Configure the Public subnet to enable auto-assign public IPv4 address

1. To modify the auto-assign IP settings for the Public subnet, do the following:
    - Select the Public subnet
    - Click on the Actions button
    - Choose Edit subnet settings from the options.

2. Check the option Enable auto-assign public IPv4 address under Auto-assign IP settings.

3. Click on Save button and modification is done now.

Task 6: Create a Route Table for the Public subnet

In this task, we are going to create public route tables and associate it with the subnet.

1. Go to Route Tables from the left menu and click on Create route table button.
    - Name: Enter *PublicRouteTable*
    - VPC: Select MyVPC from the list.
    - Click on Create route table button.

2. Repeat the same steps to create a route table for the Private subnet.

- Name: Enter *PrivateRouteTable*
    - VPC: Select MyVPC from the list.
    - Click on Create route table button.
3. Now we will associate the subnets to the route tables.
4. Select the PublicRouteTable and go to the Subnet Associations tab.
    - Click on Edit subnet associations.
    - Select MyPublicSubnet from the list.
    - Click on Save associations button.

5. Select the PrivateRouteTable and go to the Subnet Associations tab.
    - Click on Edit subnet associations.
    - Select MyPrivateSubnet from the list.
    - Click on Save associations button.
6. Make sure not to associate any subnets with the Main Route Table.
7. PublicRouteTable: Add a route to allow Internet traffic to the VPC.
    - Select PublicRouteTable.
    - Go to Routes tab, click on Edit routes and on the next page, click on Add route button.
    - Specify the following values:
        - Destination: Enter *0.0.0.0/0*
        - Target: Select Internet Gateway from the dropdown menu to select MyInternetGateway.
        - Click on Save changes button.

Task 7: Create Security groups

1. In this lab, we will create two security groups, the first one will be used for Bastion host and the second one for private instance having access to VPC Endpoint for S3.
2. To get started with creating security groups, click on the Security groups , present in the SECURITY section in the left sidebar.
3. Click on the Create Security group button
4. Fill in the below details under Basic details:

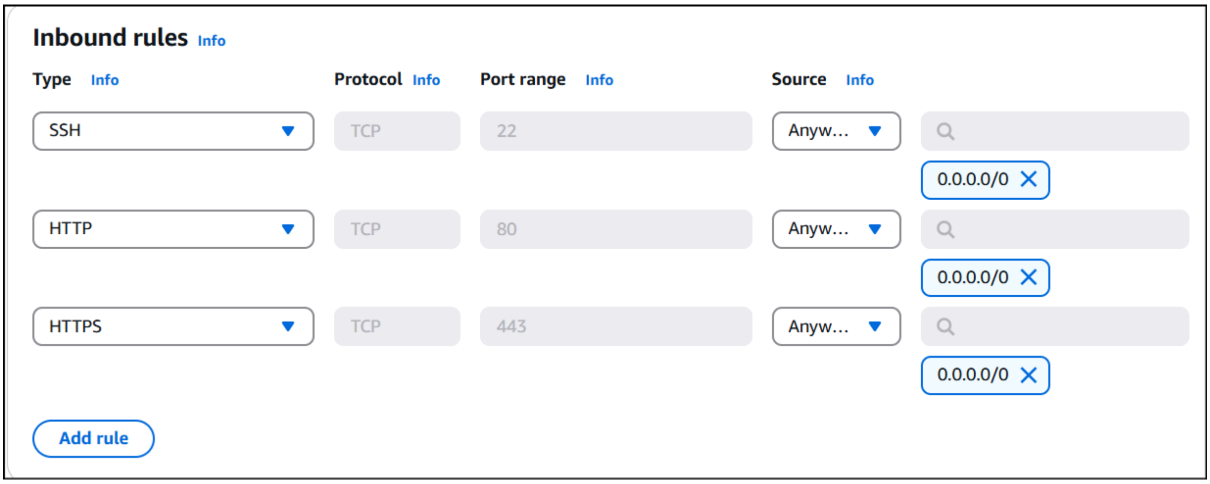
- Enter Security group name as *Bastion-SG*
- Enter Description as *Security group for the bastion host*
- Select the MyVPC under the VPC field.

5. By default, no inbound rule will be allowed, and when you check in the outbound rules, there is only one rule present that has a Type with All traffic because Security groups are Stateful in nature, when inbound is allowed outbound is also allowed.

6. To add the Inbound rules for the same, click on the Add rule button.

7. We will add 3 rules for the Bastion host security group i.e. SSH, HTTP, and HTTPS.

- For the first rule, Select the Type as *SSH*, Source as Anywhere-IPv4 and enter *0.0.0.0/0*
- For the second rule, click on the Add rule button. Select the Type as *HTTP*, Source as Anywhere-IPv4 and enter *0.0.0.0/0*
- For the third rule, click on the Add rule button. Select the Type as *HTTPS*, Source as Anywhere-IPv4 and **enter *0.0.0.0/0***



8. Finally, click on the Create Security group button.

9. The security group for the Bastion host is now created.

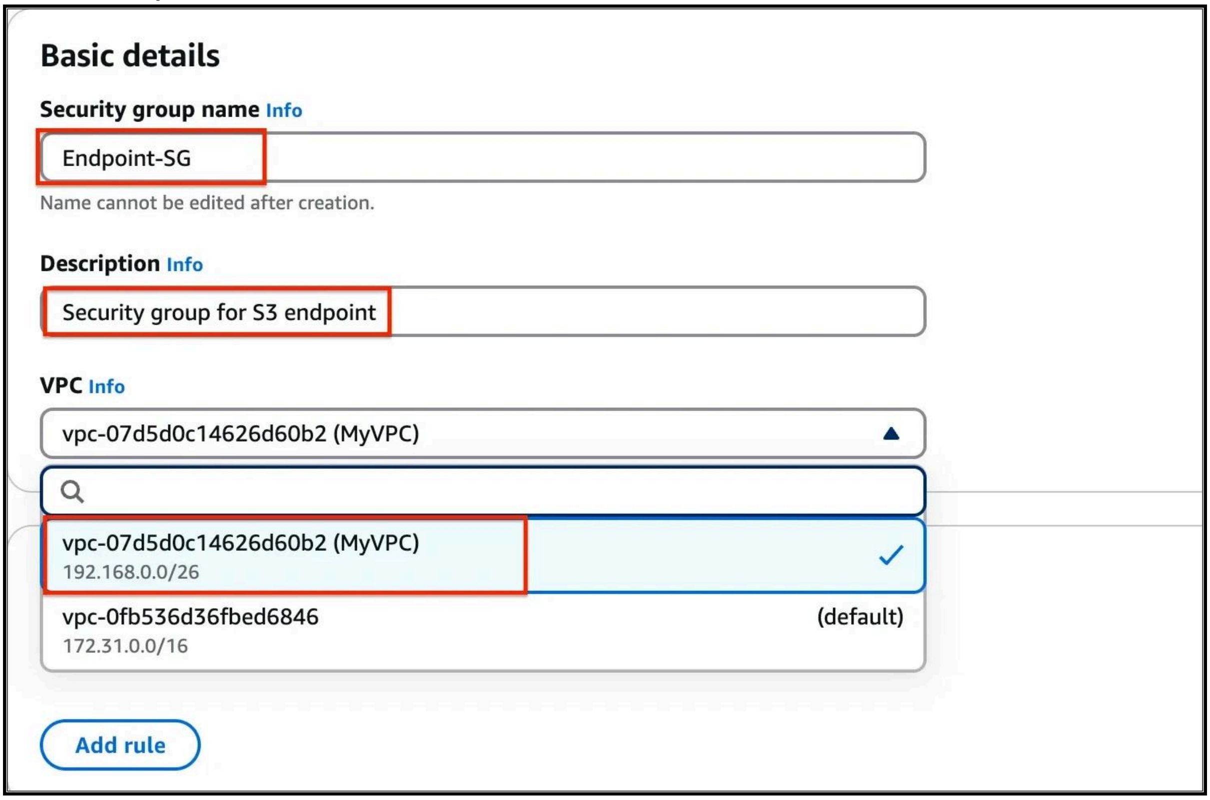
10. Click on the Security groups button, present in the SECURITY section in the left sidebar.

11. To create the second security group, click on the Create Security Group button.

12. Fill in the below details under Basic details:

- Enter Security group name as *Endpoint-SG*
- Enter Description as *Security group for S3 endpoint*

- Select the MyVPC under the VPC field.

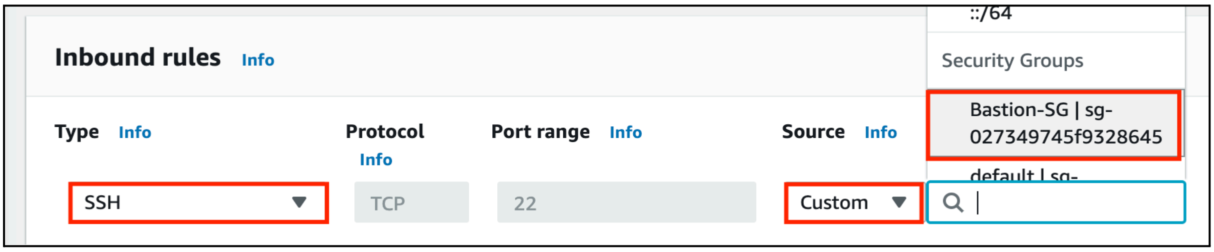


13. To add the Inbound rules for the same, click on the Add rule button.

14. We will add 1 rule for the S3 endpoint security group i.e. SSH *only*. But here our source will be Bastion host security group,

you will select the ID of Bastion-SG security group.

- For the rule, select the Type as *SSH*, Source as Custom, and type *Bastion*, Bastion hosts security group will be shown, select that Security group.



15. Finally, click on the Create Security group button.

16. The security group for the Bastion host is now created. And, it will be listed there.

Task 8: Create a Bastion host (Publicly accessible EC2 Instance)

1. Navigate to EC2 by clicking on the Services menu at the top, then click on EC2 in the Compute section.

2. Navigate to Instances on the left panel and click on Launch instances.

3. Name : Enter *Bastion-host.*

4. For Amazon Machine Image (AMI): Search for Amazon Linux 2023 AMI in the search box and click on the Select button

5 . Note: if there are two AMI's present for Amazon Linux 2023 AMI, choose kernel-6.1 AMI.

6. For Instance Type: select *t2.micro*

7. For Key pair: Select Create a new key pair Button

- Key pair name: WhizKey
- Key pair type: RSA
- Private key file format: .pem

8. Select Create key pair Button.

9. In Network Settings Click on Edit Button:

- VPC: Choose MyVPC
- Subnet : Choose Public Subnet
- Auto-assign public IP: Enable
- Select existing security group

- Security group name : Choose Bastion-SG

10. Keep rest thing default and Click on Launch Instance button.

11. Select View all Instances to view Instance you created

12. Launch Status: Your instance is now launching, Click on the instance ID and wait for complete initialization of the instance till status changes to Running.

Task 9: Create an Endpoint instance (Privately accessible EC2 instance)

1. Navigate to EC2 by clicking on the Services menu at the top, then click on EC2 in the Compute section.

2. Navigate to Instances on the left panel and click on Launch instances.

3. Name : Enter *Endpoint-instance.*

4. For Amazon Machine Image (AMI): Search for Amazon Linux 2023 AMI in the search box and click on the Select button.

5 . Note: if there are two AMI's present for Amazon Linux 2023 AMI, choose kernel-6.1 AMI.

6. For Instance Type: select *t2.micro*

7. For Key pair: Select the key pair made during the previous task.

8. In Network Settings Click on Edit Button:

- VPC: Choose MyVPC
- Subnet : Private Subnet
- Select existing security group
- Security group name : Choose Endpoint-SG

9. Keep rest thing Default and Click on Launch Instance button.

10. Select View all Instances to view Instance you created

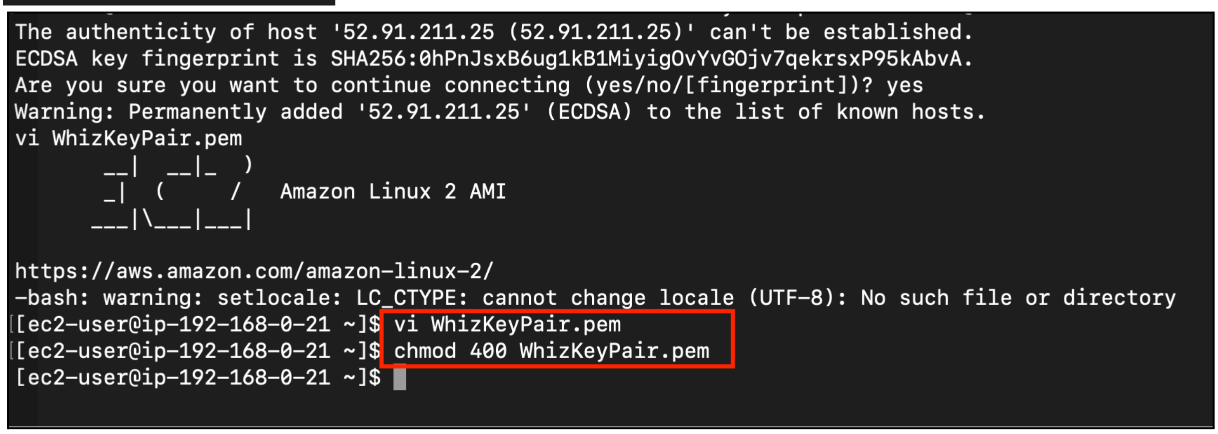
11. Launch Status: Your instance is now launching, Click on the instance ID and wait for complete initialization of the instance till status changes to Running.

12. Navigate to Instances and wait for 1-2 minutes (until the Endpoint-instance's status changes from pending to running state)

Task 10: SSH into Endpoint instance (Privately accessible) through Bastion host

1. SSH into the Bastion instance using the Bastion PEM key: WhizKey.pem
2. To SSH into Endpoint instance via the Bastion instance, we need the WhizKey.pem to be present on the Bastion instance.
3. Open the WhizKey.pem file on your local system and then copy the text content.
4. Navigate to the Bastion Instance and create a file named WhizKey.pem using the below command:

5. Press **i** and paste the content of WhizKey.pem. Save it by pressing Esc key and type :wq and hit Enter.
6. Make sure you have changed the permission of the key file to 400. You can change the permission using the below command:



7. Now you can log into the web servers using the private key copied to the bastion server with the help of the below commands.

- Note: You don't have public IP's for the Endpoint instance since we created them in a private subnet.
- Syntax : ssh -i WhizKey.pem ec2-user@<Endpoint instance's Private IP>
- Example: ssh -i WhizKey.pem ec2-user@192.168.0.55.
- When asked for confirmation type: yes

```
[ec2-user@ip-192-168-0-21 ~]$ ssh -i "WhizKeyPair.pem" ec2-user@192.168.0.55
The authenticity of host '192.168.0.55 (192.168.0.55)' can't be established.
ECDSA key fingerprint is SHA256:KY08k/NrcUOpQ4WbfekA0UQDPYWBnK7whCY+N5uX/cM.
ECDSA key fingerprint is MD5:82:a0:02:05:13:57:4c:3f:35:f6:a0:2b:79:d4:11:19.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added '192.168.0.55' (ECDSA) to the list of known hosts.

       __|  __|_  )
       _|  (     /   Amazon Linux 2 AMI
      ___|\___|___|

https://aws.amazon.com/amazon-linux-2/
[ec2-user@ip-192-168-0-55 ~]$
```

- Enter the following command **aws configure**
- Access Key: Paste the access key provided to you
- Secret Key: Paste the secret key provided to you
- Default region name: us-east-1
- Default output-format: Enter [**ENTER**]
  - **Note**: Though the assigned IAM role is having access for S3 Read, listing the bucket through AWS CLI command got failed, saying, connection timeout on S3's endpoint.

```
[ec2-user@ip-192-168-0-55 ~]$ aws s3 ls

Connect timeout on endpoint URL: "https://s3.amazonaws.com/"
[ec2-user@ip-192-168-0-55 ~]$
```

8. As, this instance's security group is only allowed to do SSH, running any other command, will fail.
9. Let's add the permission to access the S3 endpoints using the VPC Endpoint for S3.

Task 11: Create a VPC Endpoint for S3, attach it to the Private subnet's Route table

1. Navigate to VPC by clicking on the Services menu at the top, then click on VPC in the Networking and Content Delivery section.
2. Click on Endpoints present under **PrivateLink and Lattice.**
3. Click on the Create endpoints button.

4. Make sure the Service category is selected for AWS services. In the Service name search bar, type *s3*, and press enter

5. The endpoint of Type, Gateway with Service name as com.amazonaws.us-east-1.s3 will be listed.

6. Change the VPC, and select MyVPC.

7. Check the option for Route Table having name as PrivateRouteTable.

8. Finally, click on the Create endpoint button.

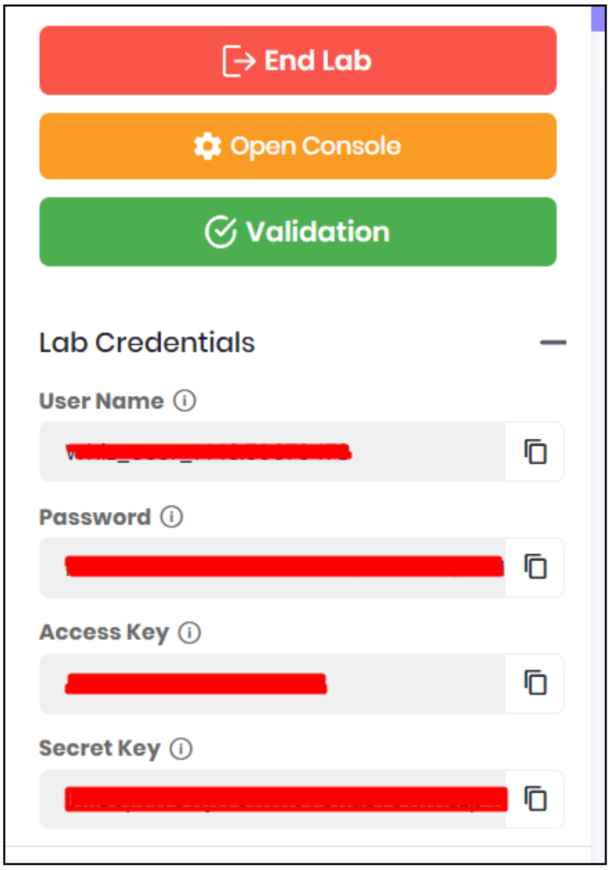
9. An endpoint will be created.

10. Click on the Close button, and within a few moments, you will see the endpoint will be listed.

11. (Optional) To check whether the endpoint is associated with the custom route table (RT for Private subnet) or not.
12. Go to the Route tables, Select the custom route table and click on the Routes options below, you will see an entry of S3.

Task 12: List all the S3 bucket and it's objects

1. Now SSH into the **Endpoint instance** from your bastion instance as mentioned in **task 10**.
   Enter aws configure.
   - Access Key: Paste the access key provided to you.
   - Secret Key: Paste the secret key provided to you.



- Default region name: us-east-1
- Default output-format: Enter [ENTER]

2. List all the bucket's using the following AWS CLI command:

```
[ec2-user@ip-192-168-0-55 ~]$ aws s3 ls
2019-12-07 09:09:02 athenacheck
```

3. List the objects of the S3 bucket starting with name whizlabs..

4. Replace the S3 bucket name for the below command

LAB : VPC Peering

Task 1: Sign in to the AWS Management Console

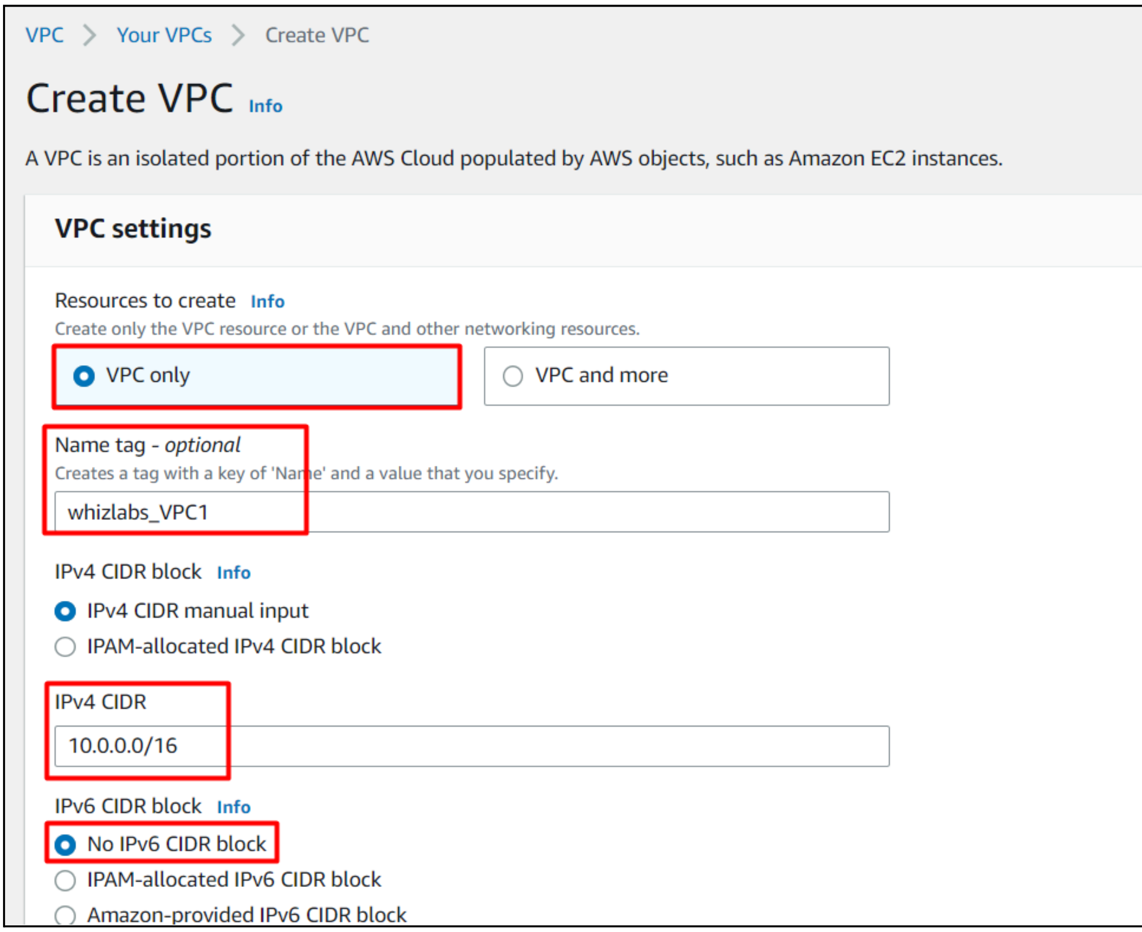
1. Click on the Open Console button, and you will get redirected to AWS Console in a new browser tab.
2. On the AWS sign-in page,

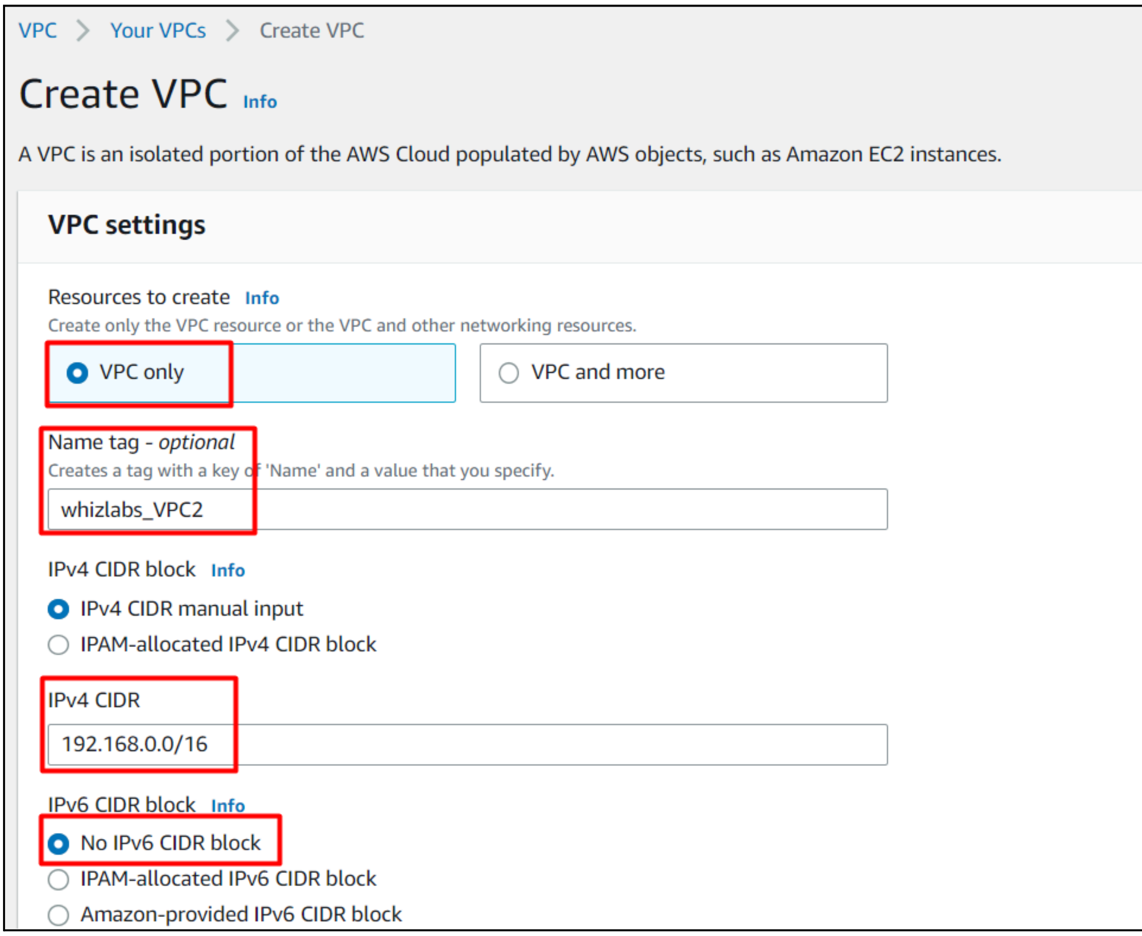
- Leave the Account ID as default. Never edit/remove the 12-digit Account ID present in the AWS Console. otherwise, you cannot proceed with the lab.
- Now copy your User Name and Password in the Lab Console to the IAM Username and Password in the AWS Console and click on the Sign in button.

3. Once Signed In to the AWS Management Console, Make the default AWS Region as US East (N. Virginia) us-east-1.

Task 2: Create two VPCs' in N. Virginia region

1. Make sure you are in the N. Virginia region.
2. Navigate to VPC by clicking on the Services menu at the top, then click on VPC in the Network and Content Delivery section.
3. Navigate to Your VPCs on the left panel and click on the Create VPC button.
    - Resources to create: Select VPC only
    - Name tag: Enter whizlabs_VPC1
    - IPv4 CIDR block: Enter 10.0.0.0/16

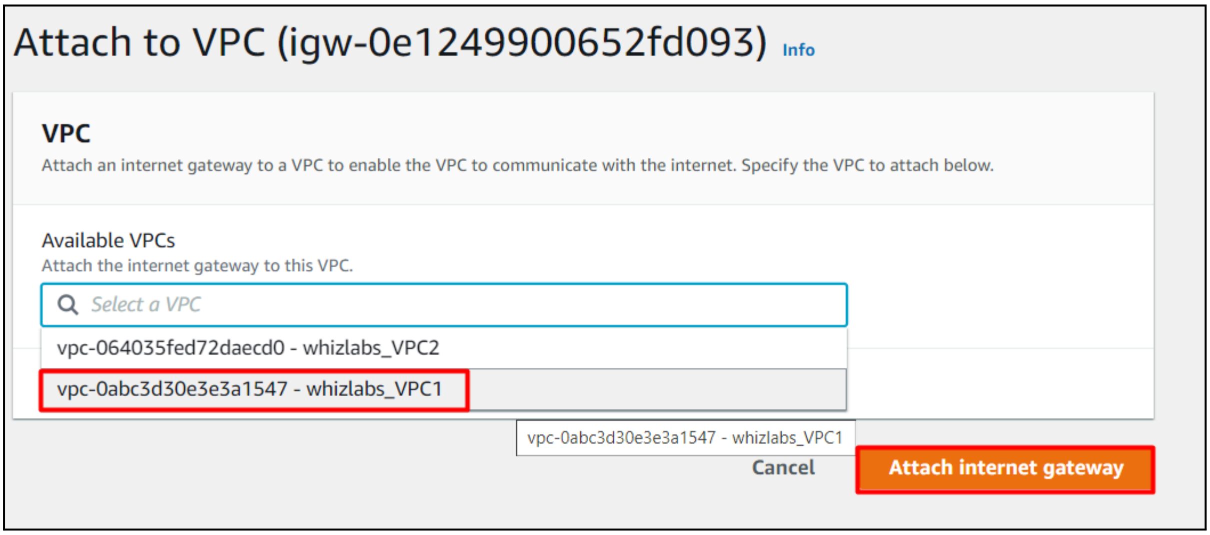


4. Click on Create VPC. To create the 2nd VPC, click on the Create VPC at the top right.
    - Resources to create: Select VPC only
    - Name tag: Enter whizlabs_VPC2
    - IPv4 CIDR block: Enter 192.168.0.0/16
        - Click on the Create VPC button.
    - 

Note: Kindly note that there will be a Default VPC and don't edit or delete the default VPC.

Task 3: Create and attach an Internet gateway

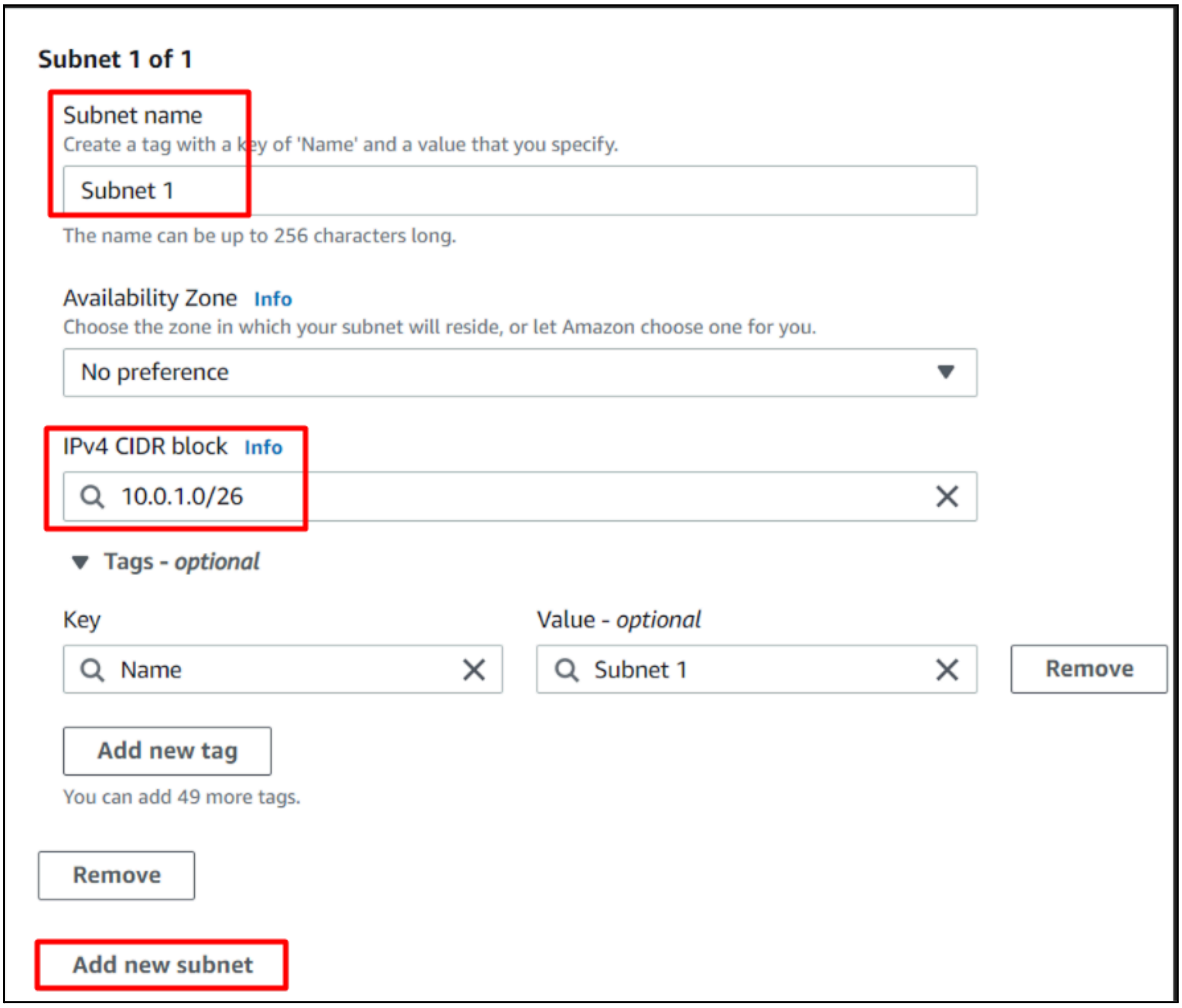
1. Navigate to the Internet Gateways from the left side menu and click on the Create internet gateway button.
    - Name tag: Enter whizlabs_IGW
2. Click on the Create internet gateway
3. Now click on the Action button and select Attach to VPC.
    - Available VPCs: Select whizlabs_VPC1 from the list.



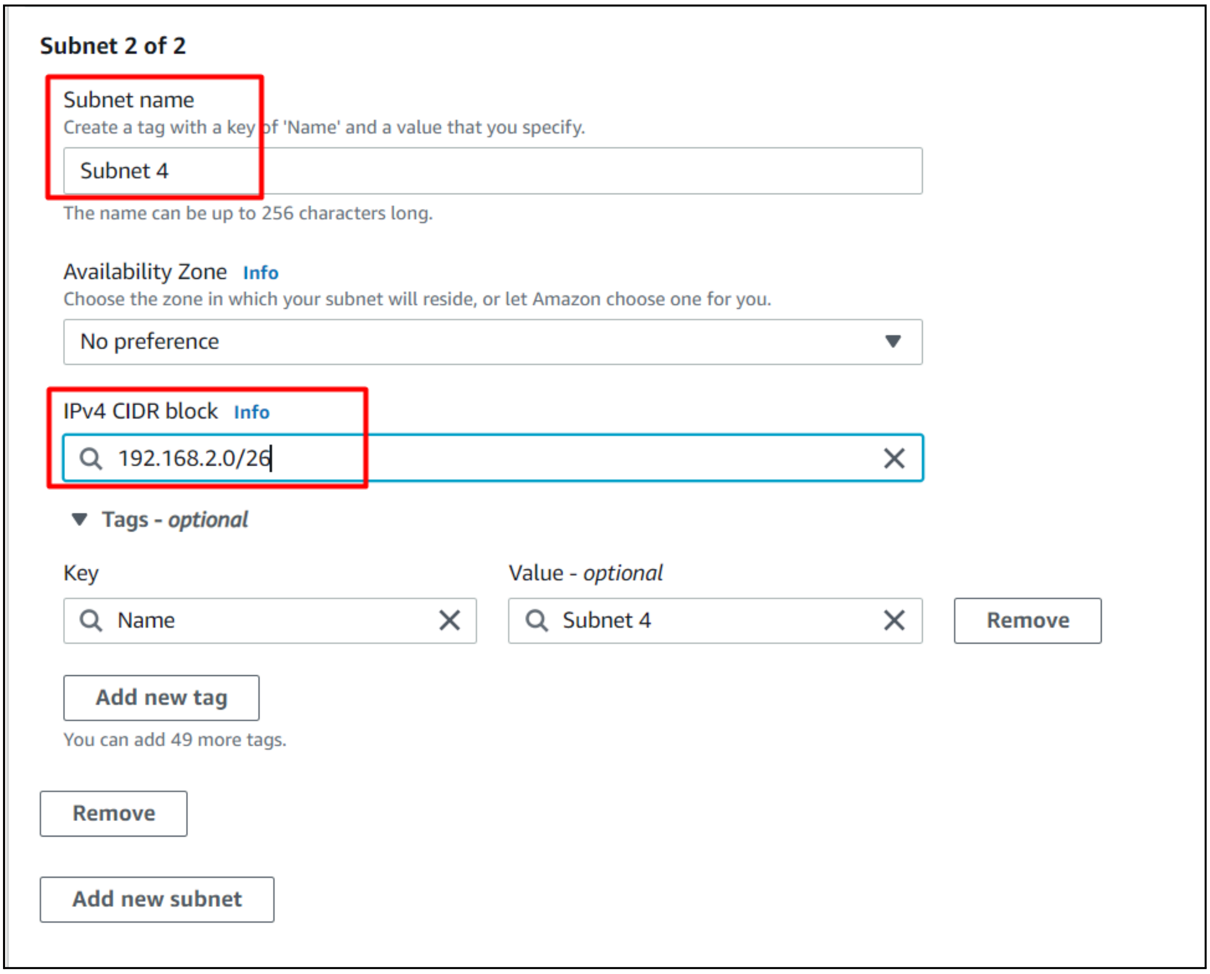
4. Now click on the Attach internet gateway.

Task 4: Creating subnets for both the VPCs

1. Navigate to Subnets from the left side menu and click on Create Subnet.
   - VPC ID: Select the whizlabs_VPC1 VPC from the list.
   - Subnet name: Enter Subnet 1
   - Availability Zone: Leave as No Preference
   - IPv4 CIDR block: Enter 10.0.1.0/26

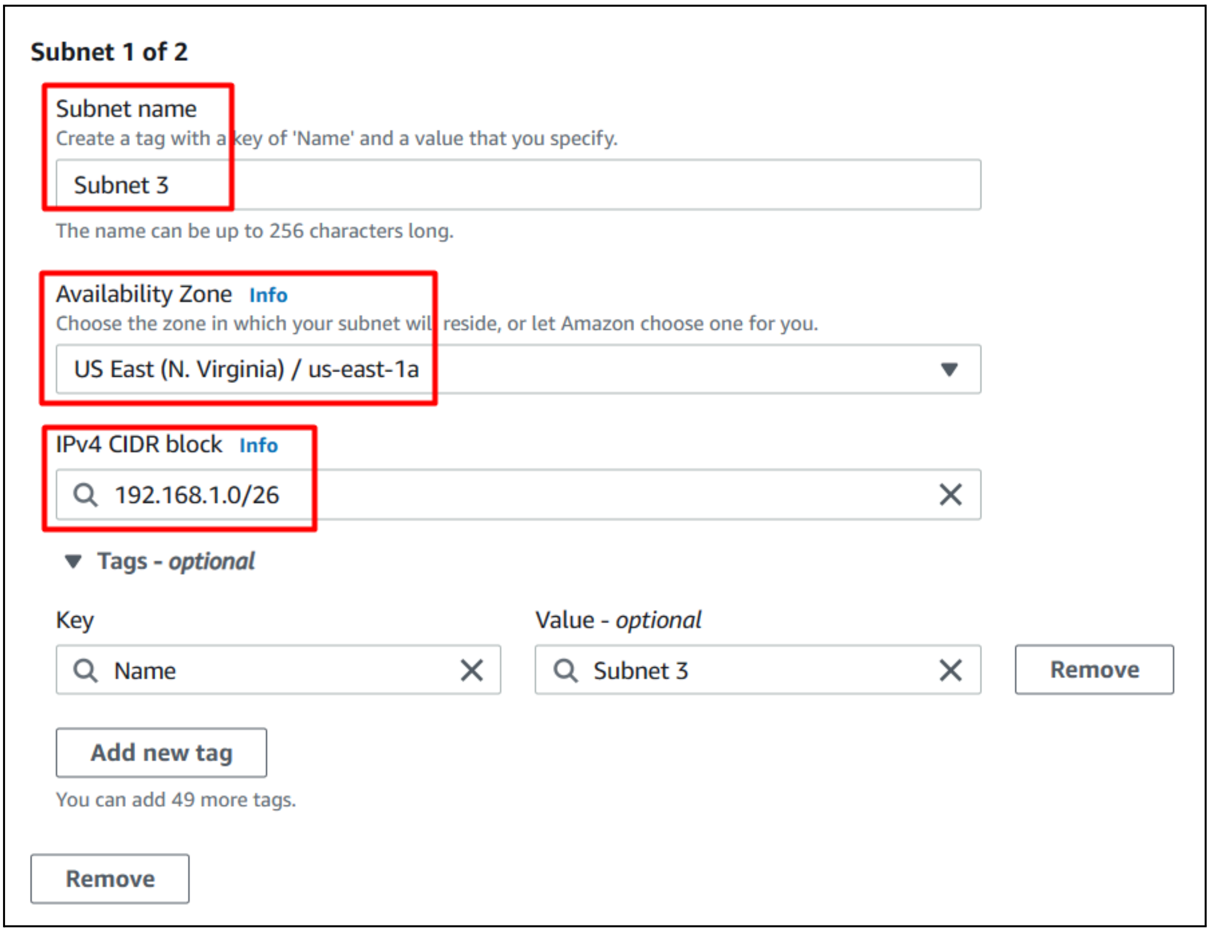


2. Click on Add new subnet again to create one more subnet
- VPC ID: Select the whizlabs_VPC1 VPC from the list.
- Subnet name: Enter Subnet 2
- Availability Zone: Leave as No Preference
- IPv4 CIDR block: Enter 10.0.2.0/26



3. Now click on the Create subnet button.
4. Similarly, Create 2 Subnets in the VPC named whizlabs_VPC2 which can be used either for public or private resources.
5. For Subnet 1:
   - VPC ID: Select the whizlabs_VPC2 VPC from the list.
   - Subnet name: Enter Subnet 3
   - Availability Zone: Select us-east-1a

- IPv4 CIDR block: Enter 192.168.1.0/26

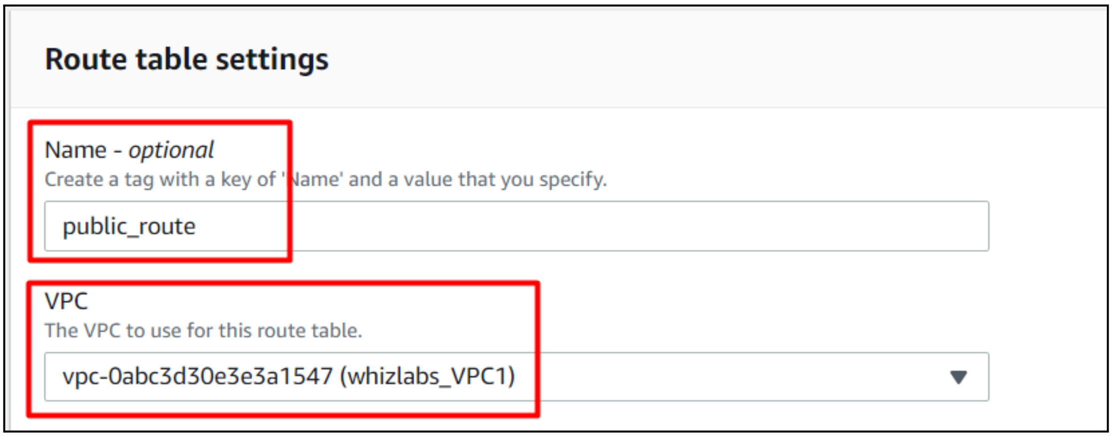


6. Click on Add new subnet to create one more subnet
- VPC ID: Select the whizlabs_VPC2 VPC from the list.
- Subnet name: Enter Subnet 4
- Availability Zone: Select us-east-1b
- IPv4 CIDR block: Enter 192.168.2.0/26
7. Click on the Create Subnet button.

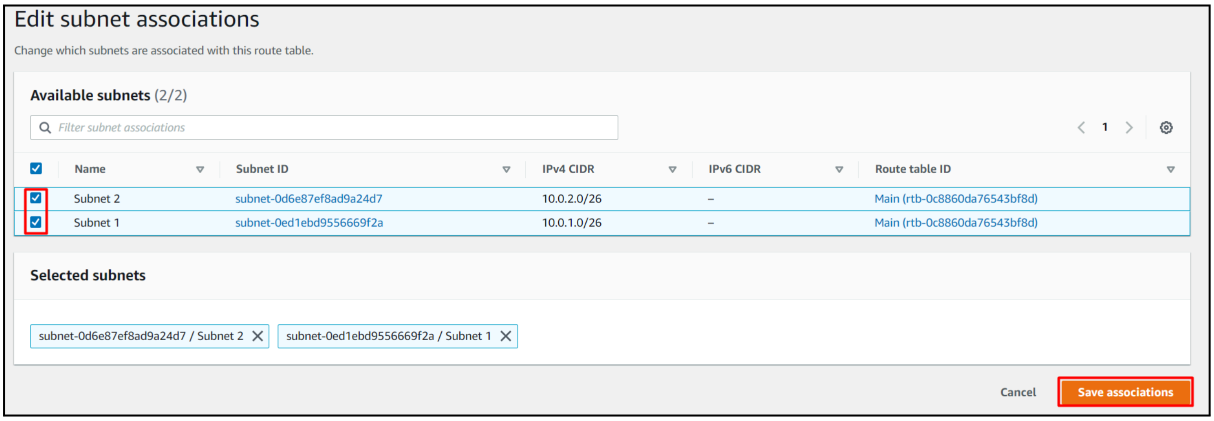
Note: There will be 6 default subnets available in the N.Virginia region don't delete or edit the subnets.

Task 5: Creating Route tables and configuring routes and associating subnets

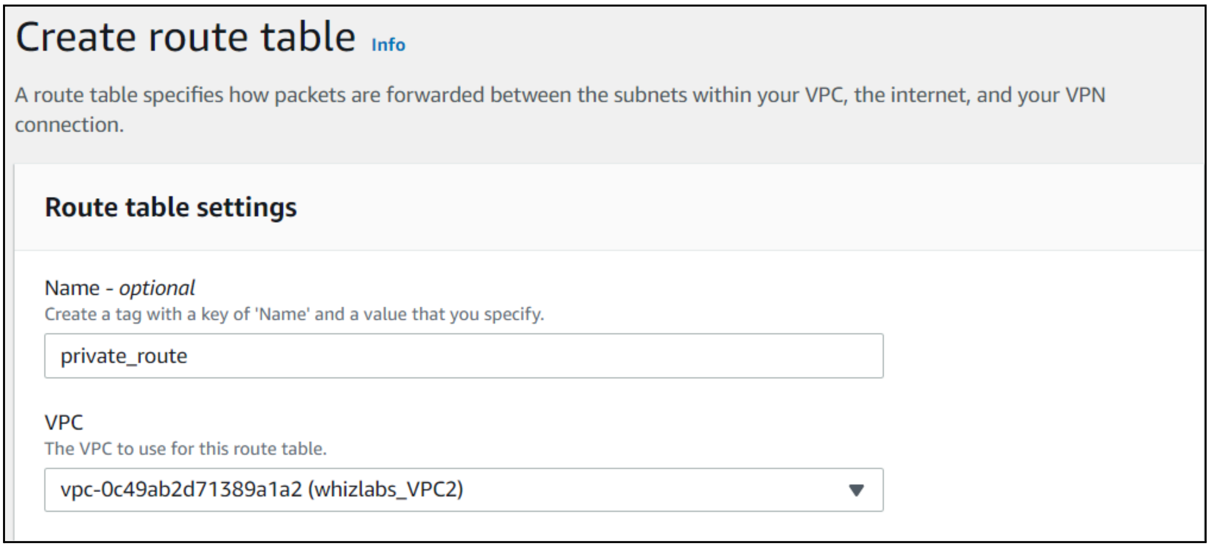
1. Create 2 route tables one for the Public route and another for the private route in different VPCs
2. Navigate to Route Tables on the left side panel and click on the Create route table button.
- Name tag: Enter public_route
- VPC: Select the whizlabs_VPC1 from the list.



3. Now click on the Create route table button.
4. Select the public_route from the list and go to the Subnet Associations tab in below.
5. Click on the Edit subnet associations tab.
6. Now Select the subnet Subnet 1 and Subnet 2 and click on the Save associations button.

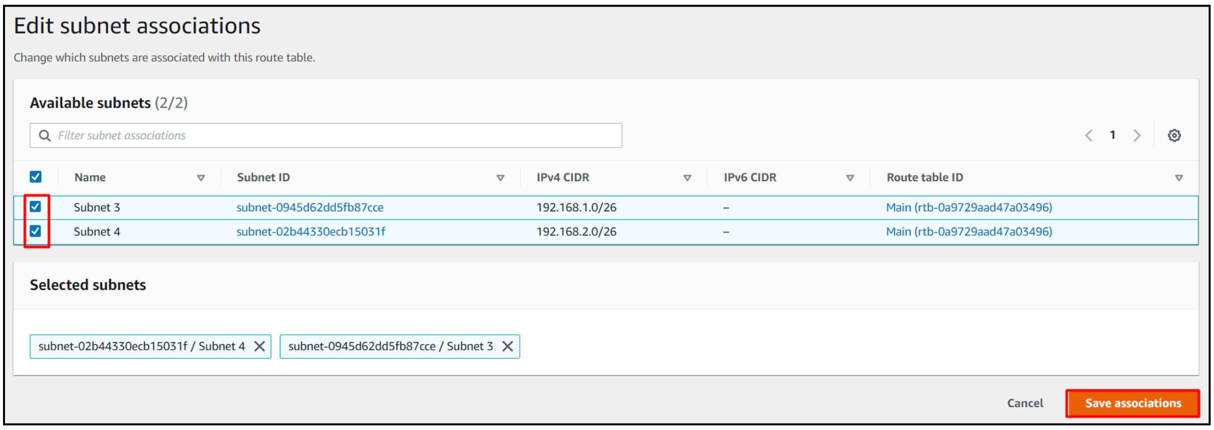


7. Click on the Create route table button again.
   - Name tag: Enter private_route
   - VPC: Select the whizlabs_VPC2 from the list.

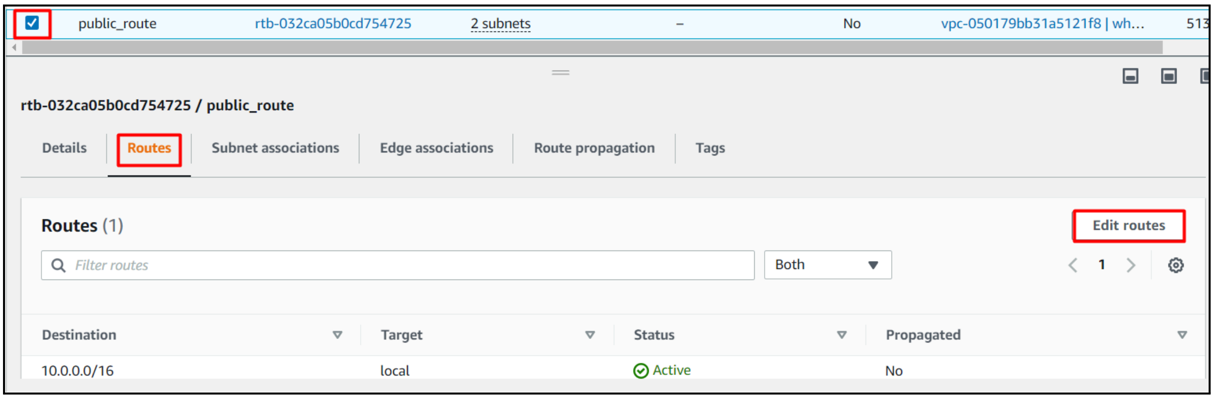


8. Now click on the Create route table button.

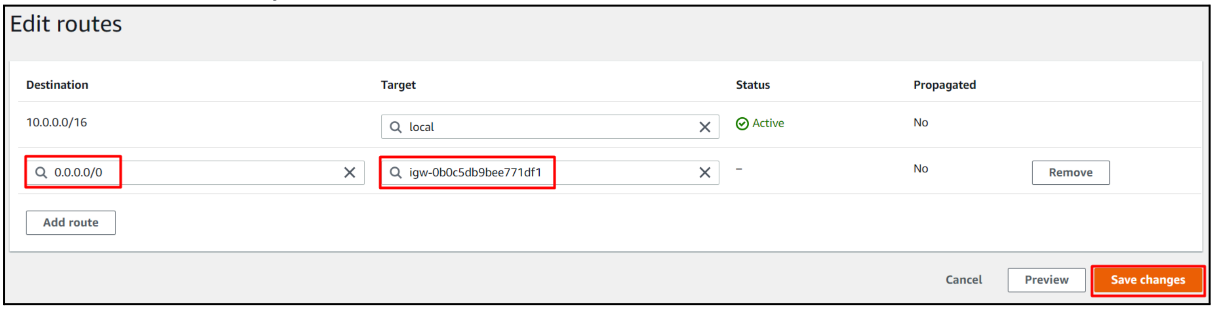
9. Select the private_route from the list and go to the Subnet Associations tab in below.
10. Click on the Edit subnet associations button.
11. Now select Subnet 3 and Subnet 4 and click on the Save associations button.



12. Select the public_route table. Go to the Routes tab below and click on the Edit Routes button.



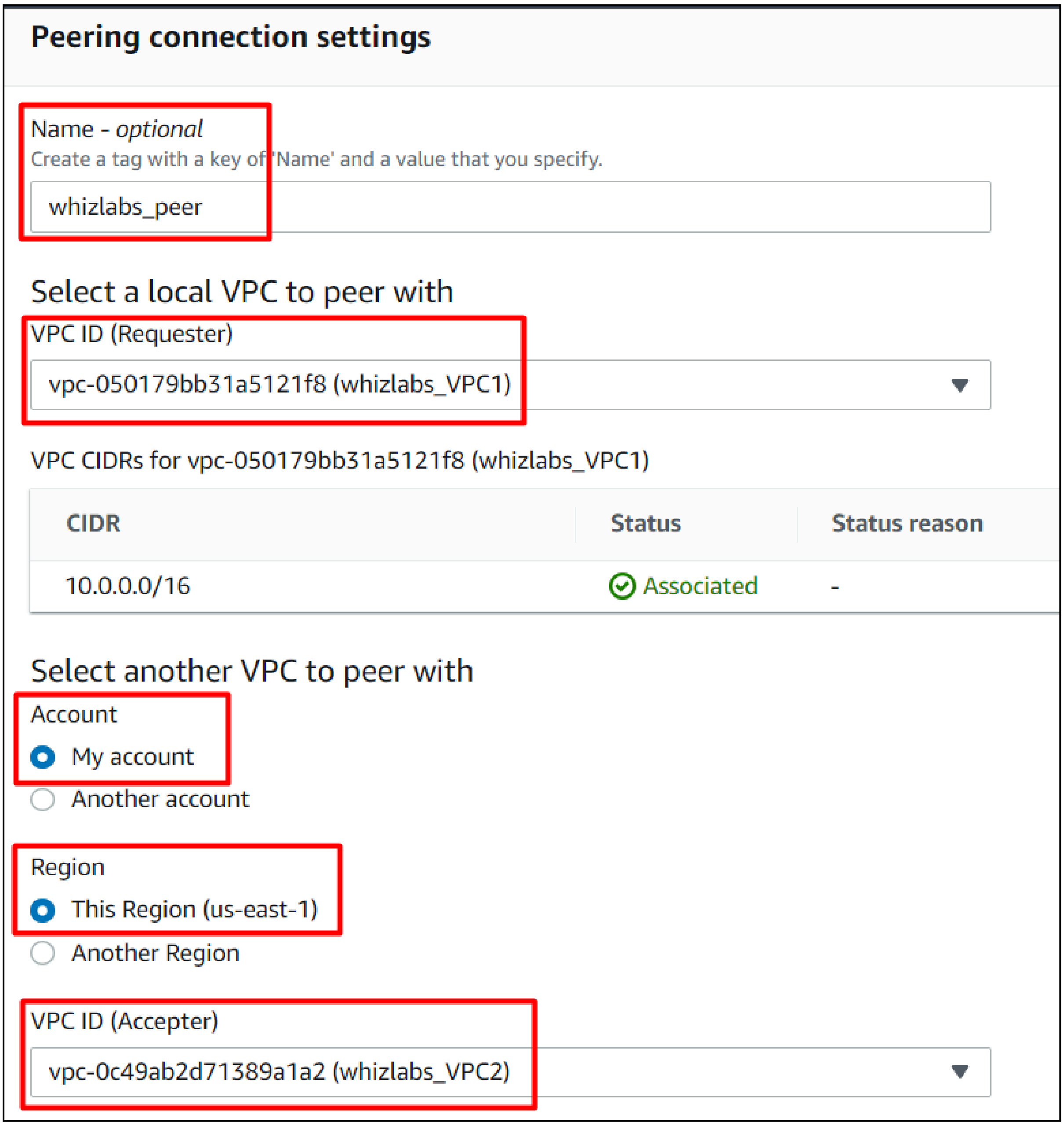
13. Now click on the Add route button.
- Destination: Enter 0.0.0.0/0
- Target: Select Internet Gateway whizlabs_IGW and then select the Internet Gateway id.



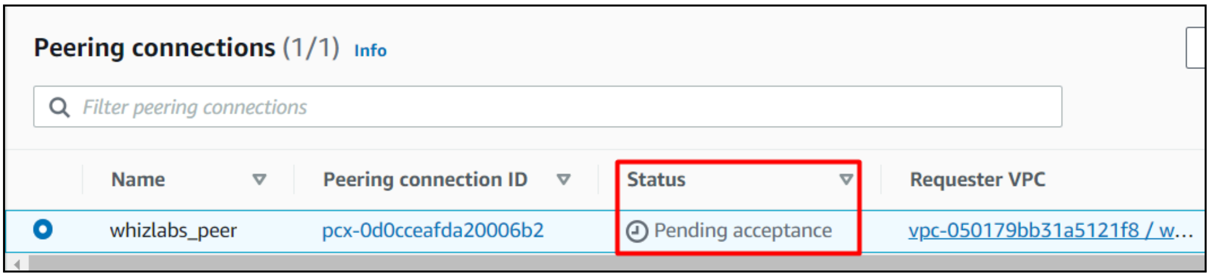
14. Click on the Save changes button.
Note: Here we haven't created any routes for the private_route table. Don't edit or delete the default Route table.

Task 6: Creating VPC Peering Connection

1. Now Navigate to your N.Virginia region VPC page.
2. On the left side menu, select Peering Connections and click on Create Peering connection
    - Peering connection name tag: Enter whizlabs_peer
    - VPC ID (Requester): Select whizlabs_VPC1
    - Account leave as default
    - Region: Select This region (us-east-1)
    - VPC ID (Accepter): whizlabs_VPC2

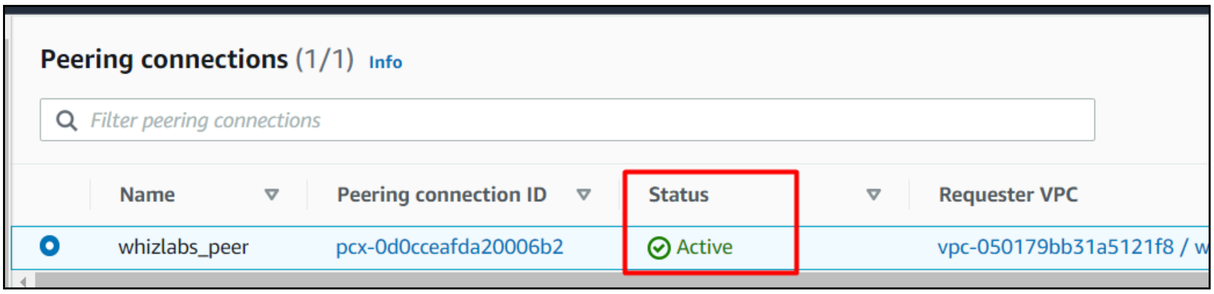


3. Click on the Create peering connection button.
4. Now the status of the peering will be in Pending Acceptance.



5. Select the peering connection and click on Action and then click on Accept Request.
6. Click on the Accept request in the popup message.

7. Once this is done, the Status changes to Active. If not please refresh the page.

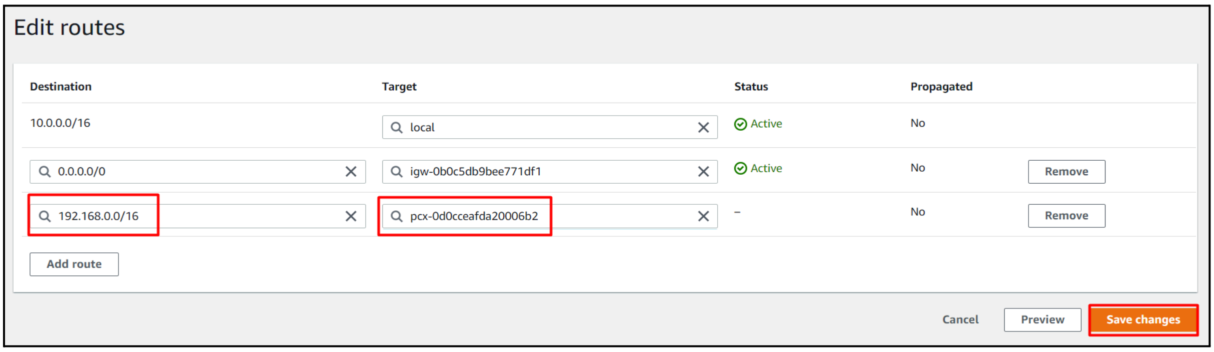


8. Now you need to add peering rules in route tables in both the VPC's.
9. This is because, without this route, the traffic will not pass the route table and reach the EC2s.

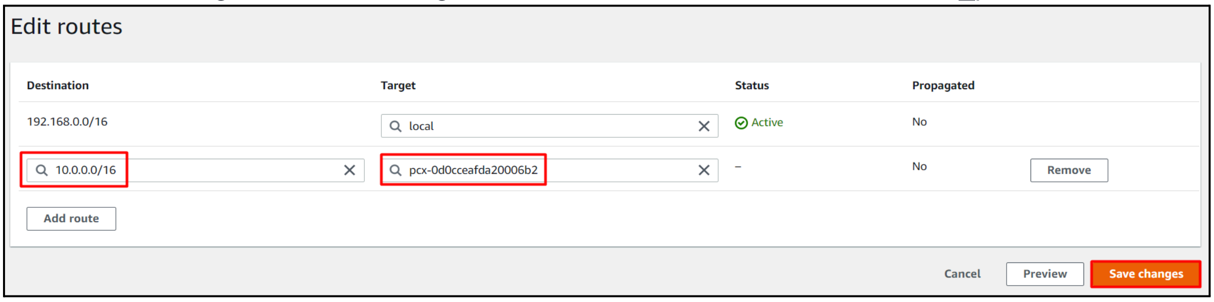
Note: We can also establish a peering connection between different AWS accounts in that case you need to Choose Another account and have to provide another accounts Account ID. Peering connections can be established between Cross Region for this you need to select Another Region and have to Choose a region from the dropdown menu.

Task 7: Editing routes in the Route table

1. Once the peering connection is established you need to edit the route tables that you have created earlier to include the route of CIDR block of other VPC.
2. You need to add routes to the route table that you have created to include the VPC CIDR range of the other VPC you need to peer.
3. Go to Services, and click on VPC. Choose Route Tables and select public_route table.
4. Select the public_route table. Go to the Routes tab below and click on the Edit routes button.
5. Now click on the Add route button.
- Destination: Enter 192.168.0.0/16
- Target: Select Peering Connection and then select the whizlabs_peer.



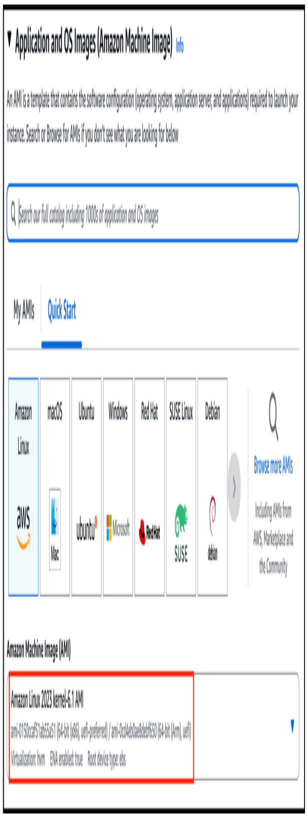
6. Click on the Save changes button.
7. Now Select the private_route table. Go to the Routes tab below and click on the Edit routes button.
8. Click on the Add route button.
- Destination: Enter 10.0.0.0/16
- Target: Select Peering Connection and then select the whizlabs_peer.



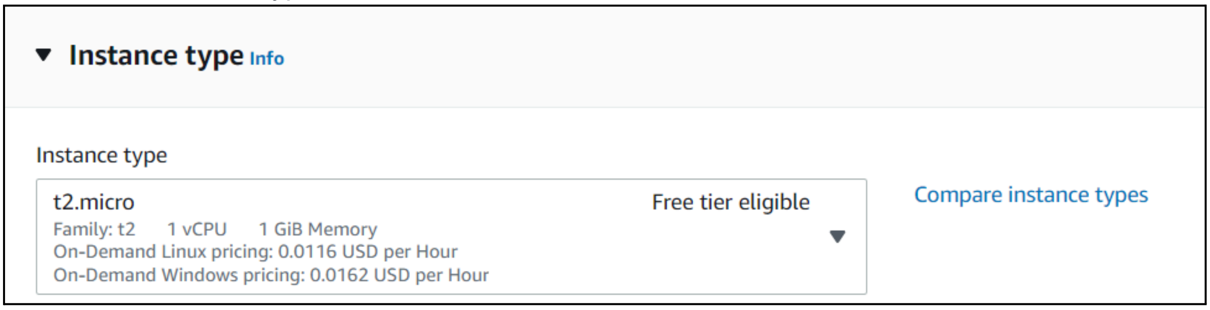
Task 8: Creating an EC2 Instance

1. Make sure you are in the N.Virginia Region.
2. Navigate to EC2 by clicking on the Services menu at the top, then click on EC2 in the Compute section.
3. Navigate to Instances from the left side menu and click on the Launch Instance button.
4. Enter Name as VPC_peering_EC2.
5. Choose an Amazon Machine Image (AMI): Select Amazon Linux 2023 kernel-6.1 AMI in the drop-down.

- Choose architecture as 64-bit(x86)

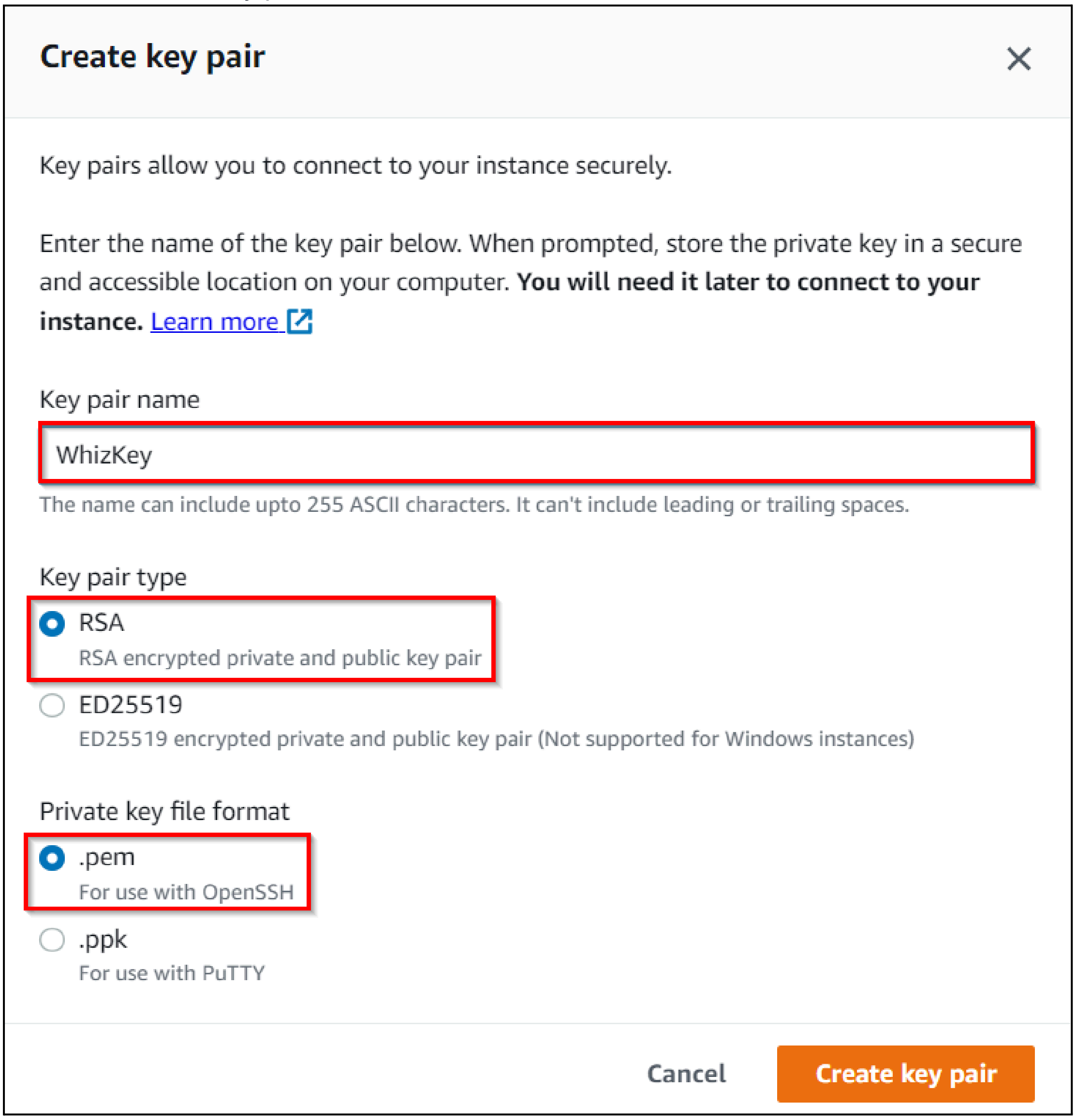


6. Choose an Instance Type: Select t2.micro.



7. For Key pair: Select Create a new key pair Button
   - Key pair name: Enter WhizKey
   - Key pair type: Select RSA
   - Private key file format: Select .pem

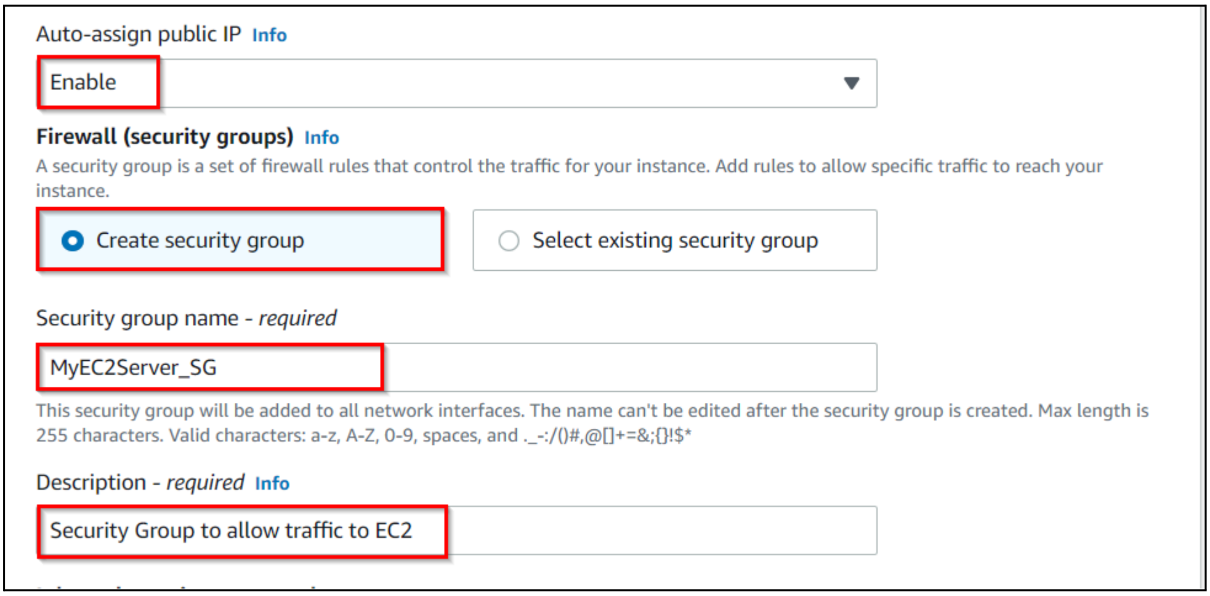
8. Select the Create key pair Button.



9. In Network Settings Click on Edit Button:

- VPC - required: Select whizlabs_VPC1
- Subnet name: Select any Subnet
- Auto-assign public IP: Enable
- Select Create new Security group
- Security group name: Enter MyEC2Server_SG

- Description: Enter Security Group to allow traffic to EC2

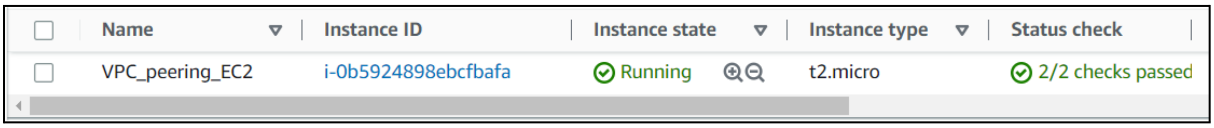


- To add SSH,
  - Choose Type: SSH
  - Source: Select Anywhere

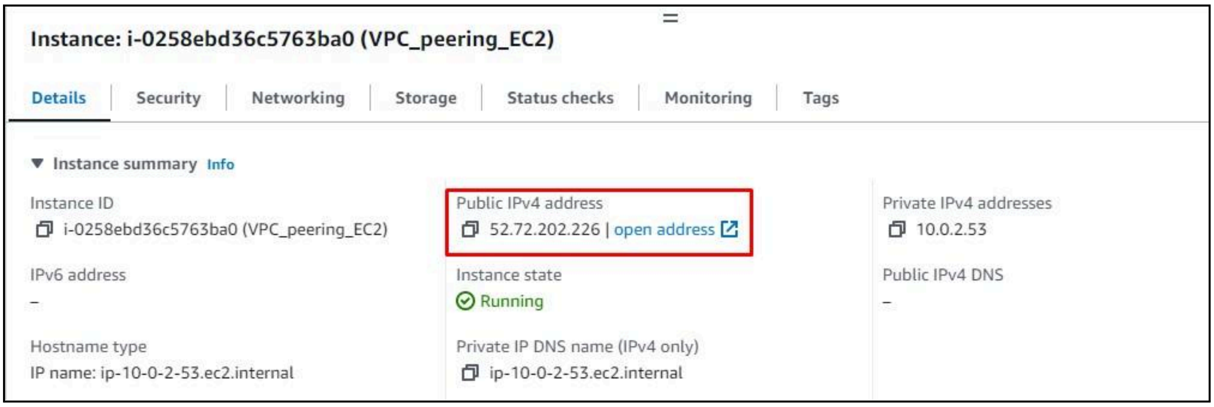
10. Now under Advanced Details, scroll down to User Data, and copy and paste the script:

```
#!/bin/bash
sudo su
dnf update -y
dnf install mariadb105 -y
```

11. Keep the Rest thing Default and Click on the Launch Instance Button.
12. Launch Status: Your instance is now launching, Click on the instance ID and wait for the complete initialization of the instance till the status changes to running.

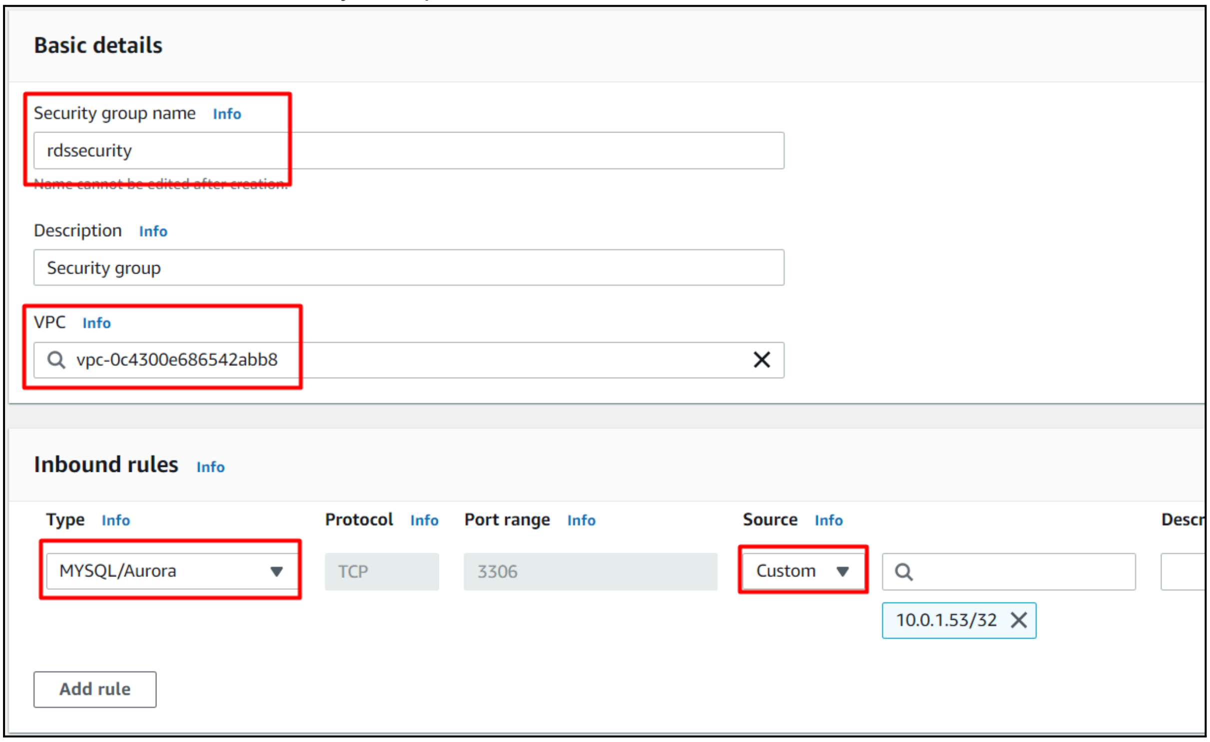


13. Note down the sample IPv4 Public IP Address of the EC2 instance. A sample is shown in the screenshot below.



Task 9: Creating a Security Group for the RDS instance

1. Create a security group for the RDS instance which will provide access for the EC2 instance to connect with the RDS.
2. Navigate to VPC by clicking Services on the top menu. Click on Security Groups in the left navigation panel. Click on the Create Security Group button and then provide the following details
   - Security group name: Enter rdssecurity
   - Description: Security group for RDS instance
   - VPC: whizlabs_VPC2 (select from the dropdown)
3. Now you need to add Inbound rules to the Security group that you have created, Click on Add rule under Inbound Rules
   - Type: MYSQL/Aurora (select from the drop-down)
   - Source: Custom: Copy and paste the EC2 instance's private IP from the instance description page along with a CIDR block range /32 at the end (in my case it is 10.0.1.61/32)
   - Click on the Create Security Group button.

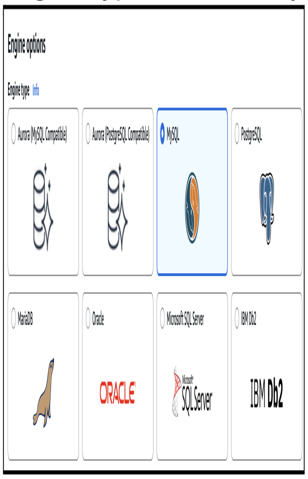


Task 10: Create a RDS Instance

1. Navigate to RDS available under the database section of the Services menu.
2. Make sure you are in the N.Virginia Region.
3. Click on Databases in the left panel.
4. Click on Create Database.
5. Let’s configure the database in the Create Database Page

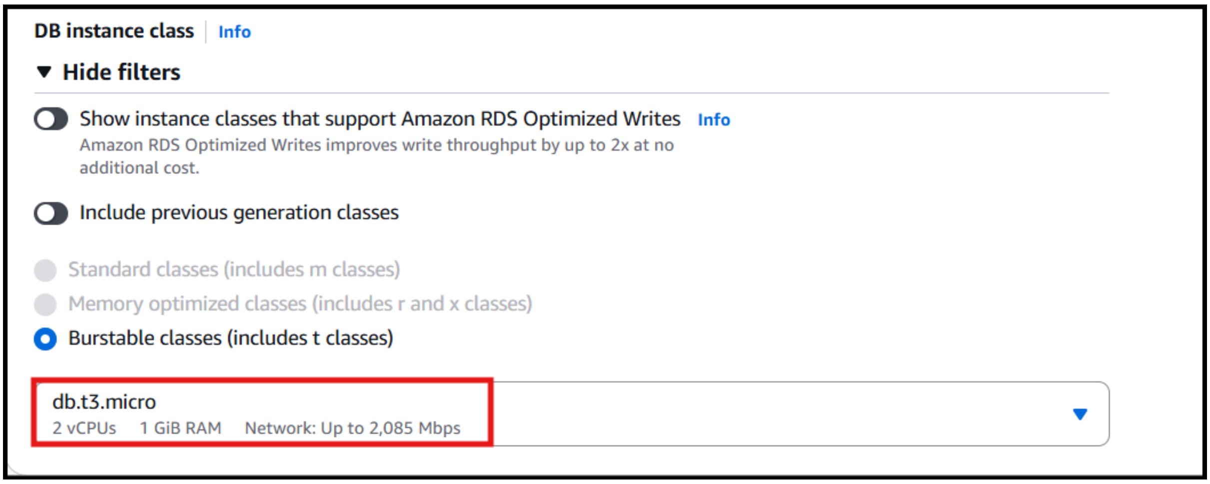
- In Choose a Database Creation Method: Select **Full Configuration**

- In Engine options
  - Engine type: Choose MySQL

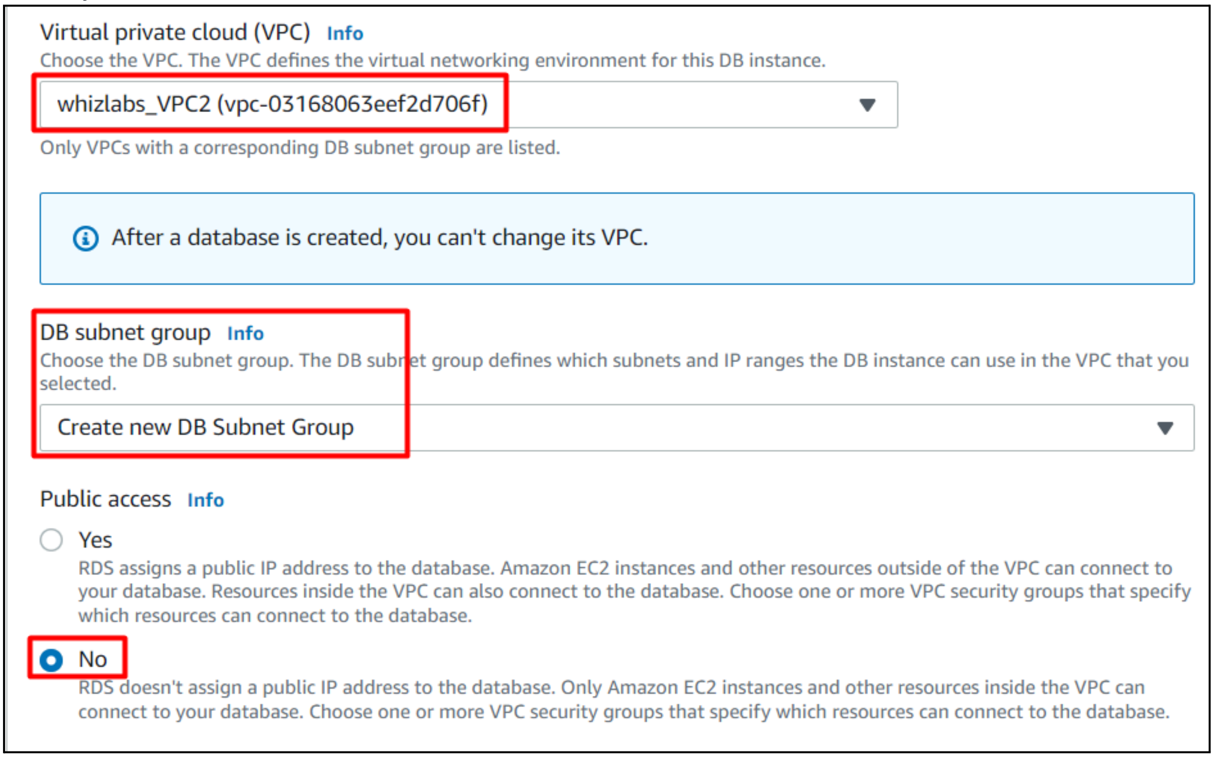


- In Templates: Choose Sandbox

- In Settings
  - DB cluster identifier: Specify cluster identifier name as whizlabs-identifier
- In credential settings
  - Master Username: Enter whizlabs_admin
  - Master password: Enter Whizlabs123
  - Confirm password: Enter Whizlabs123
  - Note: These are the username and password used to log on to your database. Please make note of them.
- In DB instance size:
- DB instance class: Choose Burstable classes (includes t classes)
- Select db.t3.micro from the available list

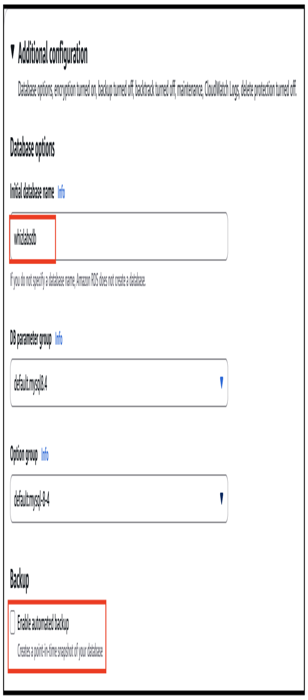


- Uncheck the Enable storage autoscaling checkbox.
- Connectivity
- Virtual Private Cloud: Choose whizlabs_VPC2
- Subnet group: Create New subnet group
- Publicly accessible: **No**

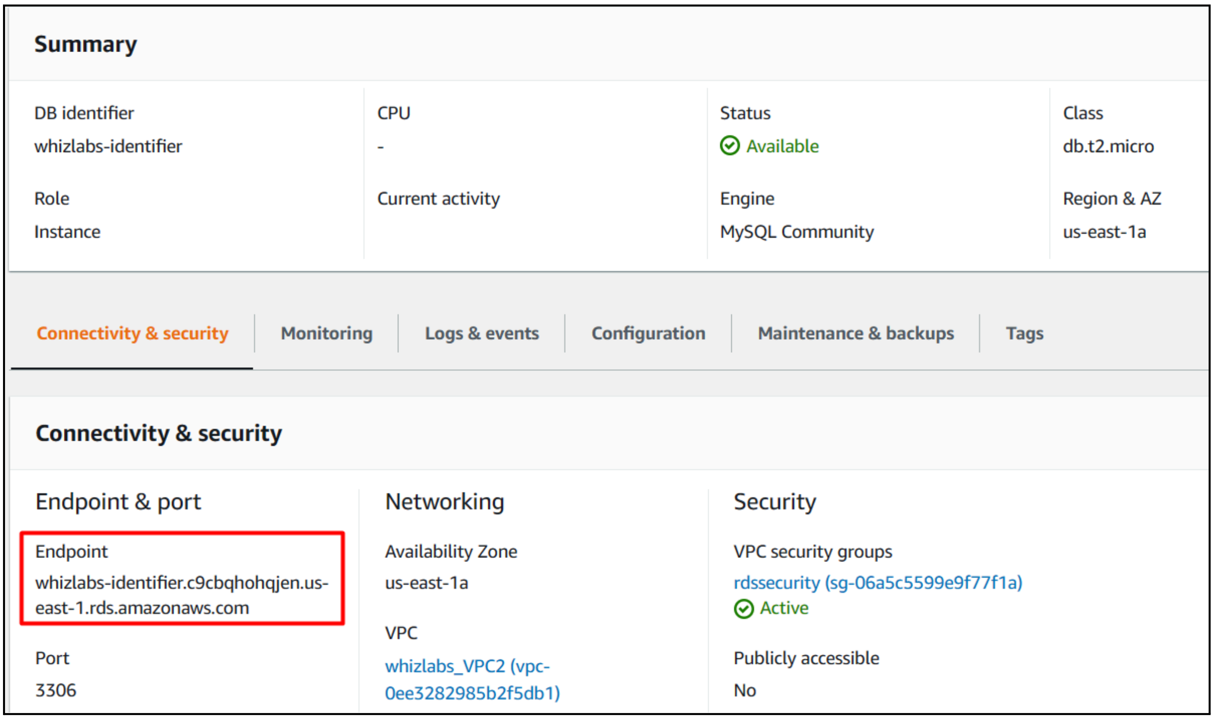


- VPC security group: Select Choose existing
- Existing VPC Security group: rdssecuritygroup (remove the default and select it from the drop-down)
- Availability Zone: No preference
- Database port: **3306**
- Click on **Additional Configuration.**
- Database options
    - Initial database name: Enter whizlabsdb
- Uncheck the Enable automated backups checkbox.

- Uncheck the Enable auto minor version upgrade checkbox



6. Once filled in all the necessary then click on Create database.
7. It will take around 10-15 minutes for the database to be created. Please wait until the database status changes from creating to available.
8. Once the database is created, you can see the Status as Available.
9. To connect the RDS instance we need the Endpoint (DNS name for RDS instance).
10. Click on the RDS instance created and then navigate to Connectivity & security to find the endpoint of your RDS instances with which you can connect to your DB instance.



11. My RDS endpoint: whizlabs-cluster.cdegnvsebaim.us-east-1.rds.amazonaws.com

Task 11: SSH into EC2 and Connect to Your Database

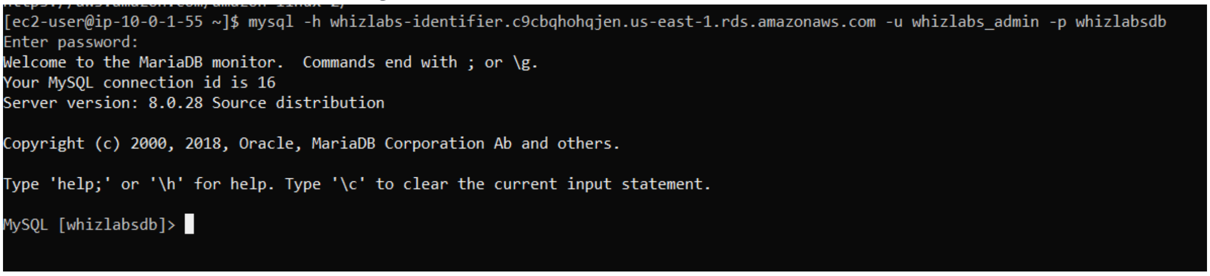
1. SSH into the EC2 instance using the following steps in SSH into EC2 Instance.
2. Switch to root user using the command:

```
sudo
-s
```

3. Now login to the RDS instance using the below command

```
mysql -h <YOUR_RDS_ENDPOINT> -u whizlabs_admin -p
whizlabsdb
```

- Hostname: whizlabs-cluster.cdegnvsebaim.us-east-1.rds.amazonaws.com (In the above command you have to remove the RDS endpoint and use your RDS's endpoint)
- User name: whizlabs_admin
- Enter the Password: Whizlabs123
- Database name: whizlabsrds
- You can now able to log in to the database as shown below:



```
[ec2-user@ip-10-0-1-55 ~]$ mysql -h whizlabs-identifier.c9cbqhohqjen.us-east-1.rds.amazonaws.com -u whizlabs_admin -p whizlabsdb
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 16
Server version: 8.0.28 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [whizlabsdb]>
```

- EC2 instance can able to establish a connection with the RDS instance if it is in the same VPC but in our case, we can able to connect the RDS instance from the EC2 instance even though they are in different VPCs. This can only be possible if a peering connection is established between the VPC's.

Lab : Introduction to ECS

Task 1: Sign in to AWS Management Console

1. Click on the Open Console button, and you will get redirected to AWS Console in a new browser tab.
2. On the AWS sign-in page,
   - Leave the Account ID as default. Never edit/remove the 12-digit Account ID present in the AWS Console. Otherwise, you cannot proceed with the lab.
   - Now copy your User Name and Password in the Lab Console to the IAM Username and Password in AWS Console and click on the Sign-in button
3. Once Signed In to the AWS Management Console, make the default AWS Region as US East (N. Virginia) us-east-1.

Task 2: Create a Key Pair for the EC2 instances, inside the ECS Cluster

1. Make sure you are in the N.Virginia Region. Navigate to EC2 Service by clicking on the Services menu in the top, then click on EC2 Service in the Compute section.
2. In the left navigation pane (scroll down) within Network & Security, click on the Key pairs
3. To create a new key pair, click on the Create key pair button.
4. Fill in the details below:
   - Name: Enter *WhizKeyPair*
   - Key pair type : RSA
   - File format: pem (Linux & Mac Users) or ppk (Windows users)
   - Leave other options as default.
   - Click on the Create key pair button.

Task 3: Launching an ECS Cluster

1. Make sure you are in the N.Virginia Region. Navigate to Elastic Container Service by clicking on the Services menu in the top, then click on Elastic Container Service in the Containers section.
2. On the left sidebar, click on the Clusters option present under the Amazon ECS section, then Click on the Create cluster button.

3. In Cluster configuration :
   - Cluster name: enter **whizcluster**

4. In Infrastructure Section :
   - Select **Fargate and Self-managed instances**.
     - Auto Scaling group (ASG): Select Create new Auto Scaling Group
     - Provisioning model: select **On-demand**
       - Container instance AMI : Select Amazon Linux 2023
     -
     - EC2 instance type : Select t2.micro

- EC2 instance role: Choose the existing IAM role **ECS_Role.** Desired capacity :
  - Minimum : Enter 1
  - Maximum : Enter 2
- SSH Key pair: Select WhizKeyPair

5. In Networking Section :
   - VPC: Select Default VPC
   - Subnets: Leave it as default.
6. Security group :
   - Select **create new security group**
     - Security group name : Enter **MySG**
     - Security group description : Enter **MySG**
   - Inbound rules for security groups:
     - Type : Select **SSH**
     - Source : Select **Anywhere**

7. Keep rest things as default and click on Create button
8. ECS Cluster will be created
9. It will take few minutes to provision the ECS Instance.

Task 4: Create Task Definitions

In this task, we are going to create a task definition for the cluster. A task definition provides a blueprint for creating tasks, which are the basic unit of work in ECS. Each task definition can define one or more containers that are run together on the same underlying EC2 instances or Fargate tasks.

1. On the left sidebar, click on the Task Definitions option present under the Amazon ECS section.
2. Click on the Create new task definition button.

3. In Task definition configuration:

- Task definition family: Enter *ecs-demo*

4. For Infrastructure requirements :
   - Launch Type : select Amazon EC2 Instances
   - Task size :
     - CPU : Enter .25 vCPU
     - Memory : Enter .5 GB
       - In Task Role: Choose none
       - In task execution role: **Choose none**
5. Under Container - 1:
   - Name: Enter *httpd*

- Image URL: Enter *httpd:2.4*

6. Under Port mappings :
    - Container port : Enter 80
    - Protocol : Select TCP
    - App protocol : Select HTTP

7. Keep rest things as default and click on the Create button

8. Task Definition ecs-demo is now created.

Task 5: Create a service and start HTTPD container in ECS

In this task, we are going to create a service and start the HTTPD container. In AWS ECS (Elastic Container Service), a service is a long-running task that ensures that a specified number of instances of a task definition are running and maintained in an ECS cluster. Services allow you to define the desired state of your tasks and automatically handle task placement, scaling, and recovery.

1. On the left sidebar, click on the Task definitions option present under the Amazon ECS section.
2. Select ecs-demo and click on Deploy and click Create service button.

3. In Service details: Choose Task definition revision: select the latest revision.

4. Service name: Enter **httpd**

5. Cluster will get selected by default.

6. In Compute configuration:

- Choose Compute option as **Launch type** and select the Launch type as **EC2**

- In **Deployment configuration**, Scheduling strategy: Select **REPLICA.**
- Desired tasks : Enter **1**

7. In Networking section:

- VPC: select the **default vpc**
- Subnets: leave it as default
- Security Group: click on **Use an existing security group** and select **MySG**.

8. Keep other options as default, and click on the Create button.

9. The service will be created, wait till the Deployments and tasks changes to 1/1 tasks running

10. Now navigate to Infrastructure tab, and scroll down to the Container Instances

11. Click on the EC2 Instance ID, and you will be redirected to the running EC2 instance.

Task 6: SSH into the underlying EC2 instance and run Docker commands

1. Please follow the steps in SSH into EC2 Instance
2. Please note that the Ec2 instance connect will not work for the ECS instance, so please try other SSH methods.
3. Get the root access using the following command:

```
sudo su
```

5. Now run the updates using the following command:

```
sudo dnf update -y
```

7. Check the Docker version by running the following command:

```
docker version
```

9. Check all the docker processes running in the ECS Cluster

```
docker ps
```

- Default ECS agent and httpd container is running in the underlying EC2 instance

Lab : **Introduction to Simple Queuing Service (SQS)**

Task 1: Sign in to AWS Management Console

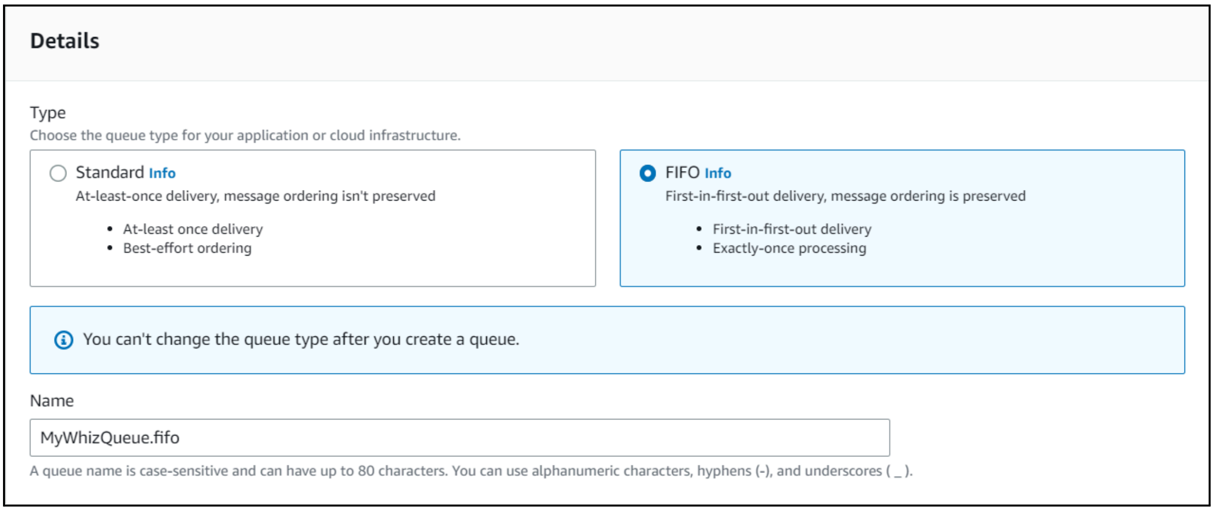
1. Click on the Open console button, and you will get redirected to AWS Console in a new browser tab.
2. On the AWS sign-in page,
    - Leave the Account ID as default. Never edit/remove the 12-digit Account ID present in the AWS Console. Otherwise, you cannot proceed with the lab.
    - Now copy your User Name and Password in the Lab Console to the IAM Username and Password in AWS Console and click on the Sign in button.
3. Once Signed In to the AWS Management Console, make the default AWS Region as US East (N. Virginia) us-east-1.

Note : If you face any issues, please go through FAQs and Troubleshooting for Labs.

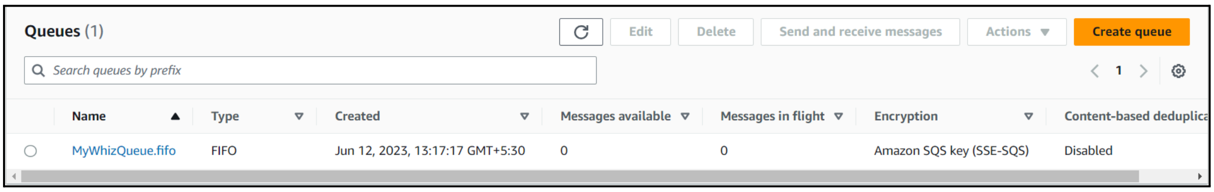
Task 2 : Create FIFO and Standard Queue using the Console

In this task, we are going to create two types of queues: FIFO (First-In-First-Out) and Standard. By creating these queues, the user can understand and experience the differences between them.

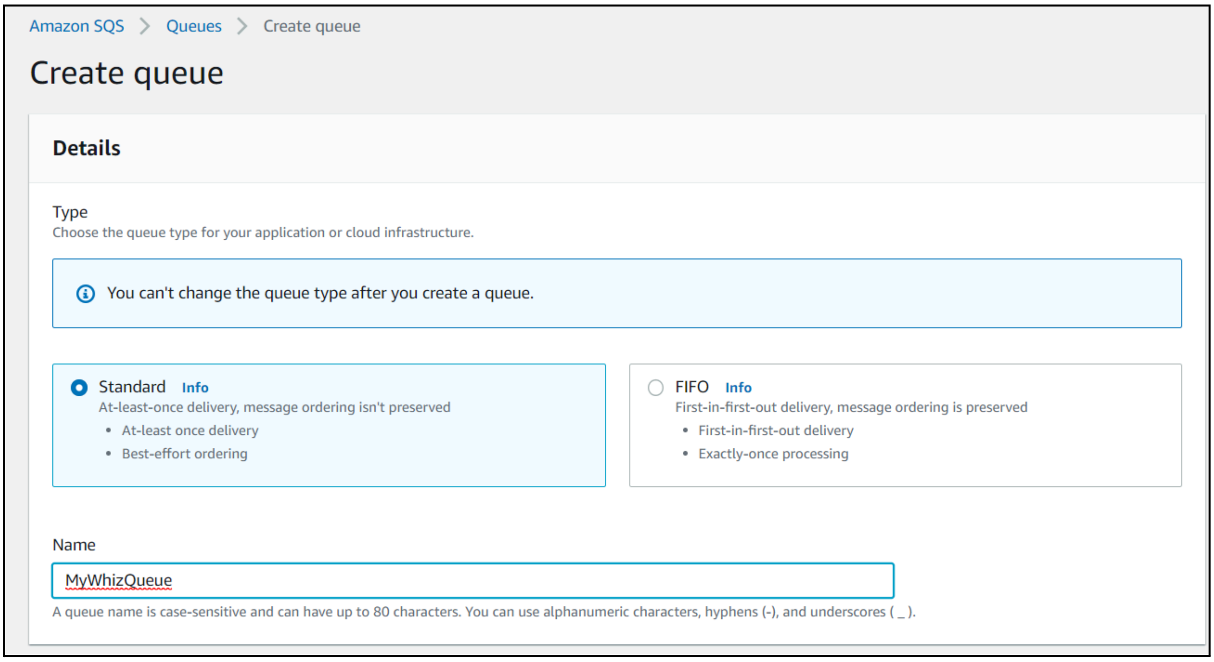
1. Make sure you are in the N. Virginia region.
2. Navigate to the Services menu on the top, search for SQS and select it.
3. Click on the Create queue button.
4. Details :
    - Type : Select FIFO
    - Name : Enter MyWhizQueue.fifo
- Note: The name of a FIFO queue must end with the .fifo suffix.
5. Leave everything as default and click on Create queue button



6. Once you click on Create Queue you will get a success message. Go back to the queues dashboard, a FIFO queue gets created as given in the image below.

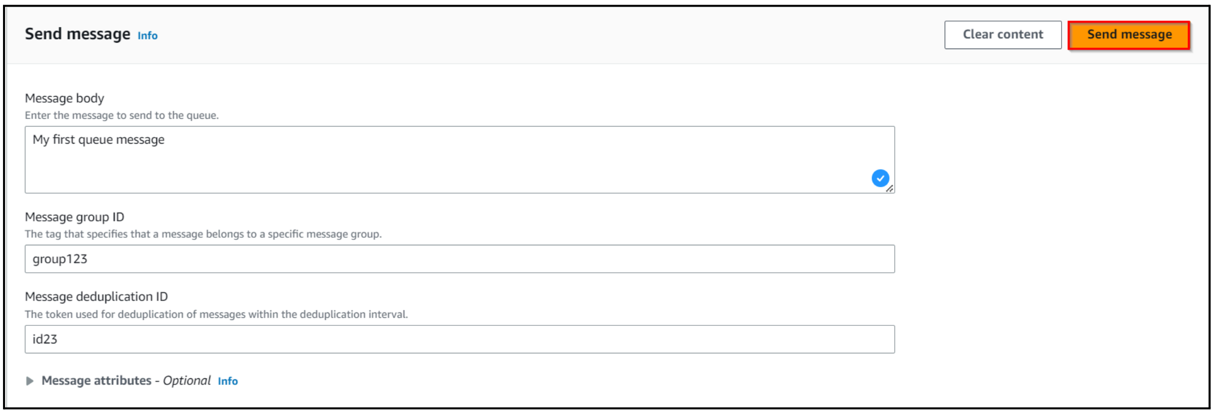


7. Now, We'll create a Standard Queue, with all the default options. The only difference is we don't provide the suffix .fifo while creating the queue.
8. Click on Create queue button.
9. Details :
    - Type : Select Standard
    - Name : Enter MyWhizQueue

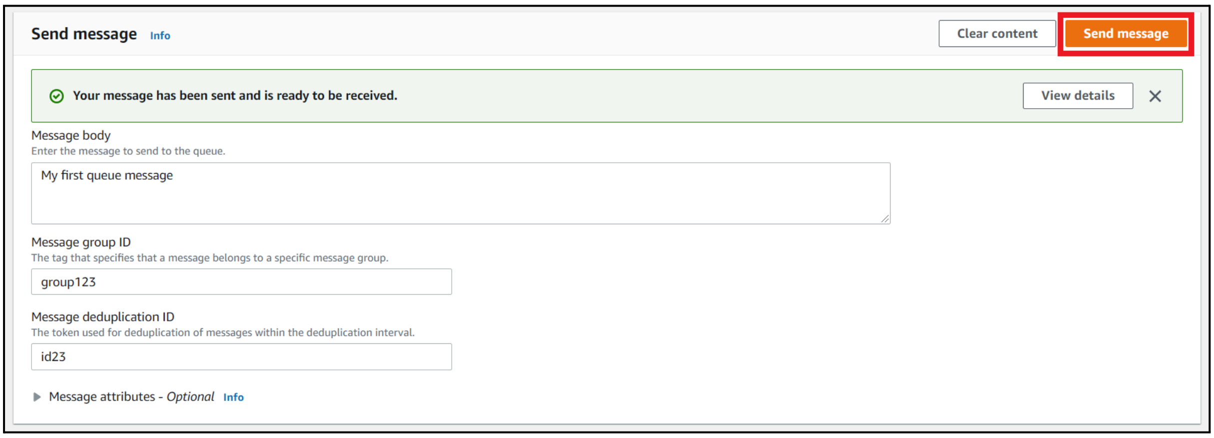


10. The Type column helps you distinguish the Standard queue from the FIFO queue at a glance. For a FIFO queue, the Content-Based deduplication column displays whether each of your messages has a unique body enabled or not.
11. Click on the MyWhizQueue.fifo to get detailed information about all the important parameters including ARN, Name and URL of the queue.
12. We'll be now sending a message from our FIFO queue.
13. Click on Send and receive messages.

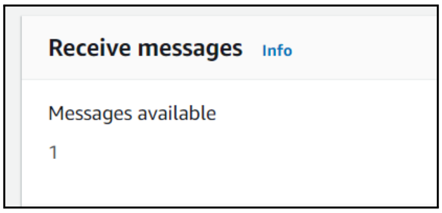
14. Send message
    - Message body : Enter My first queue message
    - Message group ID : Enter group123
    - Message deduplication ID : Enter id23
15. Now click on Send message button



16. Once you have sent a message, you get an acknowledgment.

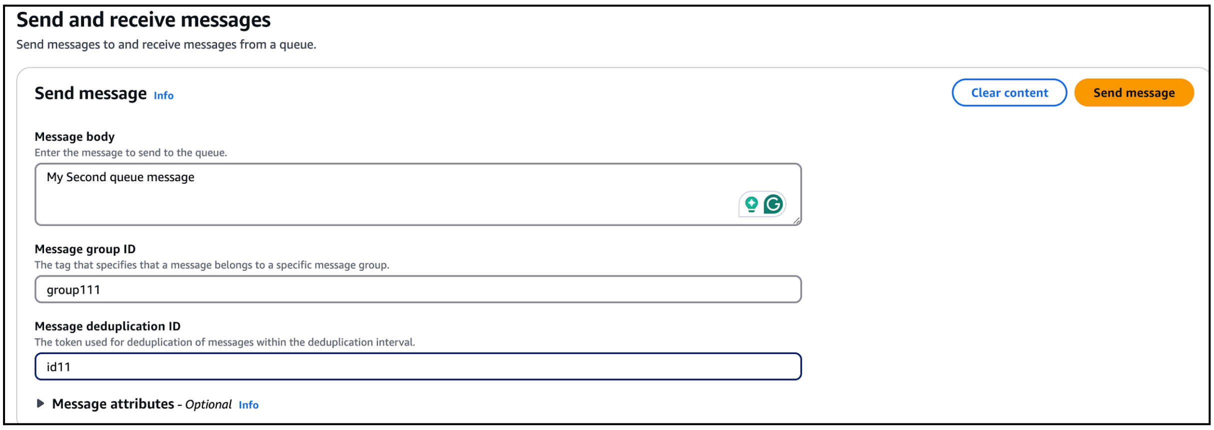


17. After sending the message, we can see that Message available changes to 1.

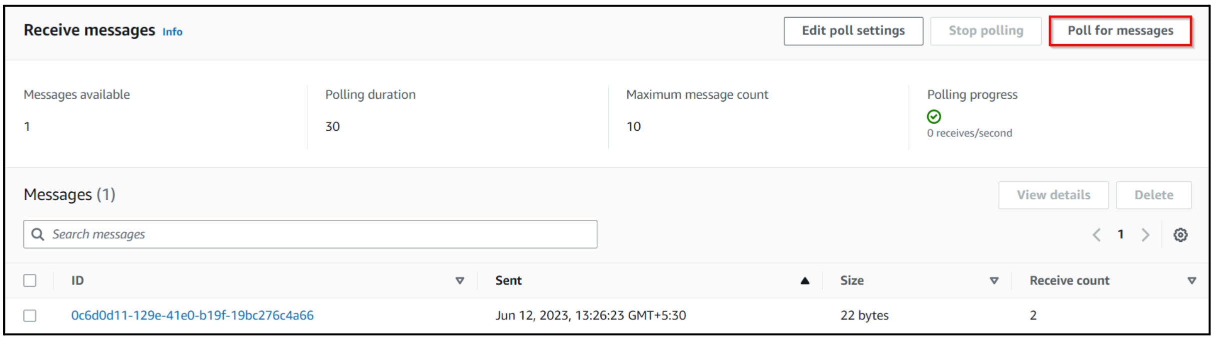


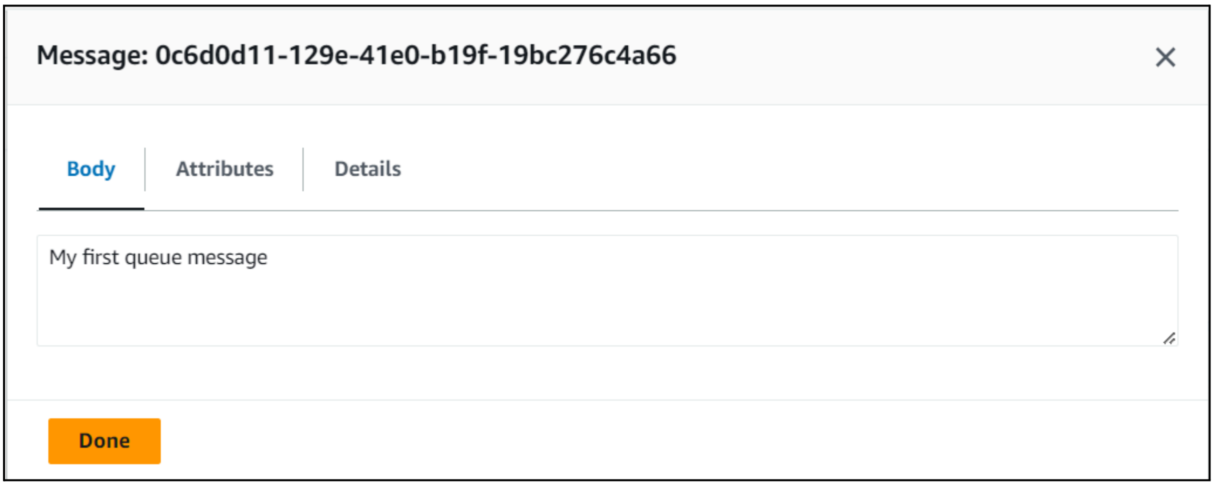
18. Similarly, send few more messages with different Message body, Message group ID and Message deduplication.

Click on Clear content button and type new message.



19. After sending the messages, Message available changes to the number of messages sent.
20. Now, to get all the messages we have sent, click on Poll for messages in the Receive messages section.
21. We can now see the message column has been filled with the messages.
22. Then click on the message to see its body.

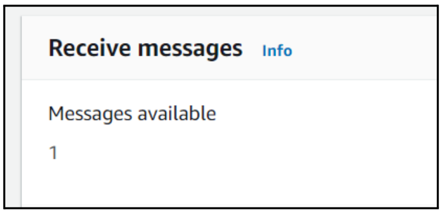




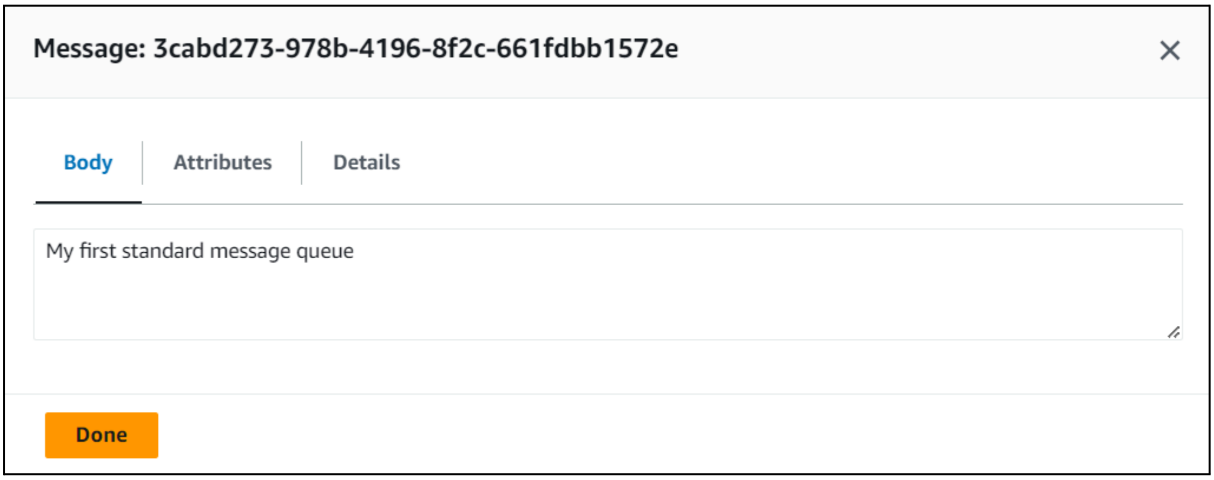
23. Hence, we have successfully sent and received messages in the console.
24. Similarly, we'll try with the Standard queue for sending the message.
25. Click on Send and receive messages.

26. Send message:
- Message body : Enter My first standard message queue
- Now click on Send message button.
27. Once the message is delivered, you'll receive an acknowledgment for successful delivery.

28. After sending the message, we can see that Message available changes to 1.



29. Similarly, send few more messages with different Message body.
30. After sending the messages, Message available changes to the number of messages sent.
31. Now, to get all the messages we have sent, click on Poll for messages in the Receive messages section.
32. Then click on the messages to see its body.



33. Hence, we have successfully sent and received messages in the console.

34. Let's try to understand the other parameters like Long Polling and how it works in SQS. Before moving to the lab, try to focus on what exactly the Long polling is and when it should be used.

Task 3 : What is Long Polling & Configuring Long Polling

In this task, we are going to introduce the concept of long polling and its benefits in Amazon SQS. Long polling is an alternative to short polling, where the client waits for a response from the server only when there are messages available in the queue. By configuring long polling, the user can reduce the number of empty responses and save computational resources, resulting in cost savings.

1. Long Polling: Let's try to understand how long polling works and why we should use it.
   - For example, if our application requires SQS messages, it will call the RecieveMessage function in the background. ReceiveMessage will check the presence of any messages in the queue and return immediately, with or without messages.
   - Calling a ReceiveMessage function in our application is fine as per the requirement, but what if the SQS client repeatedly checks for the message in the queue for any new messages. This is a problem, as continuous calls for the ReceiveMessage function will require lots of CPU cycles and tie up a thread. In this situation, we'll be using long polling. The only modification that we have to make is to update our WaitTimeSeconds argument to 1-20 seconds.
   - Now, if the queue is empty, the call will wait up to WaitTimeSeconds for a message to come in the queue before returning. If messages come in before the timeout, the call will return the message right away.
2. Remember, if the wait time for the ReceiveMessage API action is greater than 0, long polling is in effect. Long polling is cost-effective because you're not constantly polling an empty queue.
3. By default, Amazon SQS uses short polling, querying only a subset of its servers (based on a weighted random distribution) to determine whether any messages are available for a response.
4. Let’s try to make this work in our existing queue by configuring Long Polling. Select any queue (standard or FIFO) from the list and click on Edit to make changes in our queue. We’ll be selecting the Standard queue as an example.

5. Once you have selected the Edit Queue, update the Receive Message Wait Time parameter. It can be any value between 0 to 20 seconds. In our example, we have changed it to 10 seconds. This will make our long polling come into effect. If we keep the value as 0 or default, it is considered short polling. Once you have made the change, click on Save button.

Task 4 : What is Visibility Timeout & Configuring Visibility TimeOut

In this task, we are going to explain the concept of visibility timeout in Amazon SQS. Visibility timeout is the duration during which a message remains invisible to other components after being retrieved by a consumer. By configuring the visibility timeout, the user can control how long a message remains hidden from other consumers, allowing sufficient time for processing without the risk of duplicate processing.

1. Visibility timeout is a time duration in which AWS SQS avoids the other components from receiving and processing the messages.
2. Case Study:
    - Let's try to understand the definition by an example. So if you keep your visibility timeout to 1 minute for a big data job, what's going to happen is the message is going to come back into the queue because it's not going to complete in 1 minute.
    - Let's say it takes five minutes to actually process a large amount of data. The message is going to become visible in the queue and then another EC2 instance will pick it up. Because of this, you could be delivering your messages multiple times because your visibility timeout is too low.
3. Default Visibility timeout is 30 seconds.
4. Increase it if your task takes > 30 seconds.
5. The Maximum it can go up to is 12 hours.
6. Let’s try to make this work in our existing queue by configuring the visibility timeout. Select any queue (standard or FIFO) from the list and click on Edit to make changes in our queue. We’ll be selecting the Standard queue as an example.
7. Once you have selected the Edit, update the Default Visibility Timeout parameter. It can be any value between 0 seconds to 12 hours. In our example, we have changed it to 5 mins.

8. Once you have made the change click on Save at the bottom to bring changes in effect.

Task 5 : What is Delay Queue & Configuring Delay Queue

In this task, we are going to introduce the concept of a delay queue in Amazon SQS. A delay queue allows the user to postpone the delivery of messages for a specified duration. By configuring the delay queue, the user can simulate scenarios where messages need to be held back before being processed, ensuring proper sequencing and synchronization between different components.

1. Delay queue allows us to delay/postpone the delivery of a new message for a given period of time. We can easily turn any queue into a delay queue by configuring SetQueueAttributes to set the queue’s DelaySeconds attribute.
    - If we create a delay queue, any message that has been sent to this queue will remain invisible to the consumer for the configured delay time.
    - To create a delay queue set DelaySeconds attribute to a value between 0 to 900 seconds.
2. Case Study:
    - Let's try to understand the definition by an example. Consider a case where an application is trying to insert millions of data into a database. The application will send a message about the availability of this new data that has been recently inserted to other subsystems, which in turn process this message and subsequently make updates to the same row. Now there is a dependency that until the first batch completes its job and commits after the updates the next batch should not get triggered. If the message gets triggered before the updates are complete, it would fail the next batch. In this case, delayed delivery helps.
3. Let’s try to make this work in our existing queue by configuring the Delivery Delay. Select any queue (standard or FIFO) from the list and click on Edit to make changes in our queue. We’ll be selecting the Standard queue as an example.

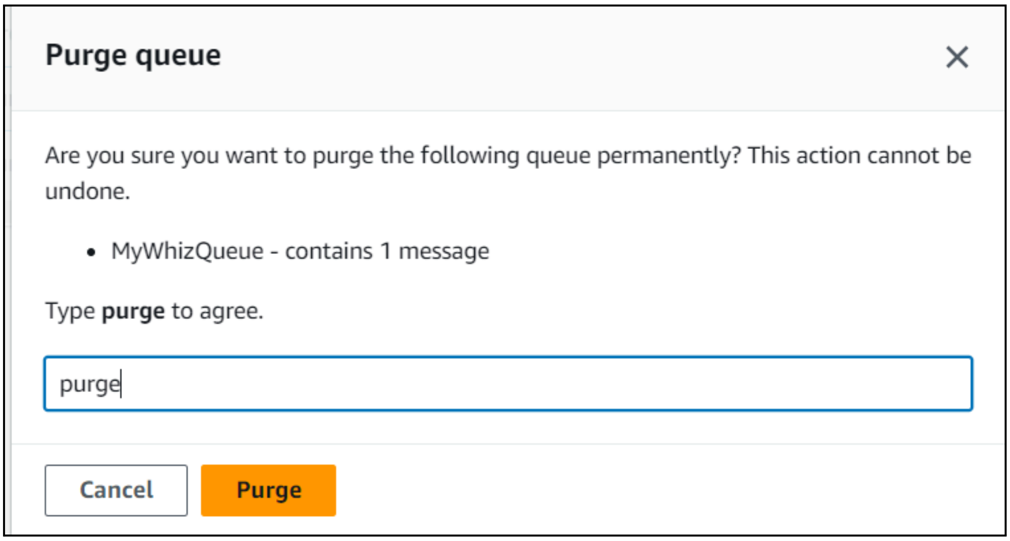
4. Once you have selected the Edit, update the Delivery Delay parameter. It can be any value between 0 seconds to 15 Mins. In our example, we have changed it to 60 secs.
5. Once you have changed click on Save at the bottom to bring changes into effect.

Task 6 : Purge Queue and Verify the same

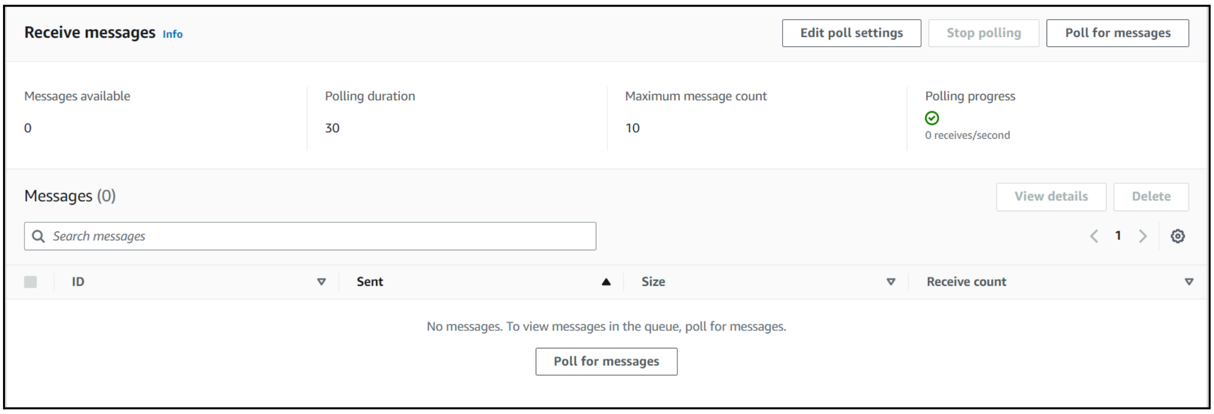
In this task, we are going to demonstrate the ability to purge a queue in Amazon SQS. Purging a queue means deleting all the messages in the queue. This task allows the user to understand the consequences of purging a queue and verify the successful deletion of messages. It also serves as a precautionary measure to ensure the queue is clean before performing further operations.

1. In this topic, we'll try to purge the queue. Before we jump into purging, let's understand first what happens if we purge a queue.
   - Purge Queue option allows us to delete the messages in the queue.
   - The message deletion process can take up to 60 sec, depending upon the size of the queue.
   - Note: Once you call the Purge Queues action, messages cannot be retrieved from the queue.
2. Let's try to make this change to our existing queue. Click on Queue Option and under Actions select Purge.

3. Once you select the purge option it will ask for a confirmation. Enter the phrase purge and click on purge.



4. To verify if purge has occurred on the standard queue, click on send and receive messages and then click on Poll for messages.
5. It will show no messages which verify that the purge is successful.



6. Similarly, repeat the purge on the standard queue and verify the same.

Task 7 : SQS points to remember

In this task, we are going to highlight and summarize important points about Amazon SQS that the user should remember. These points include details about inflight messages, maximum number of messages in a queue, message size limits, retention period, and the pull-based nature of SQS.

1. The basic difference between Delay Queue and Visibility Timeout is Delay Queue hides the message when it is first added in the queue whereas visibility timeout hides a message only after the message is retrieved from the queue.
2. Inflight messages are messages in SQS that have been received by a consumer but not yet deleted.
3. Max 120,000 messages can stay in the queue.
4. Message size is **1024** KiB.
5. It has a Default Retention period.
6. SQS is pull-based not push-based.

Lab : SNS

Task 1: Sign in to AWS Management Console

1. Click on the Open Console button, and you will get redirected to AWS Console in a new browser tab.
2. On the AWS sign-in page,
   - Leave the Account ID as default. Never edit/remove the 12-digit Account ID present in the AWS Console. Otherwise, you cannot proceed with the lab.
   - Now copy your UserName and Password in the Lab Console to the IAM Username and Password in AWS Console and click on the Sign-in button.
3. Once Signed In to the AWS Management Console, make the default AWS Region as US East (N. Virginia) us-east-1.

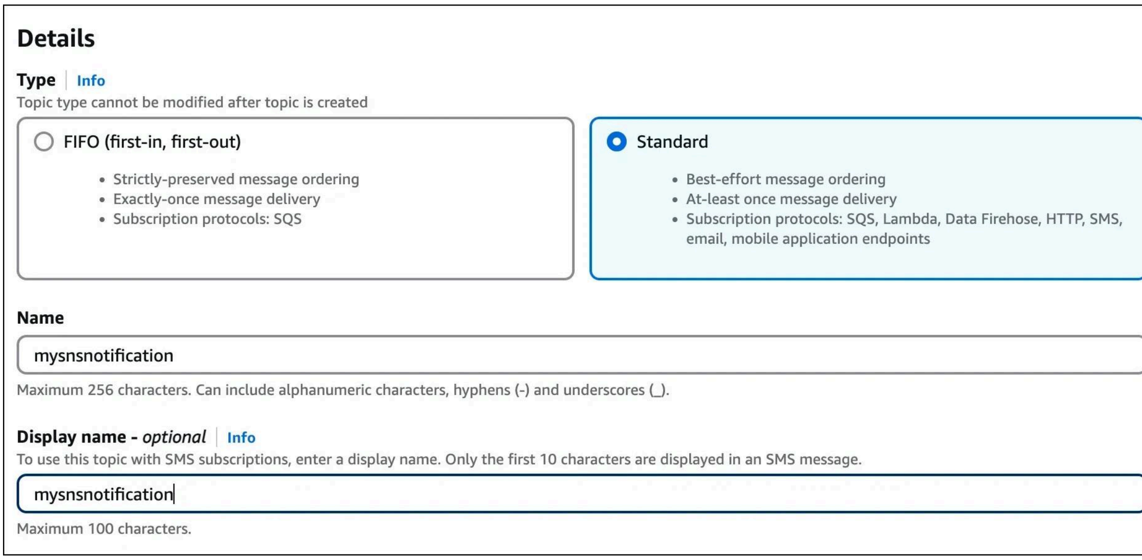
Note : If you face any issues, please go through FAQs and Troubleshooting for Labs.

Task 2: Create SNS Topic

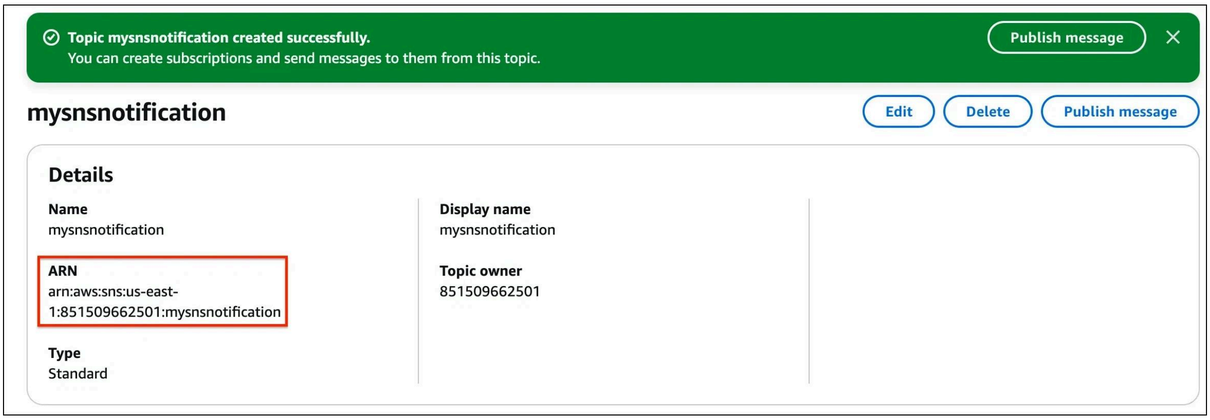
1. Make sure you are in the US East (N. Virginia) us-east-1 Region.
2. Navigate to SNS by clicking on the Services menu available under the Application Integration section.
3. Click on Topics in left panel. Click Create topic

5. Select the Type as Standard.
6. Under Details:
   - Name : Enter *mysnsnotification*

- Display name : Enter *mysnsnotification*



7. Leave other options as default and click on Create topic.
8. An SNS topic is now created.



9. Copy the ARN and save it for later.

Task 3: Subscribe to SNS Topic

In this task, we are going to subscribe an email address to the SNS Topic created in the previous task

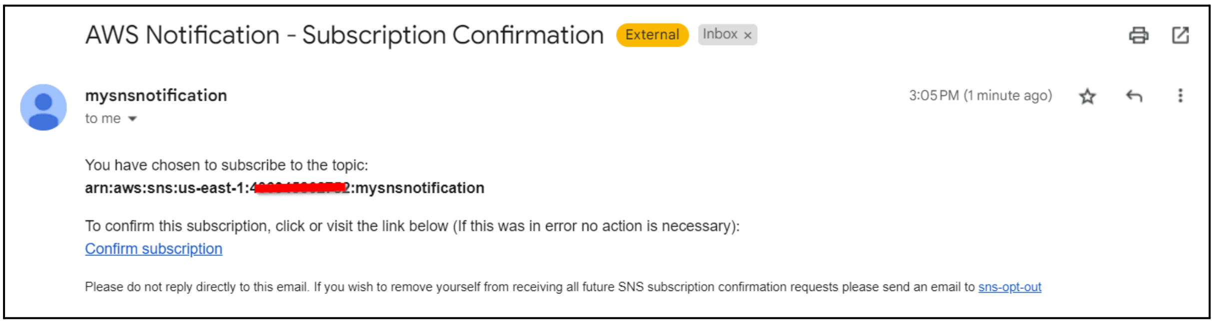
1. Once the SNS topic is created, scroll down below and click on **Create Subscription**

2. Under Details:
    - Protocol : Select Email
    - Endpoint : Enter <your Mail ID>

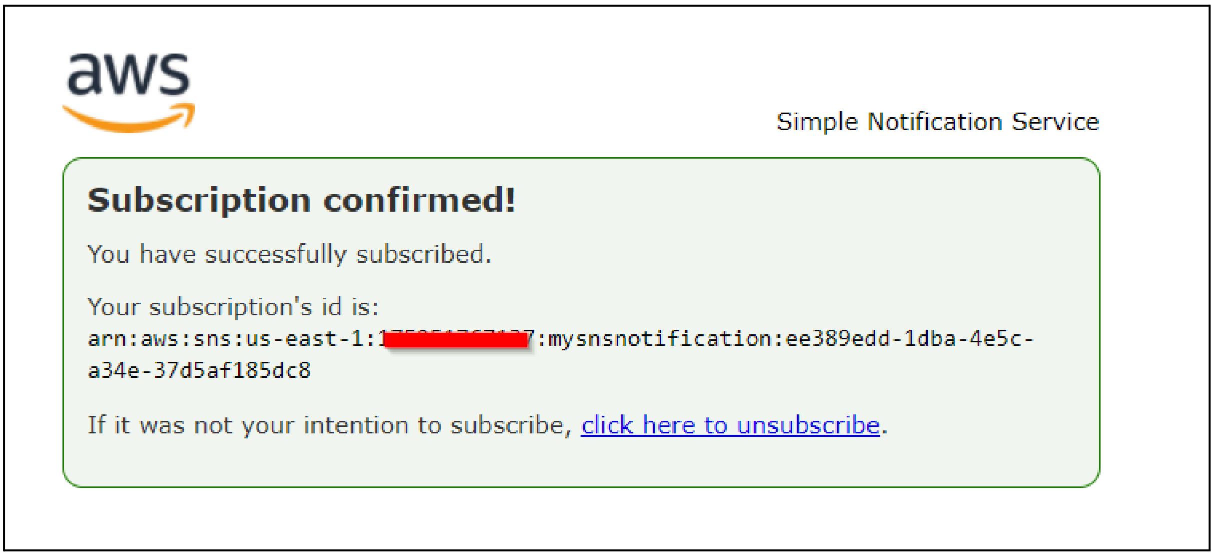
- Leave other settings as default and click on create subscription.

- Note: Make sure you give a valid email address, as you will receive an SNS notification to this email address.
- Check your Spam box if you don't see the email in your Inbox.

3. You will receive an email confirming your subscription to your email.



4. Click on Confirm subscription.



5. Your email address is now subscribed to the SNS Topic mysnsnotification.

Task 4: Create an S3 Bucket

1. Navigate to AWS S3 by clicking on Services in the top left corner. S3 is available under Storage.

2. In the S3 dashboard, click on the Create Bucket button and fill in the bucket details.

3. In the General Configuration:
    - Select Bucket type : General purpose
    - Bucket name: Enter *mys3buckettestingsns-{yourname}*
    - Example : *mys3buckettestingsns-sam*
      (Note: The Bucket Name must be unique across all existing bucket names in Amazon S3)
    - Region: Select US East (N. Virginia) us-east-1 from dropdown.

4. Object ownership: Select ACLs disabled (recommended) option
5. Leave all other settings as default and click on Create bucket.
6. Select the created bucket and click Copy ARN on the top. Save the ARN for later use.

Task 5: Update SNS Topic Access Policy

In this task, we are going to update the Access Policy of the SNS topic to enable it to send notification events based on S3 bucket event

1. Navigate back to the SNS page.
2. Click on Topics.
3. Click on *mysnsnotification.*
4. Click on Edit in the top right corner to edit the Access Policy of the SNS topic.
5. Expand Access Policy.
6. Update the SNS policy as shown below:

- Note: Here we need to update two things after pasting the below policy.
- Remove the old SNS policy and add the new policy to the SNS topic
- SNS Topic ARN in the Resources section below
- S3 bucket ARN in the Condition section below.

```
{

   "Version": "2008-10-17",

   "Id": "__default_policy_ID",

   "Statement": [

       {

           "Sid": "__default_statement_ID",

           "Effect": "Allow",

           "Principal": {

               "AWS": "*"

           },

           "Action": [

               "SNS:GetTopicAttributes",

               "SNS:SetTopicAttributes",

               "SNS:AddPermission",

               "SNS:RemovePermission",

               "SNS:DeleteTopic",

               "SNS:Subscribe",
```

```
                "SNS:ListSubscriptionsByTopic",

                "SNS:Publish",

                "SNS:Receive"

            ],

            "Resource": "<Your_SNS_Topic_ARN>",

            "Condition": {

                "ArnLike": {

                    "aws:SourceArn": "<Your_Bucket_ARN>"

                }

            }

        }

    ]

}
```

7. Note: Make sure to update the bucket ARN and SNS topic in the above policy.
8. Click on Save Changes.
9. Now, your SNS topic has access to send notification events based on S3 bucket events.

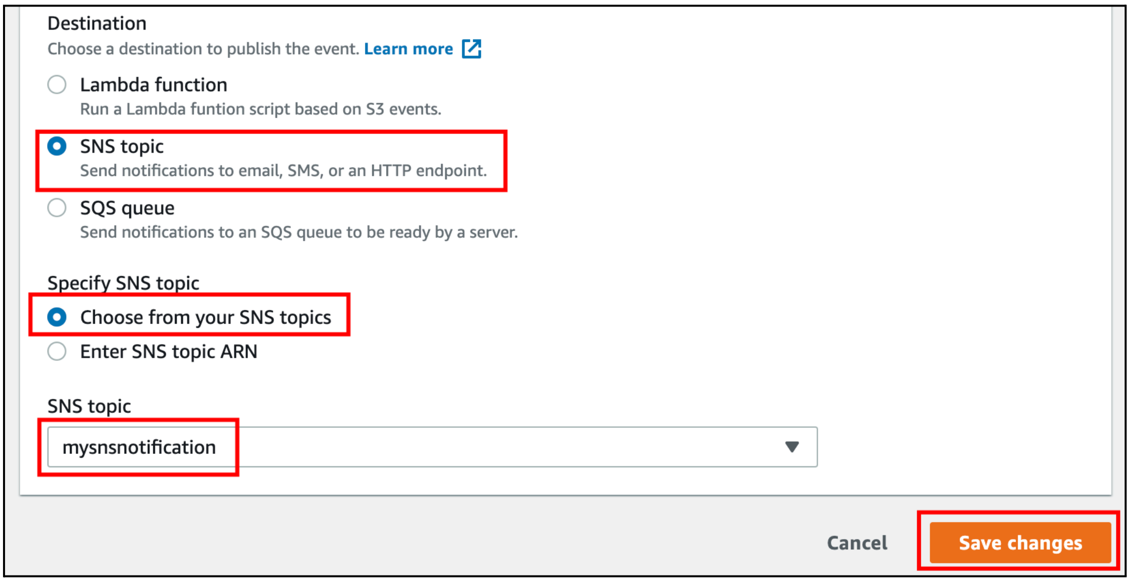
Task 6: Create S3 Event

In this task, we are going to enable event notifications in the S3 bucket that was created in a previous step

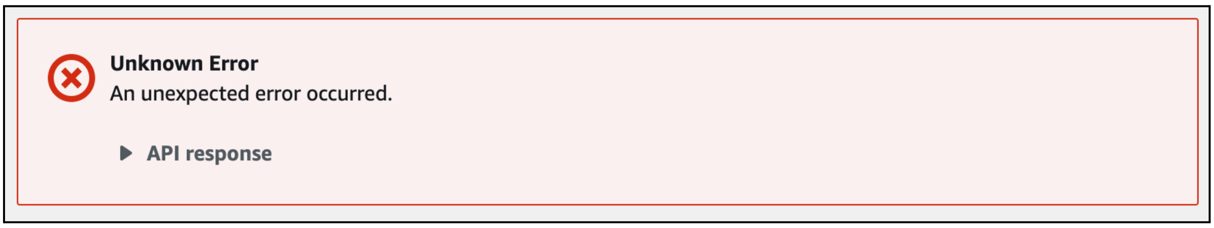
1. Navigate back to the S3 page.
2. Click on the S3 bucket that you have created in the above step.
3. Go to the Properties tab and scroll down to Event notifications.
4. Click on Create event notification button.

- Event name : Enter *myemaileventforput*
- Event types : Check PUT

- Under Destination, select SNS Topic
- Under SNS topic, select topic name created earlier.
- Click on Save changes.



5. Note: If you are getting the below error, make sure to check that you have added PUT as condition above and completed Task 5 i.e. Updating the SNS Access Policy.

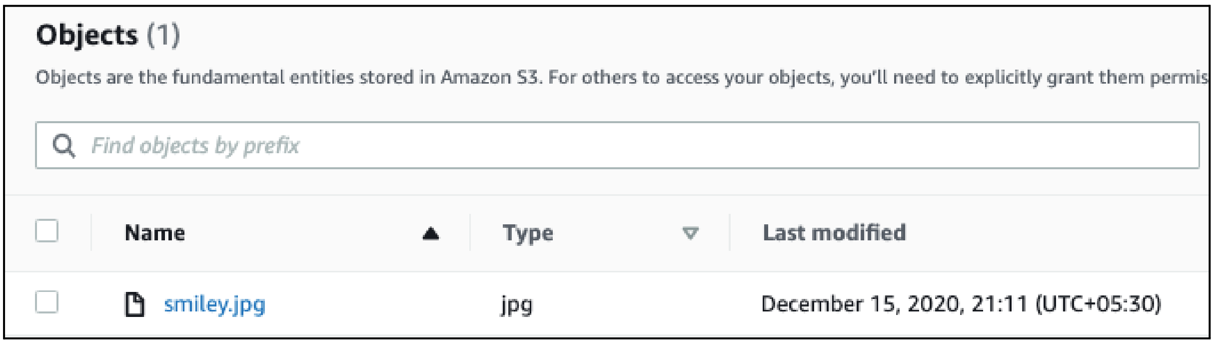


6. Now the S3 bucket has been enabled event notifications for putting new objects through our SNS topic.

Task 7: Testing the SNS Notification

In this task, we are going to test the setup of event notifications and verify that notifications are being sent successfully

1. Open your S3 bucket mys3buckettestingsns.
2. In the bucket, under Objects, click on Upload
3. Now click on Add Files and upload an image from your local system.
4. Once the image is successfully uploaded to the S3 bucket, click on Close. Now you can see the uploaded image under Objects.



5. You have successfully received an SNS notification based on the PUT object event in S3 bucket.

6. Go to your mailbox. You should have received an email from SNS.

Lab : Cloudfront

Task 1: Sign in to AWS Management Console

1. Click on the Open Console button, and you will get redirected to AWS Console in a new browser tab.
2. On the AWS sign-in page,
   - Leave the Account ID as default. Never edit/remove the 12 digit Account ID present in the AWS Console. otherwise, you cannot proceed with the lab.
   - Now copy your User Name and Password in the Lab Console to the IAM Username and Password in AWS Console and click on the Sign in button.
3. Once Signed In to the AWS Management Console, Make the default AWS Region as US East (N. Virginia) us-east-1.

Task 2: Create S3 Bucket

In this task, we are going to create a new S3 bucket in the US East (N. Virginia) region with a unique name enabling ACLs, and allowing public access.

1. Make sure you are in the US East (N. Virginia) us-east-1 Region.
2. Navigate to the Services menu at the top. Click on S3 in the Storage section.
3. In the S3 dashboard, click on the Create bucket and fill in the bucket details.
   - Bucket name: Enter *whizlabs1234567*
     - Note: S3 Bucket names are globally unique, choose a name that is available.
   - Object Ownership: Select ACLs enabled option and choose Object writer as Object owner

   - Scroll down to Block Public Access settings for this bucket and Uncheck the Block all Public Access and acknowledge the change.

- No need to change anything further, just click on the Create bucket button.

Task 3: Upload a file to an S3 bucket

1. Click on the bucket name you just created and you can see that there are no objects created in the bucket.
2. You can upload any image from your local machine or you can download our test image from Download me
3. To upload a file to our S3 bucket,
    - Click on the Upload button.

    - Click on Add files.
    - Browse for image we provided and select it. {**NOTE** : use the same image provided to ignore the validation failure}
    - Click on the Upload button.
    - You can watch the progress of the upload from within the transfer panel at the bottom of the screen.
    - Once your file has been uploaded, click on Close and you can see an object in the bucket.

Task 4: Creating Custom Error Pages

In this task, we will learn how to create customized error pages for CloudFront. These pages will be displayed in the event that an origin returns an HTTP 4xx or 5xx error. To do this, we must ensure that the error pages are stored in a location that CloudFront can access. In this case, we will use the same S3 bucket that we created previously.

1. To set up a custom error page, access the S3 bucket by clicking on it.
2. Click on Create Folder button to create a folder.

3. Enter the folder name as **CustomErrors**. For Server-side encryption Select Specify an encryption key keep rest things as default.

4. Click on the Create folder button.
5. Click on the new CustomErrors folder.
6. We will create an error.html file:
    - Create an error.html file in your local system using Notepad.
    - This custom HTML page will be used for showing errors in CloudFront.
    - Sample error.html content:

```
<html><h1>This is Error Page</h1></html>
```

7. Use the Upload button to upload the error.html file in the folder.
8. We will create a block.html file:
    - Create a block.html file in your local using Notepad.
    - This custom HTML page will be used for showing geo-restrictions of your content in CloudFront.
    - Sample block.html content:

```
<html><h1>This content is blocked in your location!!!</h1></html>
```

9. Use the Upload button to upload the block.html file in the folder.

Task 5: Making the objects public

1. Navigate back to the bucket and click on the image name. You can see the image details like Owner, size, link, etc.
2. Copy the Object URL and paste it into a new tab.
3. Example: https://whizlabs1234567.s3.amazonaws.com/whizlabs_logo_58_32.png
    - You will see the AccessDenied message, meaning the object is not publicly accessible.**whizlabs_logo_58_32.png**

4. Go back to the Bucket and click the Permissions tab.

5. Scroll down to the Bucket Policy and click on Edit button.

6. Copy and paste the below policy and save the policy.

- Note: Change the name of the bucket ARN with your bucket ARN in both the Resource option in the code.

```
{
   "Version": "2012-10-17",
   "Statement": [
       {
           "Effect": "Allow",
           "Action": ["s3:ListBucket"],
           "Principal": {"AWS": "*"},
           "Resource": "<YOUR_BUCKET_ARN>"
       },
       {
           "Effect": "Allow",
           "Action": ["s3:GetObject", "s3:PutObject"],
           "Principal": {"AWS": "*"},
           "Resource": "<YOUR_BUCKET_ARN>/*"
       }
   ]
}
```

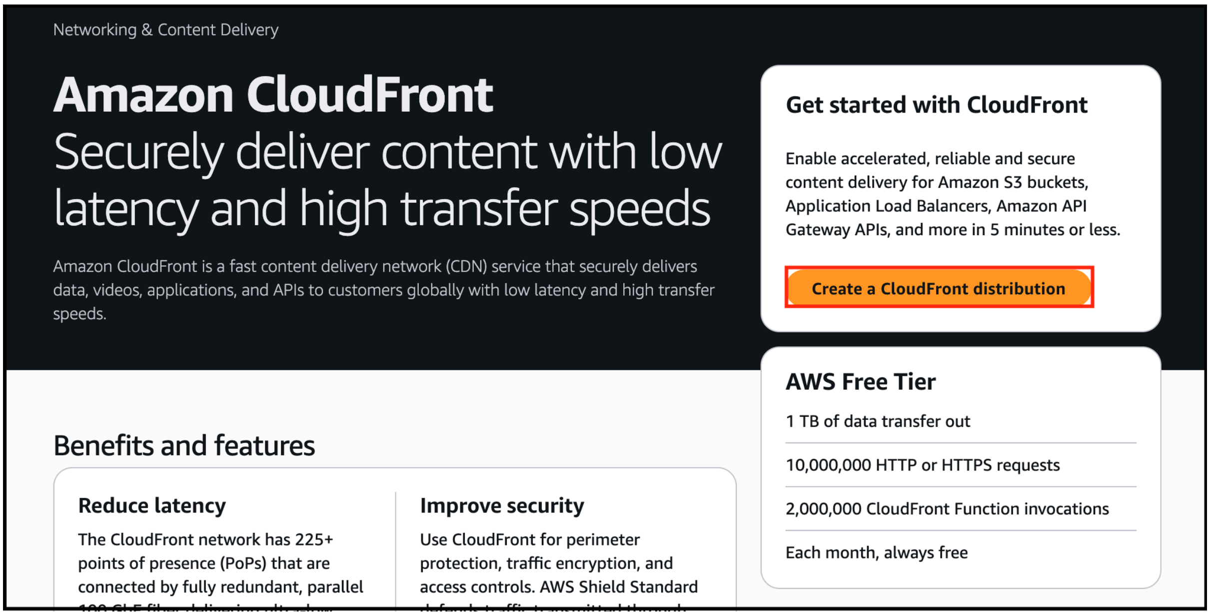
7. Open the Image Object URL again or refresh the one already open.

8. If you can see your uploaded image in the browser, it means your image is publicly accessible. If not, check your bucket policy again.

Task 6: Creating a CloudFront Distribution

1. Navigate to CloudFront by clicking on the Services menu at the top, then click on CloudFront in the Network and Content Delivery section.

2. Click on Create a CloudFront distribution button.



3. Choose a plan: Scroll down and select **Pay as you go** plan and click **Next**.

4. **Distribution name**: Enter the distribution name of your choice. Keep rest as default and click on **Next**.

5. Now Configure distribution as follows: **Specify Origin**: Select **Amazon S3** as Origin type, then click on **Browse S3** and select the bucket you have created. Leave everything as default and click **Next.**

6. In Enable security, choose Do not enable security protections under Web Application Firewall(WAF) and click **Next**.

7. In **Review and Create**, leave everything as default and click on the Create distribution button.

8. You can see that the CloudFront distribution is enabled successfully. Note: This process will take around 5-10 minutes.

9. The domain name that Amazon CloudFront assigns to your distribution appears in the list of distributions.

Task 7: Accessing Image through CloudFront

Amazon CloudFront is now pointed to Amazon S3 bucket origin and you know that the domain name is associated with the distribution. You can create a link to the image in the Amazon S3 bucket with that domain name.

1. For testing your distribution, copy your domain name and append your image name after the domain name.

- Example: https://d1jptzlydefk0d.cloudfront.net/whizlabs_logo_58_32.png

2. Open the CloudFront URL in a new tab. You can see your uploaded image.
3. You can see how much faster the CloudFront URL image loads as compared to the S3 URL. When end users request an object using a CloudFront domain name, they are automatically routed to the nearest edge location for high-performance delivery of your content.

Task 8 : Configuring Custom Error Page

1. Navigate back to CloudFront Dashboard and select the distribution created.
2. Select the Error pages tab.

   - Click on the Create custom error response button.

   - Now we need to set up our custom error page:
     - HTTP Error Code: Select 404: Not Found
     - Error Caching Minimum TTL: Enter *10*
     - Customize Error Response: Select Yes
     - Response Page Path: Enter */CustomErrors/error.html*
     - HTTP Response Code: Select 404: Not Found
     - Click on Create custom error response button.

3. Navigate back to Distributions and wait for your distribution to complete state to change Deploy.

   - Note: This process will take around **5-10 minutes.**
   - Once the state has changed to **Deploy**, you can verify the deployment by checking the Last modified column, it should display the most recent timestamp, indicating the distribution has been updated. After this, proceed to test the error page.
   - For testing your distribution, copy your domain name and append the name of an image that does not exist in your S3 bucket
   - Open the **CloudFront URL** in a new tab.

4. If you can see your HTML error page in the browser, it means you successfully set up your custom error page.

Task 9 : Restricting the Geographic Distribution of Your Content

If you need to prevent users in selected countries from accessing your content, you can specify either a whitelist (countries where they can access your content) or a blacklist (countries where they cannot) by using restrictions.

1. On the distribution settings page, select Security tab and expand **CloudFront geographic restrictions** click on Edit link near Countries.
    - Restriction Type: Select Block list
    - Select the country where you are currently and click on it to check this option.
    - Click on Save changes button.

2. Go to the distribution list and wait for your distribution to complete the state changed to deployed.
    - Once the state has been changed to deployed, we will test the restriction through CloudFront in the browser.
        - **Example: https://d1jptzlydefk0d.cloudfront.net/whizlabs_logo_58_32.png**
    - You can see the following error message:
        - 403: Error The Amazon CloudFront distribution is configured to block access from your country.

3. Let us configure a custom error page:

- Navigate back to CloudFront Dashboard and select the distribution you have created.

- On the settings page, select Error pages tab.
- Click on the Create custom error response button.

- On the settings page, select Error pages tab
- Now we need to set up our custom error page:
    - Http Error Code: Select 403: Forbidden
    - Error Caching Minimum TTL: Enter *10*
    - Customize Error Response: Select Yes
    - Response Page Path: Enter */CustomErrors/block.html*
    - HTTP Response Code: Select 403: Forbidden
    - Click on Create custom error response button.

4. Navigate back to Distributions and wait for your distribution to complete state to change Deploy.
5. Note: This process will take around 5-10 minutes.
6. Once the state has been changed to Deploy, we will test the restriction through CloudFront in the browser.
    - **Example:** *https://d1jptzlydefk0d.cloudfront.net/whizlabs_logo_58_32.png*

7. If you see the error, this means you successfully configured a custom error page and restricted image access from your country.